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1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.

2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.

3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

J40S

MLB-S

REV

ECN

DESCRIPTION OF REVISION

CK APPD
DATE

7

0001103802

ENGINEERING RELEASED

2011-04-08

Fri Apr 8 14:44:38 2011

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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-8834	1	SCHEM, MLB-S, J40S	SCH		
820-3017	1	PCBF, MLB-S, J40S	MLB		
639-1657		PCBA, MLB-S, J40S			
085-2473	1	DEV LIST, MLB-S, J40S	DEV1		DEVELOPMENT_LIST

DRAWING
TITLE=SHORTSTACK
ABBREV=DRAWING

SCH, MLB-S, J40S

Apple Inc.

051-8834

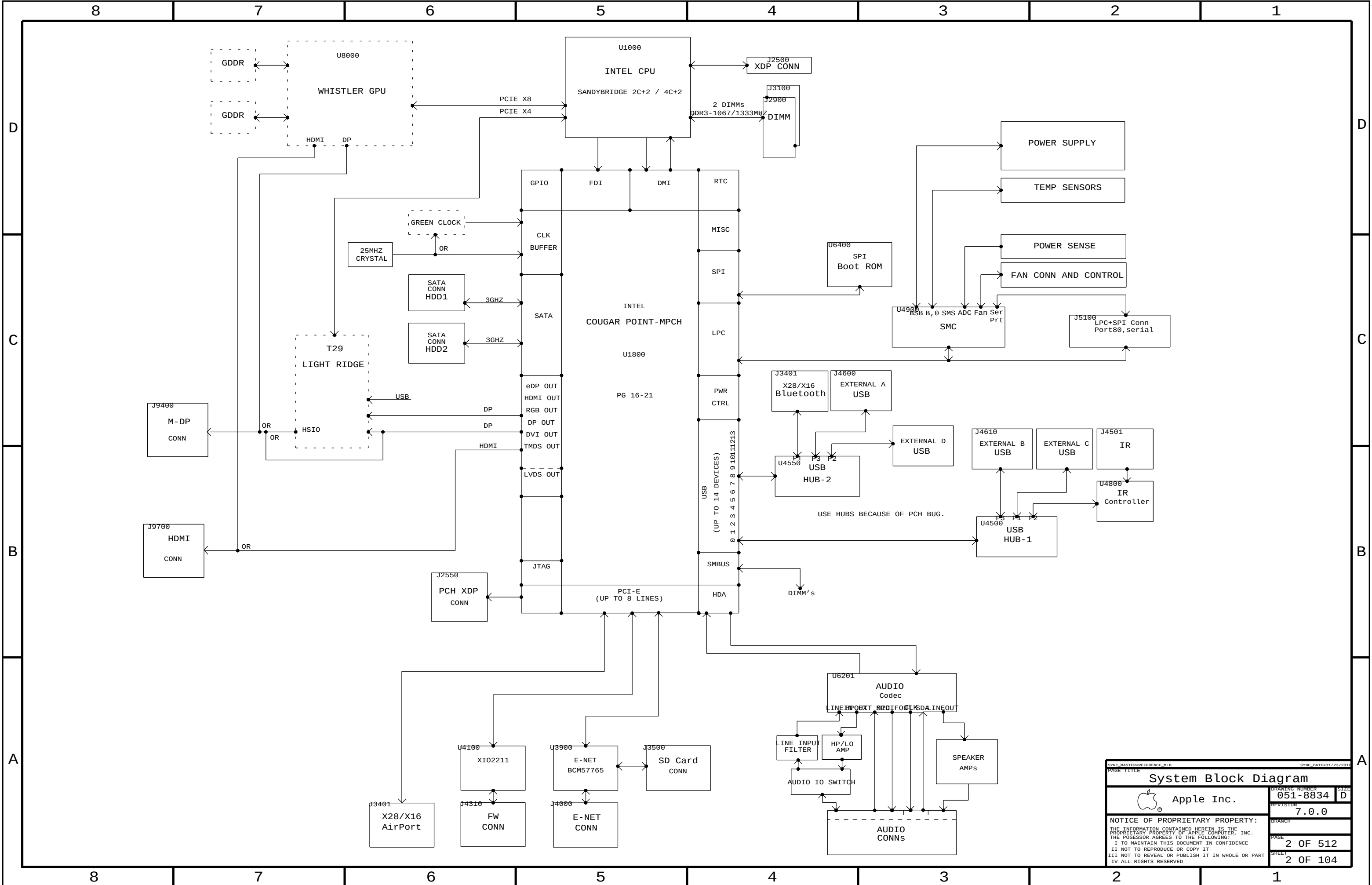
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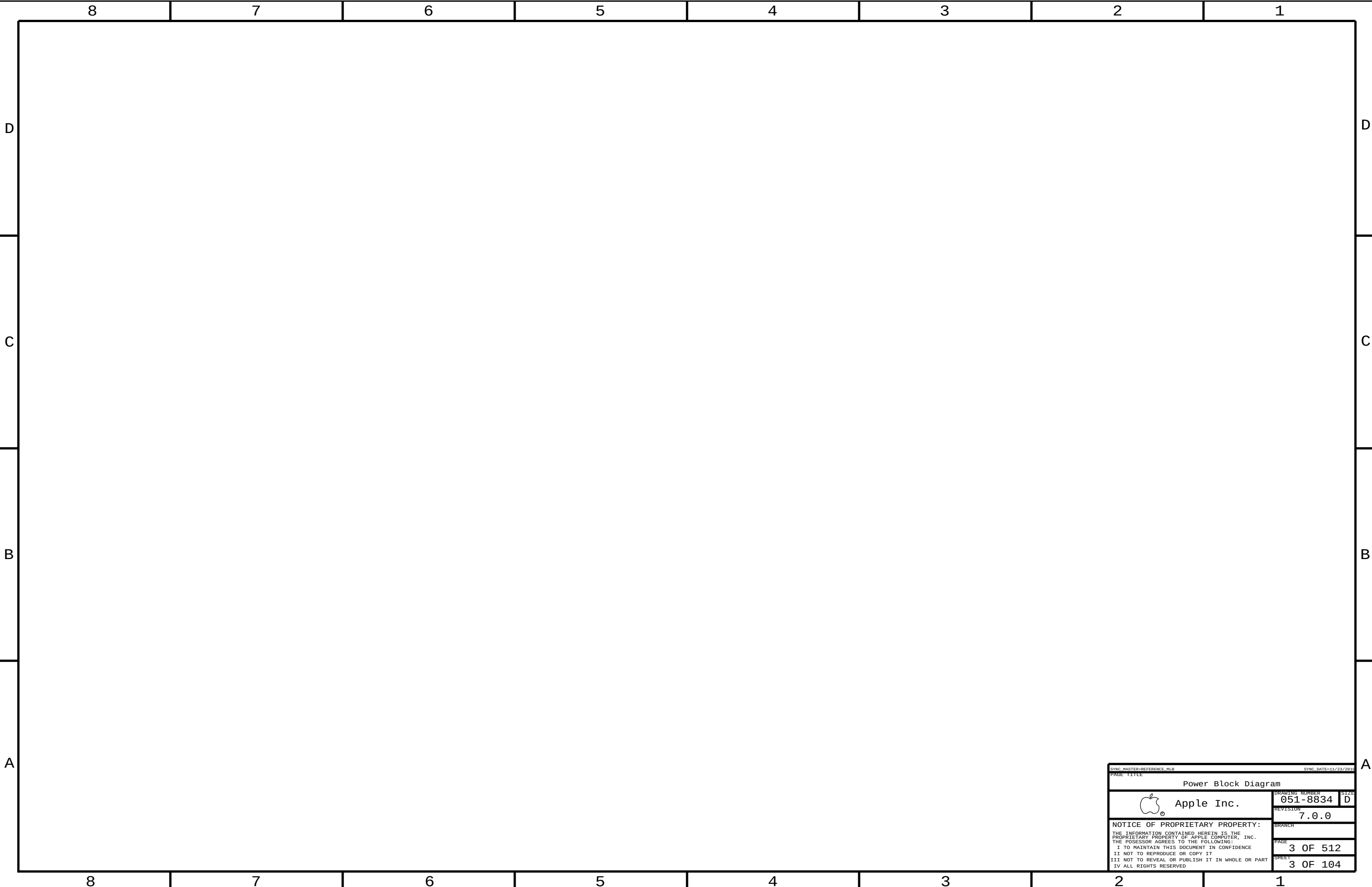
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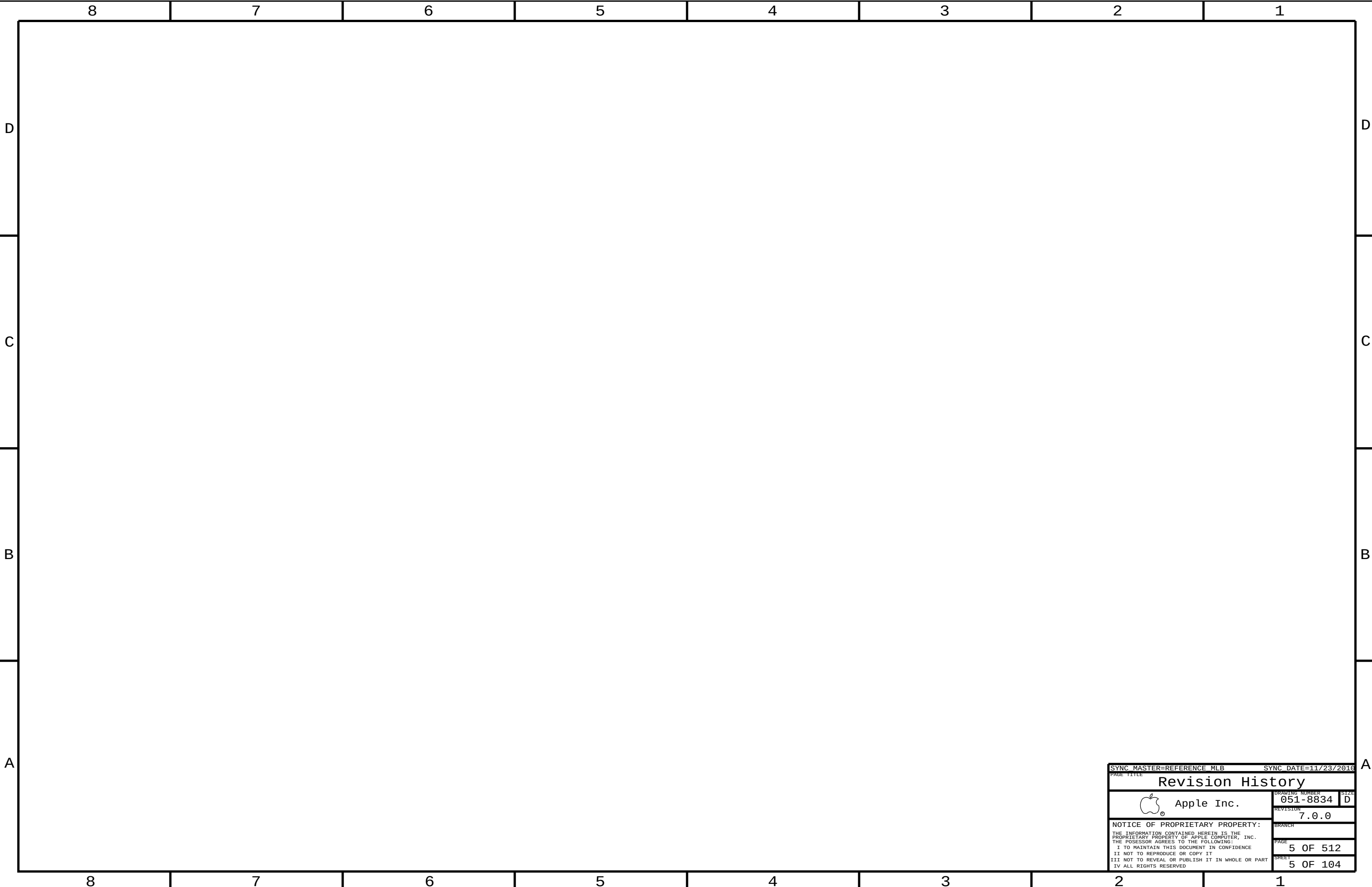
System Block Diagram		
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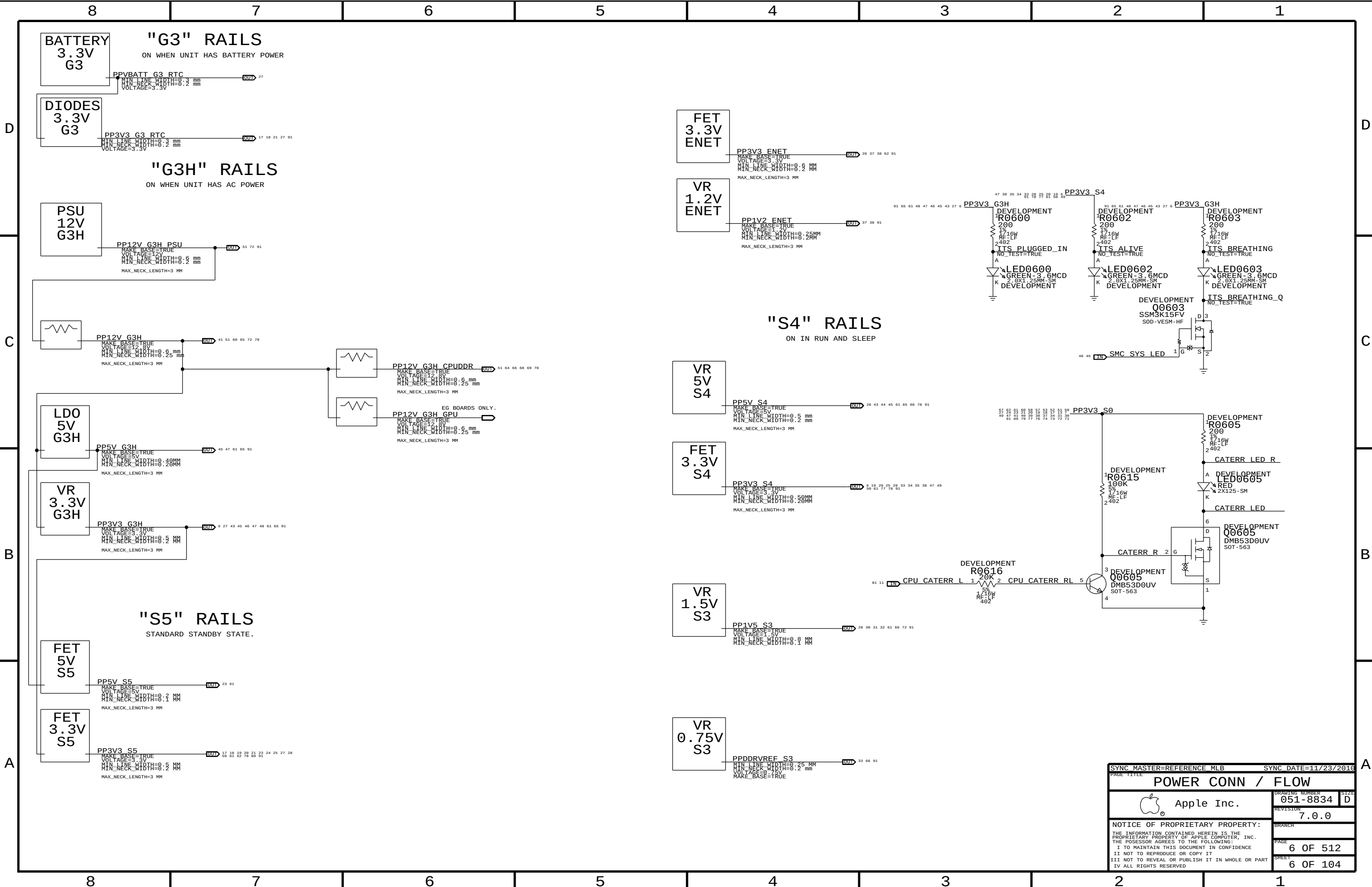
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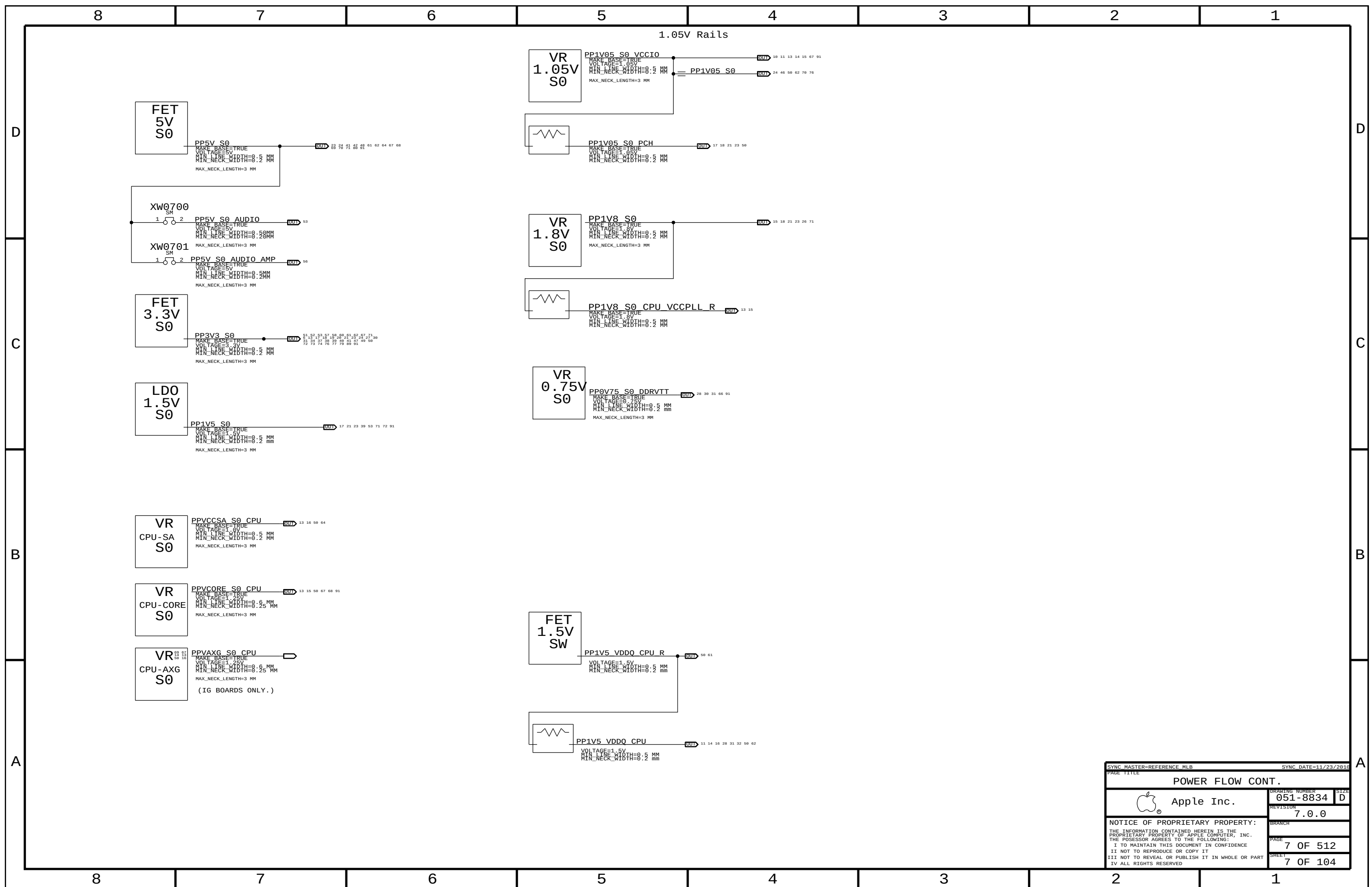
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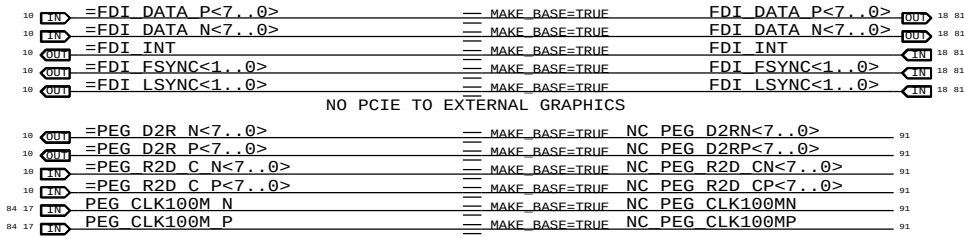


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		7.0.0	
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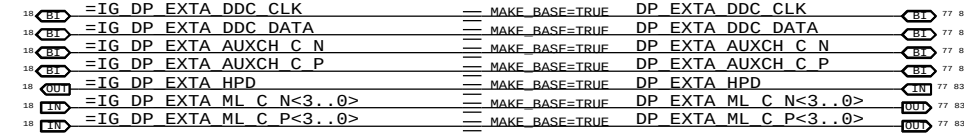


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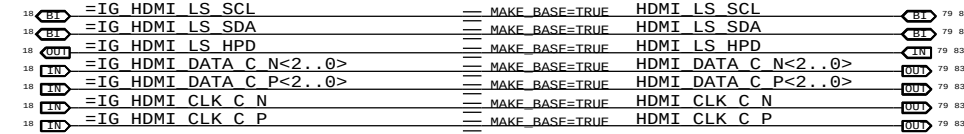
FDI BUS.



EXTERNAL DISPLAY PORT

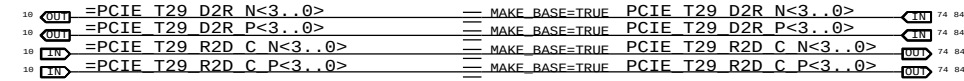


HDMI

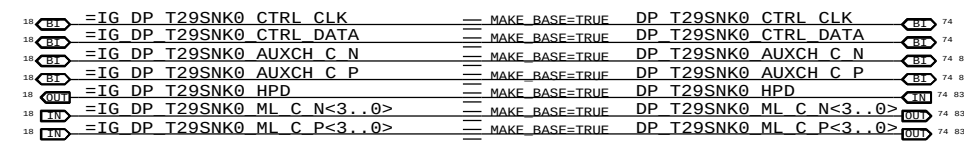


T29

PCIE



T29 DISPLAY PORT INPUT

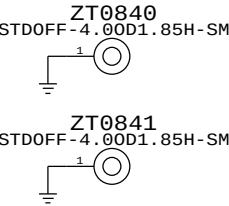


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
338S0945	1	IC, T29, SLHAJ, FCBGA, 15X15MM	U9000	CRITICAL	
335S0550	1	IC, EEPROM, SERIAL, SPI, 4KX8, 1.8V, MLP8, LF	U9090	CRITICAL	T29ROM:BLANK
341T0342	1	IC, CONF ROM, T29, J40I, S	U9090	CRITICAL	T29ROM:PROG_IS
341T0380	1	IC, CONF ROM, T29, J40G	U9090	CRITICAL	T29ROM:PROG_G
337S3997	1	IC, MCU, 32B, LPC1112A, 16KB/2KB, HVQFN25	U9330	CRITICAL	T29MCU:BLANK
341T0339	1	IC, PROGRMD, T29 PORT MCU, HVQFN25	U9330	CRITICAL	T29MCU:PROG

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S3312	1	IC, CBT104DP081A, 2:1 DP MUX, QFN	U9390	CRITICAL	T29DPMUX:NXP
353S3312	1	IC, CBT104DP081A, 2:1 DP MUX, QFN	U9390	CRITICAL	T29DPMUX:BOTH
353S3055	1	IC, PI3VEDP212, 2:1 DP MUX, QFN	U9390	CRITICAL	T29DPMUX:PERICOM

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S3055	353S3312	T29DPMUX:BOTH	?	DP MUX FROM PERICOM

USED FOR T29 IC



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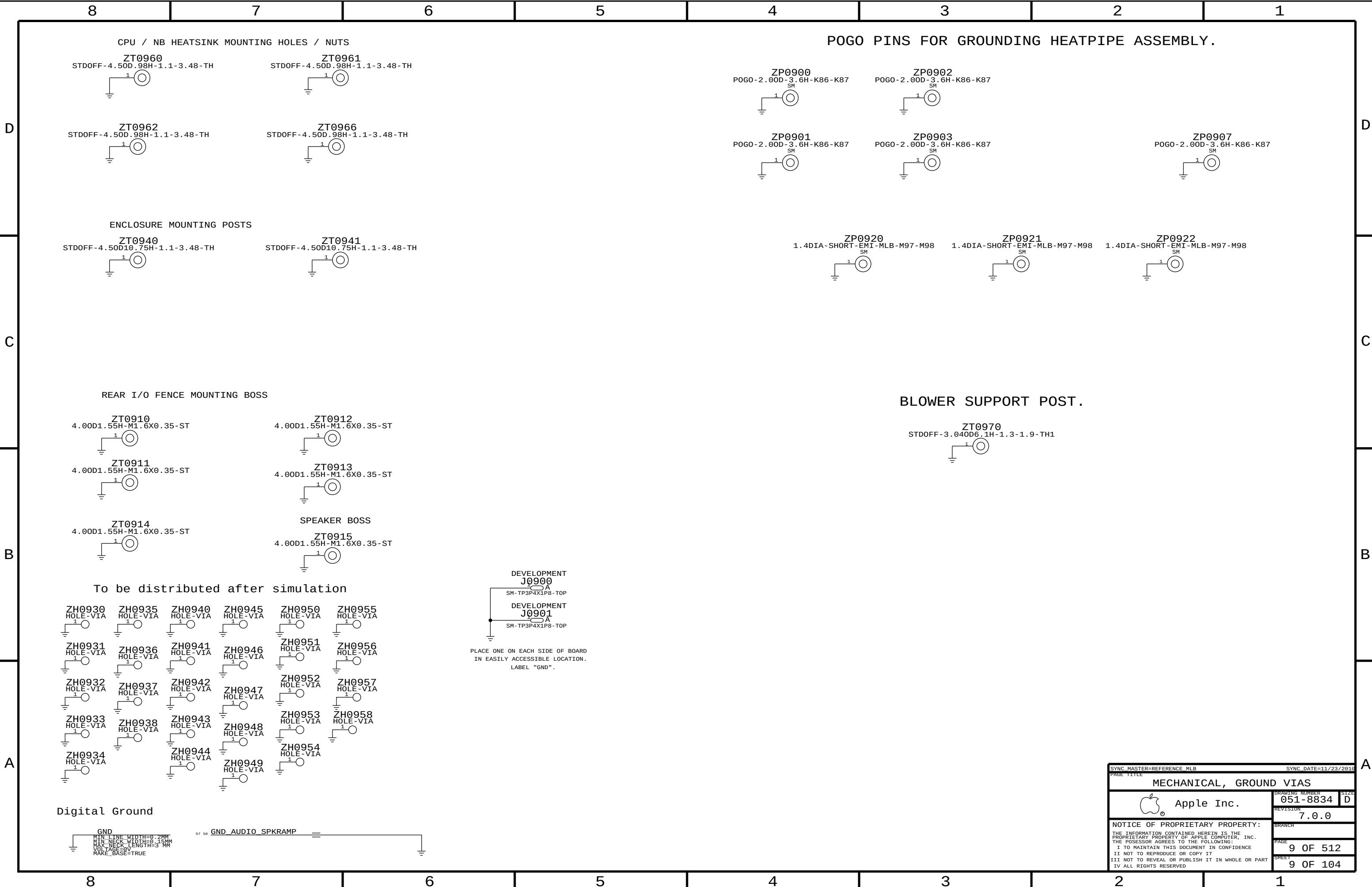
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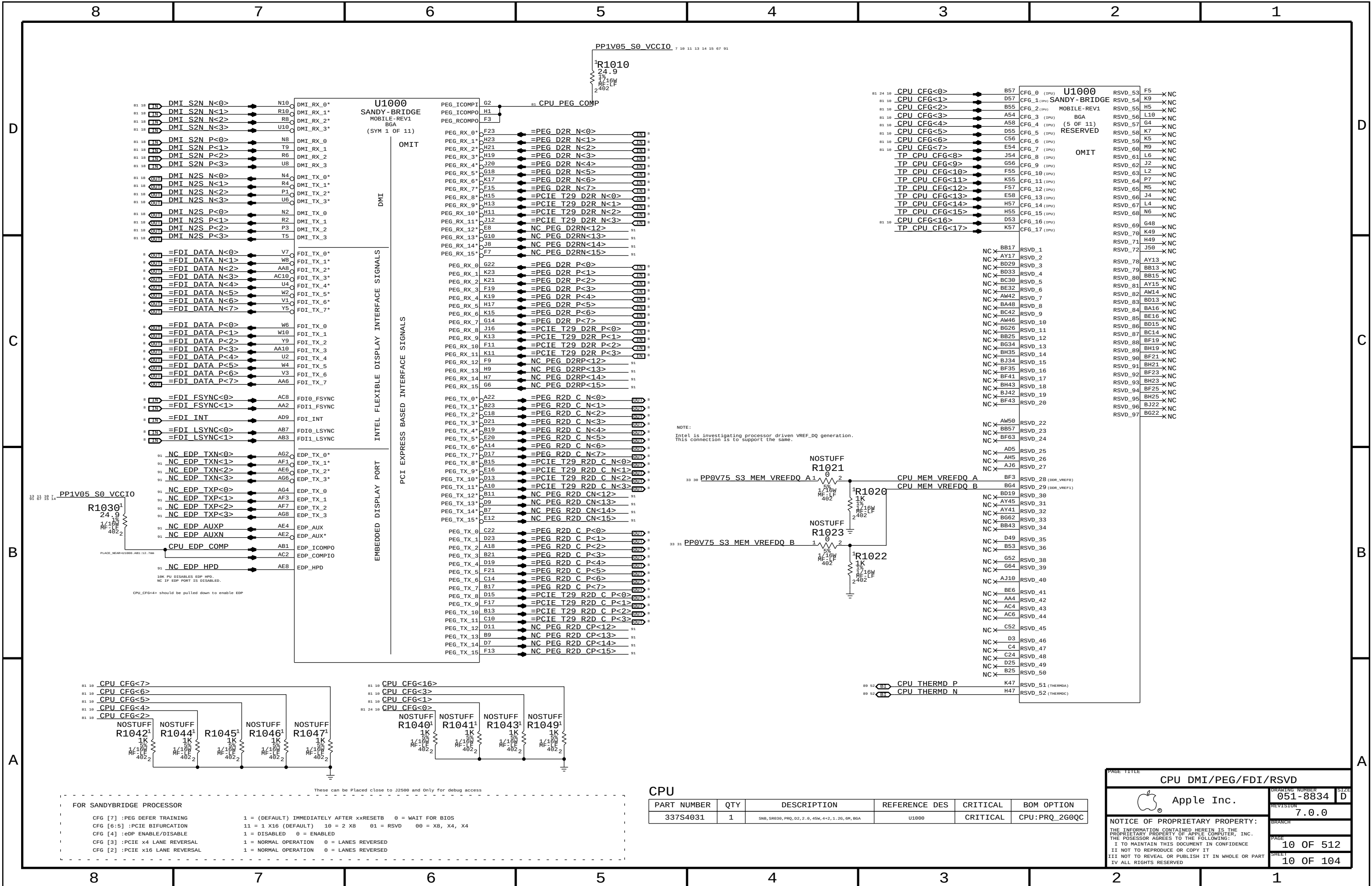
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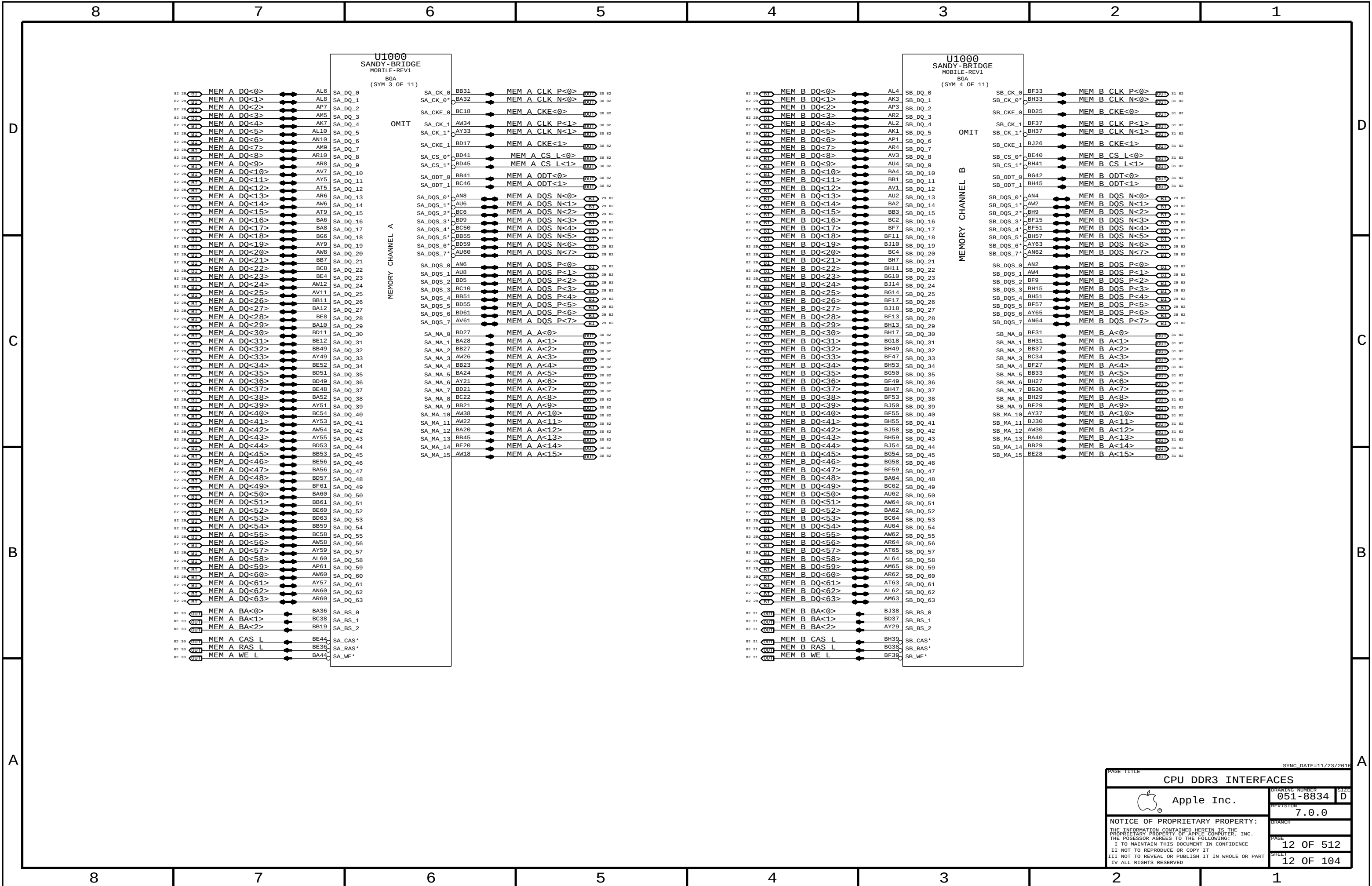
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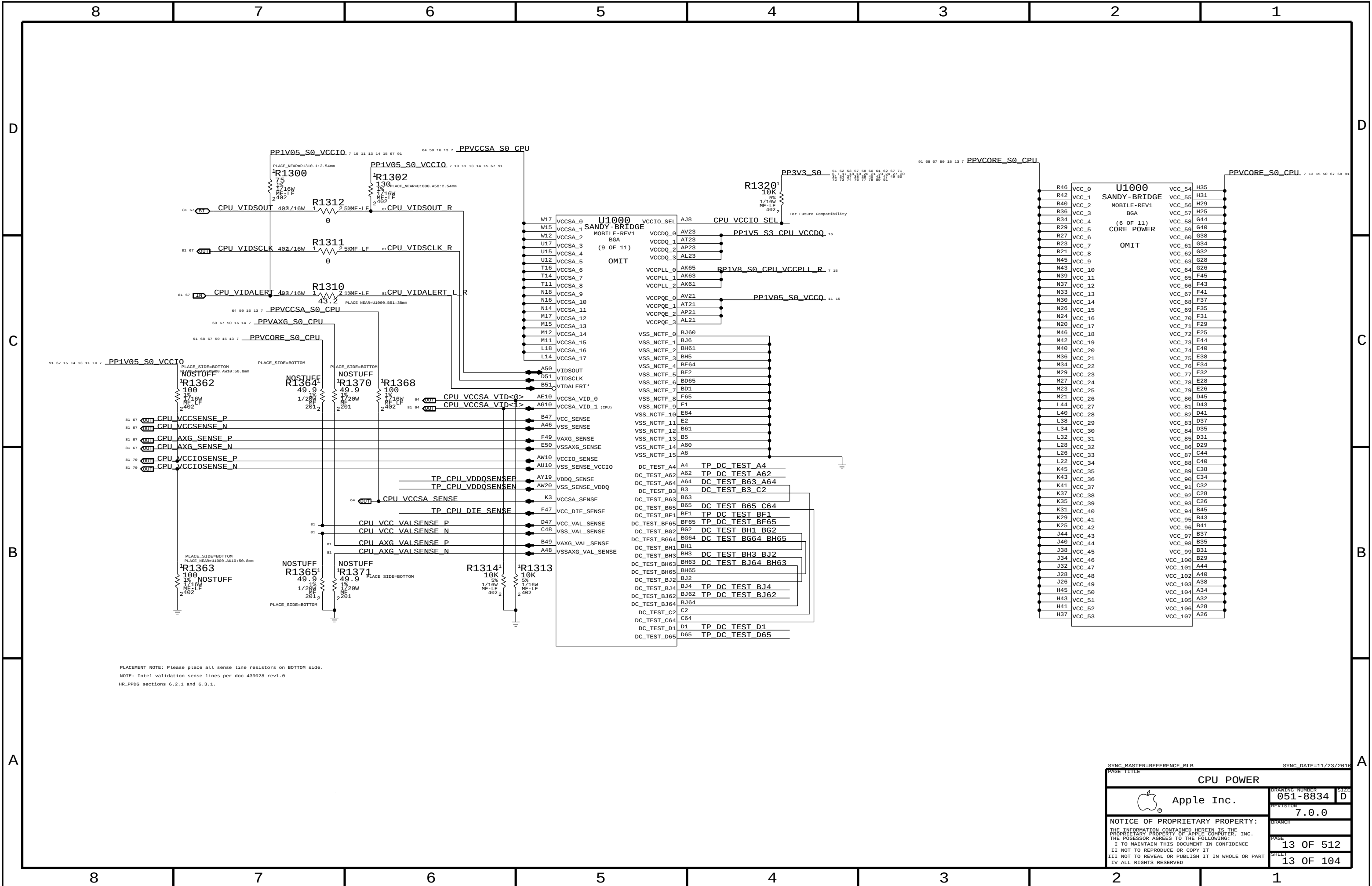


CPU

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S4031	1	SNB, SRB30, PRQ, D2, 2.0, 45W, 4+2, 1.2G, 6M, BGA	U1000	CRITICAL	CPU:PRQ_2G0QC

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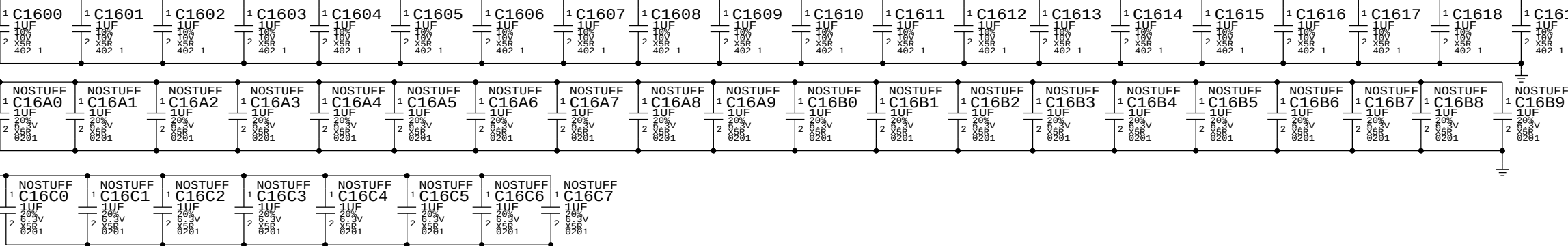


CPU Vcore DECOUPLING

Intel recommendation: 4x 470uF 4mOhm, 2x 470uF 4mOhm (NOSTUFF), 16x 22uF 0603, 4x 10uF 0603, 20x 1uF 0402, 28x 1uF 0402 (NOSTUFF)
Apple Implementation: 4x 470uF 4mOhm, 1x 470uF 4mOhm (NOSTUFF), 16x 22uF 0603, 4x 10uF 0402, 20x 1uF 0402, 28x 1uF 0201 (NOSTUFF), 4x 22uF 0603 (NOSTUFF)

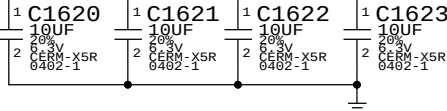
PLACEMENT_NOTE (C1600-C16C7):

Place on bottom side of U1000



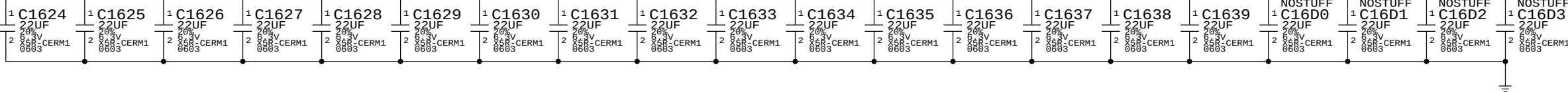
PLACEMENT_NOTE (C1620-C1623):

Place near U1000 on bottom side



PLACEMENT_NOTE (C1624-C16D5):

Place near inductors on bottom side.

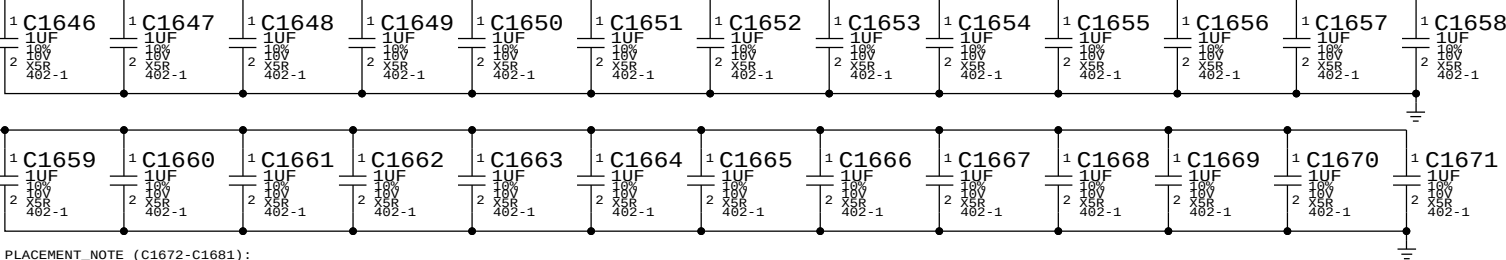


CPU VCCIO/VCCPQ DECOUPLING

Intel recommendation: 2x 330uF, 10x 10uF 0603, 26x 1uF 0402
Apple Implementation: 2x 330uF, 10x 10uF 0603, 26x 1uF 0402

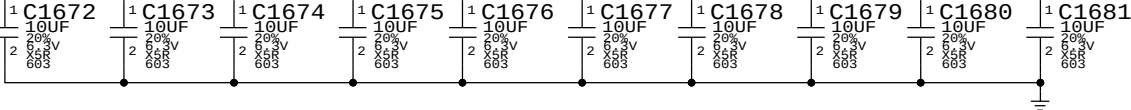
PLACEMENT_NOTE (C1646-C1671):

Place on bottom side of U1000

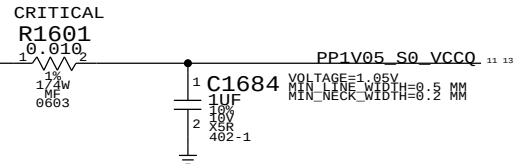


PLACEMENT_NOTE (C1672-C1681):

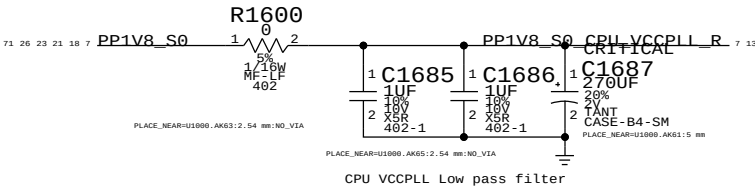
Place near U1000 on bottom side



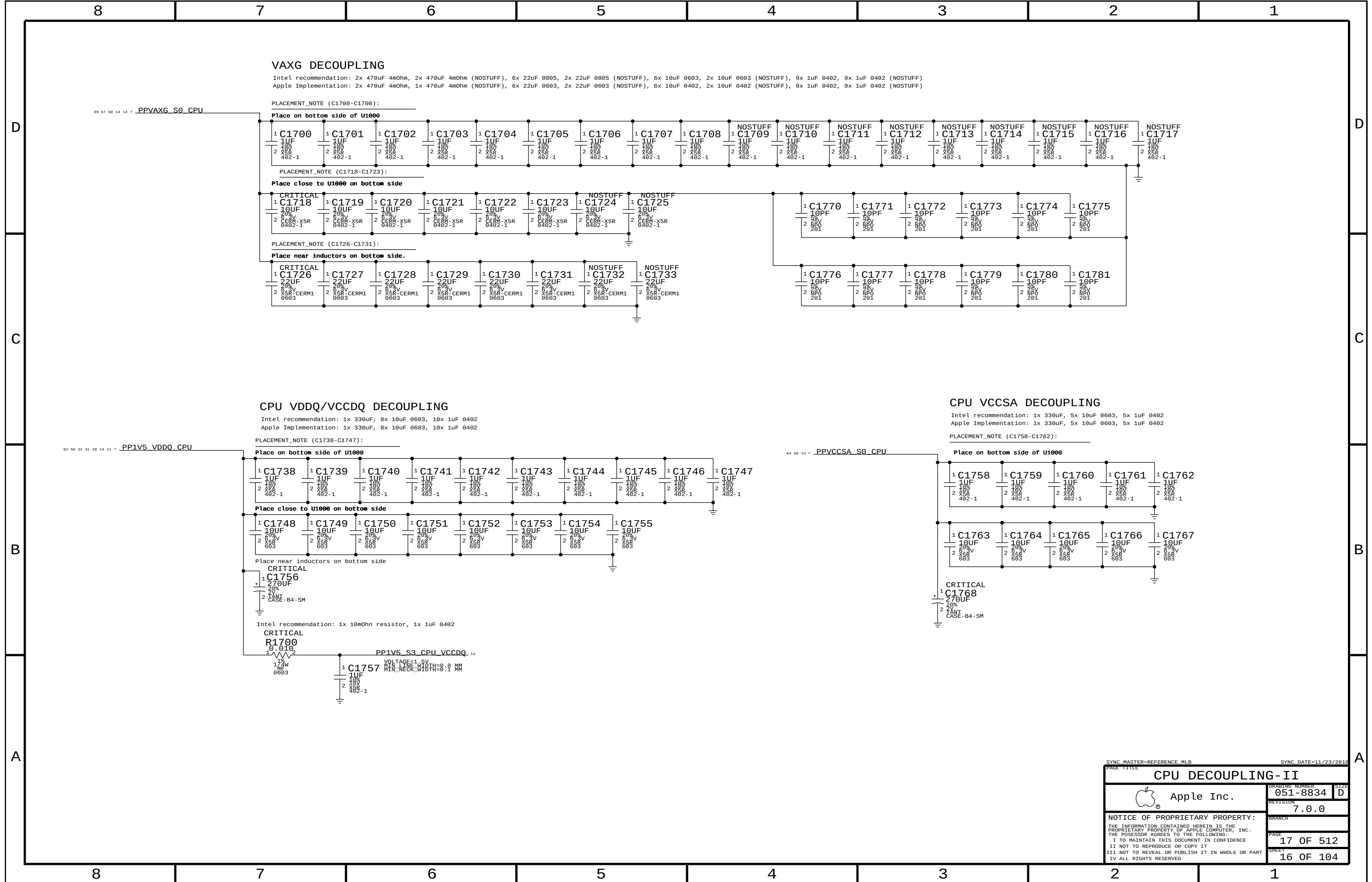
Intel recommendation: 1x 10mOhm resistor, 1x 1uF 0402



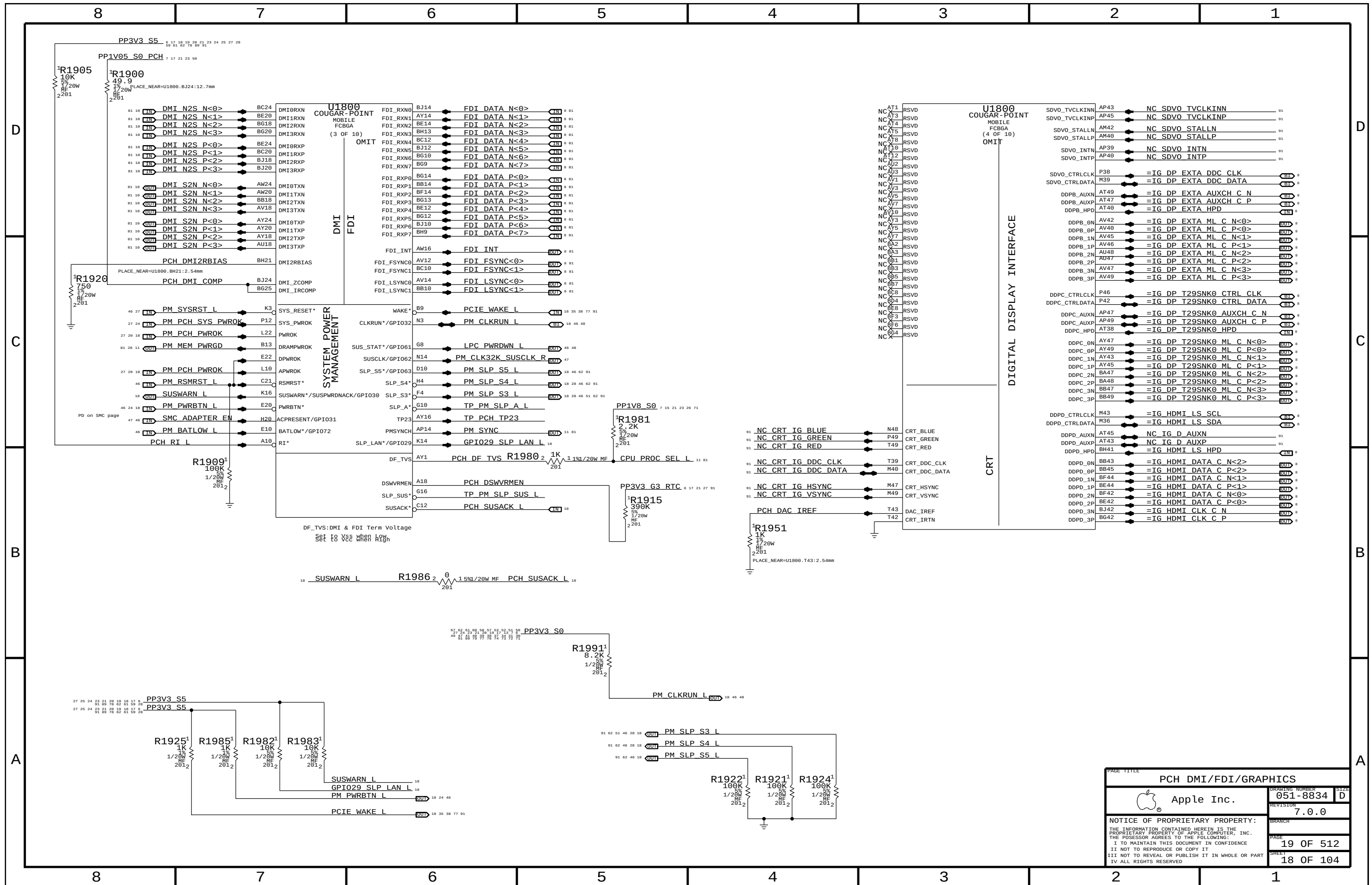
CPU VCCPLL DECOUPLING

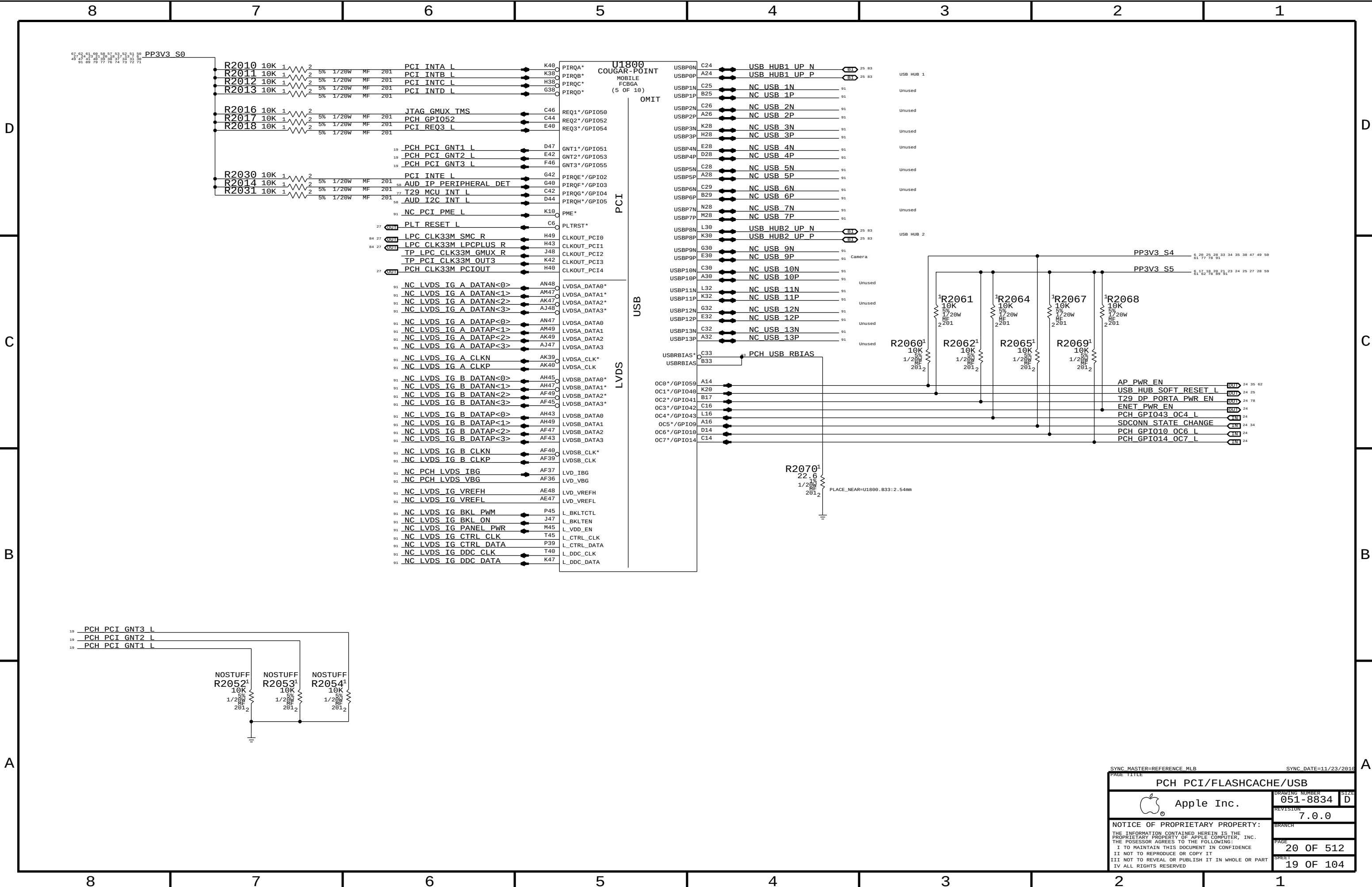


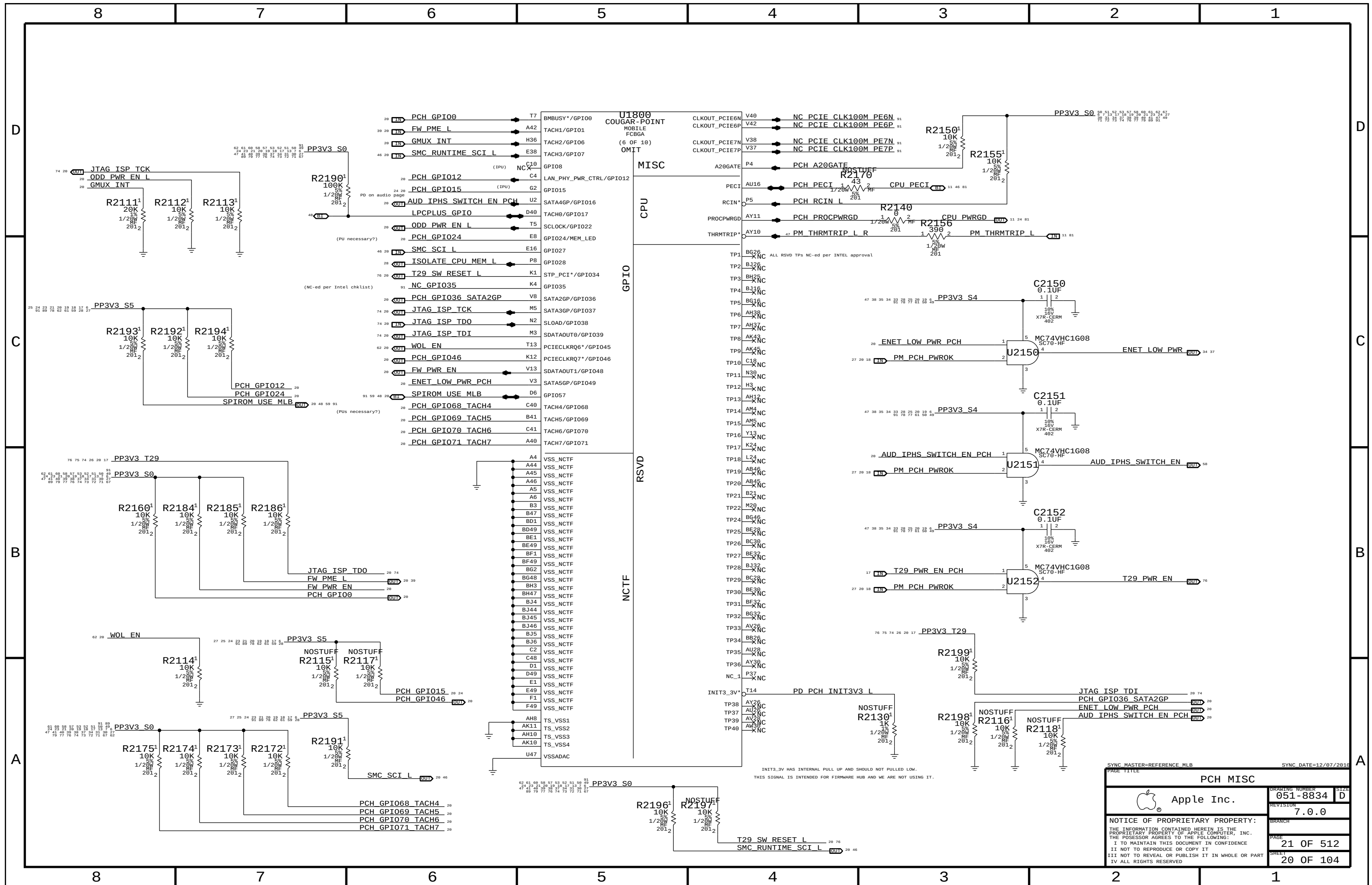
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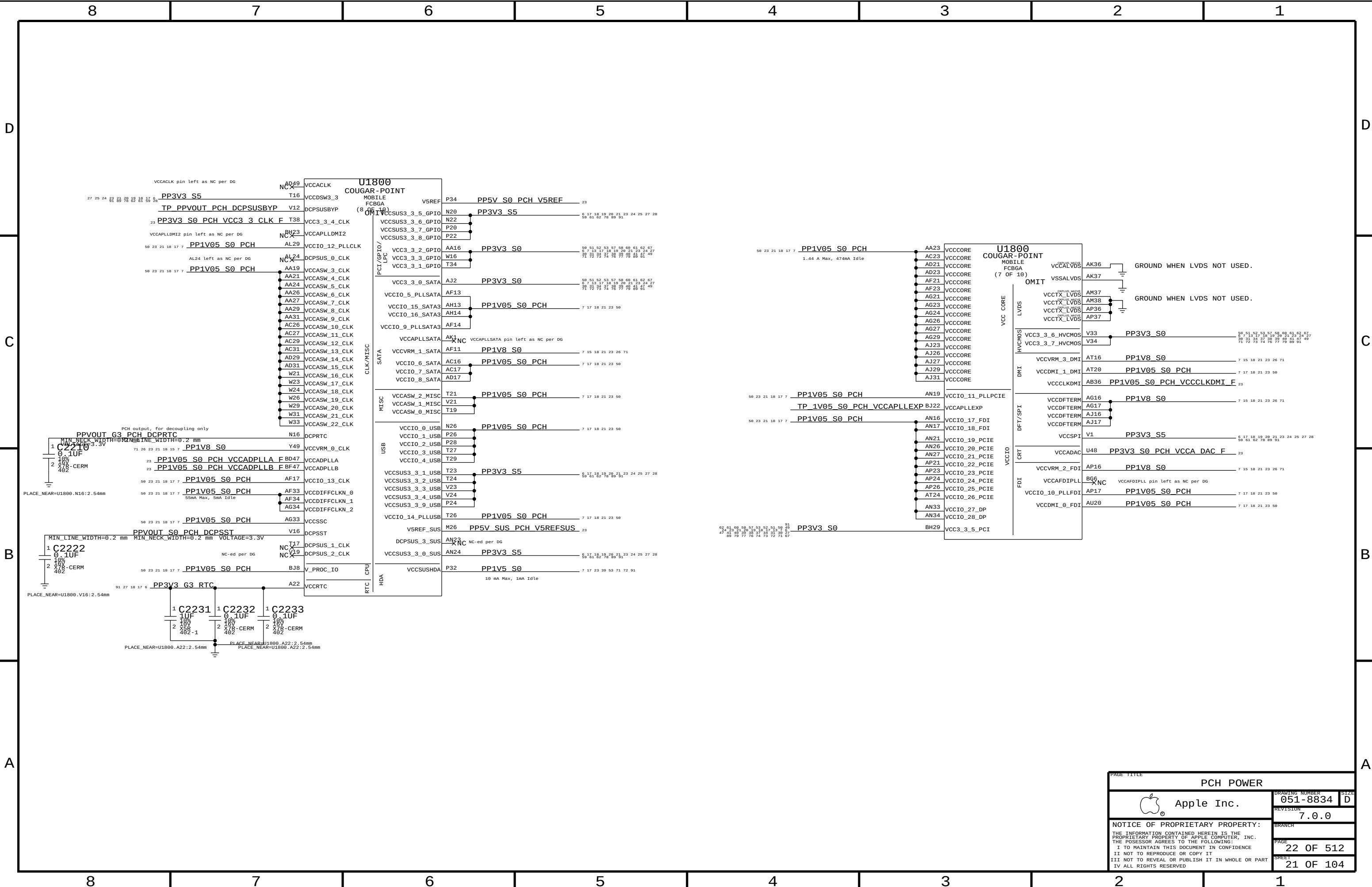


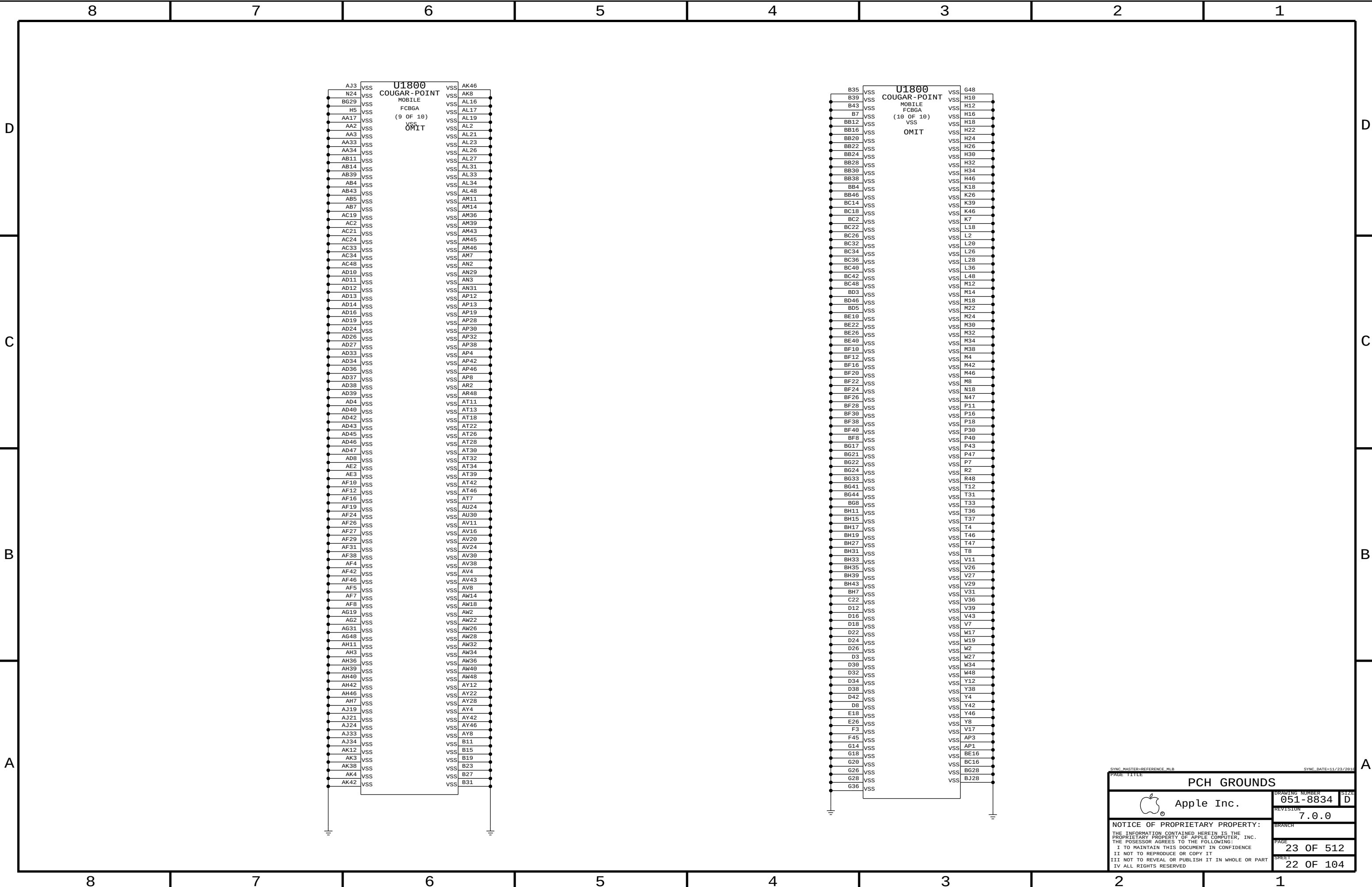
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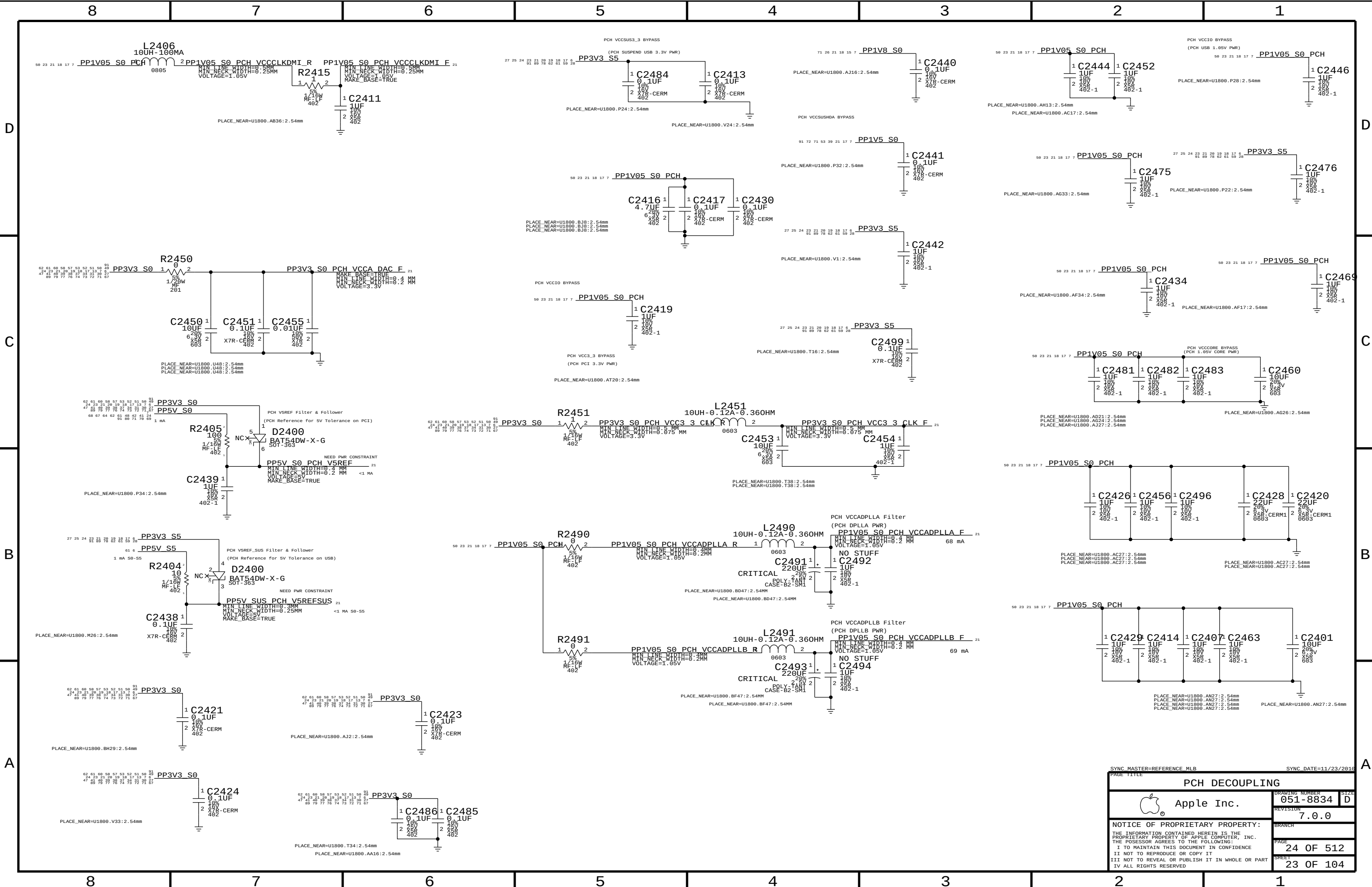


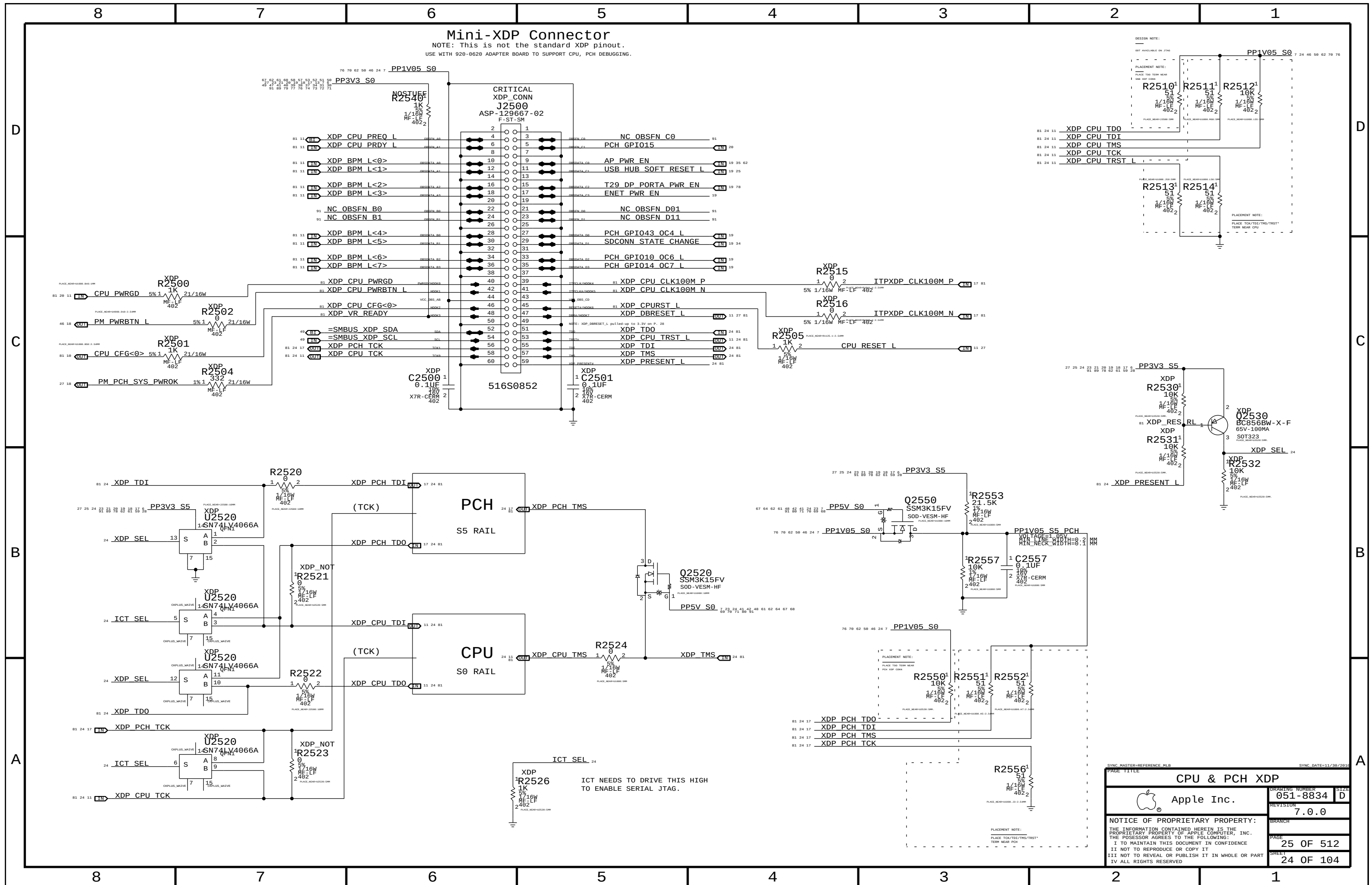


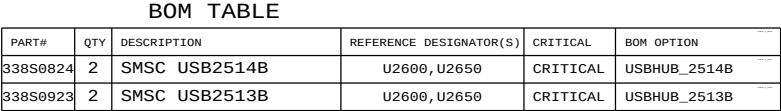
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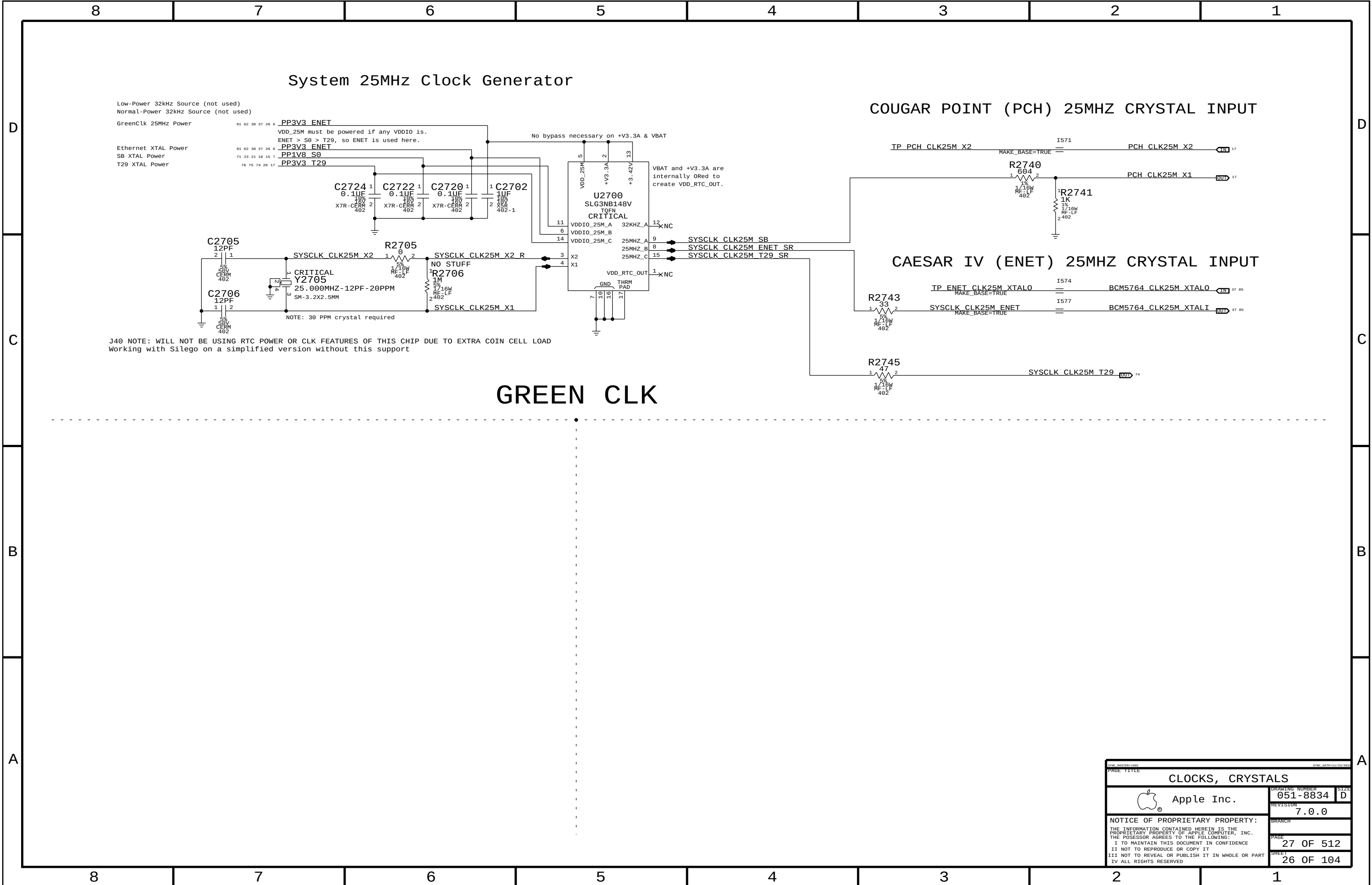
SYNC DATE=11/23/2011

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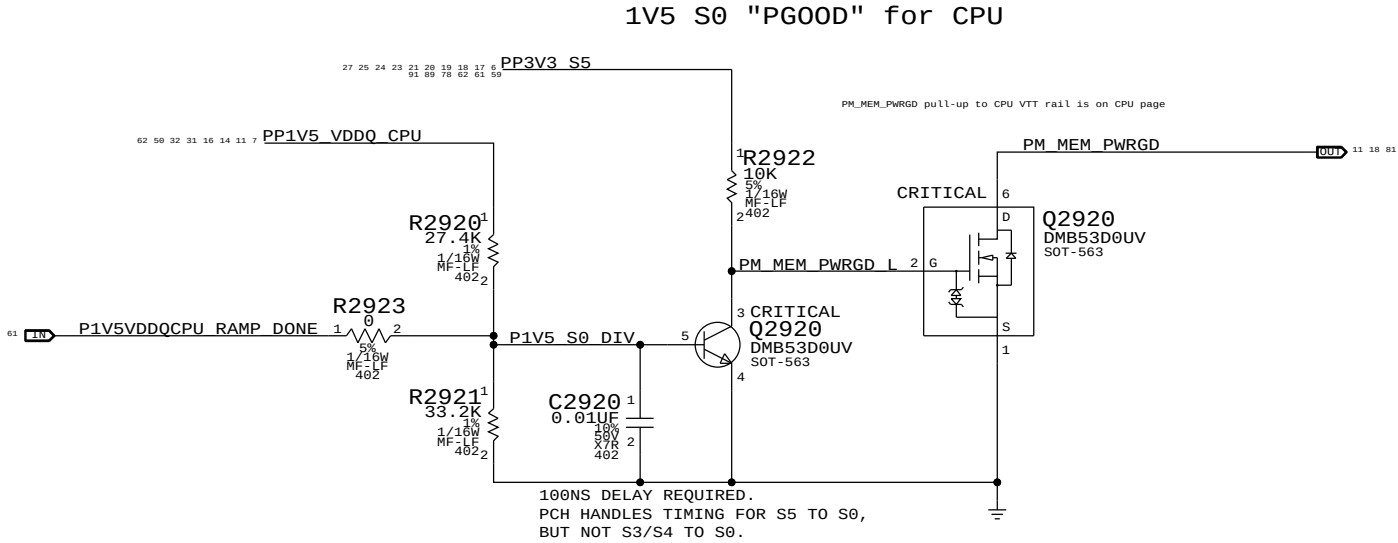
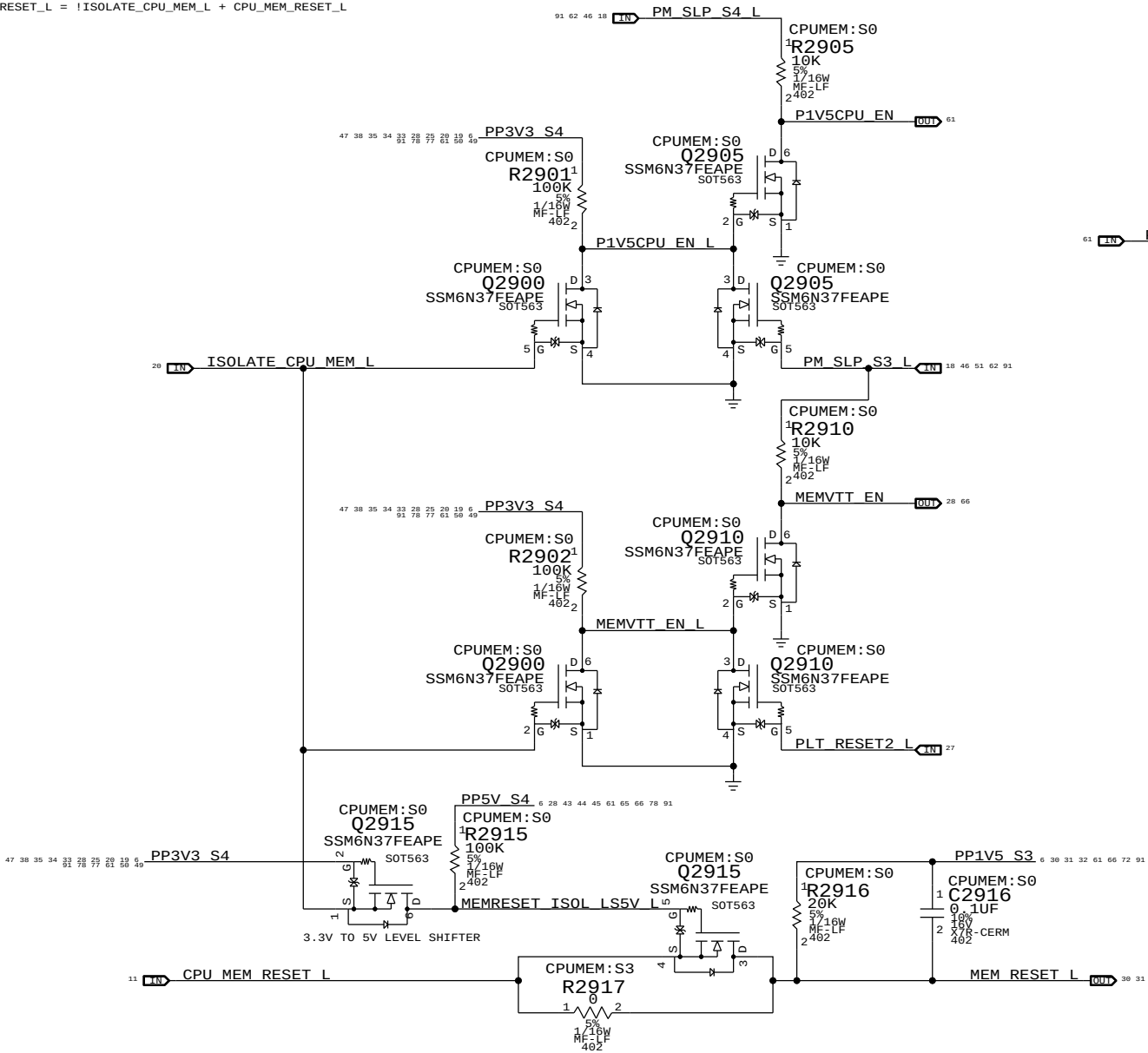
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A

The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

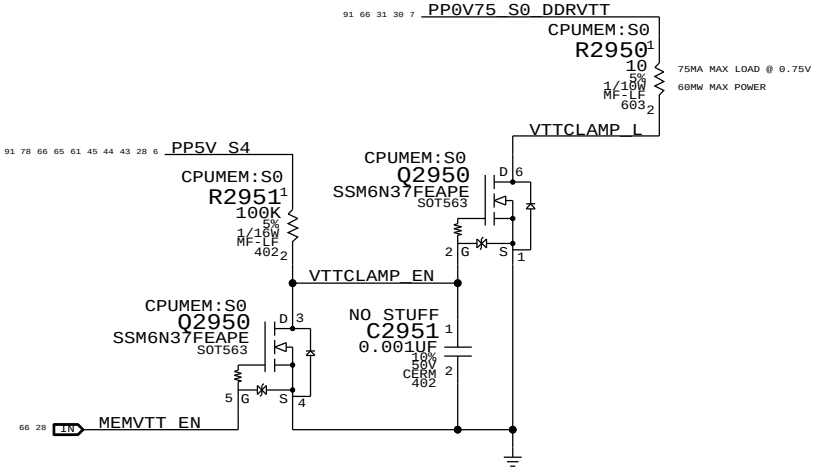
ISOLATE_CPU_MEM_L GPIO state during S3->S0 transitions determines behavior of signals.
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.
WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

P1V5CPU_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L
MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L



MEMVTT Clamp

Ensures CKE signals are held low in S3



Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	PM_SLP_S4_L	CPUMEM_RESET_L	MEM_RESET_L	MEMVTT_EN	P1V5CPU_EN
S0	0	1	1	1	1	CPUMEM_RESET_L	1	1
1	0	1	1	1	1	1	1	1
2	0	0	1	1	1	1	0	1
3	0	0	0	1	X	1	0	0
S3	4	0	0	1	1	X (*)	1	1
5	0	1	1	1	1	1	1	1
6	0	1	1	1	1	1	1	1
S0	7	1	1	1	1	CPUMEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

D

C

B

A

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SYNC DATE=11/23/2011

CPU Memory S3 Support

Apple Inc.

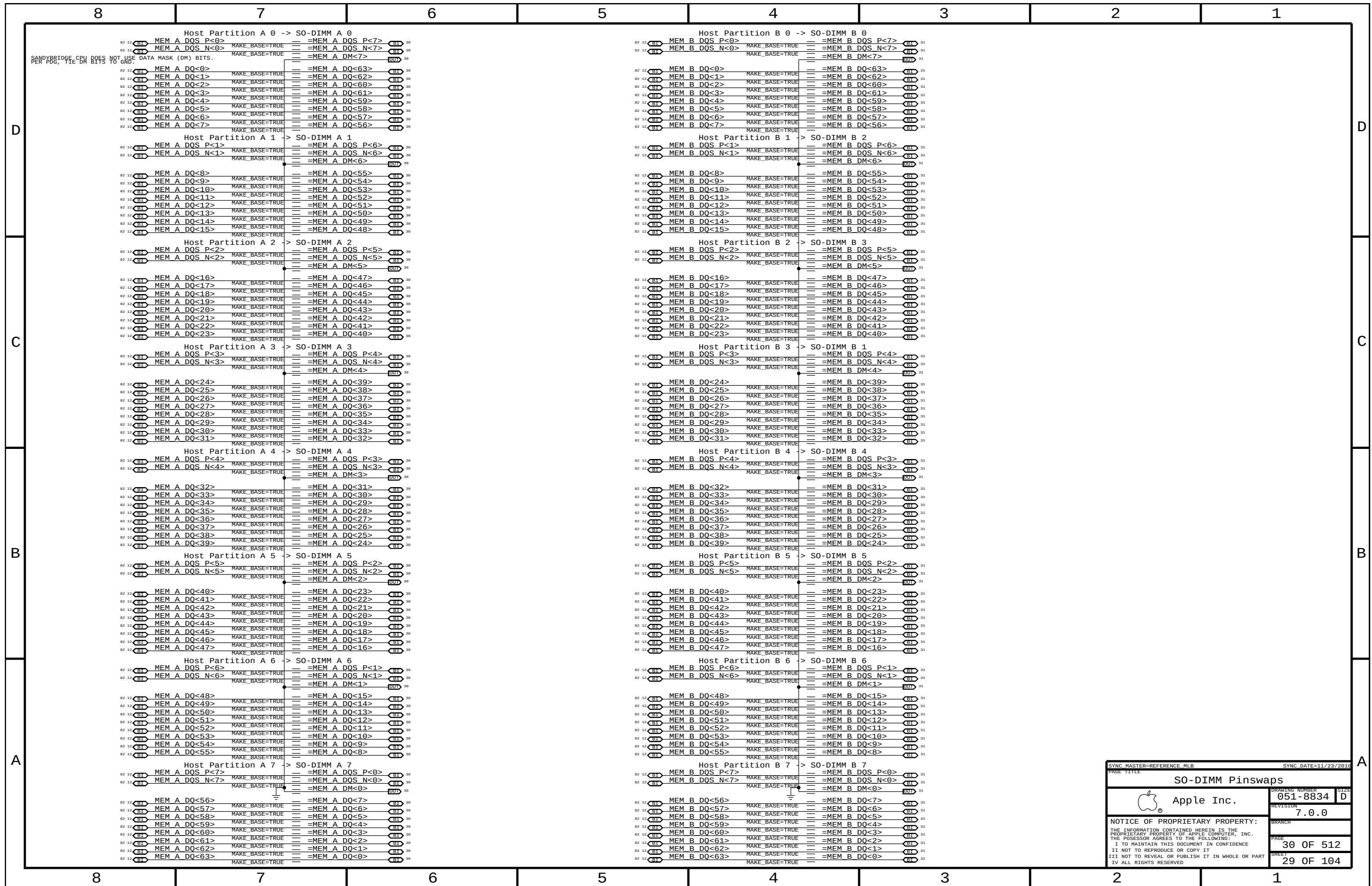
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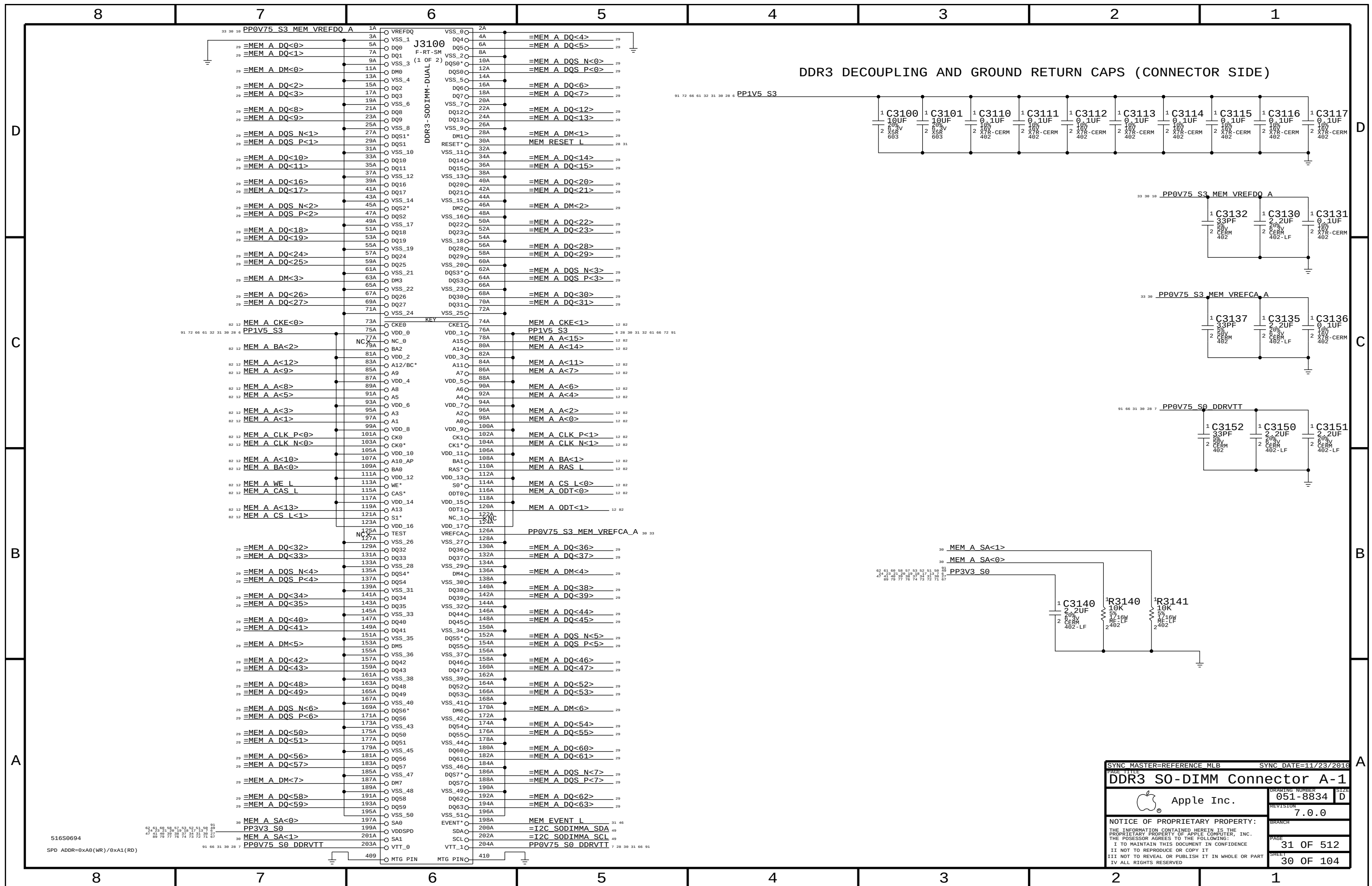
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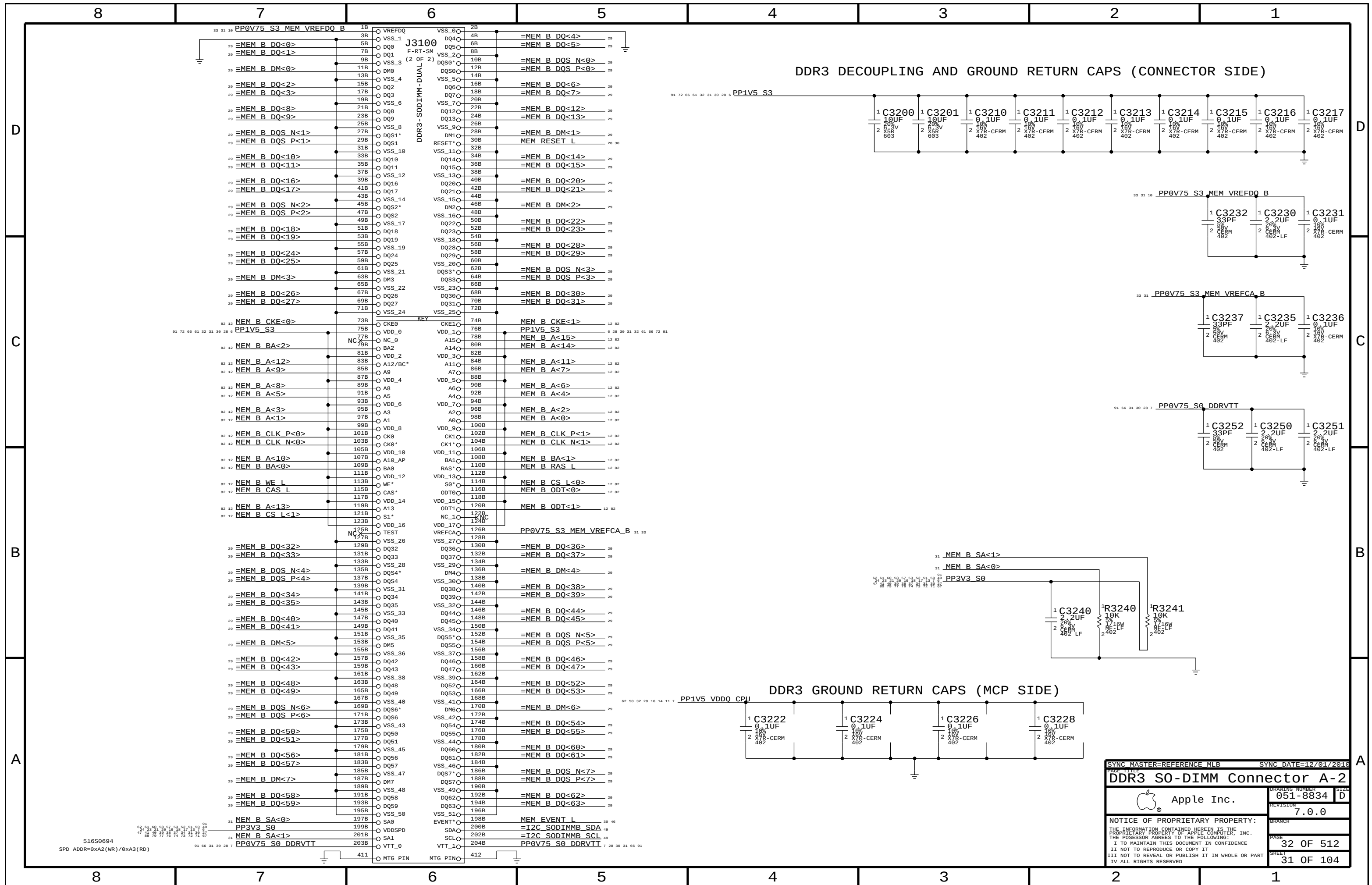
REVISION
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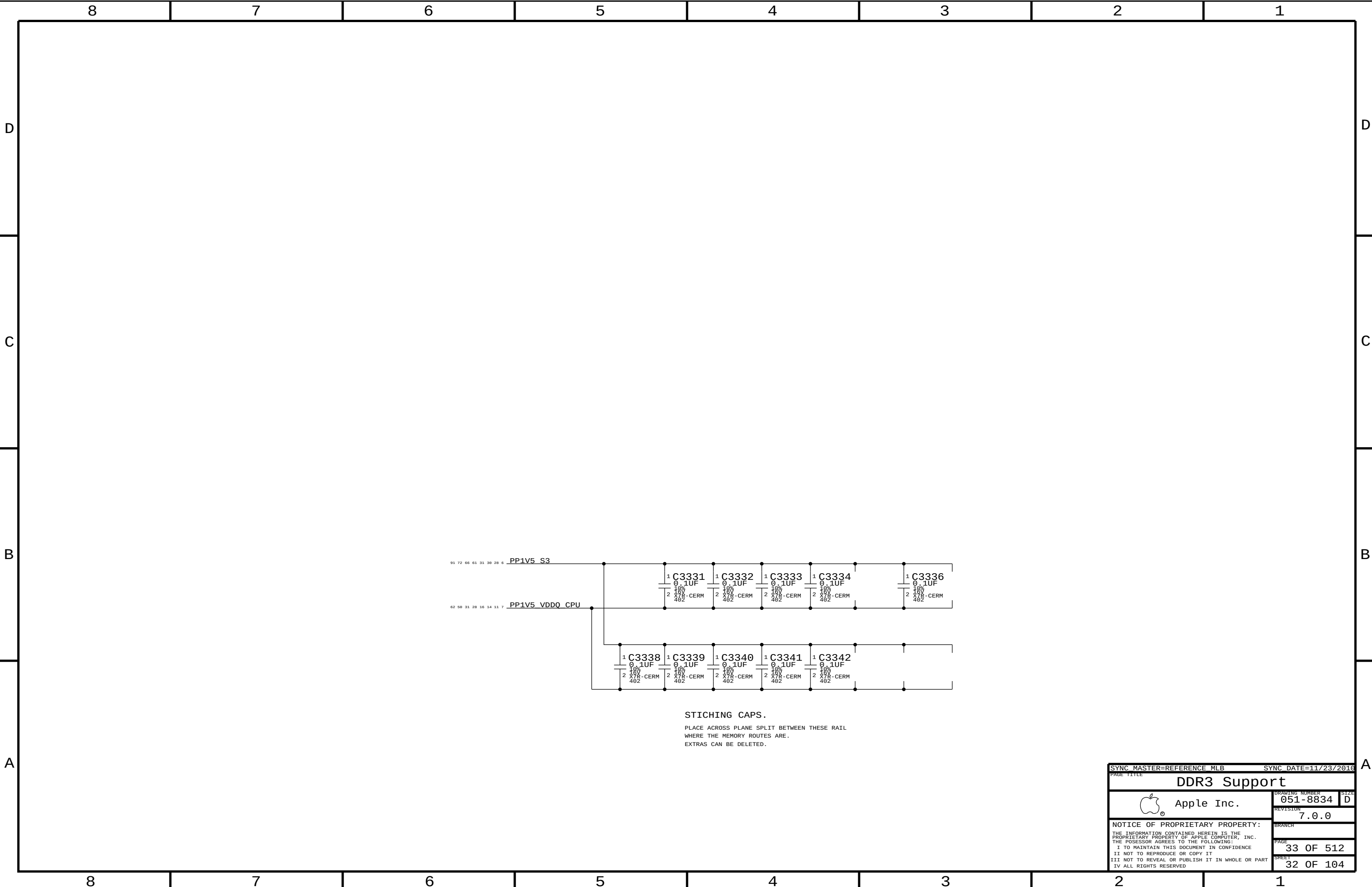
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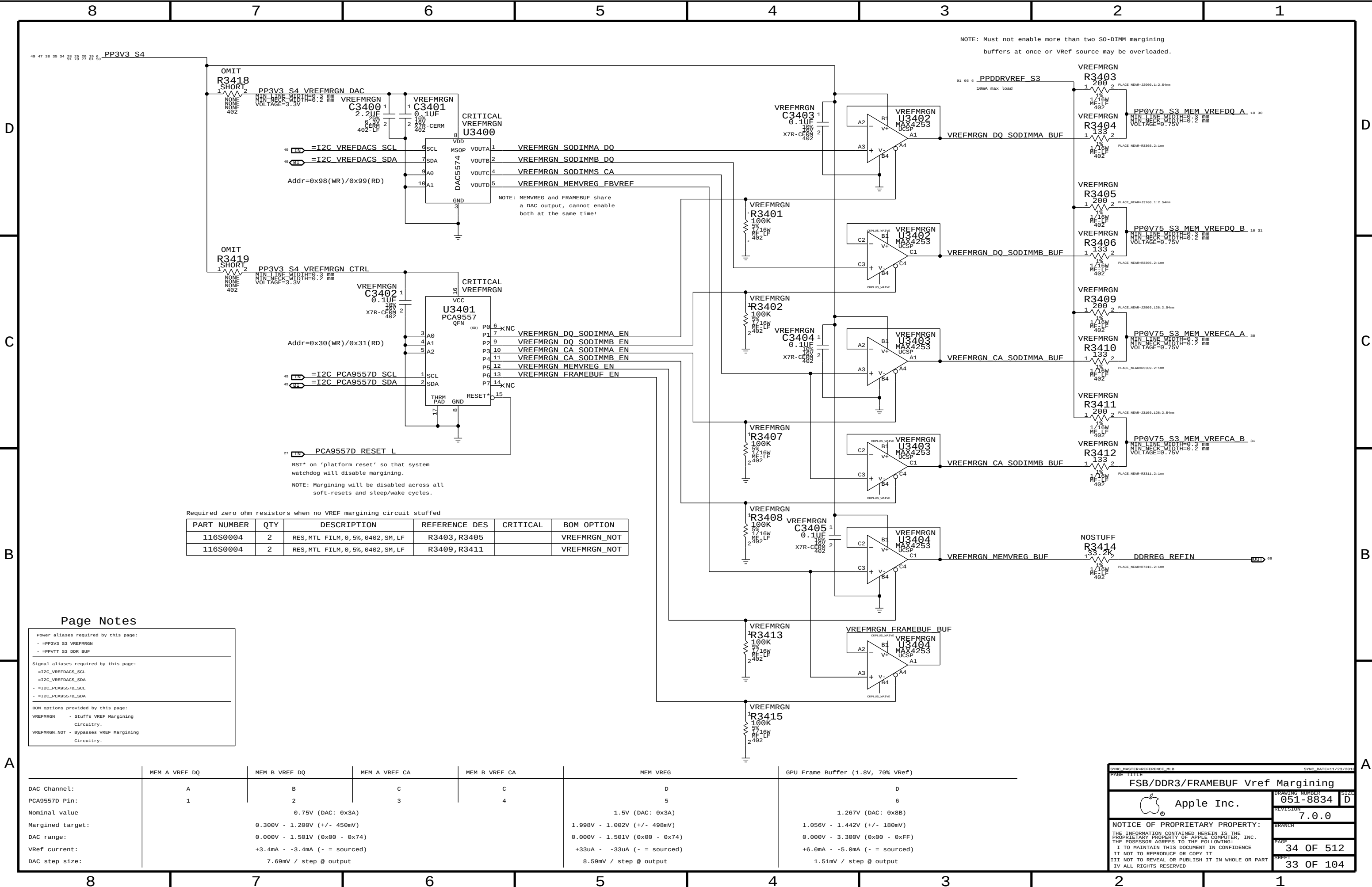








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		BRANCH	
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Page Notes

Power aliases required by this page:

- =PP3V3_S3_VREFMRGN
- =PPVTT_S3_DDR_BUF

Signal aliases required by this page:

- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:

- VREFMRGN - Stuffs VREF Margining Circuitry.
- VREFMRGN_NOT - Bypasses VREF Margining Circuitry.

	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% Vref)
DAC Channel:	A	B	C	C	D	D
PCA9557D Pin:	1	2	3	4	5	6
Nominal value		0.75V (DAC: 0x3A)			1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margin'd target:		0.300V - 1.200V (+/- 450mV)			1.998V - 1.002V (+/- 498mV)	1.056V - 1.442V (+/- 180mV)
DAC range:		0.000V - 1.501V (0x00 - 0x74)			0.000V - 1.501V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
VRef current:		+3.4mA - -3.4mA (- = sourced)			+33uA - -33uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:		7.69mV / step @ output			8.59mV / step @ output	1.51mV / step @ output

SYNC MASTER-REFERENCE: HLB

SYNC DATE: 11/23/2016

PAGE TITLE

FSB/DDR3/FRAMEBUF Vref Margining

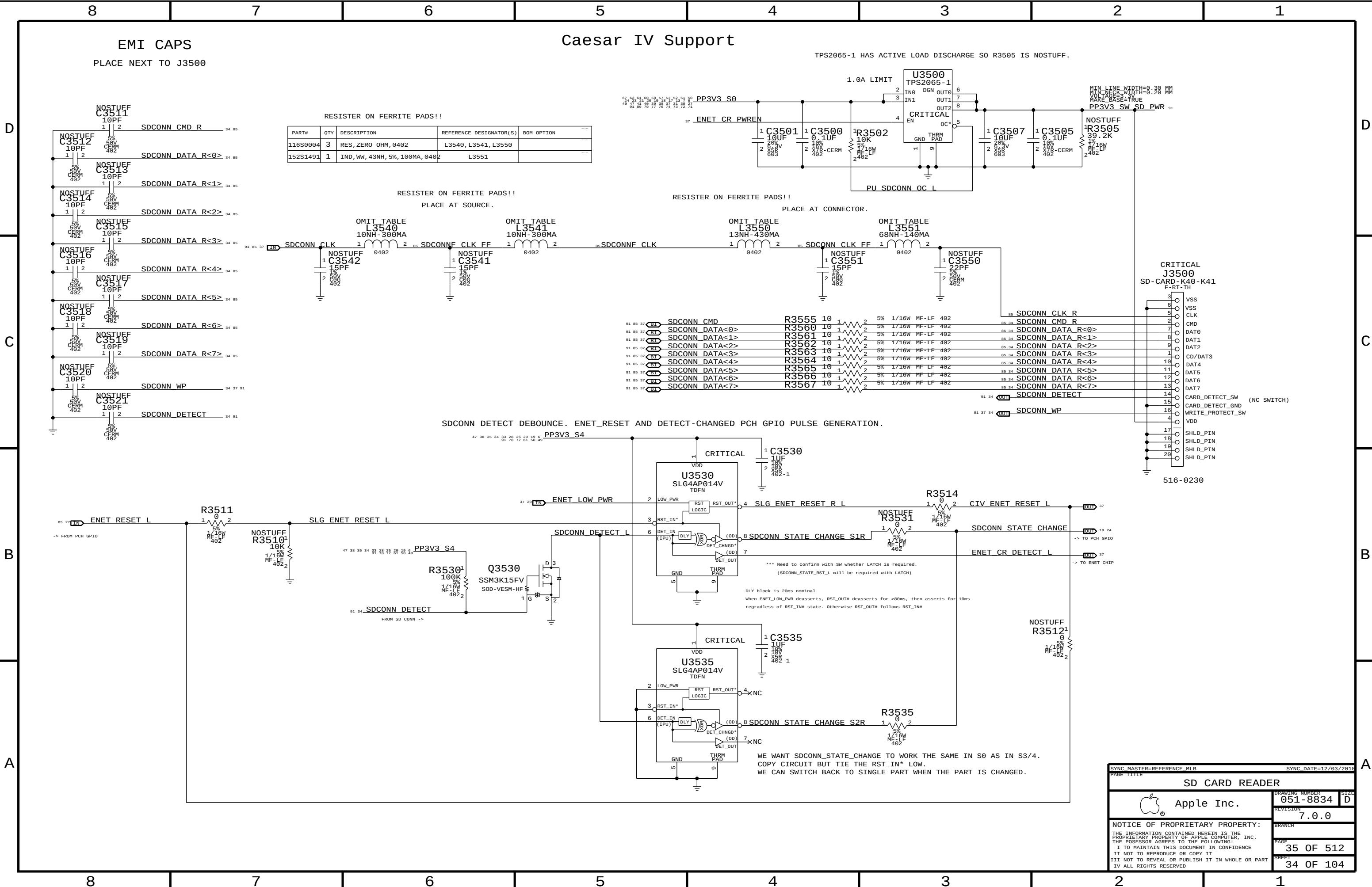
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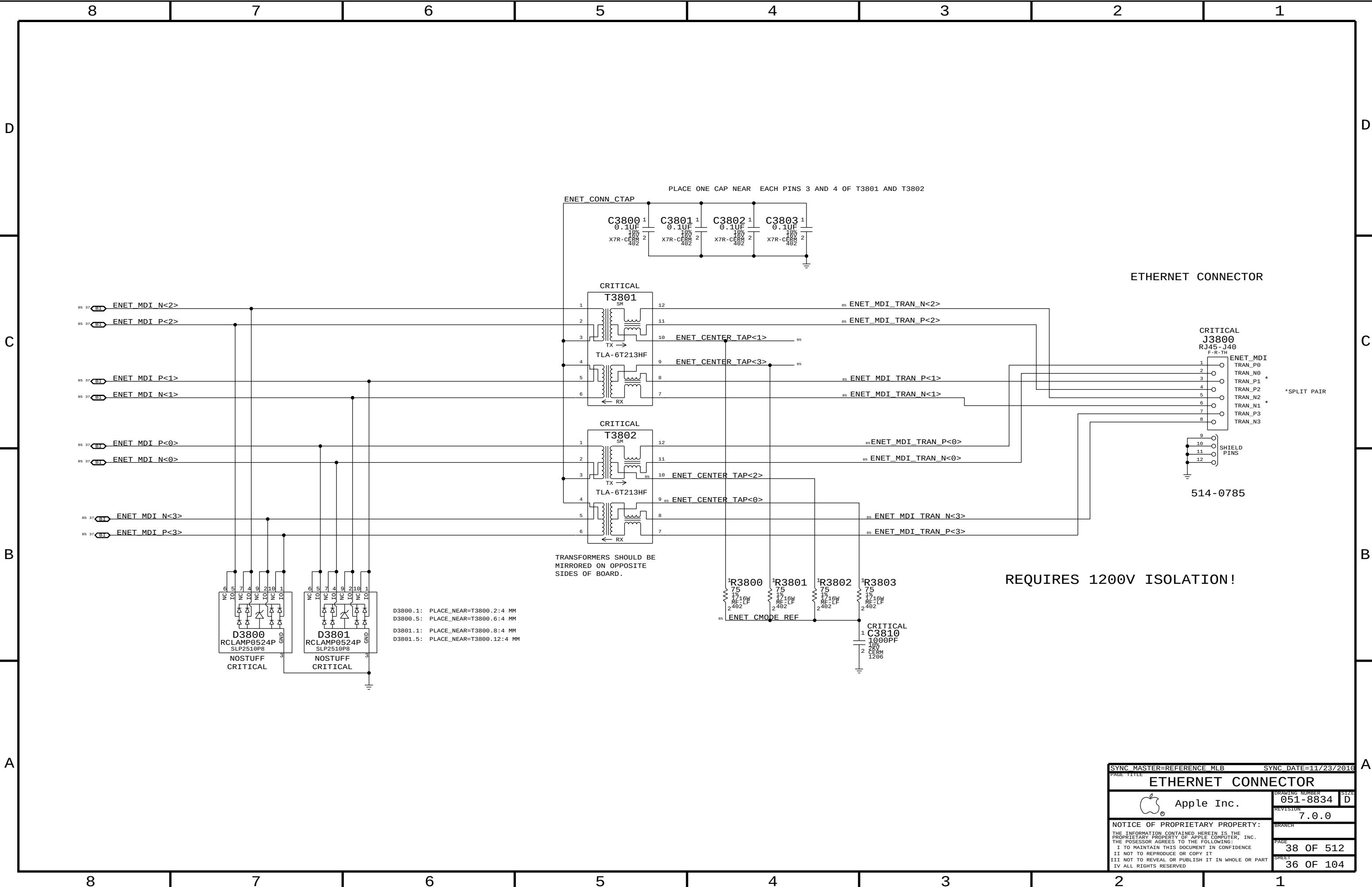
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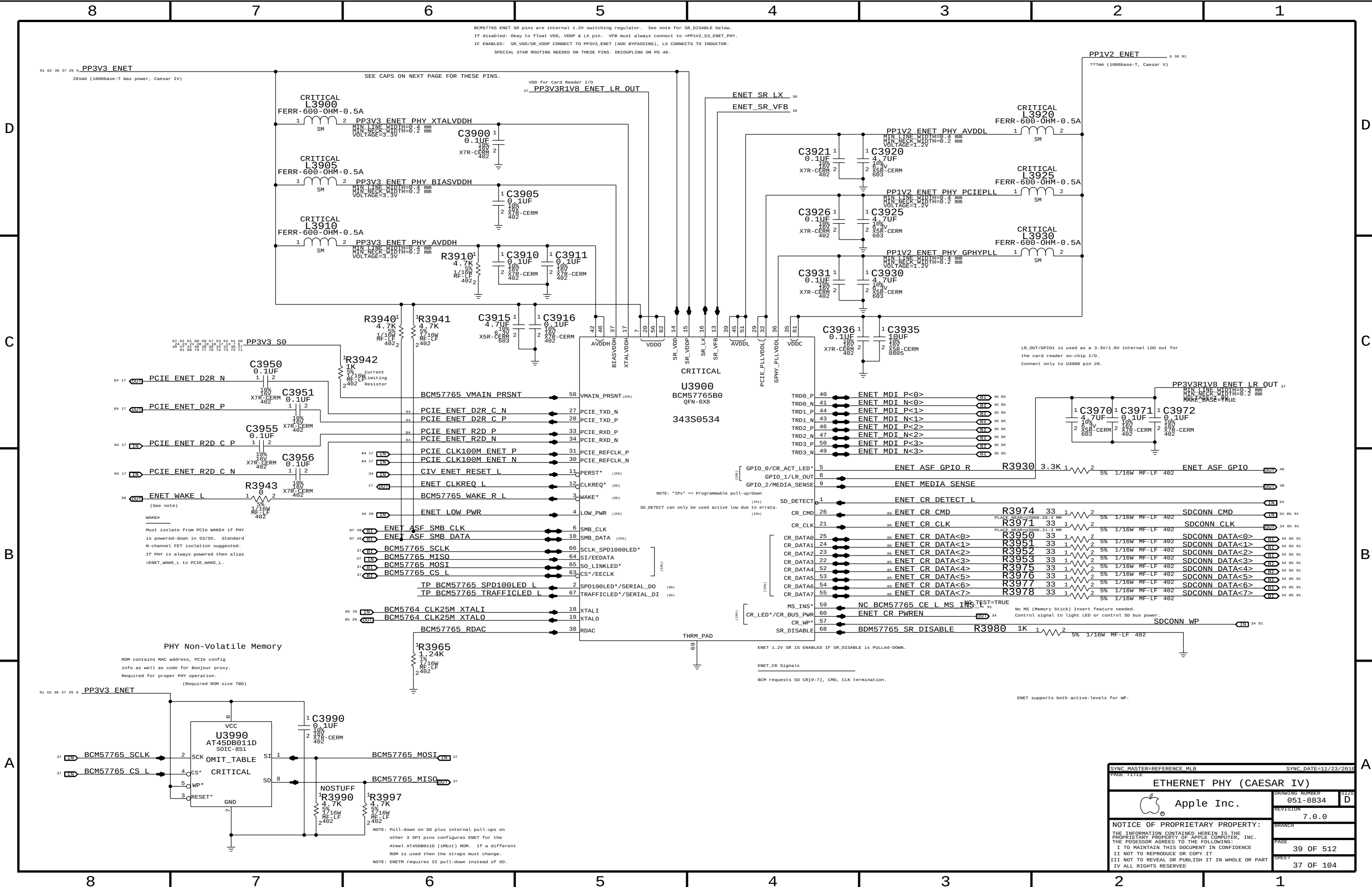


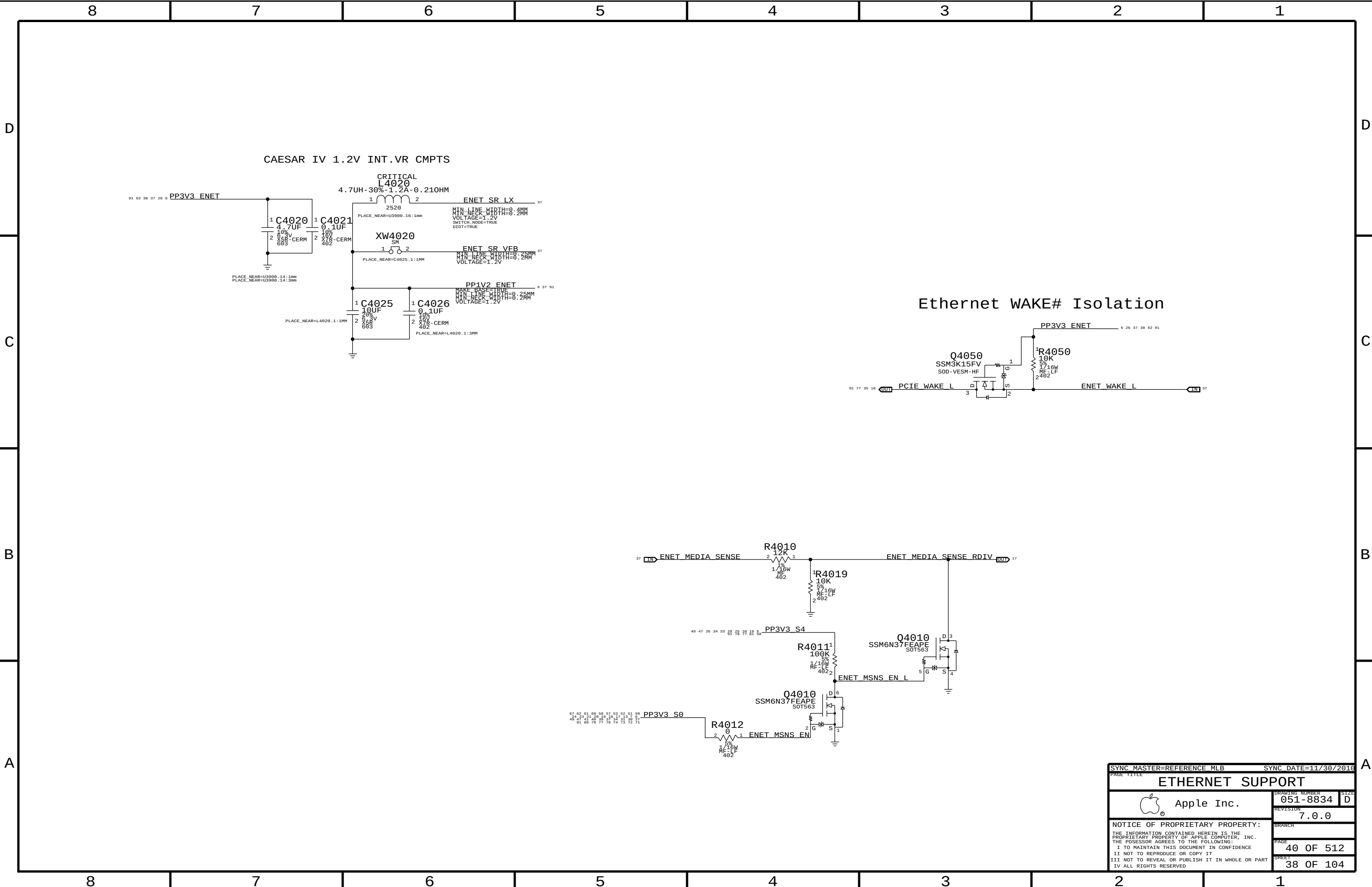
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
116S0004	3	RES,ZERO OHM,0402	L3540,L3541,L3550	
152S1491	1	IND,WV,43NH,5%,100MA,0402	L3551	

01 05 37	B1	SDCONN_CMD	R3555	10	1	2	5%	1/16W	MF-LF	402	85 34	SDCONN_CMD_R
01 05 37	B1	SDCONN_DATA<0>	R3560	10	1	2	5%	1/16W	MF-LF	402	85 34	SDCONN_DATA_R<0>
01 05 37	B1	SDCONN_DATA<1>	R3561	10	1	2	5%	1/16W	MF-LF	402	85 34	SDCONN_DATA_R<1>
01 05 37	B1	SDCONN_DATA<2>	R3562	10	1	2	5%	1/16W	MF-LF	402	85 34	SDCONN_DATA_R<2>
01 05 37	B1	SDCONN_DATA<3>	R3563	10	1	2	5%	1/16W	MF-LF	402	85 34	SDCONN_DATA_R<3>
01 05 37	B1	SDCONN_DATA<4>	R3564	10	1	2	5%	1/16W	MF-LF	402	85 34	SDCONN_DATA_R<4>
01 05 37	B1	SDCONN_DATA<5>	R3565	10	1	2	5%	1/16W	MF-LF	402	85 34	SDCONN_DATA_R<5>
01 05 37	B1	SDCONN_DATA<6>	R3566	10	1	2	5%	1/16W	MF-LF	402	85 34	SDCONN_DATA_R<6>
01 05 37	B1	SDCONN_DATA<7>	R3567	10	1	2	5%	1/16W	MF-LF	402	85 34	SDCONN_DATA_R<7>

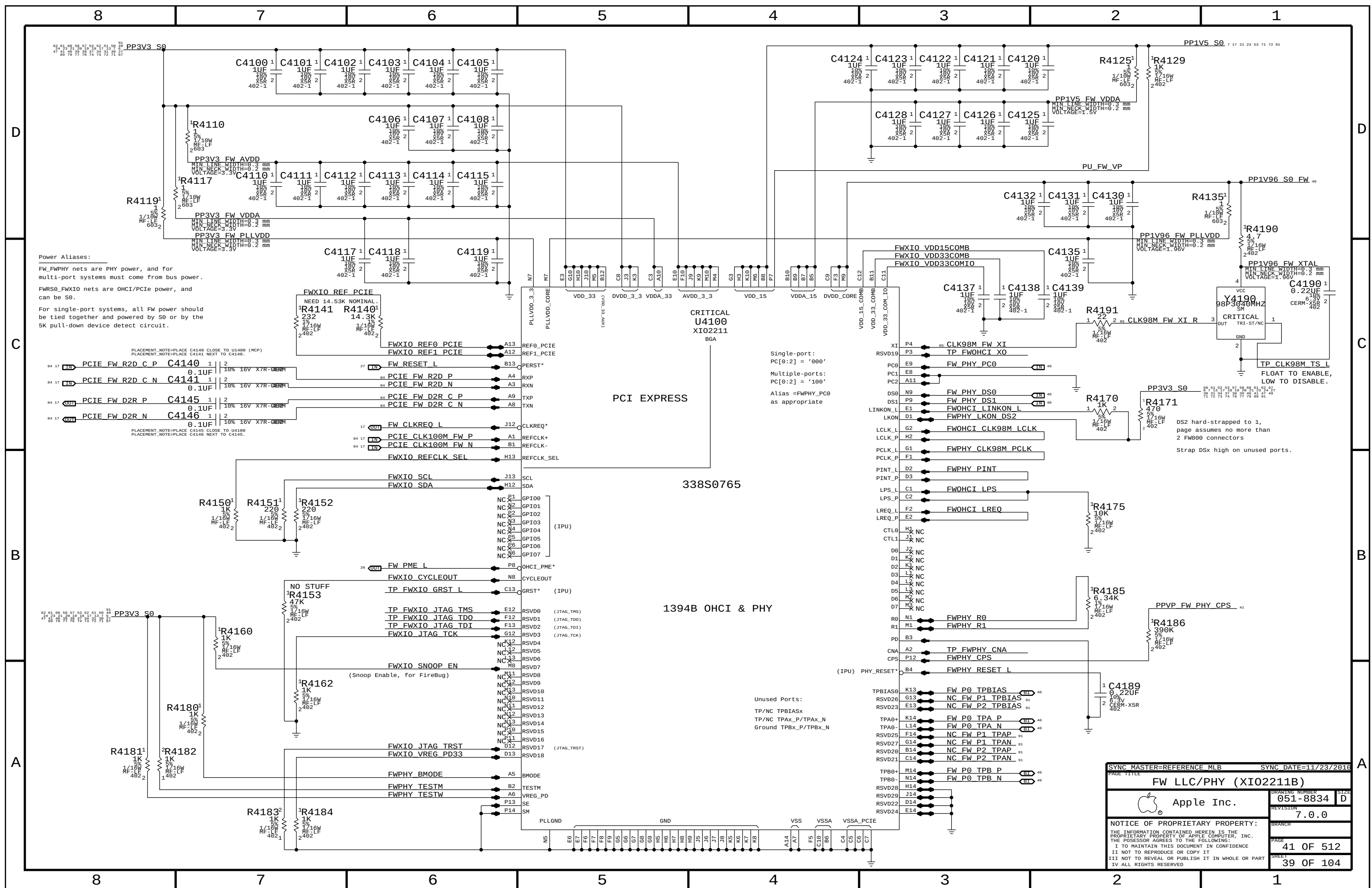
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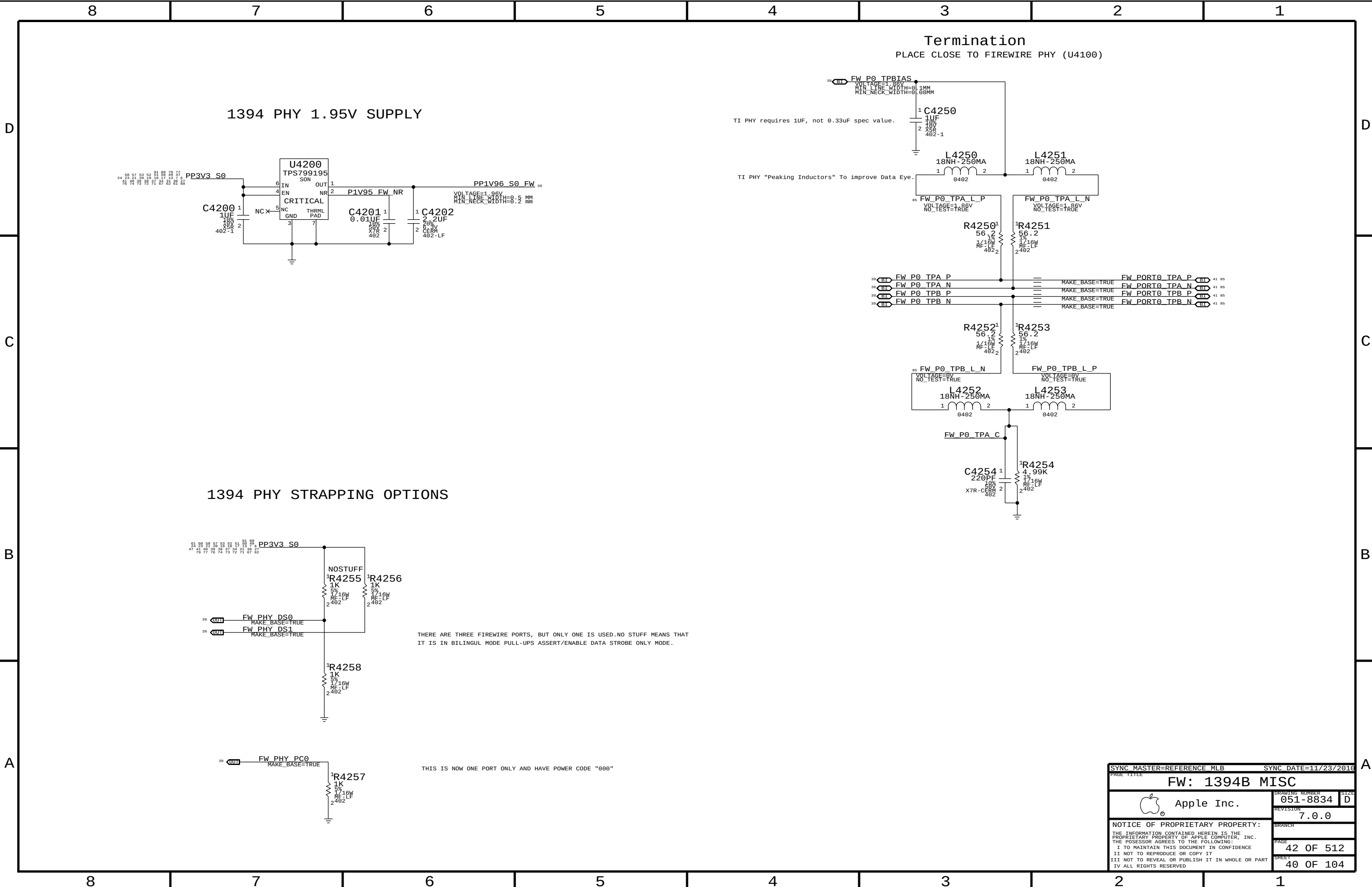






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ETHERNET SUPPORT		051-8834		D	
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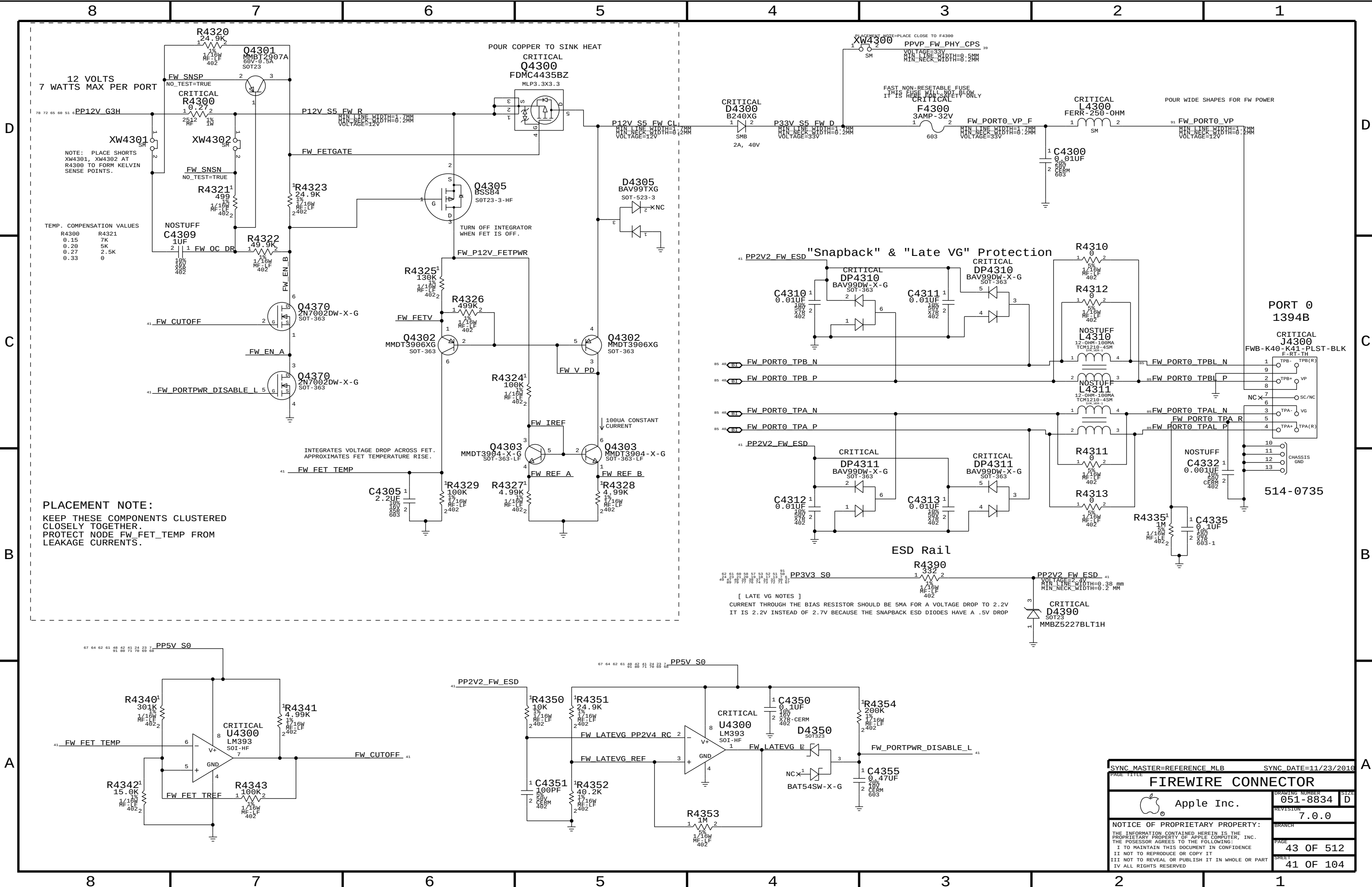


1394 PHY 1.95V SUPPLY

Termination
PLACE CLOSE TO FIREWIRE PHY (U4100)

1394 PHY STRAPPING OPTIONS

PAGE TITLE		SYNC MASTER=REFERENCE MLB		SYNC DATE=11/23/2016	
FW: 1394B MISC		DRAWING NUMBER		SIZE	
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		PAGE		SHEET	
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12 VOLTS
7 WATTS MAX PER PORT

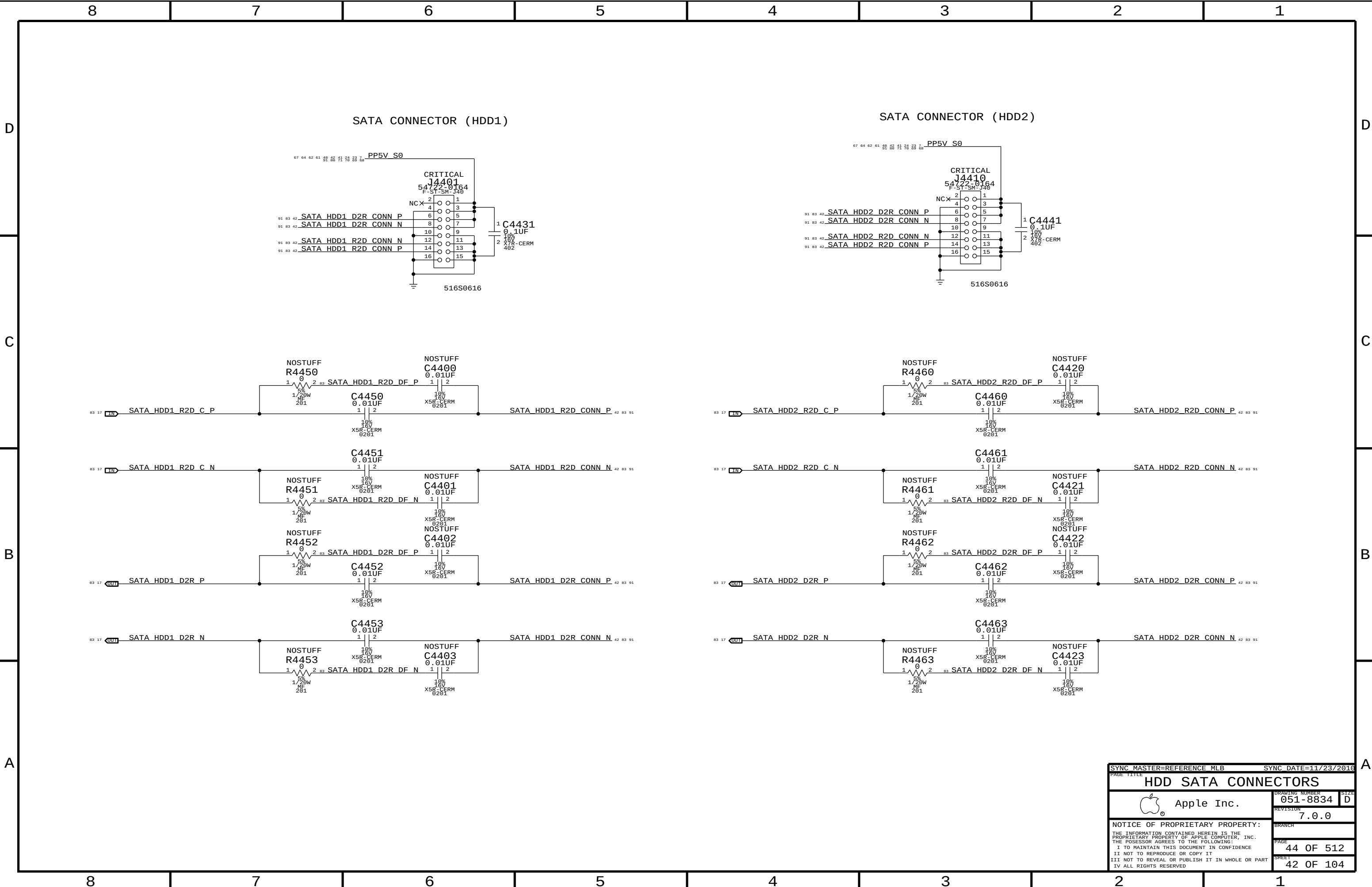
NOTE: PLACE SHORTS
XW4301, XW4302 AT
R4300 TO FORM KELVIN
SENSE POINTS.

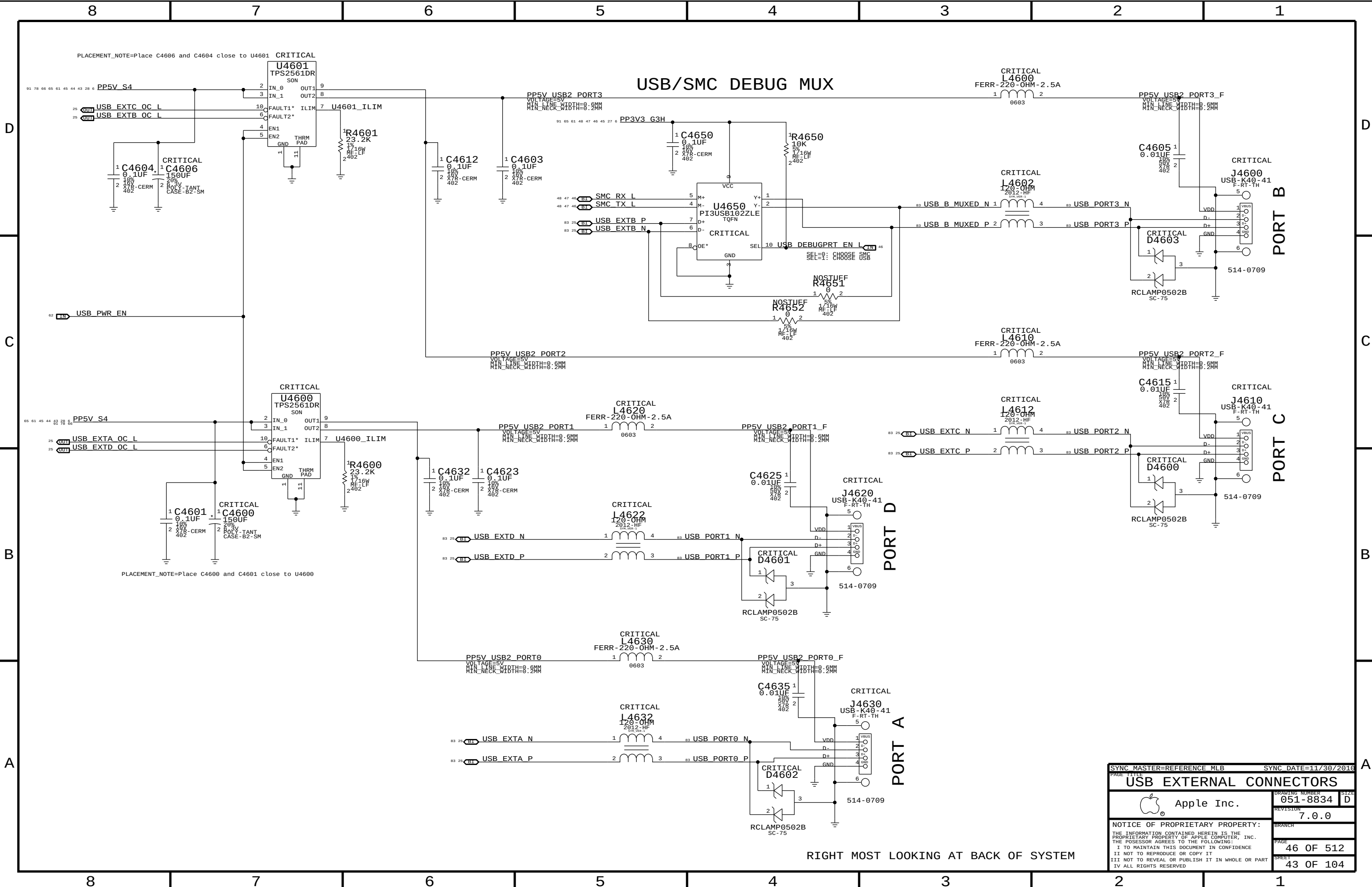
TEMP. COMPENSATION VALUES

R4300	R4321
0.15	7K
0.20	5K
0.27	2.5K
0.33	0

PLACEMENT NOTE:
KEEP THESE COMPONENTS CLUSTERED
CLOSELY TOGETHER.
PROTECT NODE FW_FET_TEMP FROM
LEAKAGE CURRENTS.

SYNC MASTER=REFERENCE MLB		SYNC DATE=11/23/2010	
PAGE TITLE		FIREWIRE CONNECTOR	
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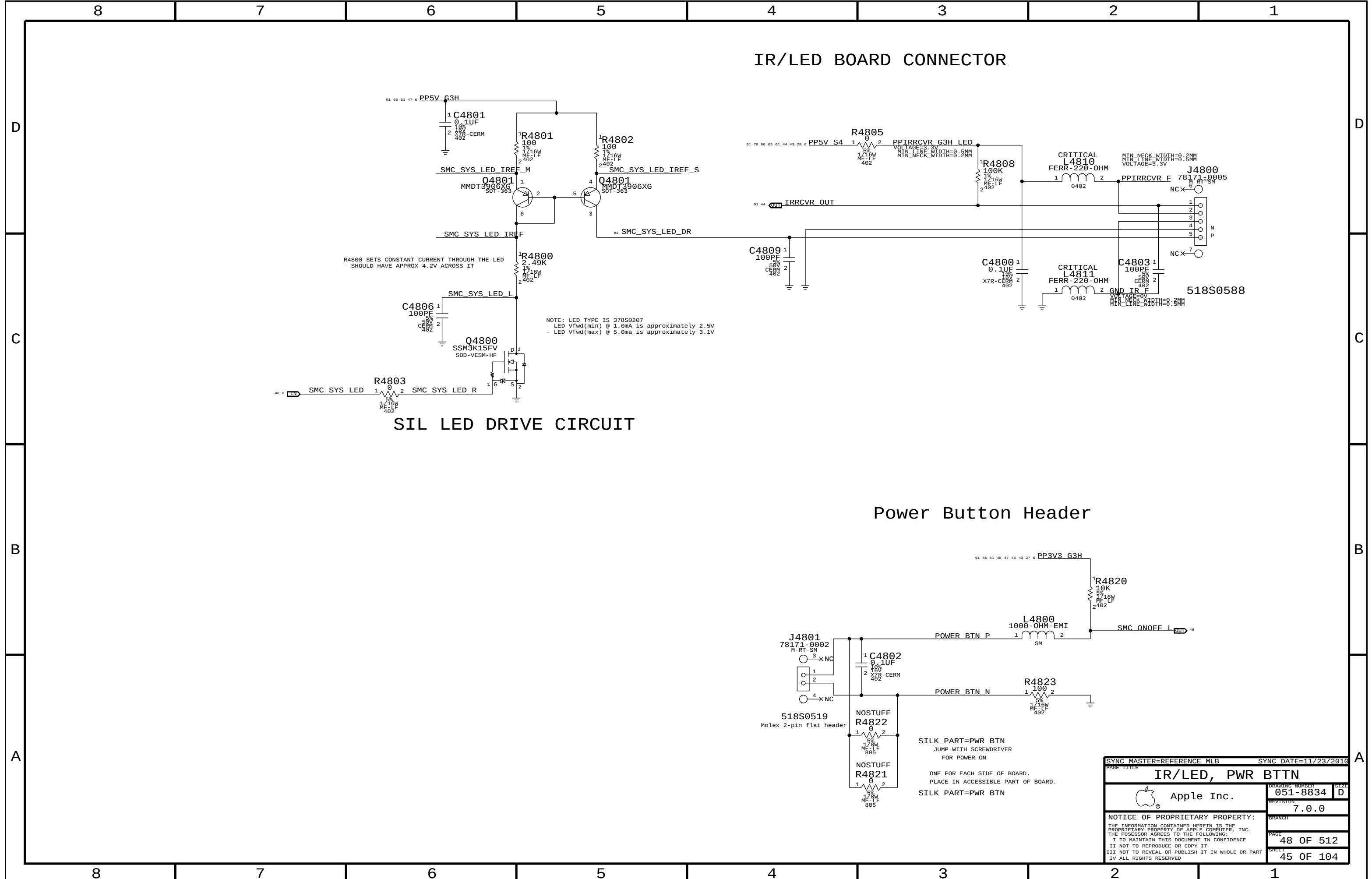


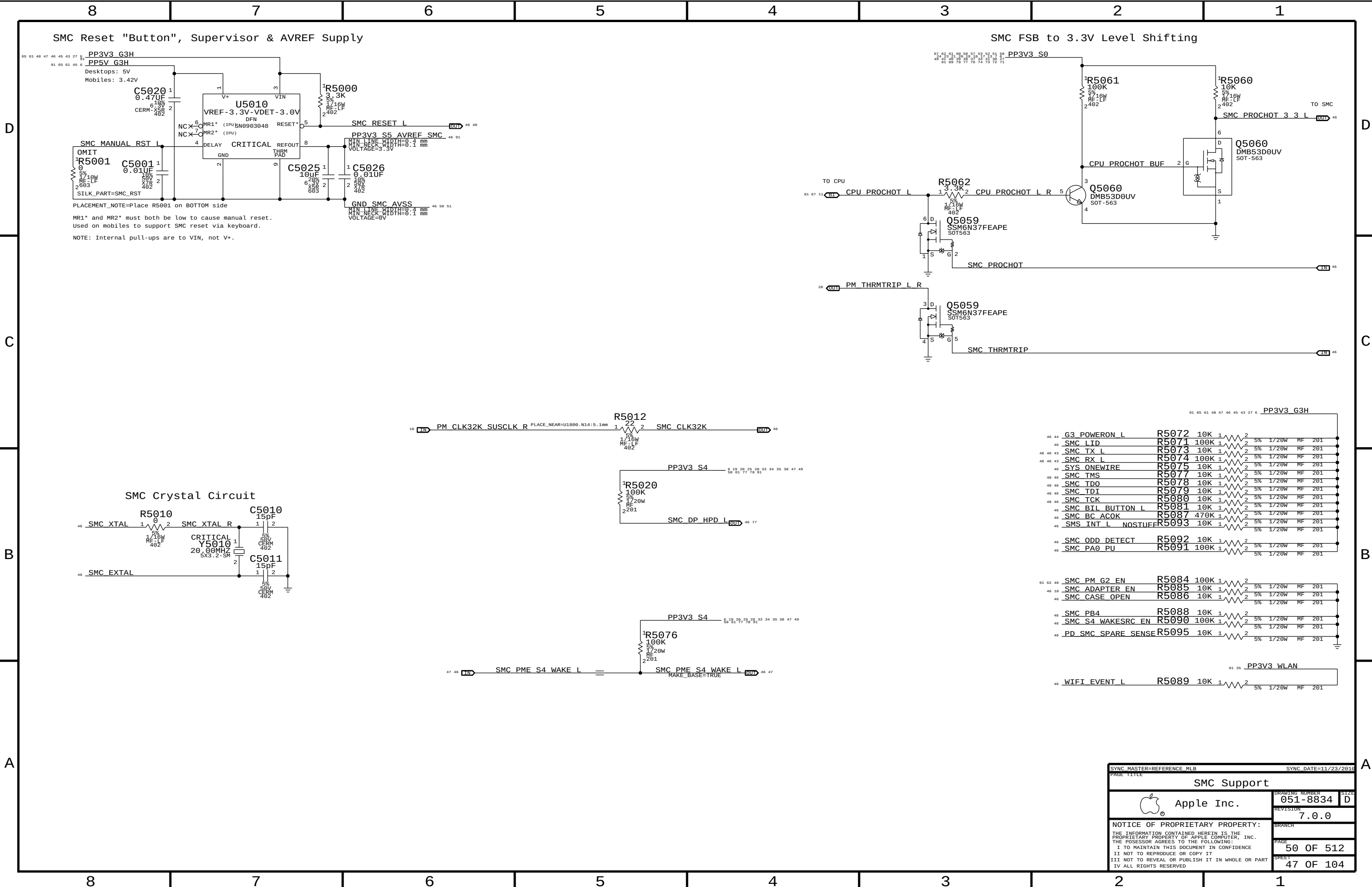
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PAGE TITLE		USB EXTERNAL CONNECTORS	
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RIGHT MOST LOOKING AT BACK OF SYSTEM

D

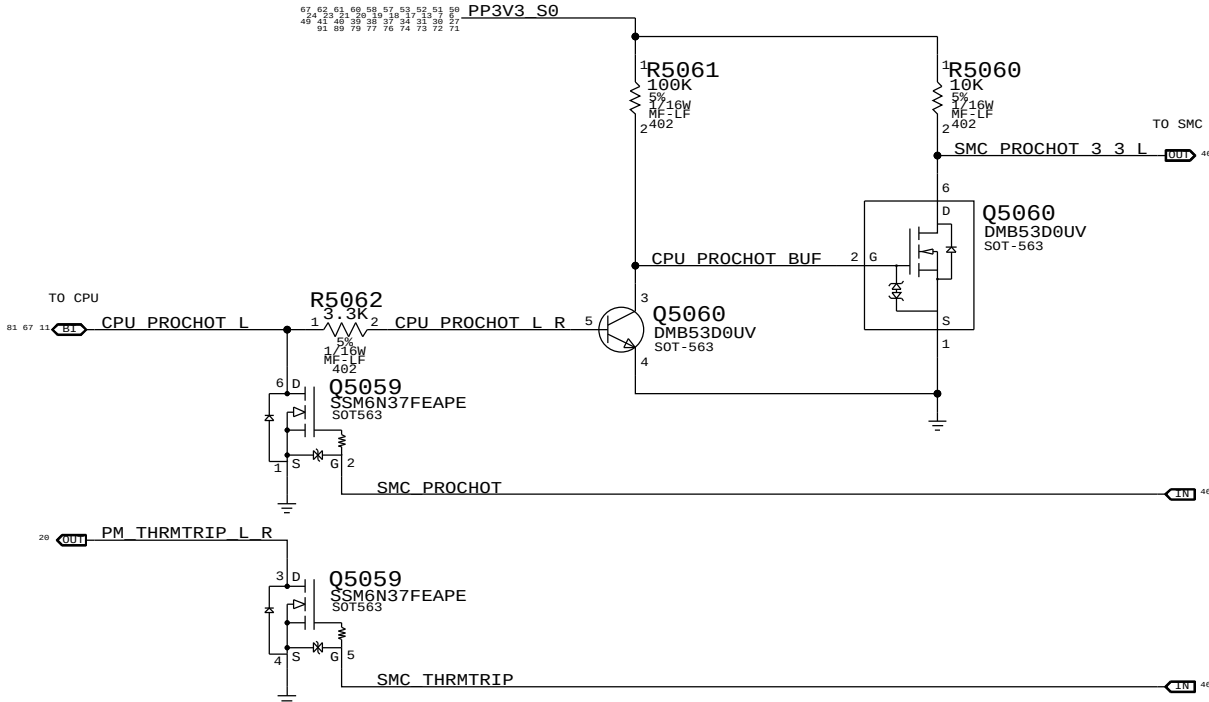
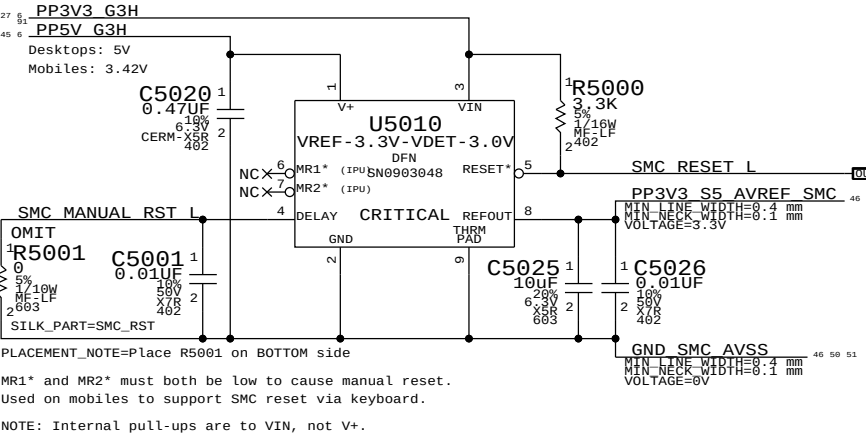




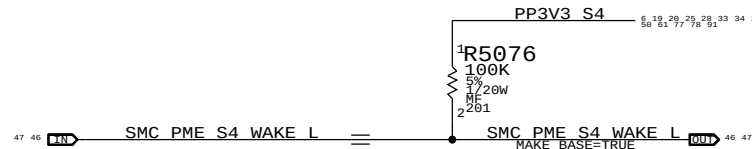
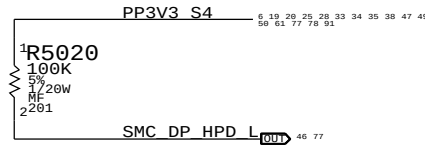
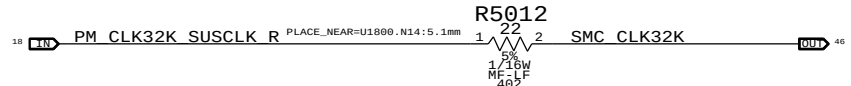
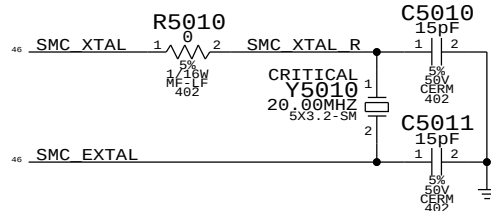


SMC Reset "Button", Supervisor & AVREF Supply

SMC FSB to 3.3V Level Shifting



SMC Crystal Circuit



PP3V3_G3H		91	65	61	48	47	46	45	43	27	6
G3_POWERON_L	R5072	10K	1	2							
SMC_LID	R5071	100K	1	2	5%	1/20W	MF	201			
SMC_TX_L	R5073	10K	1	2	5%	1/20W	MF	201			
SMC_RX_L	R5074	100K	1	2	5%	1/20W	MF	201			
SYS_ONEWIRE	R5075	10K	1	2	5%	1/20W	MF	201			
SMC_TMS	R5077	10K	1	2	5%	1/20W	MF	201			
SMC_TDO	R5078	10K	1	2	5%	1/20W	MF	201			
SMC_TDI	R5079	10K	1	2	5%	1/20W	MF	201			
SMC_TCK	R5080	10K	1	2	5%	1/20W	MF	201			
SMC_BIL_BUTTON_L	R5081	10K	1	2	5%	1/20W	MF	201			
SMC_BC_ACOK	R5087	470K	1	2	5%	1/20W	MF	201			
SMS_INT_L_NOSTUFFER	R5093	10K	1	2	5%	1/20W	MF	201			
SMC_ODD_DETECT	R5092	10K	1	2	5%	1/20W	MF	201			
SMC_PA0_PU	R5091	100K	1	2	5%	1/20W	MF	201			
PP3V3_WLAN		91	35								
WIFI_EVENT_L	R5089	10K	1	2	5%	1/20W	MF	201			

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SYNC DATE=11/23/2010

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SMC Support

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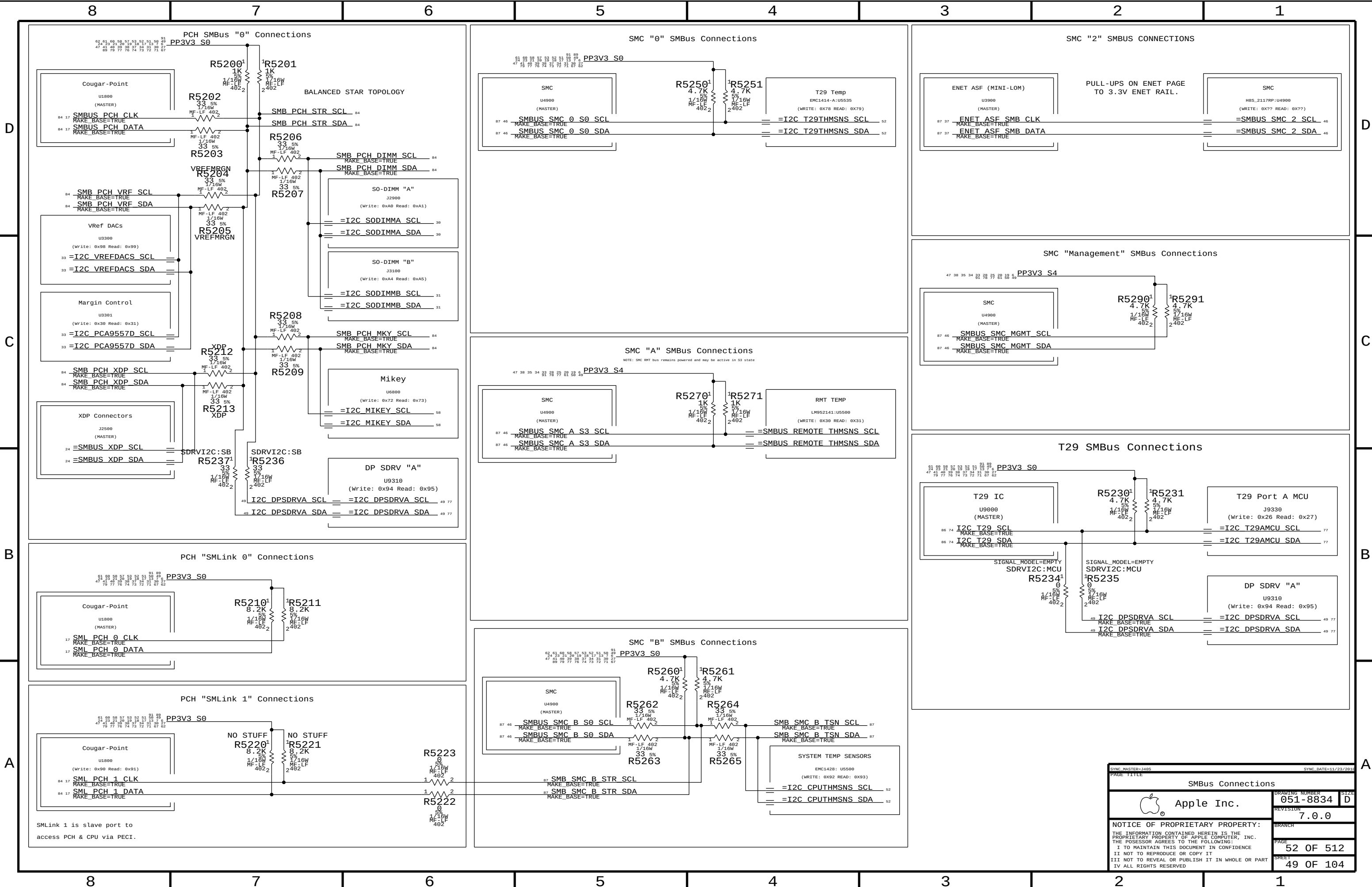
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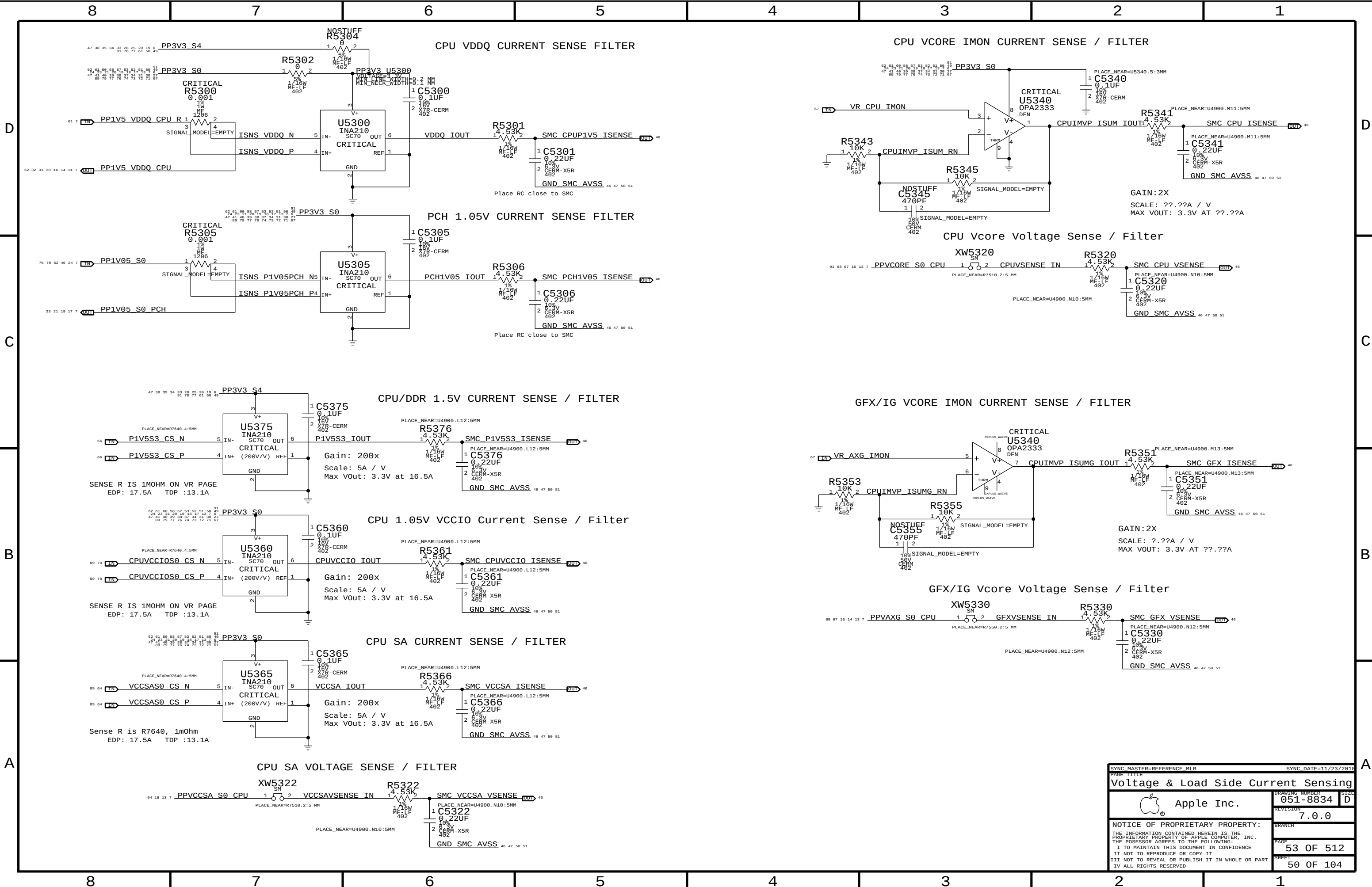


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SYNC MASTER=REFERENCE_MLB		SYNC DATE=11/23/2010	
PAGE TITLE			
Voltage & Load Side Current Sensing			
 Apple Inc.		DRAWING NUMBER	051-8834
		REVISION	7.0.0
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		PAGE	53 OF 512
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The schematic illustrates three distinct current sensing and filtering stages:

- PSU CURRENT SENSE:** Utilizes an INA214 op-amp (U5403) to monitor the power supply unit (PP12V G3H PSU). It features a critical resistor R5493 (0.002Ω) and a capacitor C5418 (0.1μF). The output is labeled DCIN IOUT.
- DCIN current Sense Filter:** Employs an INA214 op-amp (U5403) for the DC input voltage sense (PP12V G3H). It includes a critical resistor R5433 (4.53kΩ) and a capacitor C5437 (0.22μF). The output is labeled SMC DCIN ISENSE.
- CPU / DDR HIGH SIDE CURRENT SENSE / FILTER:** Uses an INA213 op-amp (U5400) for high-side current sensing on the CPU/DDR lines (PP12V G3H CPUDDR). It features a critical resistor R5400 (0.005Ω) and a capacitor C5401 (0.1μF). The output is labeled SMC CPUDDR HI ISENSE.

Each circuit also includes a MOSFET (Q5415) for switching and various passive components like resistors (R5411, R5412, R5413, R5414) and capacitors (C5414, C5416) for signal conditioning and filtering. The outputs are connected to the SMC (System Management Controller) via pins 46, 47, 50, and 51.

PSU CURRENT SENSE

CRITICAL R5493 0.002 1W 1% MF 1206

PP12V G3H PSU

SIGNAL_MODEL=EMPTY

ISNS P12VG3H N 5 IN-

ISNS P12VG3H P 4 IN+

U5403 INA214 SC70

PP3V3 S0

CRITICAL C5418 0.1UF 10V X7R-CERM 402

DCIN IOUT

R5433 4.53K 1% MF 1206

SMC DCIN ISENSE

C5437 0.22UF 10V CERM-X5R 402

GND SMC AVSS

GAIN: 100X

SCALE: 5A / V

MAX VOUT: 3.3V AT 16.5A

Place RC close to SMC

CPU / DDR HIGH SIDE CURRENT SENSE / FILTER

CRITICAL R5400 0.005 1W 1% MF 0812

PP12V G3H

ISNS HS CPUDDR N 5 IN-

ISNS HS CPUDDR P 4 IN+

U5400 INA213 SC70

PP3V3 S0

CRITICAL C5401 0.1UF 10V X7R-CERM 402

HS CPUDDR IOUT

R5403 4.53K 1% MF 1206

SMC CPUDDR HI ISENSE

C5403 0.22UF 10V CERM-X5R 402

GND SMC AVSS

Gain: 50x

Scale: 4A / V

Max Vout: 3.3V at 13.2A

Place RC close to SMC

DC-In Voltage Sense Enable & Filter

TURN ON DIVIDER ONLY IN S0.

Max Vout: 3.3V at 19.77V Input

Q5415 DMG1016V SOT-563 P-CHN

DCIN S5 VSENSE

R5413 27.4K 1% MF 1206

PLACE_NEAR=U4900.N9:5MM

RTHEVENIN = 4573 Ohms

SMC DCIN VSENSE

R5414 5.49K 1% MF 1206

C5414 0.22UF 10V CERM-X5R 402

GND SMC AVSS

R5411 100K 1% MF 1206

PP12V G3H

PDCINSENS EN L DIV

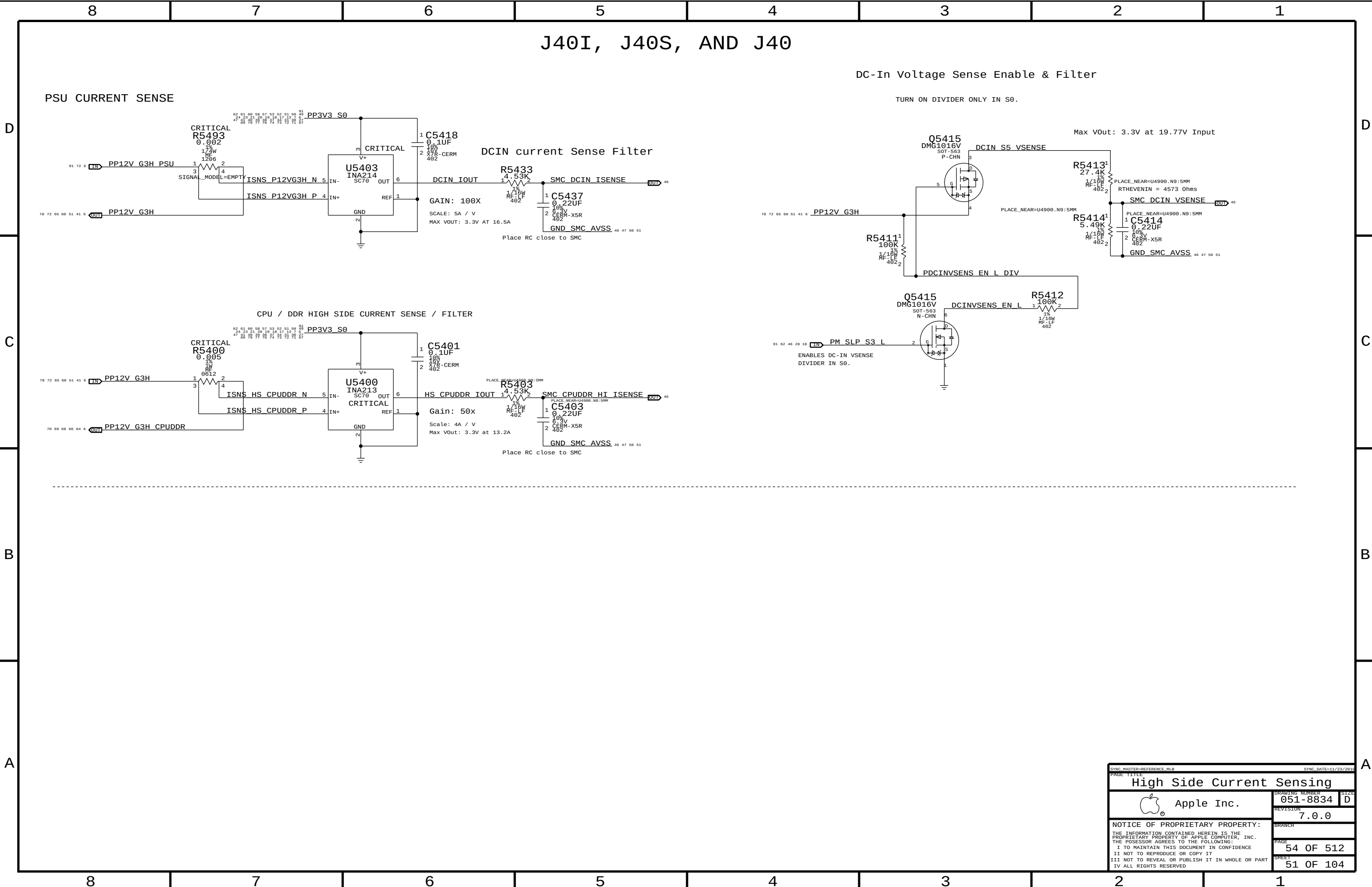
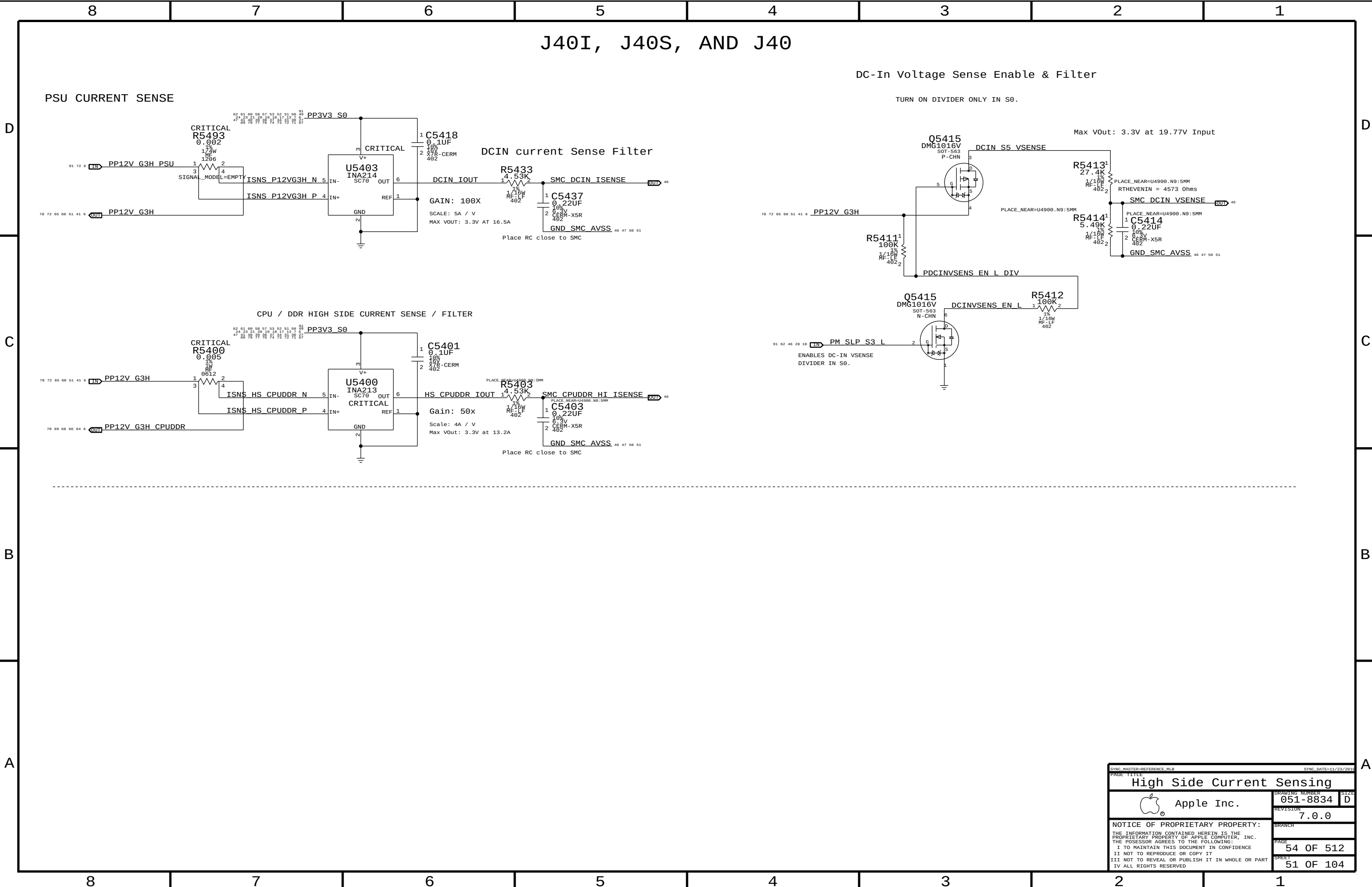
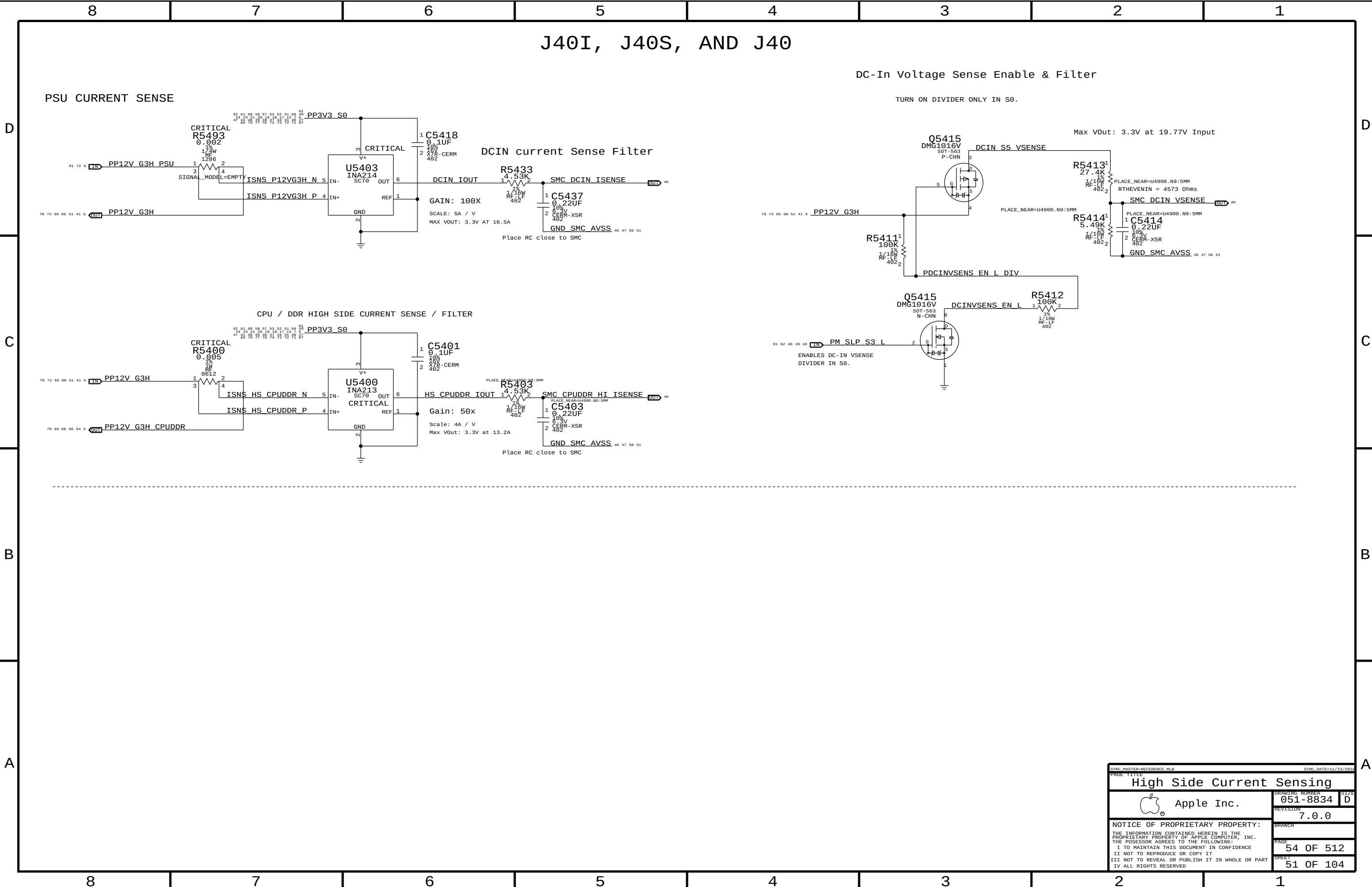
Q5415 DMG1016V SOT-563 N-CHN

DCINSENS EN L

R5412 100K 1% MF 1206

PM SLP S3 L

ENABLES DC-IN VSENSE DIVIDER IN S0.

[illegible][illegible][illegible][illegible]

J40I, J40S, AND J40

PSU CURRENT SENSE

DCIN current Sense Filter

CPU / DDR HIGH SIDE CURRENT SENSE / FILTER

DC-In Voltage Sense Enable & Filter

TURN ON DIVIDER ONLY IN S0.

Max VOut: 3.3V at 19.77V Input

ENABLES DC-IN VSENSE DIVIDER IN S0.

High Side Current Sensing

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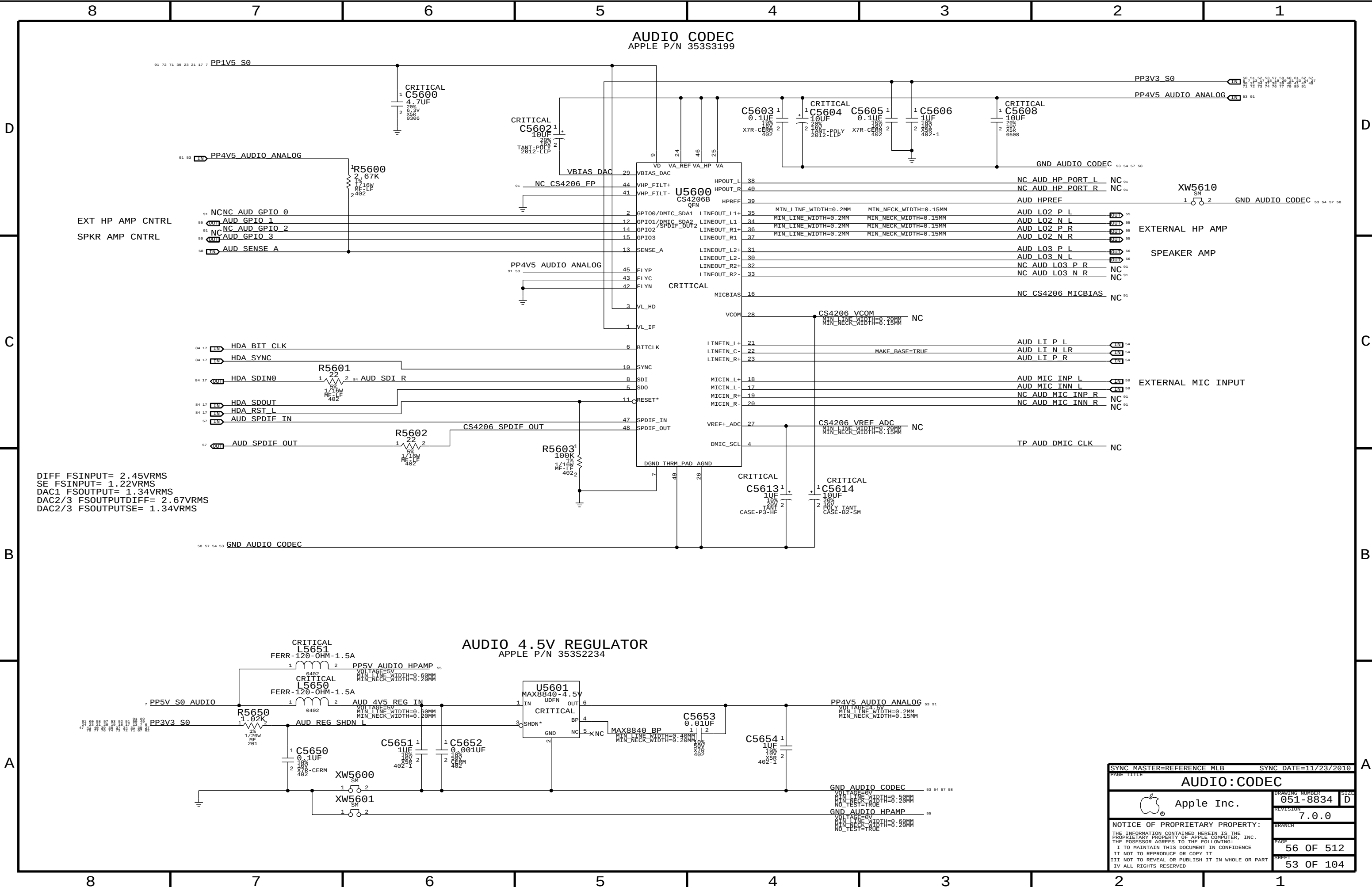
DRAWING NUMBER: 051-8834

REVISION: 7.0.0

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SHEET: 51 OF 104



DIFF FSINPUT= 2.45VRMS
SE FSINPUT= 1.22VRMS
DAC1 FSOUTPUT= 1.34VRMS
DAC2/3 FSOUTPUTDIFF= 2.67VRMS
DAC2/3 FSOUTPUTSE= 1.34VRMS

PAGE TITLE		SYNC DATE=11/23/2010	
AUDIO:CODEC		DRAWING NUMBER	051-8834
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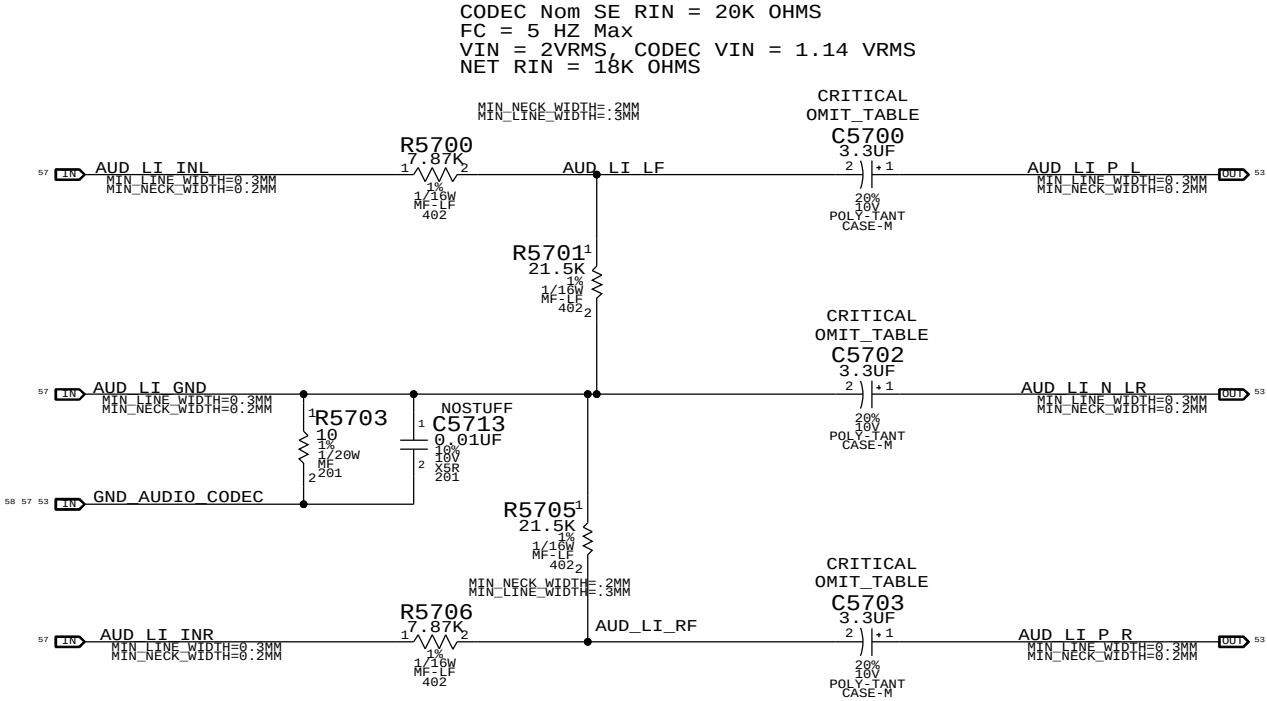
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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
128S9399	3	CAP, TANT, POLY, 4.7UF, 20%, 16V, SMD	C5700, C5702, C5703	CRITICAL	?

SYNC MASTER=REFERENCE MLB

SYNC DATE=11/23/2016

AUDIO: LINE - IN

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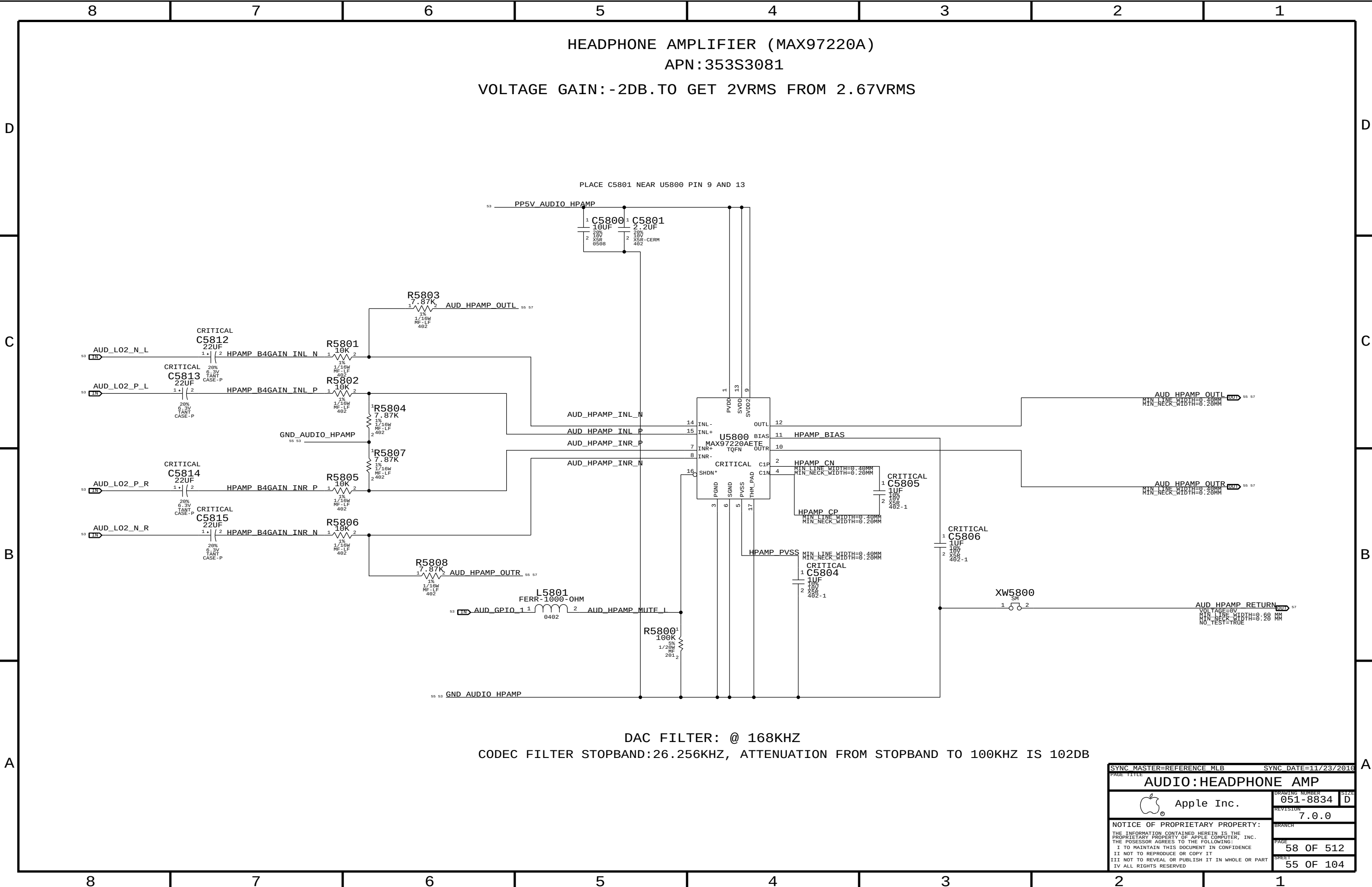
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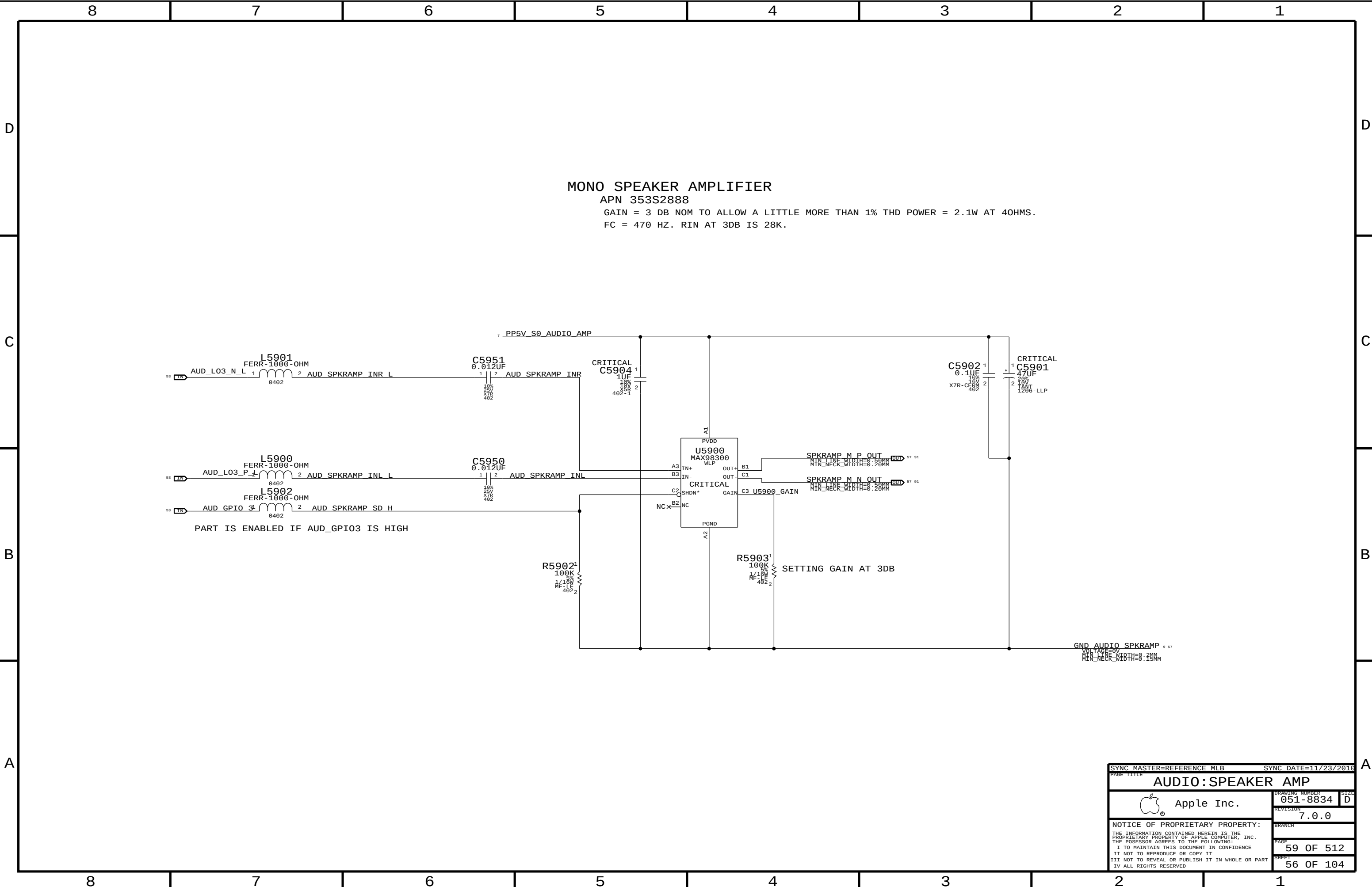
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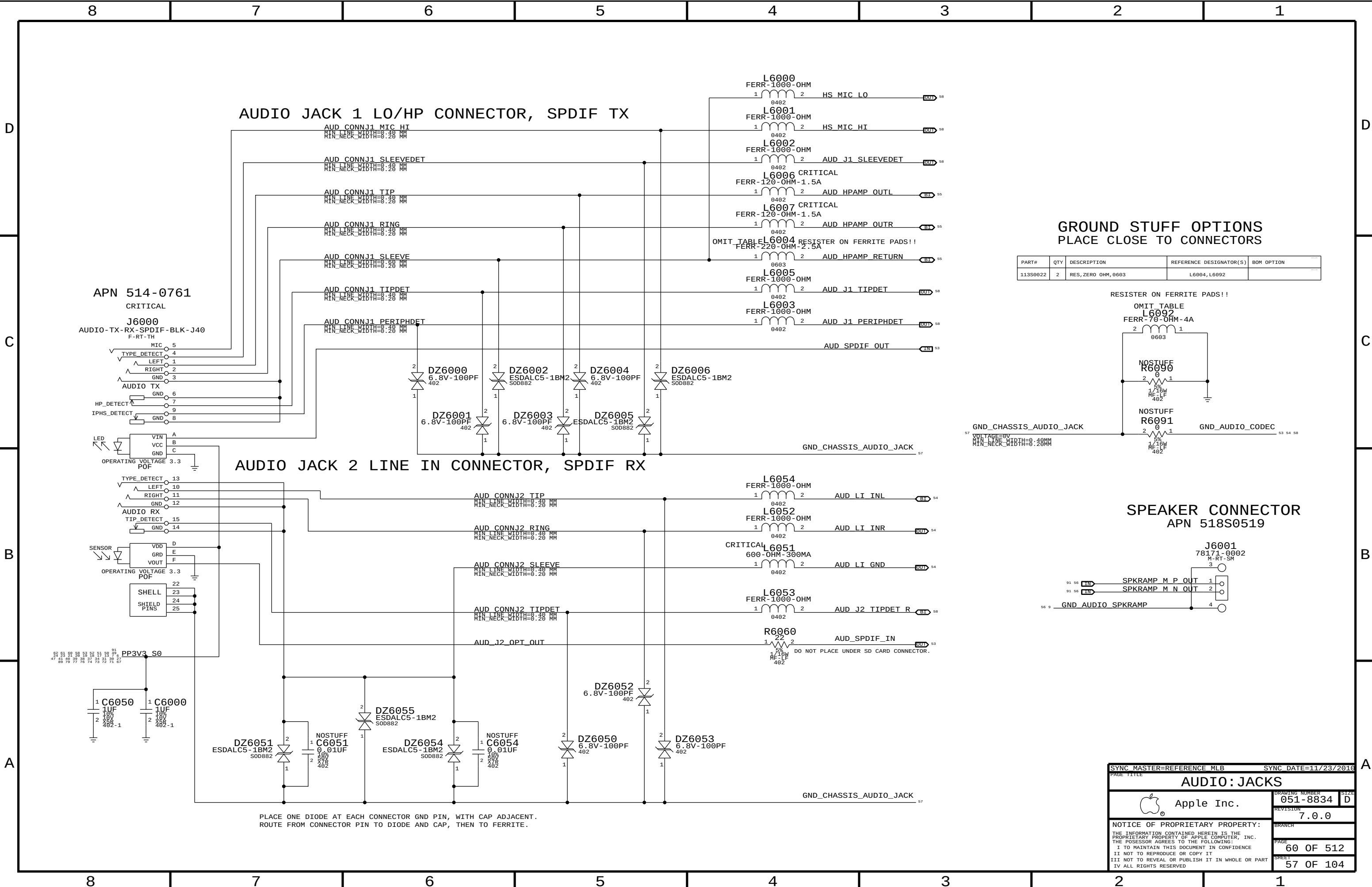
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PAGE TITLE		SYNC MASTER=REFERENCE MLB		SYNC DATE=11/23/2016	
AUDIO:HEADPHONE AMP		DRAWING NUMBER		SIZE	
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GROUND STUFF OPTIONS
PLACE CLOSE TO CONNECTORS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
113S0022	2	RES,ZERO OHM,0603	L6004,L6092	

RESISTER ON FERRITE PADS!!

OMIT TABLE
L6092

FERR-70-OHM-4A

NOSTUFF
R6090

NOSTUFF
R6091

1/10W MF-LF 402

1/10W MF-LF 402

1/10W MF-LF 402

1/10W MF-LF 402

1/10W MF-LF 402

1/10W MF-LF 402

1/10W MF-LF 402

1/10W MF-LF 402

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1/10W MF-LF 402

1/10W MF-LF 402

SPEAKER CONNECTOR
APN 518S0519

J6001

78171-0002

M-RT-SM

3

SPKRAMP M P OUT

SPKRAMP M N OUT

4

GND AUDIO SPKRAMP

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PLACE ONE DIODE AT EACH CONNECTOR GND PIN, WITH CAP ADJACENT.
ROUTE FROM CONNECTOR PIN TO DIODE AND CAP, THEN TO FERRITE.

SYNC MASTER=REFERENCE MLB

SYNC DATE=11/23/2016

AUDIO: JACKS

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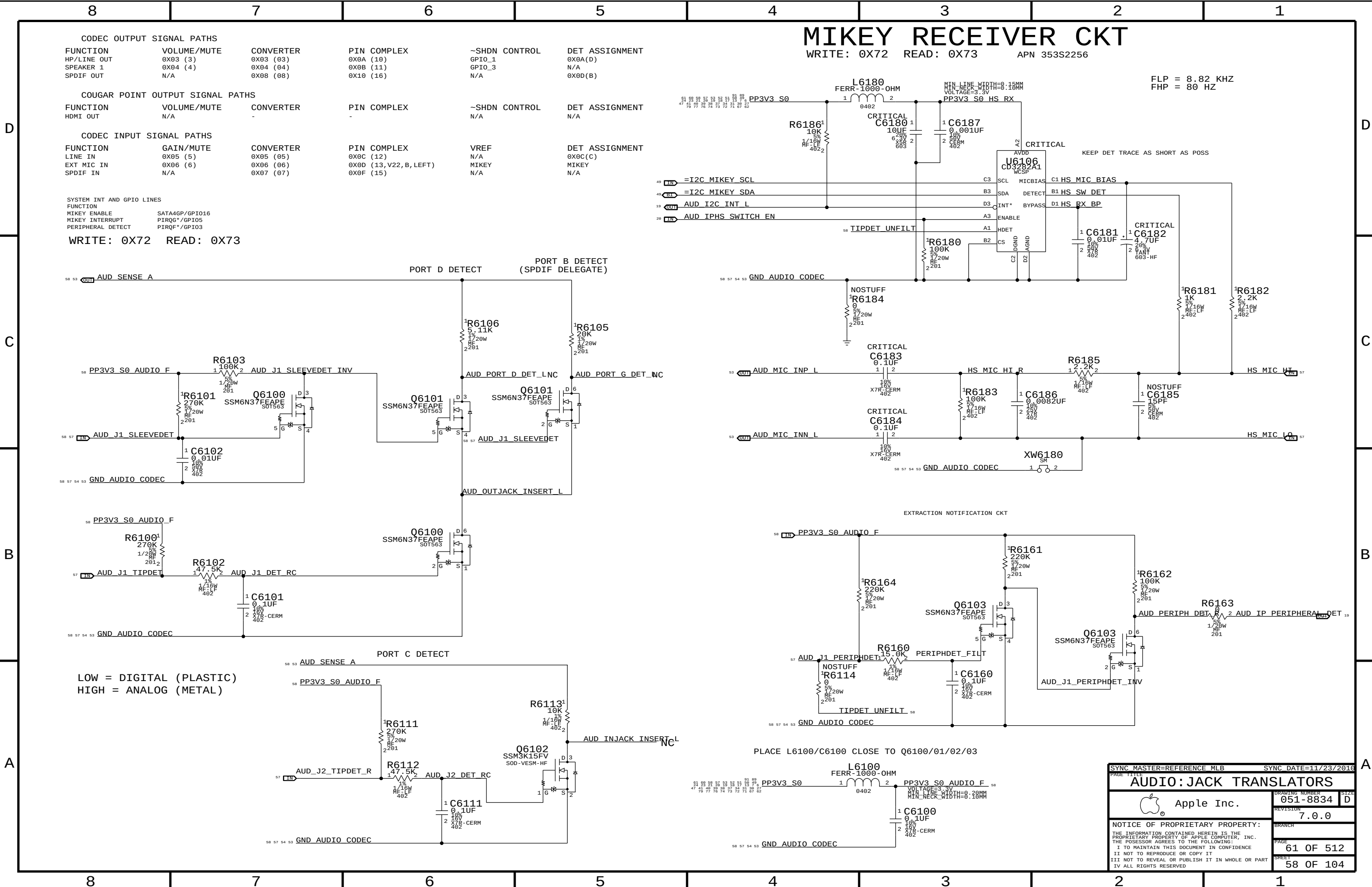
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MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S2256

FLP = 8.82 KHZ
FHP = 80 HZ

CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	~SHDN CONTROL	DET ASSIGNMENT
HP/LINE OUT	0X03 (3)	0X03 (03)	0X0A (10)	GPIO_1	0X0A(D)
SPEAKER 1	0X04 (4)	0X04 (04)	0X0B (11)	GPIO_3	N/A
SPDIF OUT	N/A	0X08 (08)	0X10 (16)	N/A	0X0D(B)

COUGAR POINT OUTPUT SIGNAL PATHS

FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	~SHDN CONTROL	DET ASSIGNMENT
HDMI OUT	N/A	-	-	N/A	N/A

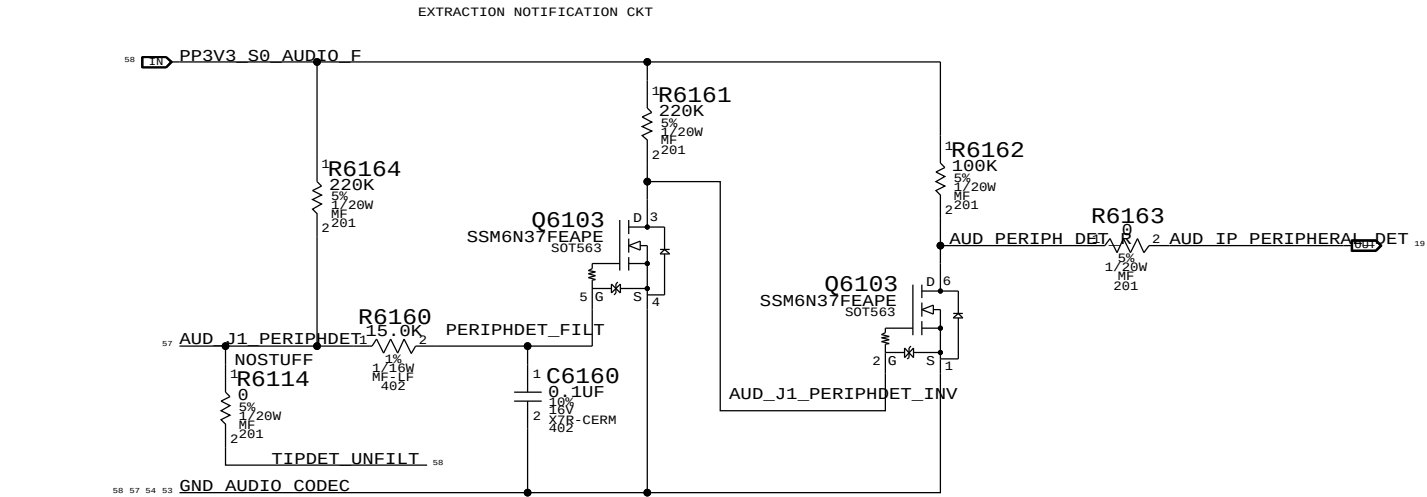
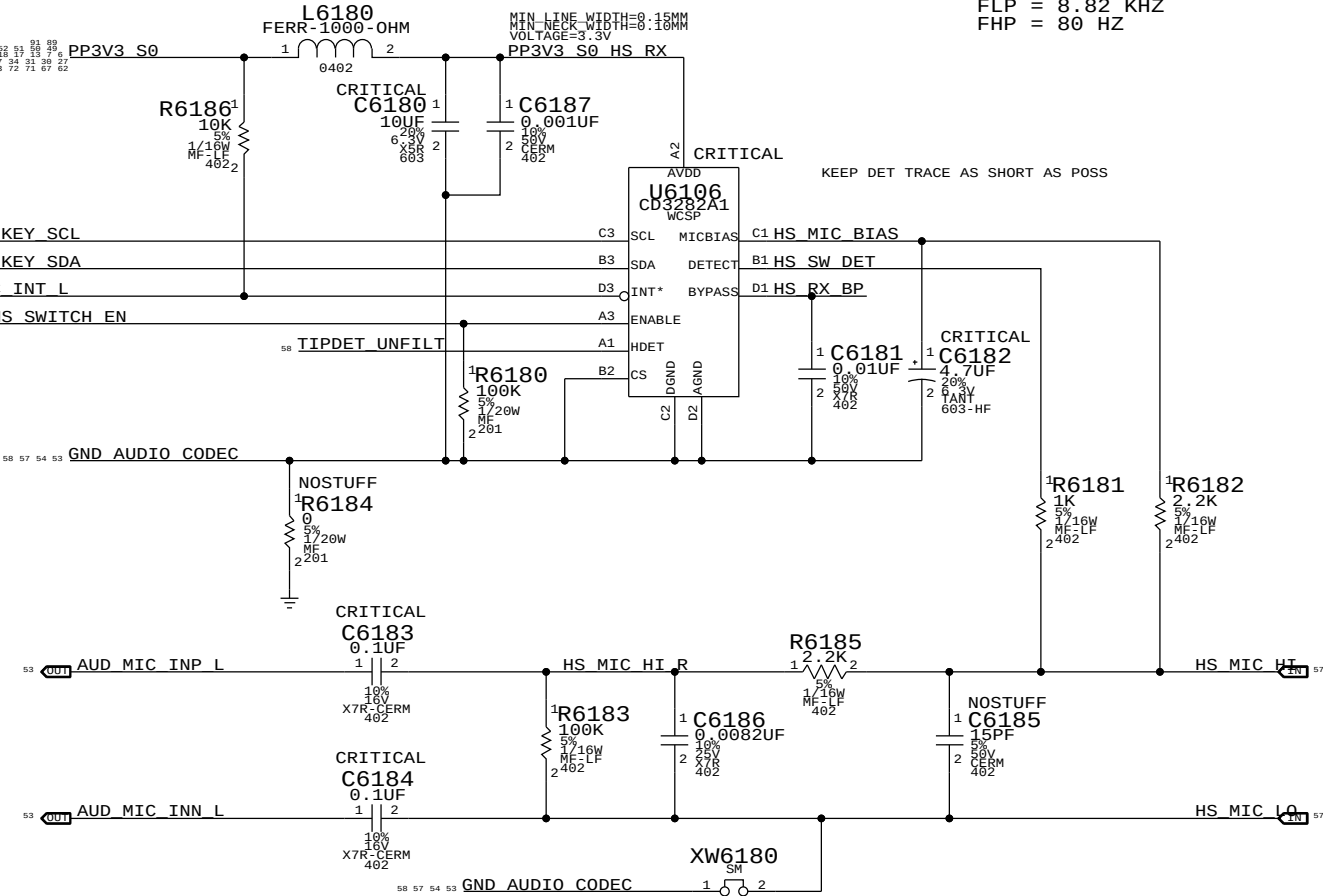
CODEC INPUT SIGNAL PATHS

FUNCTION	GAIN/MUTE	CONVERTER	PIN COMPLEX	VREF	DET ASSIGNMENT
LINE IN	0X05 (5)	0X05 (05)	0X0C (12)	N/A	0X0C(C)
EXT MIC IN	0X06 (6)	0X06 (06)	0X0D (13, V22, B, LEFT)	MIKEY	MIKEY
SPDIF IN	N/A	0X07 (07)	0X0F (15)	N/A	N/A

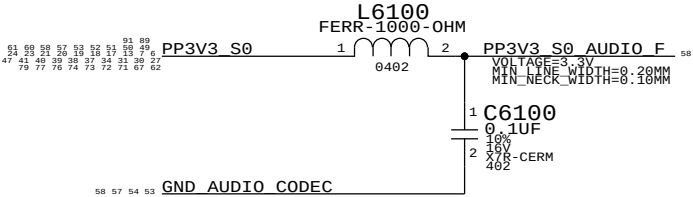
SYSTEM INT AND GPIO LINES

FUNCTION	
MIKEY ENABLE	SATA4GP/GPIO16
MIKEY INTERRUPT	PIRQG*/GPIO5
PERIPHERAL DETECT	PIRQF*/GPIO3

WRITE: 0X72 READ: 0X73



PLACE L6100/C6100 CLOSE TO Q6100/01/02/03



SYNC MASTER=REFERENCE MLB

SYNC DATE=11/23/2016

AUDIO: JACK TRANSLATORS

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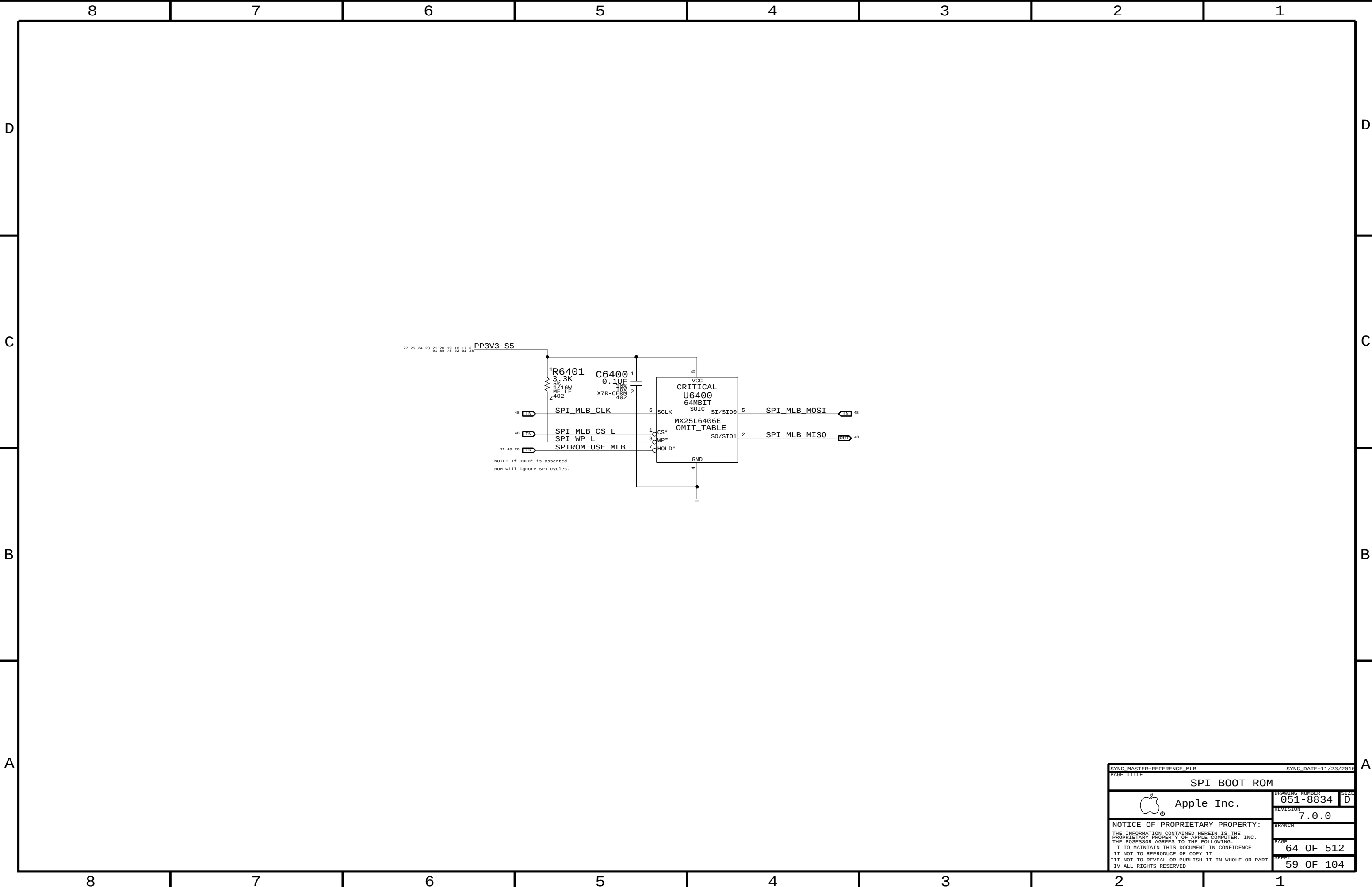
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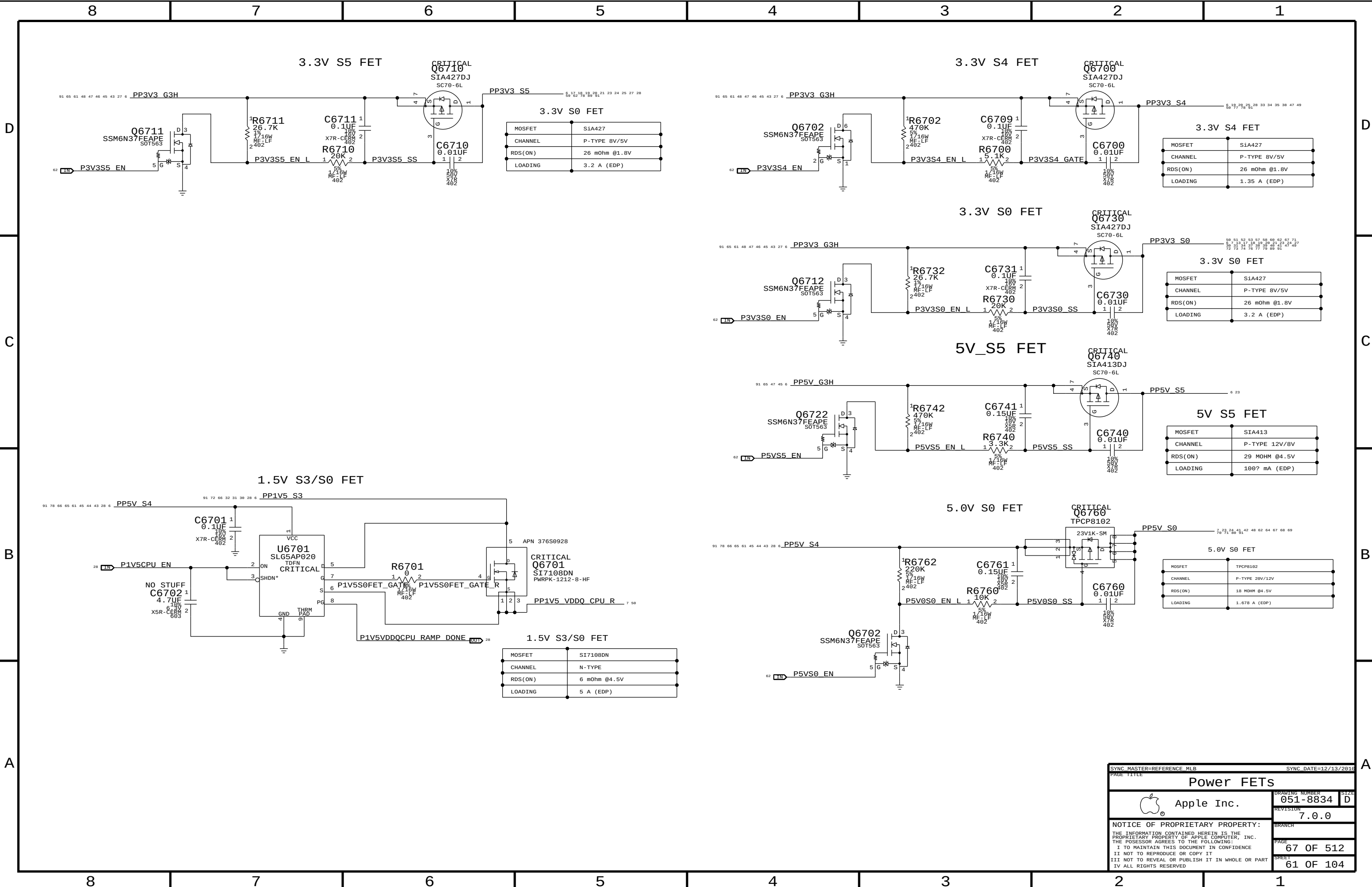
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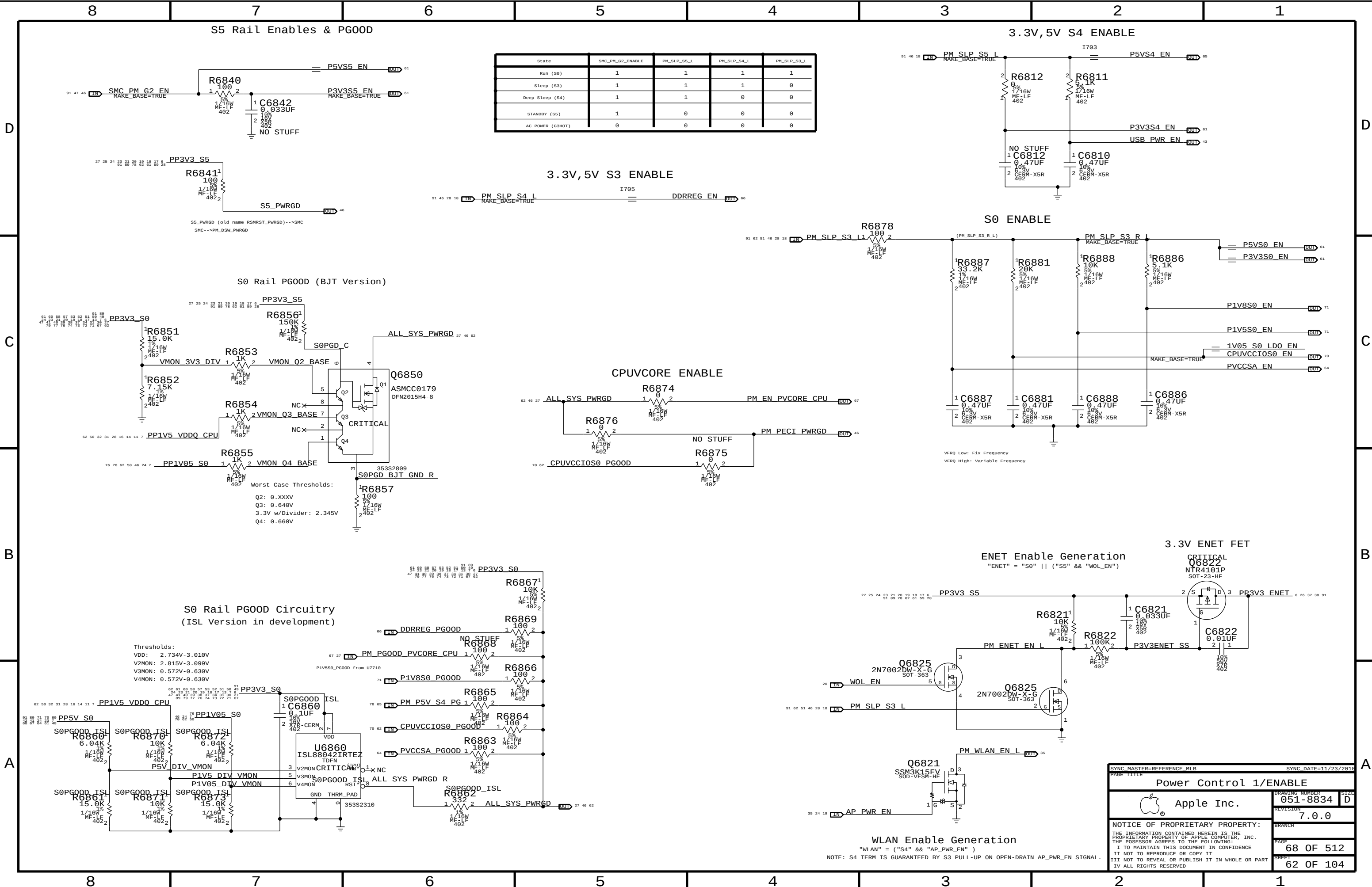
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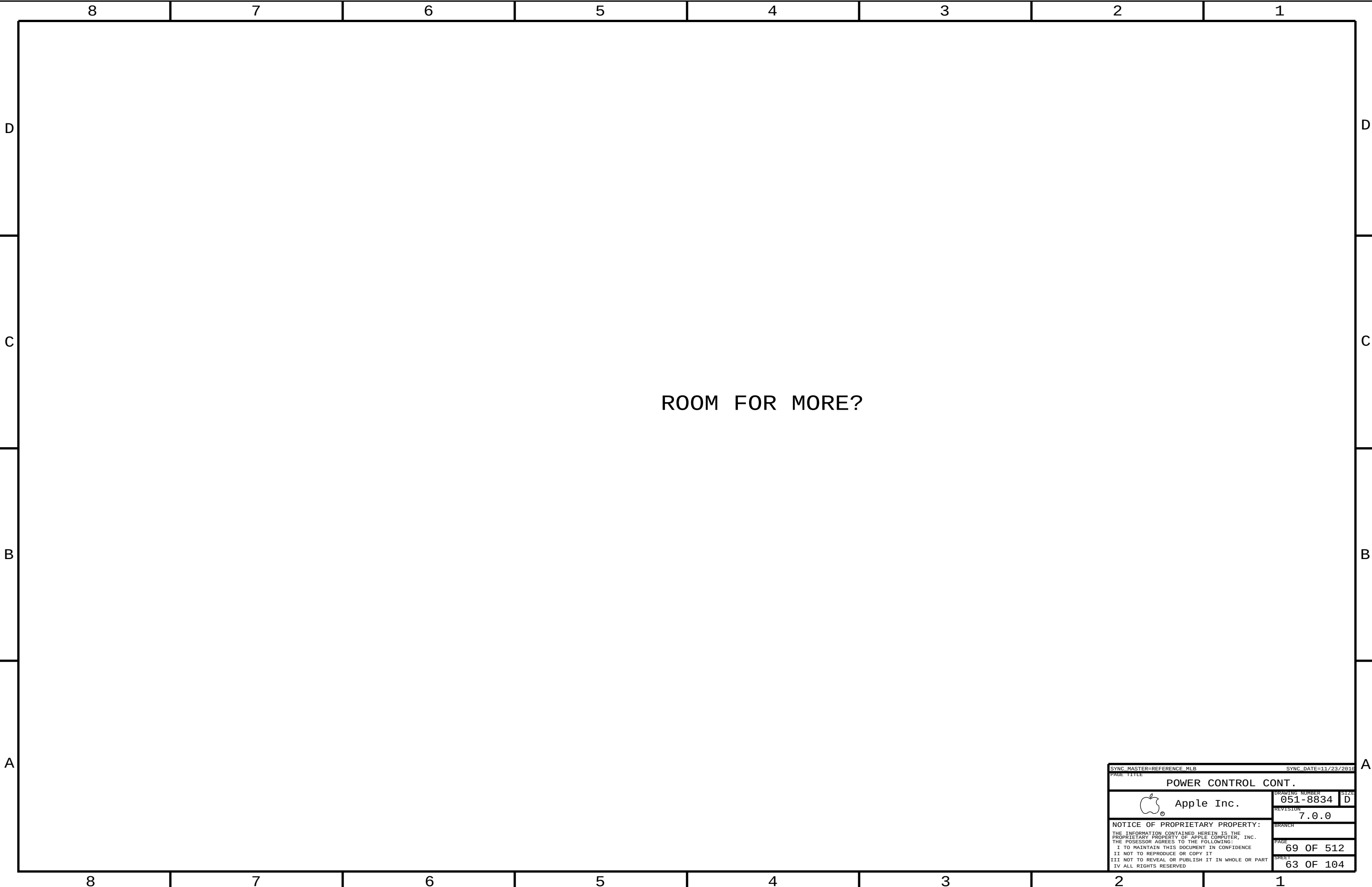
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State	SMC_PM_G2_ENABLE	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	1	1	1	1
Sleep (S3)	1	1	1	0
Deep Sleep (S4)	1	1	0	0
STANDBY (S5)	1	0	0	0
AC POWER (G3MGT)	0	0	0	0



SYNC MASTER=REFERENCE_MLB

SYNC DATE=11/23/2010

PAGE TITLE

POWER CONTROL CONT.

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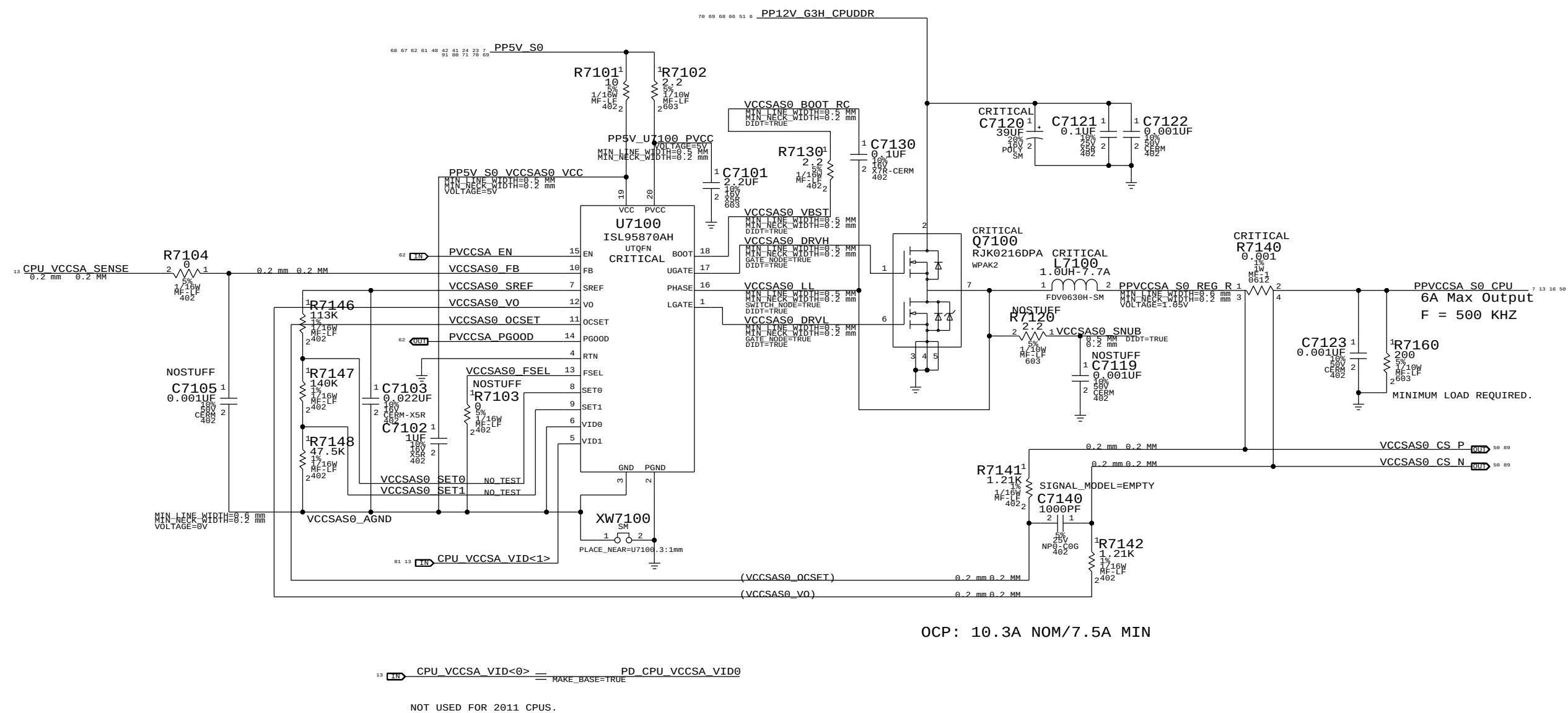
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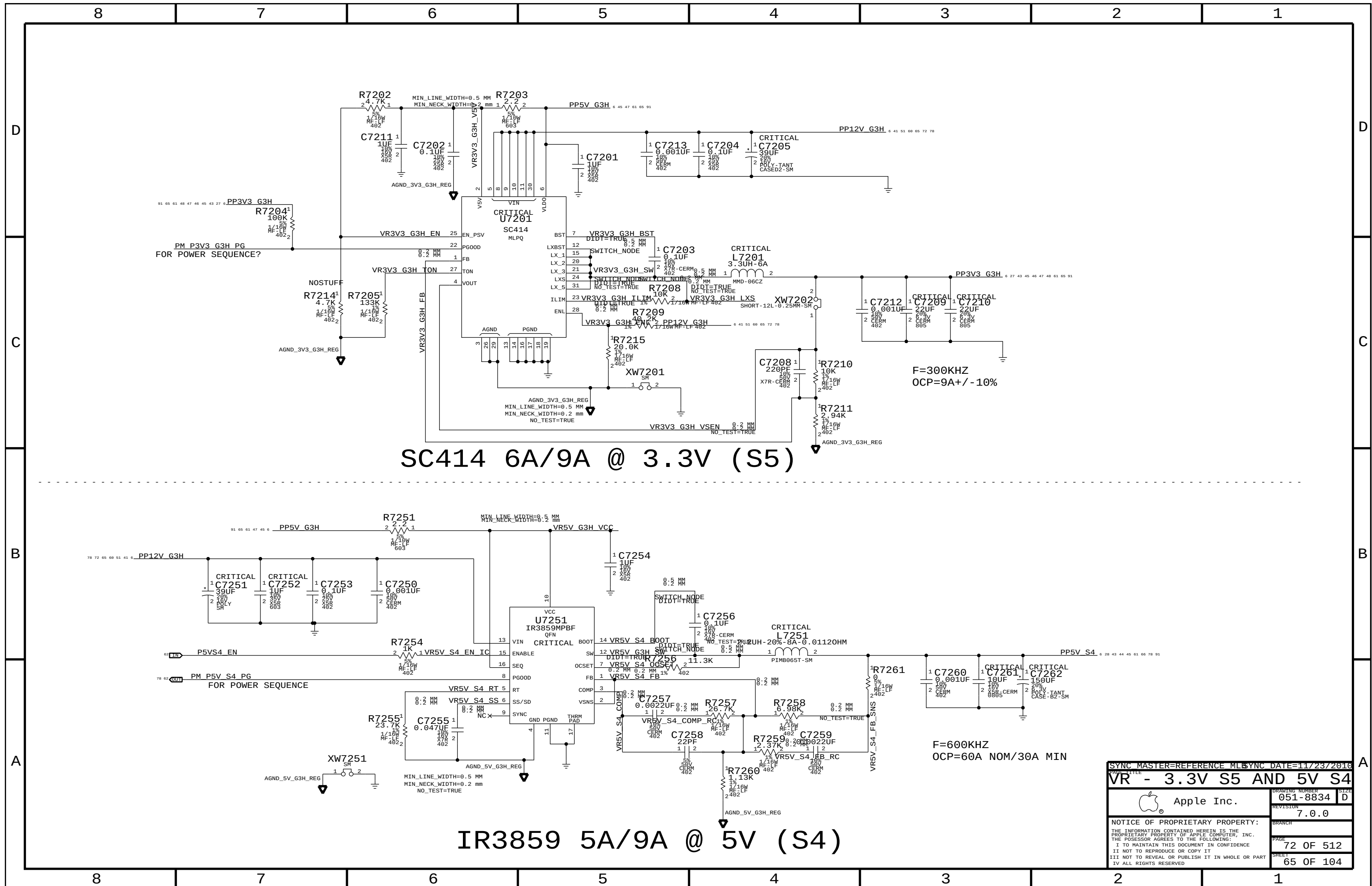
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System Agent Power Supply



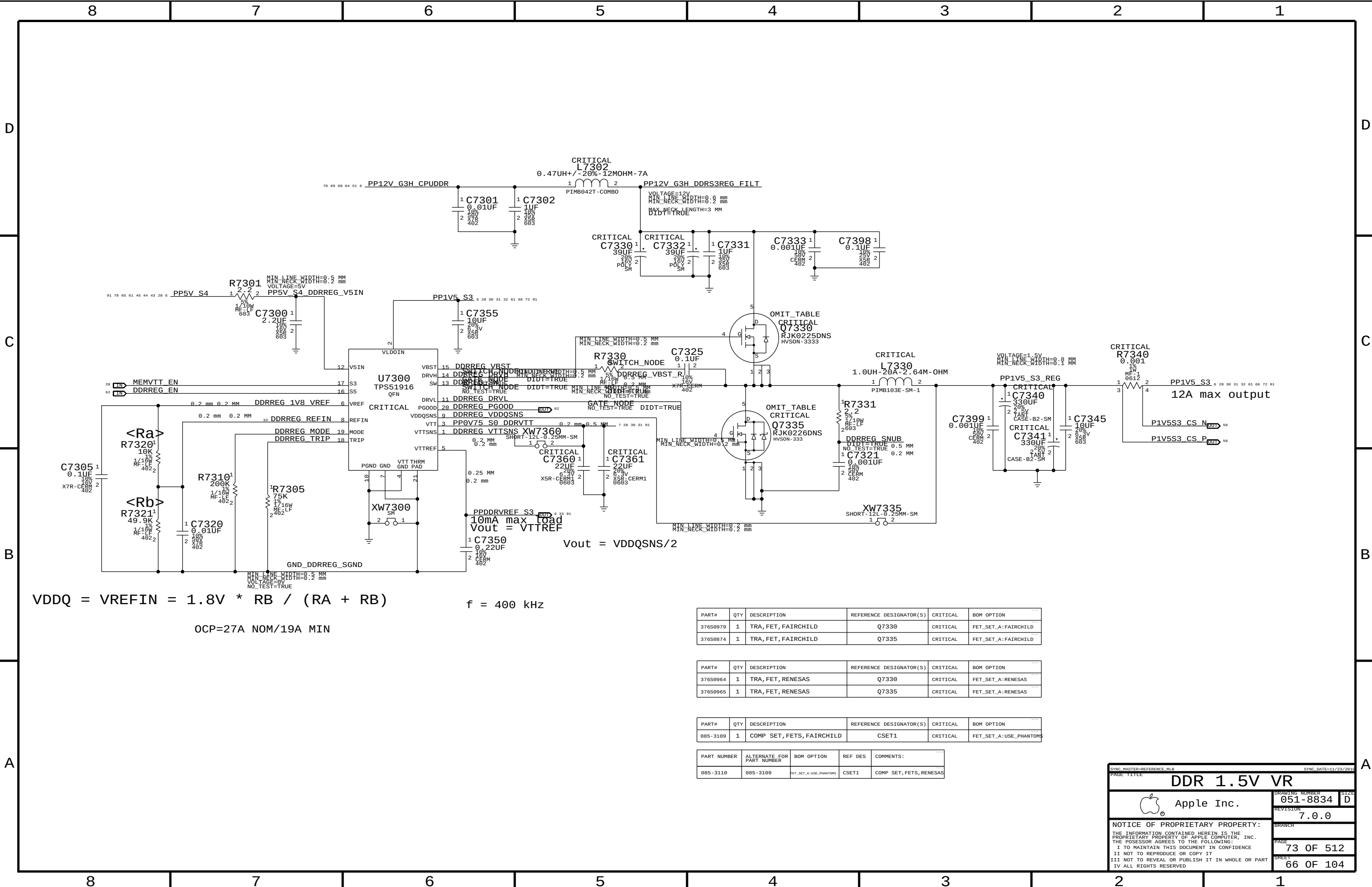
SYNC MASTER=REFERENCE MLB		SYNC DATE=11/23/2010	
PAGE TITLE			
VR - SYSTEM AGENT SUPPLY			
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SC414 6A/9A @ 3.3V (S5)

IR3859 5A/9A @ 5V (S4)

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VR - 3.3V S5 AND 5V S4		051-8834		
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$VDDQ = VREFIN = 1.8V * RB / (RA + RB)$

$f = 400 \text{ kHz}$

$OCP=27A \text{ NOM}/19A \text{ MIN}$

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
376S9979	1	TRA, FET, FAIRCHILD	Q7330	CRITICAL	FET_SET_A:FAIRCHILD
376S9874	1	TRA, FET, FAIRCHILD	Q7335	CRITICAL	FET_SET_A:FAIRCHILD

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
376S9964	1	TRA, FET, RENESAS	Q7330	CRITICAL	FET_SET_A:RENESAS
376S9965	1	TRA, FET, RENESAS	Q7335	CRITICAL	FET_SET_A:RENESAS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
085-3109	1	COMP SET, FETS, FAIRCHILD	CSET1	CRITICAL	FET_SET_A:USE_PHANTOMS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
085-3110	085-3109	FET_SET_A:USE_PHANTOMS	CSET1	COMP SET, FETS, RENESAS

SYNC MASTER-REFERENCE: HLB

SYNC DATE: 11/23/2011

DDR 1.5V VR

Apple Inc.

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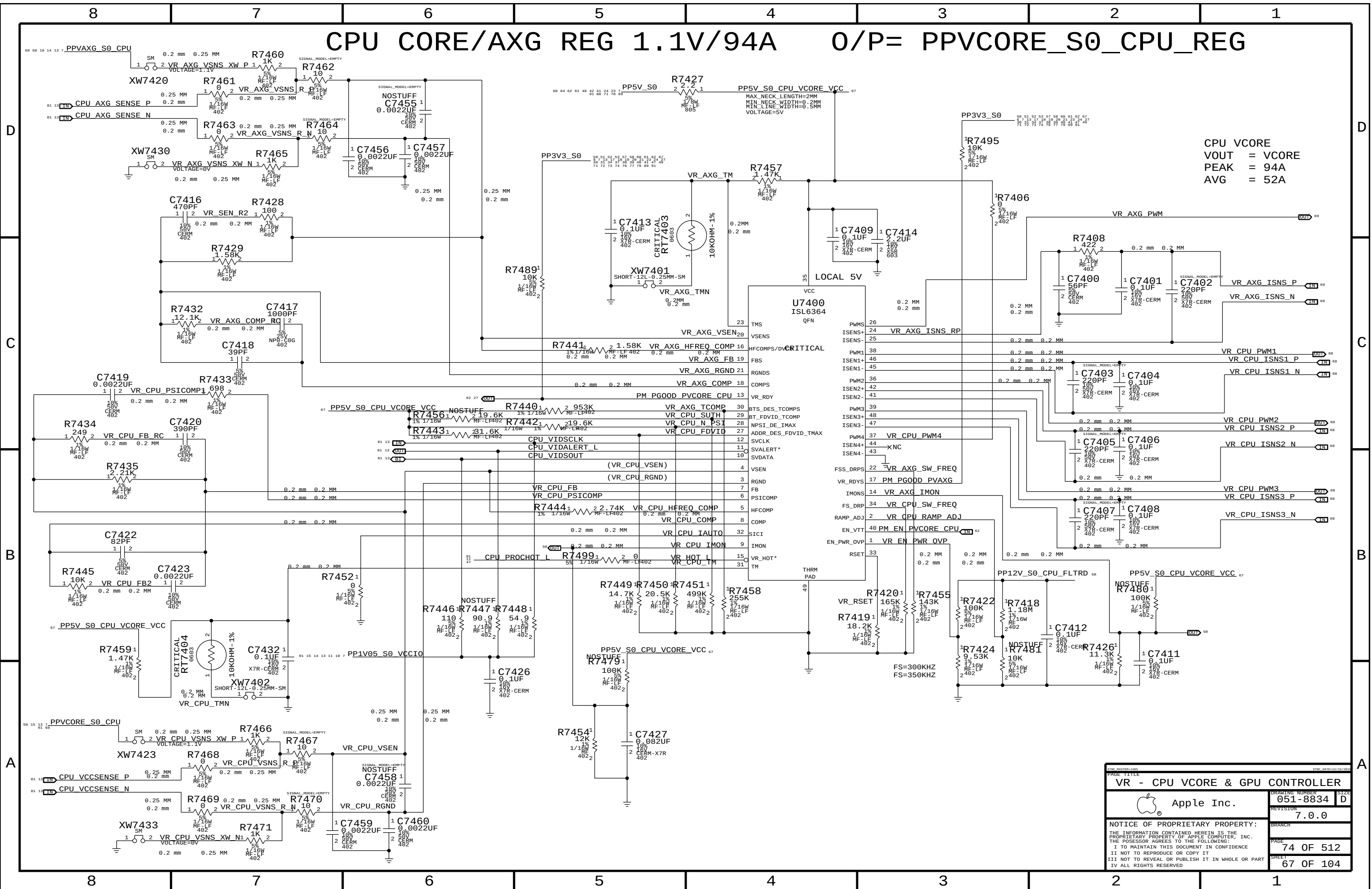
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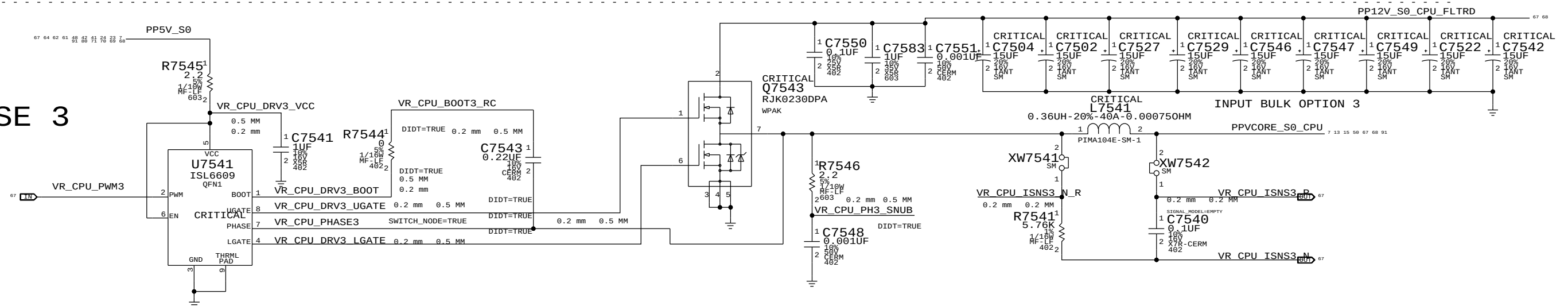
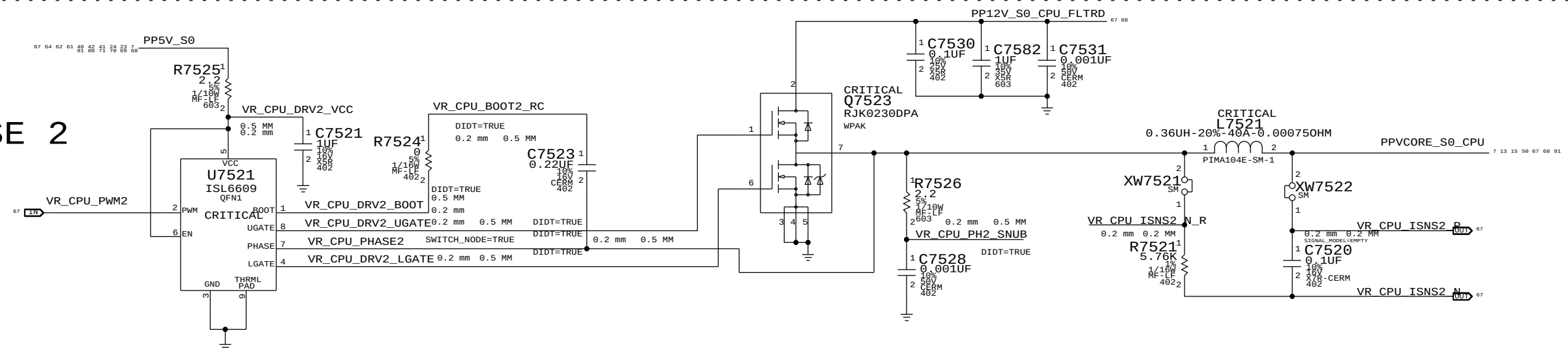
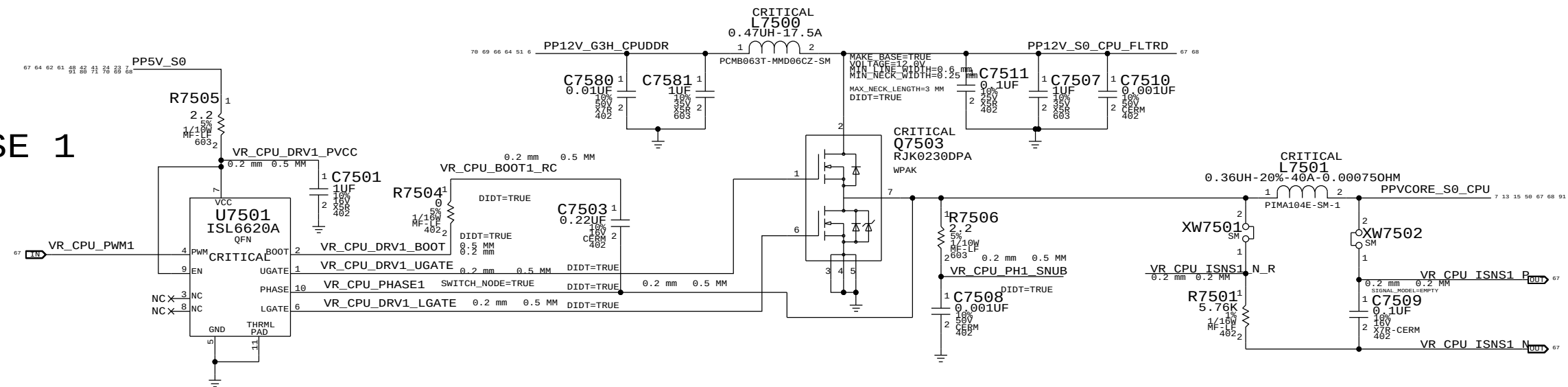
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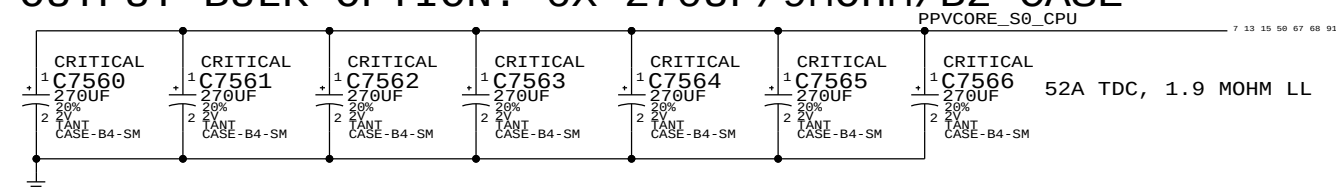
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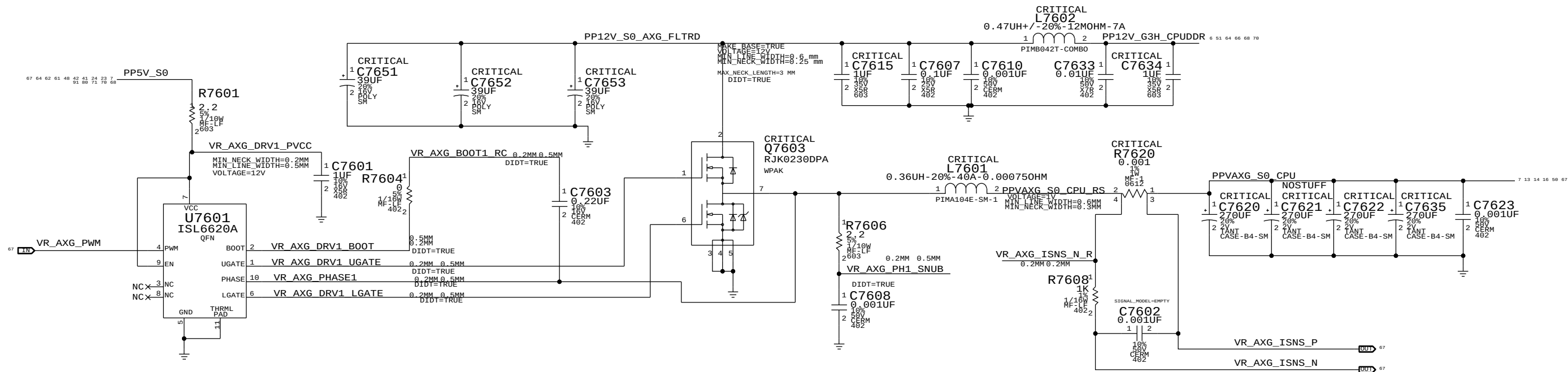


OUTPUT BULK OPTION: 6X 270UF/9MOHM/B2 CASE

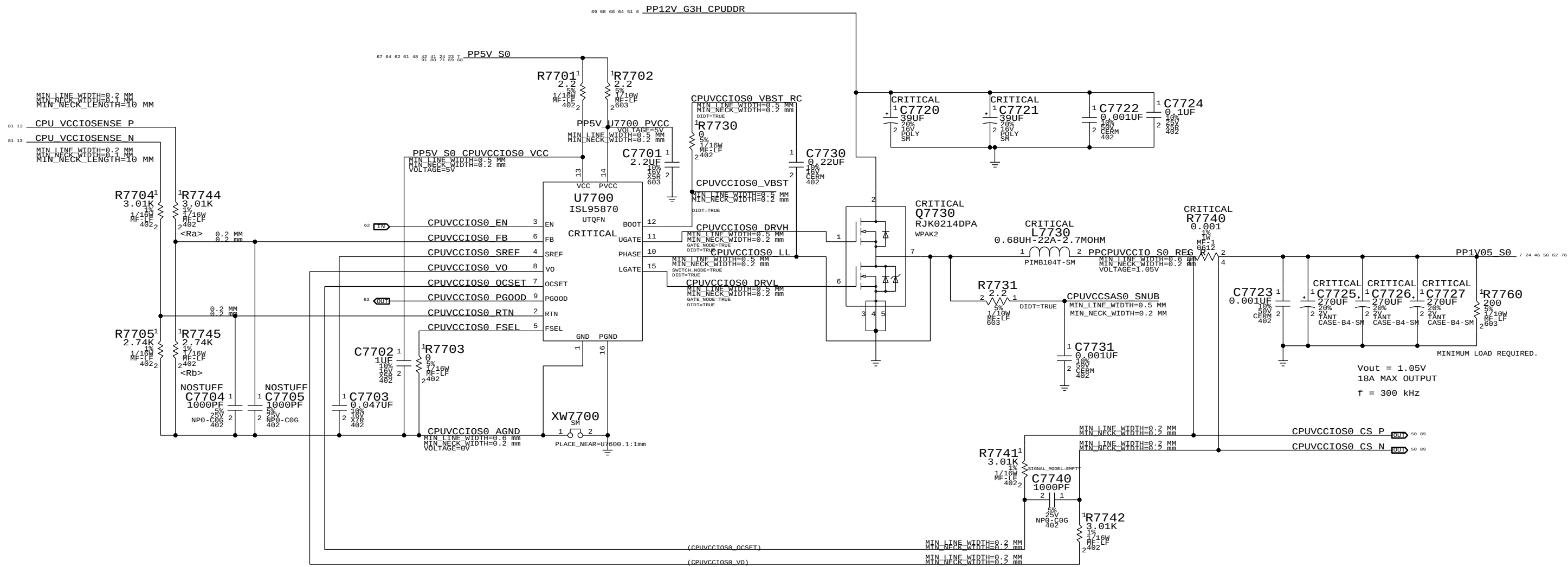


JOB NUMBER-1000 PAGE TITLE		DATE- 04/22/10 DATE- 04/22/10	
VR - CPU VCORE OUTPUT PHASES			
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AXG PHASE (22A AVG/33A PEAK, 3.9MOHM LL)

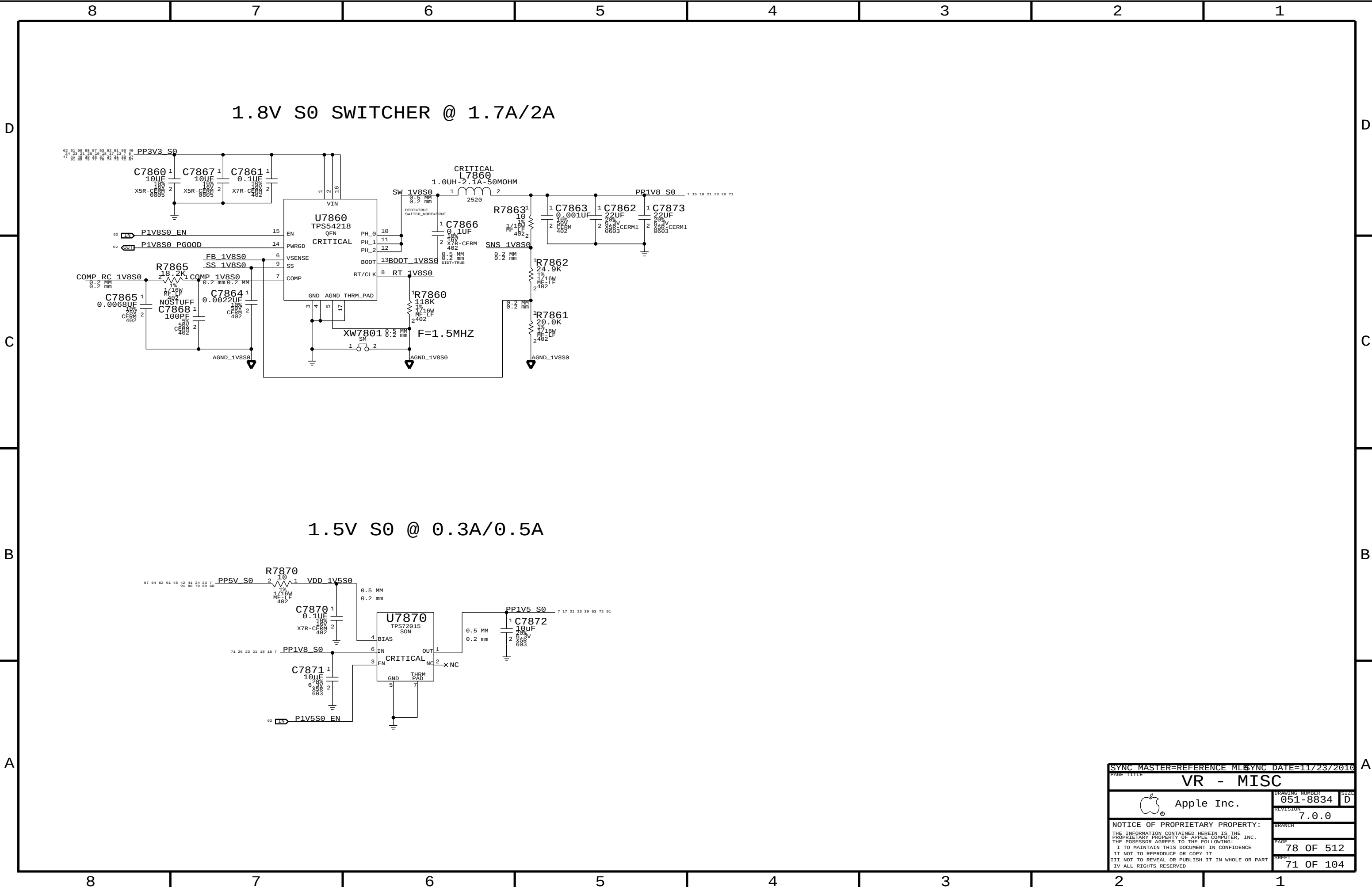


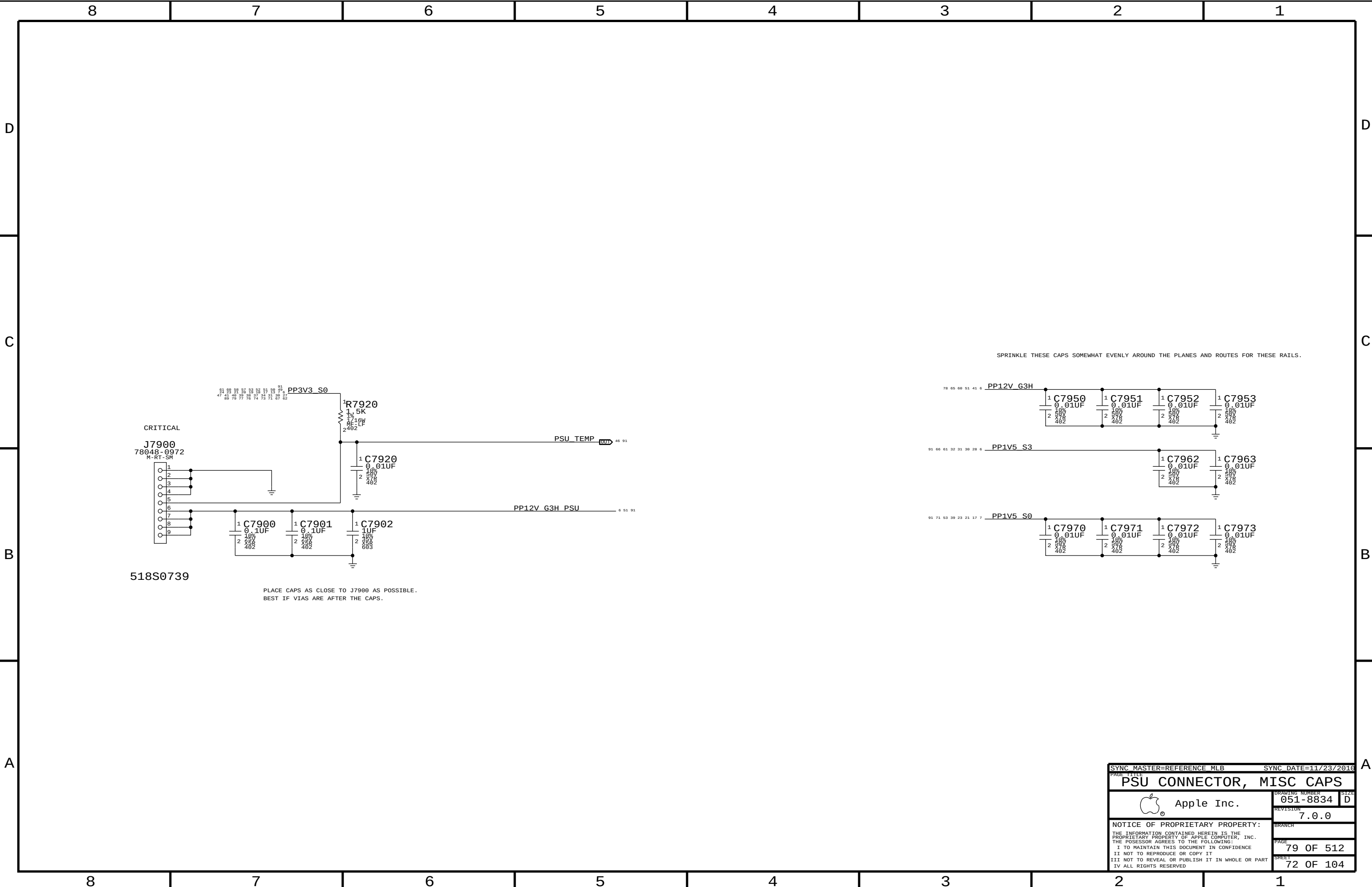
CPU VCCIO (1.05V S0) REGULATOR @ 18A/22A



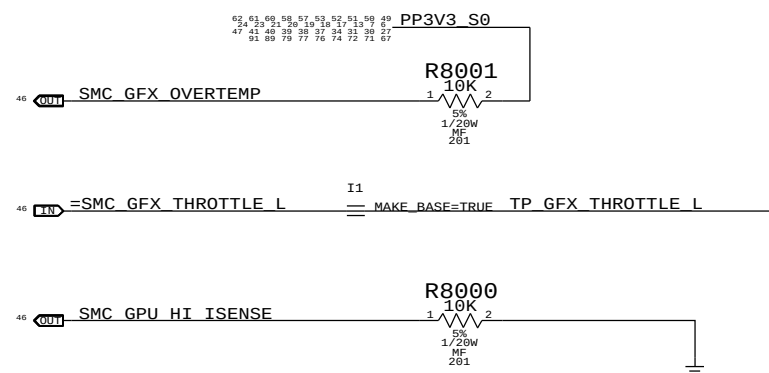
OCP = $R7641 \times 8.5\mu A / R7640$
OCP = 25.6A NOM/21.3A MIN
 $V_{out} = 0.5V \times (1 + R_a / R_b)$

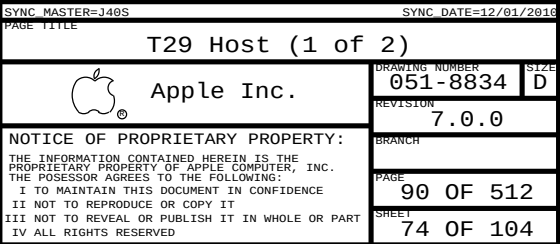
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VR - 1.05V S0			
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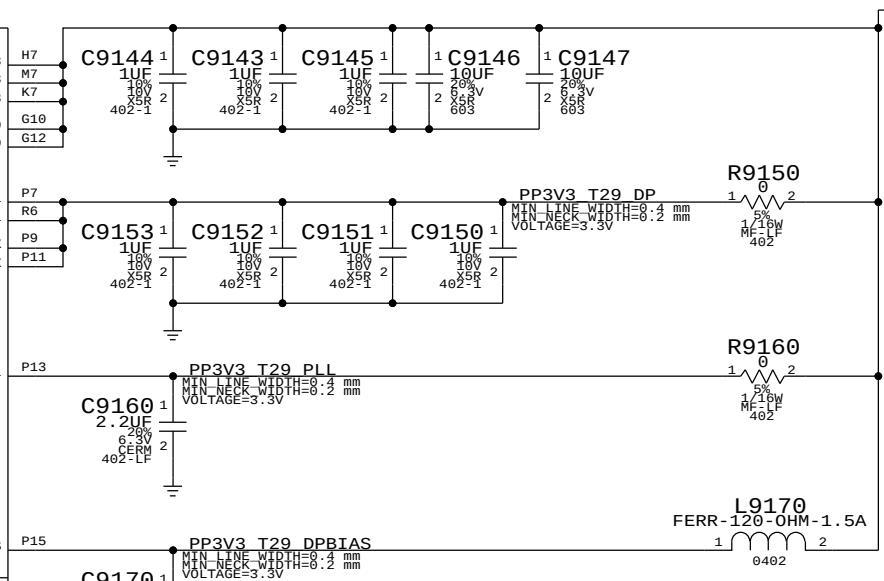
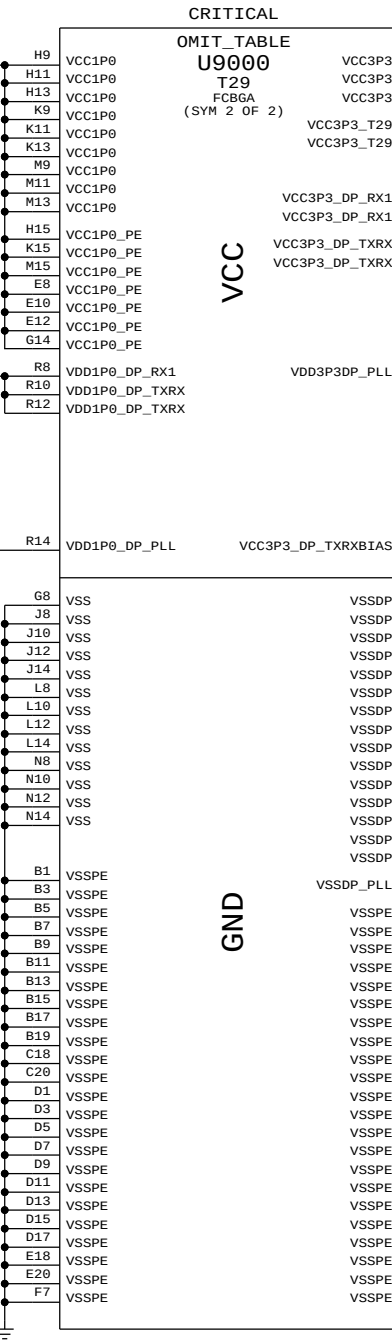
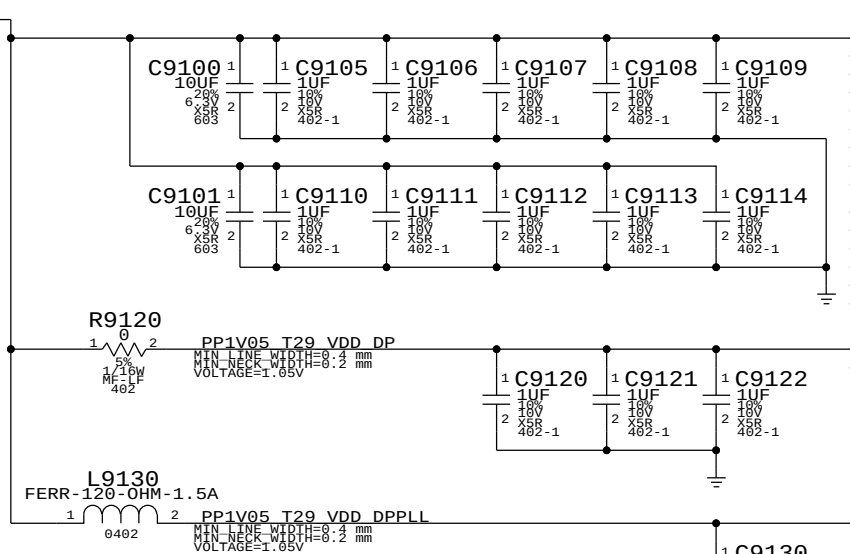
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8 7 6 5 4 3 2 1

76 PP1V05 T29
2100 mA (Single Port)
2250 mA (Dual Port)
EDP: 3000 mA



PP3V3 T29
135 mA (Single-Port)
152 mA (Dual-Port)
EDP: 200 mA

0-ohms are placeholders for now, replace with proper values after characterization.

8 7 6 5 4 3 2 1

SYNC MASTER=J46S		SYNC DATE=11/23/2010	
PAGE TITLE		T29 Host (2 of 2)	
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		PAGE	91 OF 512
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Page Notes

Power aliases required by this page:

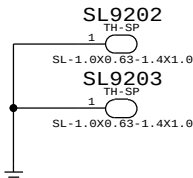
- PPVIN_SW_T29BST (8-13V Boost Input)
- PP18V_T29_REG (18V Boost Output)
- PP3V3_T29_P3V3T29FET (3.3V FET Input)
- PP3V3_T29_FET (3.3V FET Output)
- PP3V3_S0_T29PWRCTL
- PP1V05_T29_P1V05T29FET (1.05V FET Input)
- PP1V05_T29_FET (1.05V FET Output)

Signal aliases required by this page:

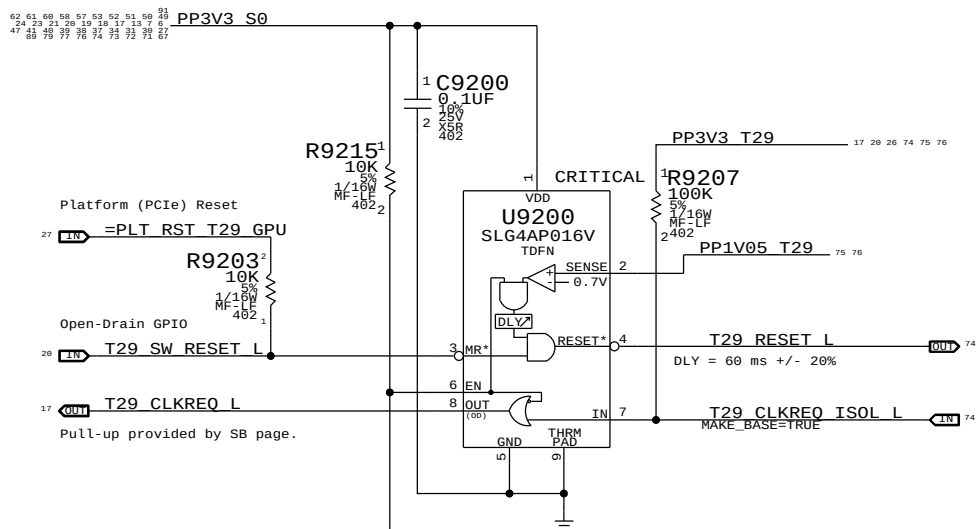
- T29_CLKREQ_L
- T29_RESET_L

BOM options provided by this page:

T29BST:Y - Stuffs 18V boost circuitry.

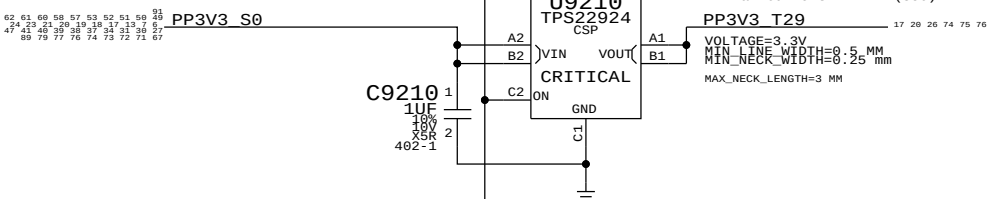


Supervisor & CLKREQ# Isolation



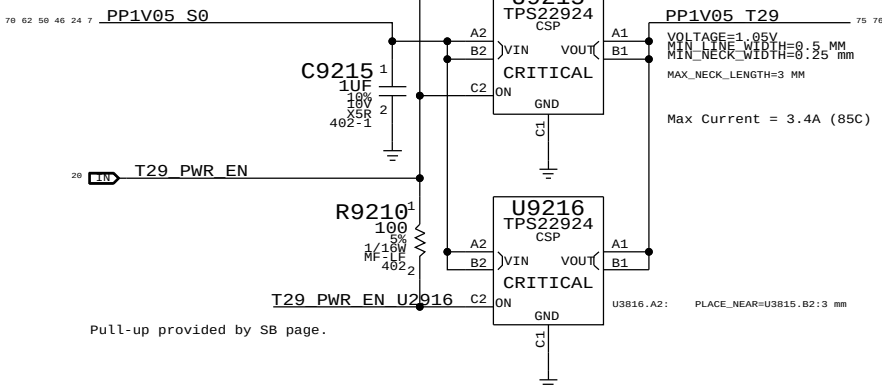
3.3V T29 Switch

Max Current = 1.7A (85C)



1.05V T29 Switch

Max Current = 3.4A (85C)



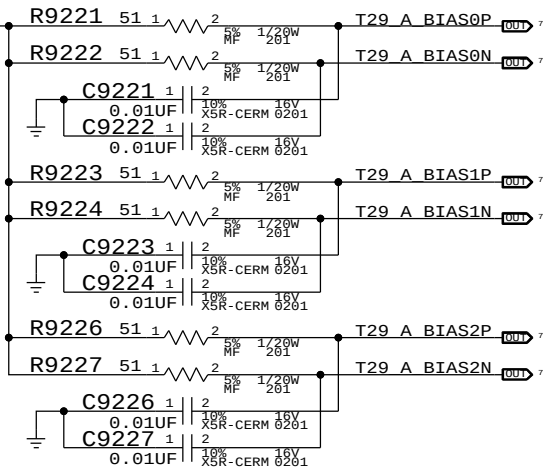
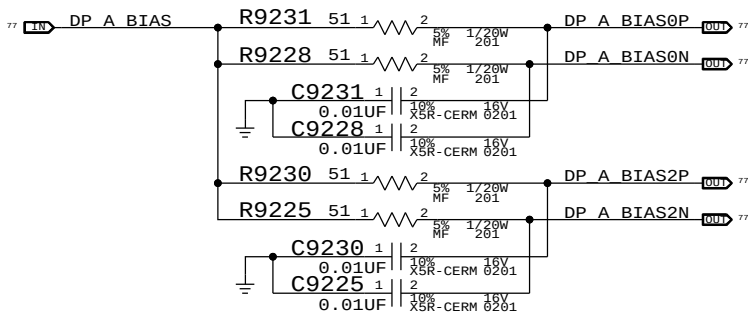
U9210 & U9215/U9216

Part	TPS22924C
Type	Load Switch
R(on)	18 mOhm Typ 50 mOhm Max

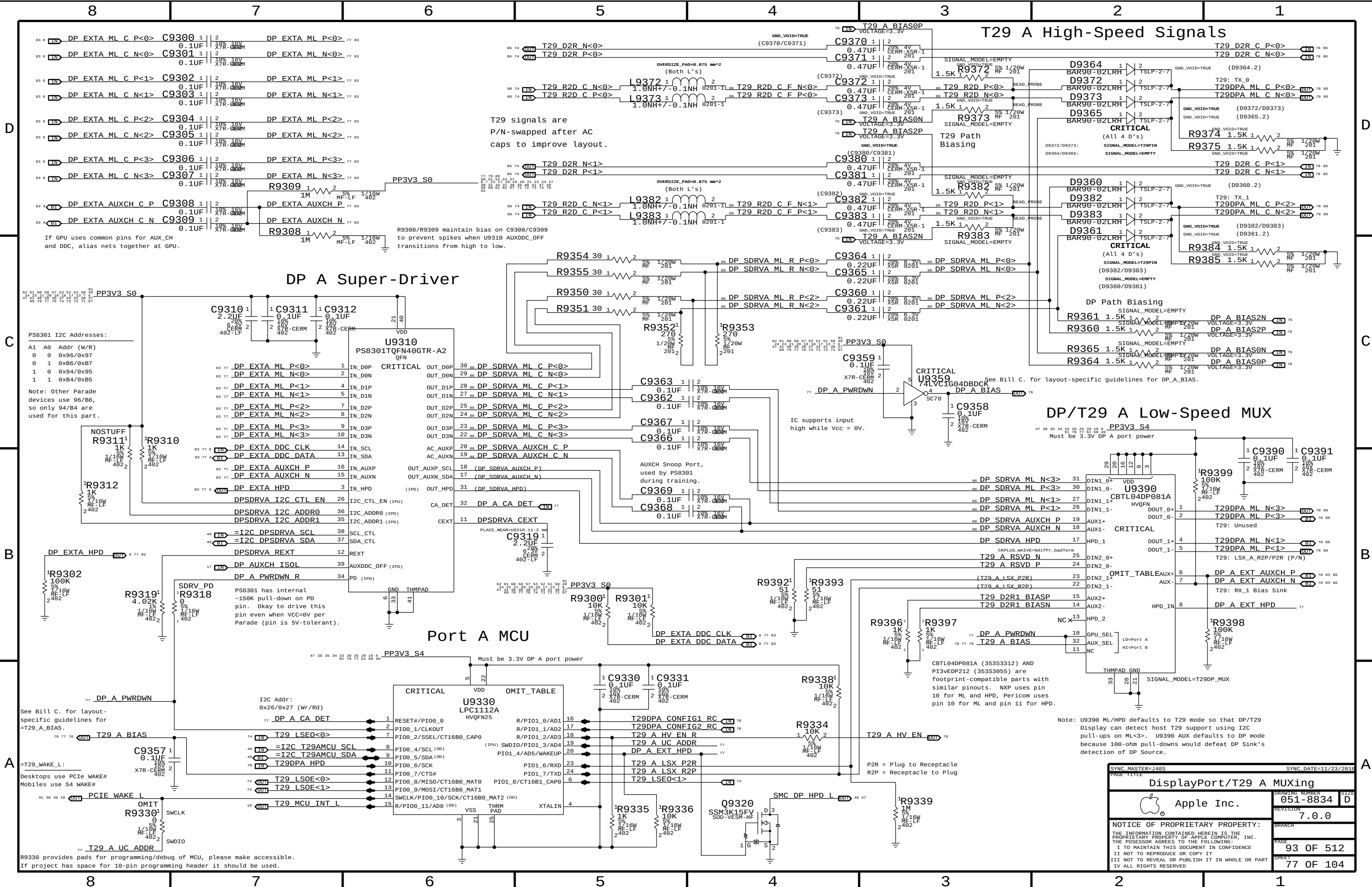
Max Output: 2A per IC

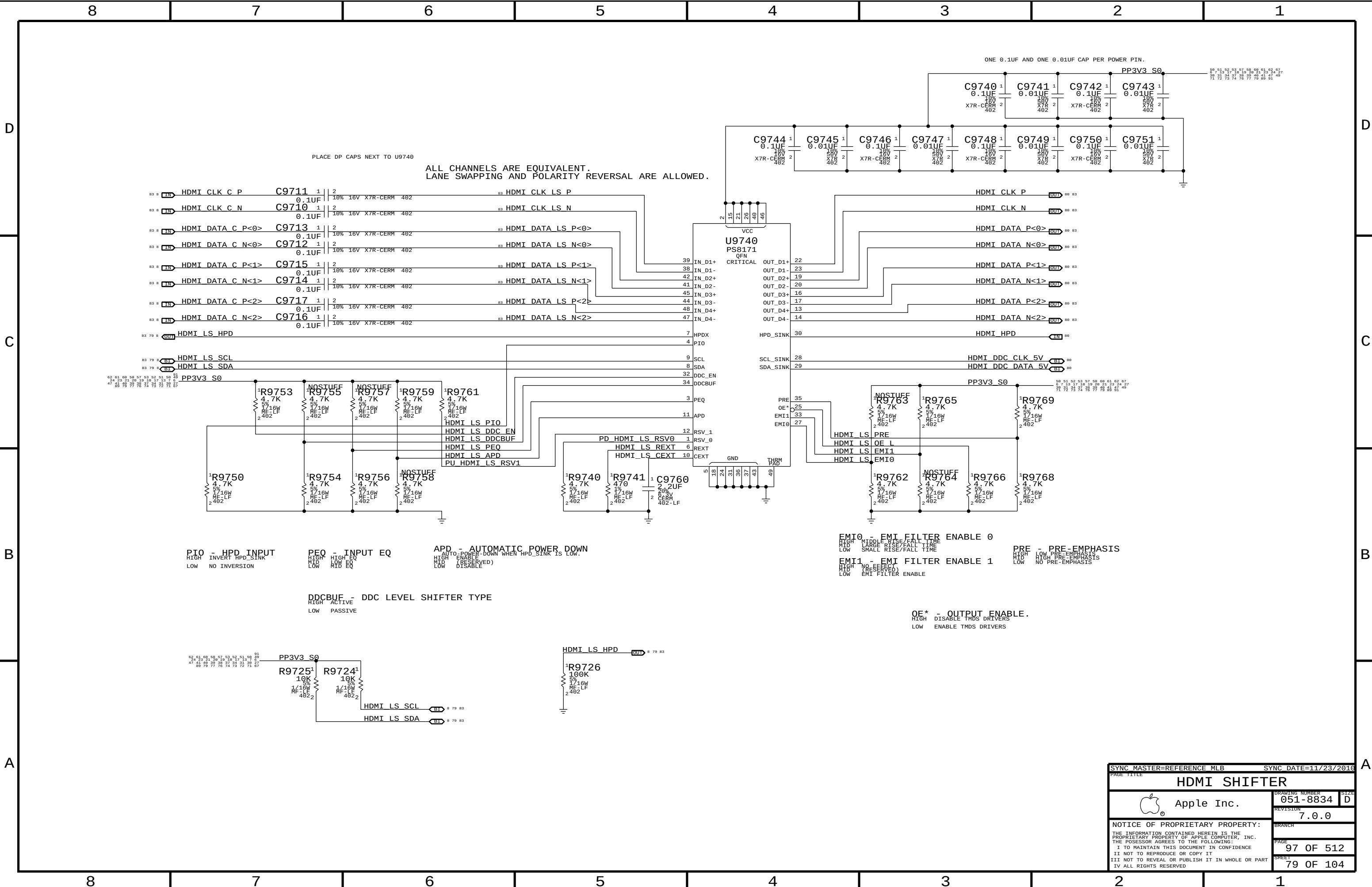
DP/T29 Bias Filters

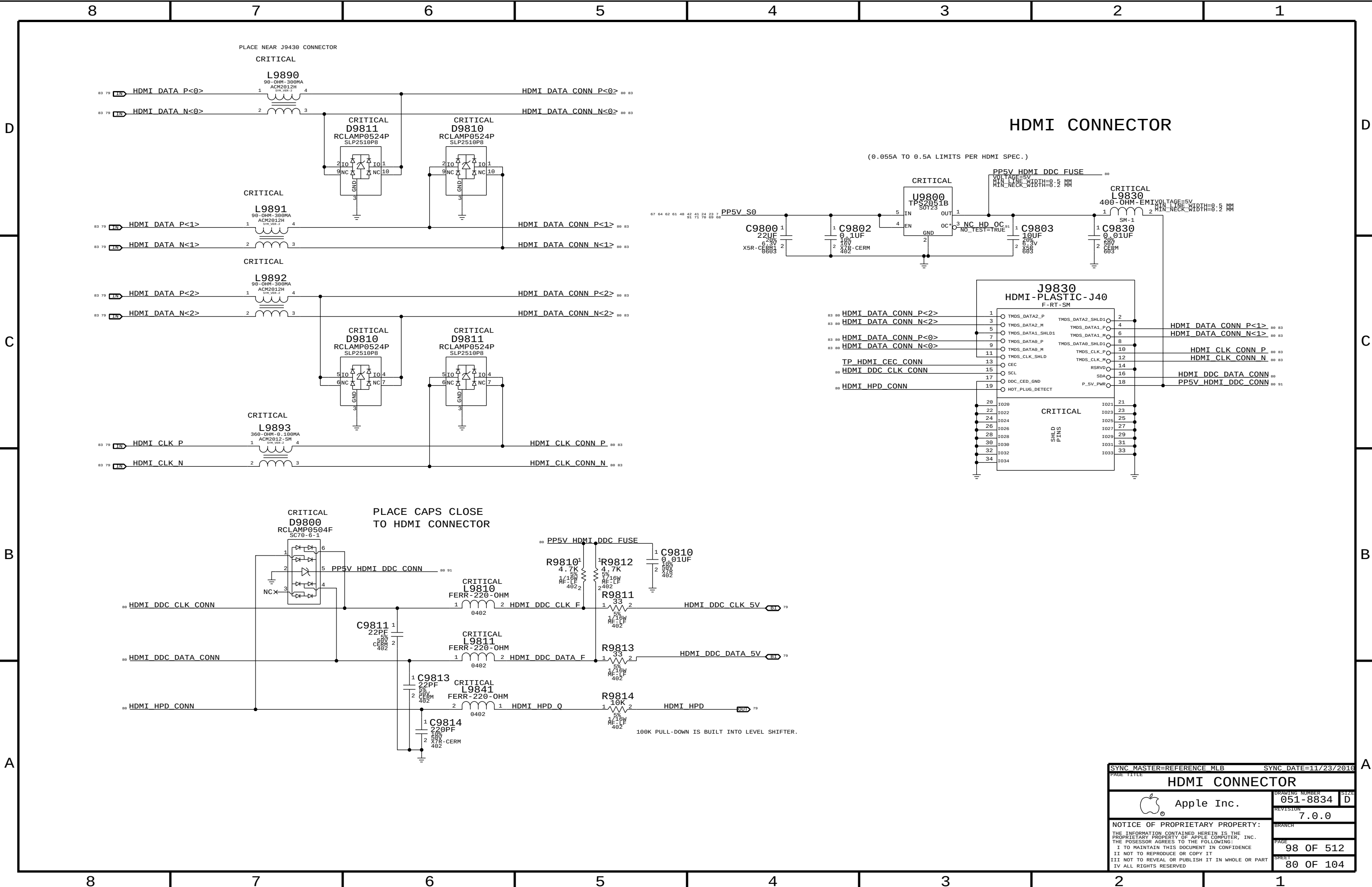
Prevents high-frequency coupling between 1.5K bias resistors.



SYNC MASTER=J46S		SYNC DATE=11/23/2010	
PAGE TITLE		T29 Power Support	
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HDMI CONNECTOR

(0.055A TO 0.5A LIMITS PER HDMI SPEC.)

CRITICAL

L9830

400-OHM-EMIV

VOLTAGE=5V

MIN-NECK-WIDTH=0.2 MM

SM-1

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CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?
CPU_8MIL	*	8 MIL	?
CPU_COMP	*	20 MIL	?
CPU_ITP	*	=2:1_SPACING	?
CPU_VCCSENSE	*	25 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	TOP, BOTTOM	=2x_DIELECTRIC	?
CPU_VID	*	0.457 MM	?

Most CPU signals with impedance requirements are 50-ohm single-ended. Some signals require 27.4-ohm single-ended impedance.

SOURCE: HR PDG section 4.1.9, PDDG section 2.5.1

PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	*	=3X_DIELECTRIC	?
CLK_PCIE	*	20 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	TOP, BOTTOM	=4X_DIELECTRIC	?

CPU Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	DMI_S2N	PCIE_85D	PCIE	DMI_S2N P<3:0>
	DMI_S2N	PCIE_85D	PCIE	DMI_S2N N<3:0>
	DMT_N2S	PCIE_85D	PCIE	DMI_N2S P<3:0>
	DMT_N2S	PCIE_85D	PCIE	DMI_N2S N<3:0>
	DMI_CLK100M	CLK_PCIE_90D	CLK_PCIE	DMI_CLK100M CPU_P
		CLK_PCIE_90D	CLK_PCIE	DMI_CLK100M CPU_N
	FDI_DATA	PCIE_85D	PCIE	FDI_DATA P<7:0>
	FDI_DATA	PCIE_85D	PCIE	FDI_DATA N<7:0>
		CPU_50S	CPU_AGTI	FDI_FSYN<1..0>
		CPU_50S	CPU_AGTI	FDI_ISYN<1..0>
		CPU_50S	CPU_AGTI	FDI_INT
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				

XDP Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	XDP_BPM	CPU1_50S	CPU1_ITP	XDP BPM L<3..0> 11 24
	XDP_BPM_L	CPU1_50S	CPU1_ITP	XDP BPM L<7..4> 11 24
	CPU_CFG	CPU1_50S	CPU1_ITP	XDP CPU_CFG<0> 24
	XDP_PRDY_L	CPU1_50S	CPU1_ITP	XDP CPU PRDY_L 11 24
	XDP_PREQ_L	CPU1_50S	CPU1_ITP	XDP CPU PREQ_L 11 24
	XDP_TDI	CPU1_50S	CPU1_ITP	XDP CPU TDI 11 24
	XDP_TDO	CPU1_50S	CPU1_ITP	XDP CPU TDO 11 24
	XDP_TMS	CPU1_50S	CPU1_ITP	XDP CPU TMS 11 24
	XDP_TCK	CPU1_50S	CPU1_ITP	XDP CPU TCK 11 24
	XDP_TRST_L	CPU1_50S	CPU1_ITP	XDP CPU TRST_L 11 24
		CPU1_50S	CPU1_ITP	XDP PCH TCK 17 24
		CPU1_50S	CPU1_ITP	XDP PCH TDI 17 24
		CPU1_50S	CPU1_ITP	XDP PCH TDO 17 24
		CPU1_50S	CPU1_ITP	XDP PCH TMS 17 24
		CPU1_50S	CPU1_ITP	
		CPU1_50S	CPU1_ITP	XDP TDI 24
		CPU1_50S	CPU1_ITP	XDP TDO 24
		CPU1_50S	CPU1_ITP	XDP TMS 24
	XDP_CPU_PWRGOOD	CPU1_50S	CPU1_ITP	XDP CPU PWRGD 24
	XDP_DBRESET_L	CPU1_50S	CPU1_ITP	XDP DBRESET_L 11 24 27
		CPU1_50S	CPU1_ITP	XDP PRESENT_L 24
		CPU1_50S	CPU1_ITP	XDP RES_RL 24
		CPU1_50S	CPU1_ITP	XDP CPU_PWRBTN_L 24
		CPU1_50S	CPU1_ITP	XDP VR_READY 24
	XDP_CLK_CPU1	CLK_PCTIE_96D	CLK_PCTIE	ITPCPU_CLK100M_P 11 17
	XDP_CLK_CPU1	CLK_PCTIE_96D	CLK_PCTIE	ITPCPU_CLK100M_N 11 17
	XDP_CLK_PCH	CLK_PCTIE_96D	CLK_PCTIE	ITPXD_P_CLK100M_P 17 24
	XDP_CLK_PCH	CLK_PCTIE_96D	CLK_PCTIE	ITPXD_P_CLK100M_N 17 24
	XDP_CLK_ITP	CLK_PCTIE_96D	CLK_PCTIE	XDP_CPU_CLK100M_P 24
	XDP_CLK_ITP	CLK_PCTIE_96D	CLK_PCTIE	XDP_CPU_CLK100M_N 24

Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
DP_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TMDS_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
TMDS_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
TMDS_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=4:1_SPACING	?	DISPLAYPORT	TOP, BOTTOM	=4:1_SPACING	?
TMDS	*	=4:1_SPACING	?	TMDS	TOP, BOTTOM	=4:1_SPACING	?

SOURCE: HR PDG, TABLES 191,193, Radeon TRM section 7.7.1
PER ANIL: MODIFIED DP CONNECTOR NEEDS 90 OHMS. MEASURE PROTO BOARD AND ADJUST IF NEEDED.

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
SATA_37SE	*	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE
SATA_50SE	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	*	=5:1_SPACING	?
SATA_ICOMP	*	15 MIL	?

SOURCE: HR PLATFORM DESIGN GUIDE, TABLES 191,193

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_USB_RBIAS	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=4:1_SPACING	?
USB_RBIAS	*	15 MIL	?

SOURCE: HR PLATFORM DESIGN GUIDE, TABLES 191, 193

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
 SATA_HDD_D2R	SATA_90D	SATA	SATA HDD1 D2R CONN N 42 91
 SATA_HDD_D2R	SATA_90D	SATA	SATA HDD1 D2R CONN P 42 91
 SATA_HDD_D2R	SATA_90D	SATA	SATA HDD1 D2R N 17 42
 SATA_HDD_D2R	SATA_90D	SATA	SATA HDD1 D2R P 17 42
 SATA_HDD_R2D	SATA_90D	SATA	SATA HDD1 R2D CONN N 42 91
 SATA_HDD_R2D	SATA_90D	SATA	SATA HDD1 R2D CONN P 42 91
 SATA_HDD_R2D	SATA_90D	SATA	SATA HDD1 R2D C N 17 42
 SATA_HDD_R2D	SATA_90D	SATA	SATA HDD1 R2D C P 17 42
 SATA_HDD_D2R	SATA_90D	SATA	SATA HDD2 D2R CONN N 42 91
 SATA_HDD_D2R	SATA_90D	SATA	SATA HDD2 D2R CONN P 42 91
 SATA_HDD_D2R	SATA_90D	SATA	SATA HDD2 D2R N 17 42
 SATA_HDD_D2R	SATA_90D	SATA	SATA HDD2 D2R P 17 42
 SATA_HDD_R2D	SATA_90D	SATA	SATA HDD2 R2D CONN N 42 91
 SATA_HDD_R2D	SATA_90D	SATA	SATA HDD2 R2D CONN P 42 91
 SATA_HDD_R2D	SATA_90D	SATA	SATA HDD2 R2D C N 17 42
 SATA_HDD_R2D	SATA_90D	SATA	SATA HDD2 R2D C P 17 42
 SATA_HDD_D2R	SATA_90D	SATA	SATA HDD1 D2R DF N 42 91
 SATA_HDD_R2D	SATA_90D	SATA	SATA HDD1 D2R DF P 42 91
 SATA_HDD_D2R	SATA_90D	SATA	SATA HDD1 R2D DF P 42 91
 SATA_HDD_D2R	SATA_90D	SATA	SATA HDD2 D2R DF N 42 91
 SATA_HDD_R2D	SATA_90D	SATA	SATA HDD2 D2R DF P 42 91
 SATA_HDD_R2D	SATA_90D	SATA	SATA HDD2 R2D DF N 42 91
 SATA_HDD_R2D	SATA_90D	SATA	SATA HDD2 R2D DF P 42 91
 PCH_SATA3_ICOMP	SATA_50SF	SATA_ICOMP	PCH_SATA3COMP 17
 PCH_SATA_ICOMP	SATA_37SF	SATA_ICOMP	PCH_SATA1COMP 17
 PCH_USB_RBTAS	PCH_USB_RBTAS	USB_RBTAS	PCH_USB_RB1AS 19
 USB_HUB1_UP	USB_85D	USB	USB_HUB1_UP N 19 25
 USB_HUB1_UP	USB_85D	USB	USB_HUB1_UP P 19 25
 USB_HUB2_UP	USB_85D	USB	USB_HUB2_UP N 19 25
 USB_HUB2_UP	USB_85D	USB	USB_HUB2_UP P 19 25
 USB_EXT_A	USB_85D	USB	USB_EXT_A N 25 43
 USB_EXT_A	USB_85D	USB	USB_EXT_A P 25 43
 USB_EXT_B	USB_85D	USB	USB_EXT_B N 25 43
 USB_EXT_B	USB_85D	USB	USB_EXT_B P 25 43
 USB_EXT_C	USB_85D	USB	USB_EXT_C N 25 43
 USB_EXT_C	USB_85D	USB	USB_EXT_C P 25 43
 USB_EXT_D	USB_85D	USB	USB_EXT_D N 25 43
 USB_EXT_D	USB_85D	USB	USB_EXT_D P 25 43
 USB_PORT0_N	USB_85D	USB	USB_PORT0_N 43
 USB_PORT0_P	USB_85D	USB	USB_PORT0_P 43
 USB_PORT1_N	USB_85D	USB	USB_PORT1_N 43
 USB_PORT1_P	USB_85D	USB	USB_PORT1_P 43
 USB_PORT2_N	USB_85D	USB	USB_PORT2_N 43
 USB_PORT2_P	USB_85D	USB	USB_PORT2_P 43
 USB_PORT3_N	USB_85D	USB	USB_PORT3_N 43
 USB_PORT3_P	USB_85D	USB	USB_PORT3_P 43
 USB_IR_N	USB_85D	USB	USB_IR_N 25 44
 USB_IR_P	USB_85D	USB	USB_IR_P 25 44
 USB_IR_R_N	USB_85D	USB	USB_IR_R_N 44
 USB_IR_R_P	USB_85D	USB	USB_IR_R_P 44
 USB_BT_N	USB_85D	USB	USB_BT_N 25 35
 USB_BT_P	USB_85D	USB	USB_BT_P 2

DP Properties

ELECTRICAL_CONSTRAINT_SET		PHYSICAL	NET_TYPE	SPACING	
		DP_90D	DISPLAYPORT	DP_AUX_CH_CK_N	
		DP_90D	DISPLAYPORT	DP_AUX_CH_CK_P	
		DP_90D	DISPLAYPORT	DP_AUX_CH_C_N	
		DP_90D	DISPLAYPORT	DP_AUX_CH_C_P	
		T29DP_90D	T29DP	DP_A_EXT_AUXCH_N	77 78 86
		T29DP_90D	T29DP	DP_A_EXT_AUXCH_P	77 78 86
		DP_90D	DISPLAYPORT	DP_EXT_AUXCH_C_N	8 77
		DP_90D	DISPLAYPORT	DP_EXT_AUXCH_C_P	8 77
		DP_90D	DISPLAYPORT	DP_EXT_AUXCH_N	77
		DP_90D	DISPLAYPORT	DP_EXT_AUXCH_P	77
		DP_90D	DISPLAYPORT	DP_EXT_DDC_CLK_AUX_P	
		DP_90D	DISPLAYPORT	DP_EXT_DDC_DATA_AUX_N	
		DP_55S	DISPLAYPORT	DP_EXT_DDC_CLK	8 77
		DP_55S	DISPLAYPORT	DP_EXT_DDC_DATA	8 77
		DP_55S	DISPLAYPORT	DP_EXT_HPD	8 77
		DP_90D	DISPLAYPORT	DP_T29SNK0_AUXCH_C_N	8 74
		DP_90D	DISPLAYPORT	DP_T29SNK0_AUXCH_C_P	8 74
		DP_90D	DISPLAYPORT	DP_T29SNK0_AUXCH_N	74
		DP_90D	DISPLAYPORT	DP_T29SNK0_AUXCH_P	74
		DP_90D	DISPLAYPORT	DP_EXT_ML_C_N<3..0>	8 77
		DP_90D	DISPLAYPORT	DP_EXT_ML_C_P<3..0>	8 77
		DP_90D	DISPLAYPORT	DP_EXT_ML_N<3..0>	77
		DP_90D	DISPLAYPORT	DP_EXT_ML_P<3..0>	77
		DP_90D	DISPLAYPORT	DP_ML_CK_N<3..0>	
		DP_90D	DISPLAYPORT	DP_ML_CK_P<3..0>	
		DP_90D	DISPLAYPORT	DP_ML_C_N<3..0>	
		DP_90D	DISPLAYPORT	DP_ML_C_P<3..0>	
180D		DP_90D	DISPLAYPORT	DP_T29SNK1_AUXCH_C_N	
180D		DP_90D	DISPLAYPORT	DP_T29SNK1_AUXCH_C_P	
180D		DP_90D	DISPLAYPORT	DP_T29SNK1_AUXCH_N	
180D		DP_90D	DISPLAYPORT	DP_T29SNK1_AUXCH_P	
		DP_90D	DISPLAYPORT	DP_T29SNK0_ML_C_N<3..0>	8 74
		DP_90D	DISPLAYPORT	DP_T29SNK0_ML_C_P<3..0>	8 74
		DP_90D	DISPLAYPORT	DP_T29SNK0_ML_N<3..0>	74
		DP_90D	DISPLAYPORT	DP_T29SNK0_ML_P<3..0>	74
		DP_55S	DISPLAYPORT	DP_T29SNK0_DDC_CLK	
		DP_55S	DISPLAYPORT	DP_T29SNK0_DDC_DATA	
		DP_55S	DISPLAYPORT	DP_T29SNK0_HPD	8 74
180D		DP_90D	DISPLAYPORT	DP_T29SNK1_ML_C_N<3..0>	
180D		DP_90D	DISPLAYPORT	DP_T29SNK1_ML_C_P<3..0>	
180D		DP_90D	DISPLAYPORT	DP_T29SNK1_ML_N<3..0>	
180D		DP_90D	DISPLAYPORT	DP_T29SNK1_ML_P<3..0>	
180D		DP_55S	DISPLAYPORT	DP_T29SNK1_HPD	

Manual
Diff Pair

HDMI Properties

		NET_TYPE	
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
	TMDS_1000	TMDS	HDMI CLK CONN N 80
	TMDS_1000	TMDS	HDMI CLK CONN P 80
	TMDS_850	TMDS	HDMI CLK C N 8 79
	TMDS_850	TMDS	HDMI CLK C P 8 79
	TMDS_850	TMDS	HDMI CLK LS N 79
	TMDS_850	TMDS	HDMI CLK LS P 79
	TMDS_1000	TMDS	HDMI CLK N 79 80
	TMDS_1000	TMDS	HDMI CLK P 79 80
	TMDS_1000	TMDS	HDMI DATA CONN N<2..0> 80
	TMDS_1000	TMDS	HDMI DATA CONN P<2..0> 80
	TMDS_850	TMDS	HDMI DATA C N<2..0> 8 79
	TMDS_850	TMDS	HDMI DATA C P<2..0> 8 79
	TMDS_850	TMDS	HDMI DATA LS N<2..0> 79
	TMDS_850	TMDS	HDMI DATA LS P<2..0> 79
	TMDS_1000	TMDS	HDMI DATA N<2..0> 79 80
	TMDS_1000	TMDS	HDMI DATA P<2..0> 79 80
	TMDS_55S	TMDS	HDMI LS SCL 8 79
	TMDS_55S	TMDS	HDMI LS SDA 8 79
	TMDS_55S	TMDS	HDMI LS HPD 8 79

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	8 MIL	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		PHYSICAL	SPACING	
	LPC_AD	LPC_50S	LPC		LPC AD<3..0>	17 46 48
		LPC_50S	LPC		LPC R AD<3..0>	17
	LPC_FRAME_L	LPC_50S	LPC		LPC FRAME L	17 46 48
		LPC_50S	LPC		LPC FRAME R L	17
	LPC_RESET_L	LPC_50S	LPC		LPCPLUS RESET L	27 48
	PCH_LPC_CLK0	CLK_LPC_50S	CLK_LPC		LPC CLK33M SMC R	19 27
		CLK_LPC_50S	CLK_LPC		LPC CLK33M SMC	27 46
		CLK_LPC_50S	CLK_LPC		LPC CLK33M LPCPLUS	27 48
		CLK_LPC_50S	CLK_LPC		LPC CLK33M LPCPLUS R	19 27
		SMB_50S	SMB		SMB PCH STR_SCL	49
		SMB_50S	SMR		SMB PCH STR_SDA	49
		SMB_50S	SMR		SMB PCH DIMM_SCL	49
		SMB_50S	SMB		SMB PCH DIMM_SDA	49
		SMB_50S	SMR		SMB PCH MKY_SCL	49
		SMB_50S	SMR		SMB PCH MKY_SDA	49
		SMB_50S	SMR		SMB PCH VRF_SCL	49
		SMB_50S	SMR		SMB PCH VRF_SDA	49
	SMBUS_PCH_CLK	SMB_50S	SMR		SMBUS PCH_CLK	17 49
		SMB_50S	SMB		SMBUS PCH_DATA	17 49
	SMBUS_PCH_0_CLK	SMB_50S	SMB		SMB PCH_XDP_SCL	49
		SMB_50S	SMB		SMB PCH_XDP_SDA	49
		SMB_50S	SMB		SML_PCH_1_CLK	17 49
		SMB_50S	SMR		SML_PCH_1_DATA	17 49
	SPT_CLK	SPT_55S	SPT		SPI_CLK_R	17 48
		SPT_55S	SPT		SPI_CLK	48 91
	SPT_MOST	SPT_55S	SPT		SPI_MOSI_R	17 48
		SPT_55S	SPT		SPI_MOSI	48 91
	SPT_MISO	SPT_55S	SPT		SPI_MISO	17 48 91
	SPT_CS0	SPT_55S	SPT		SPI_CS0_R_L	17 48
		SPT_55S	SPT		SPI_CS0_L	48 91
	HDA_BIT_CLK	HDA_50S	HDA		HDA_BIT_CLK	17 53
		HDA_50S	HDA		HDA_BIT_CLK_R	17
	HDA_SYNC	HDA_50S	HDA		HDA_SYNC	17 53
		HDA_50S	HDA		HDA_SYNC_R	17
	HDA_RST_L	HDA_50S	HDA		HDA_RST_R_L	17
		HDA_50S	HDA		HDA_RST_L	17 53
	HDA_SDIN0	HDA_50S	HDA		HDA_SDIN0	17 53
		HDA_50S	HDA		AUD_SDI_R	53
	HDA_SDOUT	HDA_50S	HDA		HDA_SDOUT	17 53
		HDA_50S	HDA		HDA_SDOUT_R	17

PC95	CLK_PCTFE_90D	CLK_PCTFE	PCH_CLK96M_DOT_P	17
PC97	CLK_PCTFE_90D	CLK_PCTFE	PCH_CLK96M_DOT_N	17
PC98	CLK_PCTFE_90D	CLK_PCTFE	PCH_CLK100M_SATA_P	17
PC99	CLK_PCTFE_90D	CLK_PCTFE	PCH_CLK100M_SATA_N	17
PC100	CPU_50S	CLK_PCTFE	PCH_CLK14P3M_REFCLK	17
PC101	CPU_50S	CLK_PCTFE	PCH_CLK33M_PCIIN	17 27

			PCIE_85D	PCIE	PCIE ENET R2D P	37
			PCIE_85D	PCIE	PCIE ENET R2D N	37
		PCIE_ENET_R2D	PCIE_85D	PCIE	PCIE ENET R2D C P	17 37
			PCIE_85D	PCIE	PCIE ENET R2D C N	17 37
		PCIE_ENET_D2R	PCIE_85D	PCIE	PCIE ENET D2R P	17 37
			PCIE_85D	PCIE	PCIE ENET D2R N	17 37
			PCIE_85D	PCIE	PCIE ENET D2R C P	37
			PCIE_85D	PCIE	PCIE ENET D2R C N	37
			PCIE_85D	PCIE	PCIE AP R2D P	35 91
			PCIE_85D	PCIE	PCIE AP R2D N	35 91
		PCIE_AP_R2D	PCIE_85D	PCIE	PCIE AP R2D C P	17 35
			PCIE_85D	PCIE	PCIE AP R2D C N	17 35
		PCIE_AP_D2R	PCIE_85D	PCIE	PCIE AP D2R P	17 35 91
			PCIE_85D	PCIE	PCIE AP D2R N	17 35 91
			PCIE_85D	PCIE	PCIE FW R2D P	39
			PCIE_85D	PCIE	PCIE FW R2D N	39
		PCIE_FW_R2D	PCIE_85D	PCIE	PCIE FW R2D C P	17 39
			PCIE_85D	PCIE	PCIE FW R2D C N	17 39
		PCIE_FW_D2R	PCIE_85D	PCIE	PCIE FW D2R P	17 39
			PCIE_85D	PCIE	PCIE FW D2R N	17 39
			PCIE_85D	PCIE	PCIE FW D2R C P	39
			PCIE_85D	PCIE	PCIE FW D2R C N	39
1250			CLK_PCIE_90D	CLK_PCIE	PCIE CLK100M PCH P	17
1250			CLK_PCIE_90D	CLK_PCIE	PCIE CLK100M PCH N	17
1250	PCIE_CLK100M_T29		CLK_PCIE_90D	CLK_PCIE	PCIE CLK100M T29 P	17 74
1250			CLK_PCIE_90D	CLK_PCIE	PCIE CLK100M T29 N	17 74
	PCIE_CLK100M		CLK_PCIE_90D	CLK_PCIE	PEG CLK100M P	8 17
			CLK_PCIE_90D	CLK_PCIE	PEG CLK100M N	8 17
	PCIE_CLK100M_ENET		CLK_PCIE_90D	CLK_PCIE	PCIE CLK100M ENET P	17 37
			CLK_PCIE_90D	CLK_PCIE	PCIE CLK100M ENET N	17 37
1250	PCIE_CLK100M_AP		CLK_PCIE_90D	CLK_PCIE	PCIE CLK100M AP CONN P	35 91
1250			CLK_PCIE_90D	CLK_PCIE	PCIE CLK100M AP CONN N	35 91
1250	PCIE_CLK100M_AP		CLK_PCIE_90D	CLK_PCIE	PCIE CLK100M AP P	17 35
			CLK_PCIE_90D	CLK_PCIE	PCIE CLK100M AP N	17 35
	PCIE_CLK100M_FW		CLK_PCIE_90D	CLK_PCIE	PCIE CLK100M FW P	17 39
			CLK_PCIE_90D	CLK_PCIE	PCIE CLK100M FW N	17 39
1250	PCIE_T29_R2D	PCIE_85D	PCIE	PCIE T29 R2D C P<3...0>	8 74	
1250	PCIE_T29_R2D	PCIE_85D	PCIE	PCIE T29 R2D C N<3...0>	8 74	
1250	PCIE_T29_R2D	PCIE_85D	PCIE	PCIE T29 R2D P<3...0>	74	
1250	PCIE_T29_R2D	PCIE_85D	PCIE	PCIE T29 R2D N<3...0>	74	
1250	PCIE_T29_D2R	PCIE_85D	PCIE	PCIE T29 D2R P<3...0>	8 74	
1250	PCIE_T29_D2R	PCIE_85D	PCIE	PCIE T29 D2R N<3...0>	8 74	
1250	PCIE_T29_D2R	PCIE_85D	PCIE	PCIE T29 D2R C P<3...0>	74	
1250	PCIE_T29_D2R	PCIE_85D	PCIE	PCIE T29 D2R C N<3...0>	74	

SYNCH MASTER-REFERENCE MLB		SYNCH DATE=11/23/2010	
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ETHERNET CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_3X	*	=3:1_SPACING	?

SOURCE: Broadcom 5764-DS04-RDS Page 38

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_100D	*	=100_OHM_DIFF=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_MDI	*	0.6 MM	?	ENET_HV	*	*	2KV_SPACING
2KV_SPACING	*	0.50 MM	100	ENET_HV	ENET_HV	*	ENET_MDI
2KV_SPACING	TOP, BOTTOM	1.27 MM	100				

NOTE: 2.2kV isolation needed on all primary-side ethernet signals. Min 1.27mm separation from all other copper on the board!

SD CARD INTERFACE CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SD_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_INTERFACE	*	=3X_DIELECTRIC	?

FireWire Interface Constraints

[illegible]

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FW_TP	*	=3:1_SPACING	?

Ethernet Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
		ENET_50S	ENET_3X	BCM5764 CLK25M XTALI	26 37
		ENET_50S	ENET_3X	BCM5764 CLK25M XTALO	26 37
		ENET_50S	ENET_3X	ENET RESET L	27 34
	ENET_MDI	ENET_100D	ENET_MDI	ENET MDI P<3..0>	36 37
		ENET_100D	ENET_MDI	ENET MDI N<3..0>	36 37
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					

High-voltage isolated ethernet signals.

SD Interface Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
SD55S		SD 55S	SD INTERFACE	SDCONN CLK 34 37 91
SD55S		SD 55S	SD INTERFACE	SDCONN CLK FF 34
SD55S		SD 55S	SD INTERFACE	SDCONN CMD 34 37 91
SD55S		SD 55S	SD INTERFACE	SDCONN DATA<7..0> 34 37 91
SD55S		SD 55S	SD INTERFACE	SDCONN CLK R 34
SD55S		SD 55S	SD INTERFACE	SDCONN CMD R 34
SD55S		SD 55S	SD INTERFACE	SDCONN DATA R<7..0> 34
SD55S		SD 55S	SD INTERFACE	ENET_CR CLK 37
SD55S		SD 55S	SD INTERFACE	ENET_CR CMD 37
SD55S		SD 55S	SD INTERFACE	ENET_CR DATA<7..0> 37
SD55S		SD 55S	SD INTERFACE	SDCONN_F CLK 34
SD55S		SD 55S	SD INTERFACE	SDCONN_F CLK_FF 34

FireWire Net Properties

[illegible]

Port 0 and 2 Not Used

DisplayPort Signal Constraints

NOTE: DisplayPort Physical/Spacing Constraints provided by Chipset or GPU page.

T29 I2C Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_I2C_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_I2C	*	=2x_DIELECTRIC	?

T29 SPI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_SPI	*	=2x_DIELECTRIC	?

T29/DP Connector Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29DP_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29DP	*	=5x_DIELECTRIC	?	T29DP	TOP, BOTTOM	=7x_DIELECTRIC	?

SOURCE: Bill Cornelius's T29 Routing Notes

T29 IC Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE	
	PHYSICAL	SPACING

T29/DP Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	T29_R2D0	T29DP_90D	T29DP	T29_R2D P<0>
	T29_R2D0	T29DP_90D	T29DP	T29_R2D N<0>
	T29_R2D1	T29DP_90D	T29DP	T29_R2D P<1>
	T29_R2D1	T29DP_90D	T29DP	T29_R2D N<1>
	T29_D2R0	T29DP_90D	T29DP	T29_D2R C P<0>
	T29_D2R0	T29DP_90D	T29DP	T29_D2R C N<0>
	T29_D2R1	T29DP_90D	T29DP	T29_D2R C P<1>
	T29_D2R1	T29DP_90D	T29DP	T29_D2R C N<1>
		T29DP_90D	T29DP	T29_R2D C F P<1..0>
		T29DP_90D	T29DP	T29_R2D C F N<1..0>
		T29DP_90D	T29DP	T29DPA_D2R1_AUXCH P
		T29DP_90D	T29DP	T29DPA_D2R1_AUXCH N
		T29DP_90D	T29DP	DP_SDRVA_ML C P<3..0>
		T29DP_90D	T29DP	DP_SDRVA_ML C N<3..0>
		T29DP_90D	T29DP	DP_SDRVA_ML R P<3..0>
		T29DP_90D	T29DP	DP_SDRVA_ML R N<3..0>
		T29DP_90D	T29DP	DP_SDRVA_ML P<2>
		T29DP_90D	T29DP	DP_SDRVA_ML N<2>
	DP_SDRVA_MI	T29DP_90D	T29DP	DP_SDRVA_ML P<3>
	DP_SDRVA_ML	T29DP_90D	T29DP	DP_SDRVA_ML N<3>
		T29DP_90D	T29DP	DP_SDRVA_ML P<0>
		T29DP_90D	T29DP	DP_SDRVA_ML N<0>
	DP_SDRVA_MI	T29DP_90D	T29DP	DP_SDRVA_ML P<1>
	DP_SDRVA_MI	T29DP_90D	T29DP	DP_SDRVA_ML N<1>
	DP_SDRVA_AUXCH	T29DP_90D	T29DP	DP_SDRVA_AUXCH P
	DP_SDRVA_AUXCH	T29DP_90D	T29DP	DP_SDRVA_AUXCH N
		T29DP_90D	T29DP	DP_SDRVA_AUXCH C P
		T29DP_90D	T29DP	DP_SDRVA_AUXCH C N
		T29DP_90D	T29DP	T29DPA_ML P<3..0>
		T29DP_90D	T29DP	T29DPA_ML N<3..0>
		T29DP_90D	T29DP	T29DPA_ML C P<3..0>
		T29DP_90D	T29DP	T29DPA_ML C N<3..0>
		T29DP_90D	T29DP	DP_A_EXT_AUXCH P
		T29DP_90D	T29DP	DP_A_EXT_AUXCH N
		T29DP_90D	T29DP	T29DPA_D2R1_AUXCH_CC P
		T29DP_90D	T29DP	T29DPA_D2R1_AUXCH_CC N
		T29DP_90D	T29DP	T29_D2R_CC P<0>
		T29DP_90D	T29DP	T29_D2R_CC N<0>



PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
1T01_DIFFPAIR	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

SMC SMBus Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
 SMBUS_SMC_A_S3_SCL	SMB_50S	SMB	SMBUS_SMC_A_S3_SCL 46 49
 SMBUS_SMC_A_S3_SDA	SMB_50S	SMB	SMBUS_SMC_A_S3_SDA 46 49
 SMBUS_SMC_B_S0_SCL	SMB_50S	SMB	SMBUS_SMC_B_S0_SCL 46 49
 SMBUS_SMC_B_S0_SDA	SMB_50S	SMB	SMBUS_SMC_B_S0_SDA 46 49
 SMBUS_SMC_0_S0_SCL	SMB_50S	SMB	SMBUS_SMC_0_S0_SCL 46 49
 SMBUS_SMC_0_S0_SDA	SMB_50S	SMB	SMBUS_SMC_0_S0_SDA 46 49
 SMBUS_SMC_BSA_SCL	SMB_50S	SMB	ENET_ASF_SMB_CLK 37 49
 SMBUS_SMC_BSA_SDA	SMB_50S	SMB	ENET_ASF_SMB_DATA 37 49
 SMBUS_SMC_MGMT_SCL	SMB_50S	SMB	SMBUS_SMC_MGMT_SCL 46 49
 SMBUS_SMC_MGMT_SDA	SMB_50S	SMB	SMBUS_SMC_MGMT_SDA 46 49
 SMB_SMC_B_STR_SCL	SMB_50S	SMB	SMB_SMC_B_STR_SCL 49
 SMB_SMC_B_STR_SDA	SMB_50S	SMB	SMB_SMC_B_STR_SDA 49
 SMB_SMC_B_TSN_SCL	SMB_50S	SMB	SMB_SMC_B_TSN_SCL 49
 SMB_SMC_B_TSN_SDA	SMB_50S	SMB	SMB_SMC_B_TSN_SDA 49

SYNC_MASTER=REFERENCE_MLB

SYNC_DATE=11/23/2016

PAGE_TITLE

SMC Constraints

 Apple Inc.

DRAWING_NUMBER

051-8834

SIZE

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REVISION

7.0.0

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PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_1T01_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_1T01_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	*	=1:1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR
AUDIODIFF	*	=1:1_DIFFPAIR	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2:1_SPACING	?
THERM	*	=2:1_SPACING	?
AUDIO	*	=2:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENETCONN	*	25 MILS	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_MDI	GND	*	GND_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

PCIE	GND	*	GND_P2MM
SATA	GND	*	GND_P2MM
USB	GND	*	GND_P2MM
CLK_PCIE	SB_POWER	*	PwR_P2MM
SATA	SB_POWER	*	PwR_P2MM
USB	SB_POWER	*	PwR_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	GND	*	GND_P2MM
MEM_CMD	GND	*	GND_P2MM
MEM_CTRL	GND	*	GND_P2MM
MEM_DATA	GND	*	GND_P2MM
MEM_DQS	GND	*	GND_P2MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	2.54 MM OVERRIDE	OVERRIDE	OVERRIDE
MEM_72D OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	2.54 MM OVERRIDE	OVERRIDE	OVERRIDE
PCIE_85D OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	10 MM OVERRIDE	OVERRIDE	OVERRIDE
USB_85D OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	12.7 MM OVERRIDE	OVERRIDE	OVERRIDE
CPU_27P4S OVERRIDE	BOTTOM OVERRIDE	OVERRIDE	OVERRIDE	0.23 MM OVERRIDE	2.54 MM OVERRIDE	OVERRIDE	OVERRIDE

Breakout Constraint Overrides

Alternate diffpair width/gap through BGA fanout areas

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
CLK_PCIE_90D	BGA	100_DIFF_BGA
DP_90D	BGA	100_DIFF_BGA
TMD5_85D	BGA	TMD5_85D
ENET_100D	BGA	100_DIFF_BGA
FW_110D	BGA	100_DIFF_BGA
SATA_90D	BGA	100_DIFF_BGA
PCIE_85D	BGA	PCIE_85D
MEM_85D	BGA	MEM_85D
USB_85D	BGA	USB_85D
GDDR5_80D	BGA	GDDR5_80D

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
MEM_40S	BGA	BGA_P075
MEM_50S	BGA	BGA_P075

J40* SPECIFIC NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		SPACING	
	PHYSICAL			
 SENSE_DIFFPAIR	SENSE_1T01_55S	SENSE	VCCSAS0_CS_P	50 64
	SENSE_1T01_55S	SENSE	VCCSAS0_CS_N	50 64
 SENSE_DIFFPAIR	SENSE_1T01_55S	SENSE	CPUVCCIOS0_CS_P	50 70
	SENSE_1T01_55S	SENSE	CPUVCCIOS0_CS_N	50 70
 5372 SENSE_DIFFPAIR	SENSE_1T01_55S	SENSE	ISNS_HS_GPU_P	
 5373	SENSE_1T01_55S	SENSE	ISNS_HS_GPU_N	

J40* SPECIFIC NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE	
	PHYSICAL	SPACING
		SB_POWER PP3V3_S5 0 17 18 19 20 21 23 24 25 27 28 58 61 62 78 81
		SB_POWER PP3V3_S0 58 51 52 53 57 58 60 61 62 67 6 7 13 17 18 19 20 21 23 24 27 28 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48
		GND GND

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	SENSE_DIFFPAIR	THERM 1T01 55S	THERM	CPUTHMSNS_D2_P
		THERM 1T01 55S	THERM	CPUTHMSNS_D2_N
	SENSE_DIFFPAIR	THERM 1T01 55S	THERM	CPU_THERMD_P
		THERM 1T01 55S	THERM	CPU_THERMD_N
	SENSE_DIFFPAIR	THERM 1T01 55S	THERM	GPUTHMSNS_D_P
		THERM 1T01 55S	THERM	GPUTHMSNS_D_N
	SENSE_DIFFPAIR	THERM 1T01 55S	THERM	GPU_TDIODE_P
		THERM 1T01 55S	THERM	GPU_TDIODE_N
	SENSE_DIFFPAIR	THERM 1T01 55S	THERM	T29_THERMD_P
		THERM 1T01 55S	THERM	T29_THERMD_N

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J40* 12 LAYER BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS			BOARD AREAS			BOARD UNITS (ALL OF MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM			NO_TYPE, BGA			MM	15.5.1
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP, BOTTOM	Y	0.310 MM	0.095 MM			
27P4_OHM_SE	*	Y	0.250 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
37_OHM_SE	TOP, BOTTOM	Y	0.185 MM	0.095 MM			
37_OHM_SE	*	Y	0.155 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP, BOTTOM	Y	0.165 MM	0.095 MM			
40_OHM_SE	*	Y	0.135 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	TOP, BOTTOM	Y	0.13 MM	0.13 MM			
45_OHM_SE	*	Y	0.099 MM	0.099 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP, BOTTOM	Y	0.110 MM	0.095 MM			
50_OHM_SE	*	Y	0.090 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP, BOTTOM	Y	0.090 MM	0.090 MM			
55_OHM_SE	*	Y	0.076 MM	0.076 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
72_OHM_DIFF	*	Y	0.154 MM	0.154 MM	=STANDARD	0.200 MM	0.200 MM
72_OHM_DIFF	TOP, BOTTOM	Y	0.175 MM	0.175 MM		0.200 MM	0.200 MM
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	Y	0.105 MM	0.105 MM	=STANDARD	0.120 MM	0.120 MM
80_OHM_DIFF	TOP, BOTTOM	Y	0.135 MM	0.135 MM		0.160 MM	0.160 MM
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	Y	0.110 MM	0.090 MM	=STANDARD	0.180 MM	0.180 MM
85_OHM_DIFF	TOP, BOTTOM	Y	0.125 MM	0.090 MM		0.190 MM	0.190 MM
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	Y	0.102 MM	0.090 MM	=STANDARD	0.220 MM	0.220 MM
90_OHM_DIFF	TOP, BOTTOM	Y	0.115 MM	0.090 MM		0.230 MM	0.230 MM
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	Y	0.080 MM	0.080 MM	=STANDARD	0.200 MM	0.200 MM
100_OHM_DIFF	TOP, BOTTOM	Y	0.089 MM	0.089 MM		0.220 MM	0.220 MM
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	Y	0.065 MM	0.065 MM	=STANDARD	0.200 MM	0.200 MM
110_OHM_DIFF	TOP, BOTTOM	Y	0.075 MM	0.075 MM		0.330 MM	0.330 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_DIFF_BGA	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
100_DIFF_BGA	ISL3, ISL4	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
100_DIFF_BGA	ISL9, ISL10	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM

NOTE: 100_DIFF_BGA is 100-ohms differential impedance on outer layers and 95-ohms on inner layers.

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
BGA_P075	TOP	Y	0.110 MM	0.080 MM	10 MM		
BGA_P075	BOTTOM	Y	0.110 MM	0.073 MM	10 MM		

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P1MM	*	=DEFAULT	?
BGA_P2MM	*	=DEFAULT	?
BGA_P075MM	*	0.075 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P075MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1.5:1_SPACING	*	0.15 MM	?
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
4:1_SPACING	*	0.4 MM	?
5:1_SPACING	*	0.5 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	*	0.140 MM	?
3X_DIELECTRIC	*	0.210 MM	?
4X_DIELECTRIC	*	0.280 MM	?
5X_DIELECTRIC	*	0.350 MM	?
7X_DIELECTRIC	*	0.490 MM	?

NOTE: Based on K92 m1b stackup.

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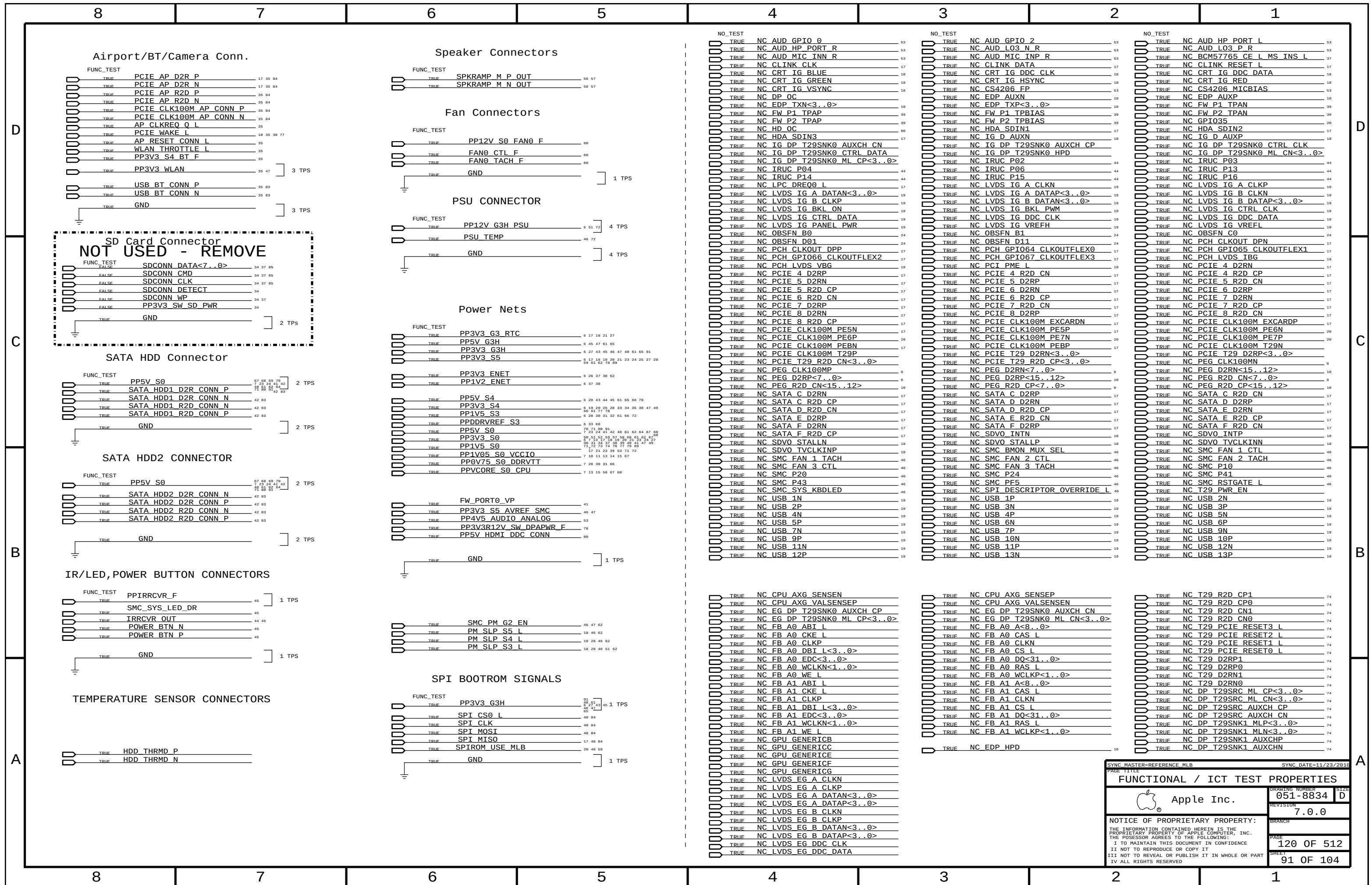
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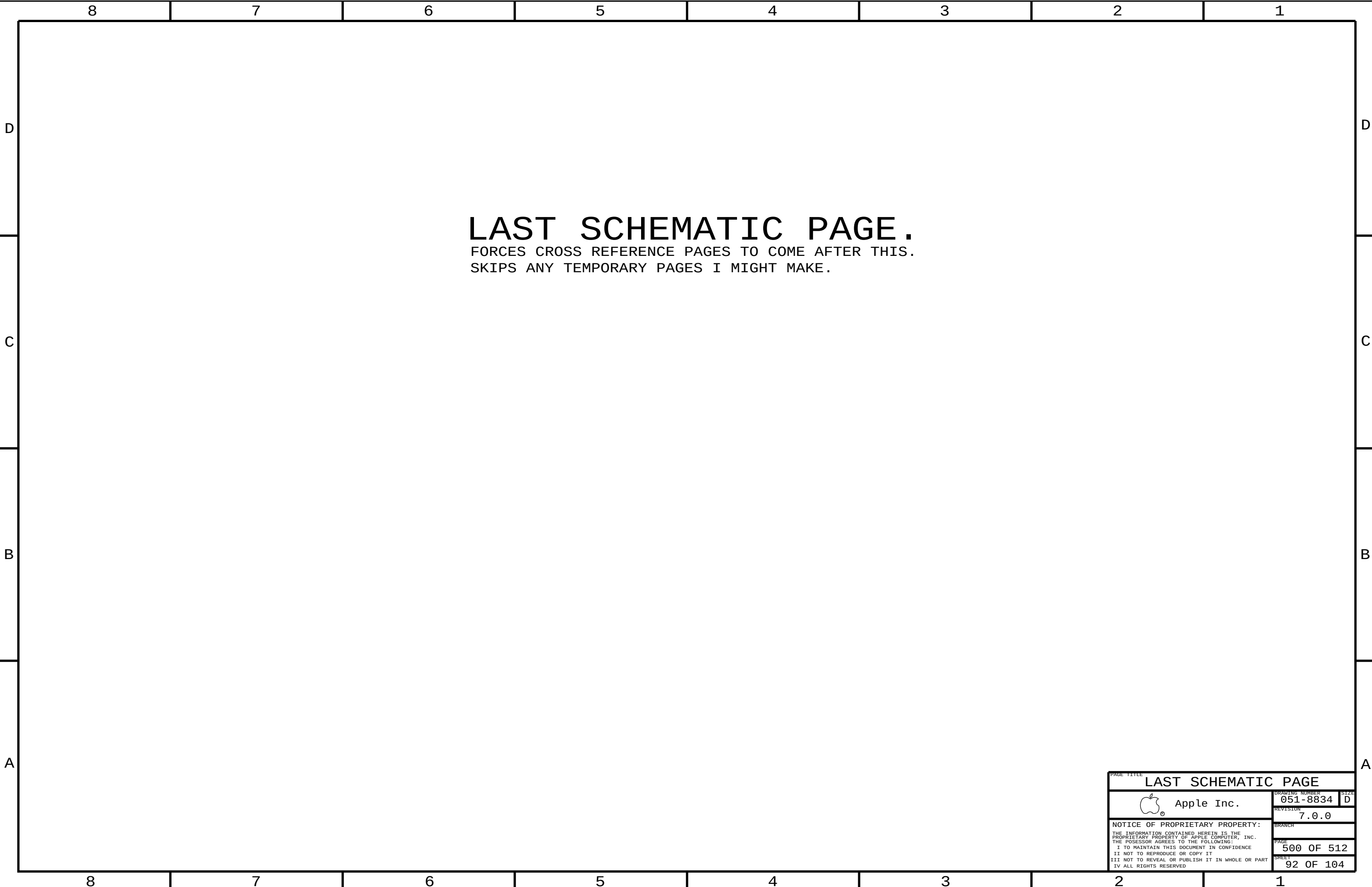
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SYNC DATE=11/23/2010





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Basenet Report

mlb

Apr 8 14:52:14 2011

Base nets and synonyms for mlb_lib.MLB(=mlb_lib.mlb(sch_1))

Base SignalSynonymsLocation([Zone][dir])

1V05_S0_L0D_EN

1V05_S0_L0D_EN - @mlb_lib.MLB

62C1

62C1 79C6

3V3R2V3_DPAPWR_ADJ

3V3R2V3_DPAPWR_ADJ - @mlb_lib.MLB

7807

7807

3V3R2V3_SS_A

3V3R2V3_SS_A - @mlb_lib.MLB

7808

7808

80101A_NC

80101A_NC - @mlb_lib.MLB

78C8

78C8

=I2C_DPSDRVA_SCL

=I2C_DPSDRVA_SCL - @mlb_lib.MLB

49B1 49B6 77B7

49B1 49B6 77B7

=I2C_DPSDRVA_SDA

=I2C_DPSDRVA_SDA - @mlb_lib.MLB

49B1 49B6 77B7

49B1 49B6 77B7

=I2C_DPSDRVA_SDA

=I2C_DPSDRVA_SDA - @mlb_lib.MLB

49B1 49B6 77B7

49B1 49B6 77B7

=IG_DP_T29SNK0_CTRL_CLK

=IG_DP_T29SNK0_CTRL_CLK - @mlb_lib.MLB

88B 18C1

88B 18C1

=IG_DP_T29SNK0_CTRL_CLK

=IG_DP_T29SNK0_CTRL_CLK - @mlb_lib.MLB

88B 18C1

88B 18C1

=IG_DP_T29SNK0_CTRL_DATA

=IG_DP_T29SNK0_CTRL_DATA - @mlb_lib.MLB

88B 18C1

88B 18C1

=PLT_RST_T29_GPU

=PLT_RST_T29_GPU - @mlb_lib.MLB

27C1 76C8

27C1 76C8

=SMC_GFX_THROTTLE_L

=SMC_GFX_THROTTLE_L - @mlb_lib.MLB

46C8 73B6

46C8 73B6

ALL_SYS_PWRGD

ALL_SYS_PWRGD - @mlb_lib.MLB

27A8 46D8 62A5 62C5 62C6

27A8 46D8 62A5 62C5 62C6

ALL_SYS_PWRGD_R

ALL_SYS_PWRGD_R - @mlb_lib.MLB

62A6

62A6

AMBIENT_THRMD_N

AMBIENT_THRMD_N - @mlb_lib.MLB

52B3 52C8

52B3 52C8

AMBIENT_THRMD_P

AMBIENT_THRMD_P - @mlb_lib.MLB

52B3 52C8

52B3 52C8

AP_CLKREQ_L

AP_CLKREQ_L - @mlb_lib.MLB

17B4 17B8 35A3

17B4 17B8 35A3

AP_CLKREQ_Q_L

AP_CLKREQ_Q_L - @mlb_lib.MLB

35A4 91D7

35A4 91D7

AP_PWR_EN

AP_PWR_EN - @mlb_lib.MLB

19C1 24D4 35A3 62A4

19C1 24D4 35A3 62A4

AP_RESET_CONN_L

AP_RESET_CONN_L - @mlb_lib.MLB

35B4 91D7

35B4 91D7

AP_RESET_L

AP_RESET_L - @mlb_lib.MLB

27D1 35B3

27D1 35B3

AUD_4V5_REG_IN

AUD_4V5_REG_IN - @mlb_lib.MLB

53A6

53A6

AUD_CONNJ1_MIC_HI

AUD_CONNJ1_MIC_HI - @mlb_lib.MLB

57D7

57D7

AUD_CONNJ1_PERIPHDET

AUD_CONNJ1_PERIPHDET - @mlb_lib.MLB

57C7

57C7

AUD_CONNJ1_RING

AUD_CONNJ1_RING - @mlb_lib.MLB

57D7

57D7

AUD_CONNJ1_SLEEVE

AUD_CONNJ1_SLEEVE - @mlb_lib.MLB

57C7

57C7

AUD_CONNJ1_SLEEVEDET

AUD_CONNJ1_SLEEVEDET - @mlb_lib.MLB

57D7

57D7

AUD_CONNJ1_TIP

AUD_CONNJ1_TIP - @mlb_lib.MLB

57C7

57C7

AUD_CONNJ1_TIPDET

AUD_CONNJ1_TIPDET - @mlb_lib.MLB

57C7

57C7

AUD_CONNJ2_RING

AUD_CONNJ2_RING - @mlb_lib.MLB

57B6

57B6

AUD_CONNJ2_SLEEVE

AUD_CONNJ2_SLEEVE - @mlb_lib.MLB

57B6

57B6

AUD_CONNJ2_TIP

AUD_CONNJ2_TIP - @mlb_lib.MLB

57B6

57B6

AUD_CONNJ2_TIPDET

AUD_CONNJ2_TIPDET - @mlb_lib.MLB

57B6

57B6

AUD_GPIO_1

AUD_GPIO_1 - @mlb_lib.MLB

53D7 55B6

53D7 55B6

AUD_GPIO_3

AUD_GPIO_3 - @mlb_lib.MLB

53C7 56B7

53C7 56B7

AUD_HPAMP_INL_N

AUD_HPAMP_INL_N - @mlb_lib.MLB

55C5

55C5

AUD_HPAMP_INL_P

AUD_HPAMP_INL_P - @mlb_lib.MLB

55C5

55C5

AUD_HPAMP_INR_N

AUD_HPAMP_INR_N - @mlb_lib.MLB

55C5

55C5

AUD_HPAMP_INR_P

AUD_HPAMP_INR_P - @mlb_lib.MLB

55C5

55C5

AUD_HPAMP_MUTE_L

AUD_HPAMP_MUTE_L - @mlb_lib.MLB

55B5

55B5

AUD_HPAMP_OUTL

AUD_HPAMP_OUTL - @mlb_lib.MLB

55C1 55C5 57D3

55C1 55C5 57D3

AUD_HPAMP_OUTR

AUD_HPAMP_OUTR - @mlb_lib.MLB

55B1 55B5 57D3

55B1 55B5 57D3

AUD_HPAMP_RETURN

AUD_HPAMP_RETURN - @mlb_lib.MLB

55B1 57C3

55B1 57C3

AUD_HPREF

AUD_HPREF - @mlb_lib.MLB

53D3

53D3

AUD_I2C_INT_L

AUD_I2C_INT_L - @mlb_lib.MLB

19D6 58D5

19D6 58D5

AUD_INJACK_INSERT_L

AUD_INJACK_INSERT_L - @mlb_lib.MLB

58A5

58A5

AUD_IPHS_SWITCH_EN

AUD_IPHS_SWITCH_EN - @mlb_lib.MLB

20B1 58D5

20B1 58D5

AUD_IPHS_SWITCH_EN_PCH

AUD_IPHS_SWITCH_EN_PCH - @mlb_lib.MLB

20A1 20B3 20D6

20A1 20B3 20D6

AUD_IP_PERIPHERAL_DET

AUD_IP_PERIPHERAL_DET - @mlb_lib.MLB

19D6 58B1

19D6 58B1

AUD_J1_DET_RC

AUD_J1_DET_RC - @mlb_lib.MLB

58B7

58B7

AUD_J1_PERIPHDET

AUD_J1_PERIPHDET - @mlb_lib.MLB

57C3 58A4

57C3 58A4

AUD_J1_PERIPHDET_INV

AUD_J1_PERIPHDET_INV - @mlb_lib.MLB

58A2

58A2

AUD_J1_SLEEVEDET

AUD_J1_SLEEVEDET - @mlb_lib.MLB

57D3 58C6 58C8

57D3 58C6 58C8

AUD_J1_SLEEVEDET_INV

AUD_J1_SLEEVEDET_INV - @mlb_lib.MLB

58C7

58C7

AUD_J1_TIPDET

AUD_J1_TIPDET - @mlb_lib.MLB

57C3 58B8

57C3 58B8

AUD_J2_DET_RC

AUD_J2_DET_RC - @mlb_lib.MLB

58A6

58A6

AUD_J2_OPT_OUT

AUD_J2_OPT_OUT - @mlb_lib.MLB

57B6

57B6

AUD_J2_TIPDET_R

AUD_J2_TIPDET_R - @mlb_lib.MLB

57B3 58A7

57B3 58A7

AUD_L1_GND

AUD_L1_GND - @mlb_lib.MLB

54C6 57B3

54C6 57B3

AUD_L1_INL

AUD_L1_INL - @mlb_lib.MLB

54D6 57B3

54D6 57B3

AUD_L1_INR

AUD_L1_INR - @mlb_lib.MLB

54C6 57B3

54C6 57B3

AUD_L1_LF

AUD_L1_LF - @mlb_lib.MLB

54D5

54D5

AUD_L1_N_LR

AUD_L1_N_LR - @mlb_lib.MLB

53C2 54C3

53C2 54C3

AUD_L1_P_L

AUD_L1_P_L - @mlb_lib.MLB

53C2 54D3

53C2 54D3

AUD_L1_P_R

AUD_L1_P_R - @mlb_lib.MLB

53C2 54C3

53C2 54C3

AUD_L1_RF

AUD_L1_RF - @mlb_lib.MLB

54C5

54C5

AUD_L02_N_R

AUD_L02_N_R - @mlb_lib.MLB

53C2 55C8

53C2 55C8

AUD_L02_N_L

AUD_L02_N_L - @mlb_lib.MLB

53C2 55C8

53C2 55C8

AUD_L02_P_L

AUD_L02_P_L - @mlb_lib.MLB

53D2 55C8

53D2 55C8

AUD_L02_P_R

AUD_L02_P_R - @mlb_lib.MLB

53D2 55B8

53D2 55B8

AUD_L03_N_L

AUD_L03_N_L - @mlb_lib.MLB

53C2 56C7

53C2 56C7

AUD_L03_P_L

AUD_L03_P_L - @mlb_lib.MLB

53C2 56B7

53C2 56B7

AUD_MIC_INN_L

AUD_MIC_INN_L - @mlb_lib.MLB

53C2 58C4

53C2 58C4

AUD_MIC_INP_L

AUD_MIC_INP_L - @mlb_lib.MLB

53C2 58C4

53C2 58C4

AUD_OUTJACK_INSERT_L

AUD_OUTJACK_INSERT_L - @mlb_lib.MLB

58B6

58B6

AUD_PERIPH_DET_R

AUD_PERIPH_DET_R - @mlb_lib.MLB

58B2

58B2

AUD_PORT_D_DET_L

AUD_PORT_D_DET_L - @mlb_lib.MLB

58C5

58C5

AUD_PORT_G_DET_L

AUD_PORT_G_DET_L - @mlb_lib.MLB

58C5

58C5

AUD_REG_SHDN_L

AUD_REG_SHDN_L - @mlb_lib.MLB

53A7

53A7

AUD_SDI_R

AUD_SDI_R - @mlb_lib.MLB

53C6 84C3

53C6 84C3

AUD_SENSE_A

AUD_SENSE_A - @mlb_lib.MLB

53C7 58A7 58C8

53C7 58A7 58C8

AUD_SPDIF_IN

AUD_SPDIF_IN - @mlb_lib.MLB

53C7 57B3

53C7 57B3

AUD_SPDIF_OUT

AUD_SPDIF_OUT - @mlb_lib.MLB

53C7 57C3

53C7 57C3

AUD_SPKRAMP_INL

AUD_SPKRAMP_INL - @mlb_lib.MLB

56B6

56B6

AUD_SPKRAMP_INL_L

AUD_SPKRAMP_INL_L - @mlb_lib.MLB

56B7

56B7

AUD_SPKRAMP_INR

AUD_SPKRAMP_INR - @mlb_lib.MLB

56C6

56C6

AUD_SPKRAMP_INR_L

AUD_SPKRAMP_INR_L - @mlb_lib.MLB

56C7

56C7

AUD_SPKRAMP_SO_H

AUD_SPKRAMP_SO_H - @mlb_lib.MLB

37B6

37B6

BCM5764_CLK25M_XTAL1

BCM5764_CLK25M_XTAL1 - @mlb_lib.MLB

26C1 37B6 85D3

26C1 37B6 85D3

BCM5764_CLK25M_XTAL0

BCM5764_CLK25M_XTAL0 - @mlb_lib.MLB

26C1 37B6 85D3

26C1 37B6 85D3

BCM5764_CLK25M_XTALO

BCM5764_CLK25M_XTALO - @mlb_lib.MLB

26C1 37B6 85D3

26C1 37B6 85D3

BCM5765_CS_L

BCM5765_CS_L - @mlb_lib.MLB

37A8 37B6

37A8 37B6

BCM5765_MISO

BCM5765_MISO - @mlb_lib.MLB

37A6 37B6

37A6 37B6

BCM5765_MOSI

BCM5765_MOSI - @mlb_lib.MLB

37A6 37B6

37A6 37B6

BCM5765_RDAC

BCM5765_RDAC - @mlb_lib.MLB

37B6

37B6

BCM5765_SCLK

BCM5765_SCLK - @mlb_lib.MLB

37A8 37B6

37A8 37B6

BCM5765_VMAIN_PRSNT

BCM5765_VMAIN_PRSNT - @mlb_lib.MLB

37C8

37C8

BCM5765_WAKE_L

BCM5765_WAKE_L - @mlb_lib.MLB

37B6

37B6

BCM5765_SR_DISABLE

BCM5765_SR_DISABLE - @mlb_lib.MLB

37B4

37B4

BOOT_1V8S0

BOOT_1V8S0 - @mlb_lib.MLB

71C6

71C6

CATERR_LED

CATERR_LED - @mlb_lib.MLB

681

681

CATERR_LED_R

CATERR_LED_R - @mlb_lib.MLB

681

681

CATERR_R

CATERR_R - @mlb_lib.MLB

682

682

CIV_ENET_RESET_L

CIV_ENET_RESET_L - @mlb_lib.MLB

34B2 37B6

34B2 37B6

CLK98M_FW_XI

CLK98M_FW_XI - @mlb_lib.MLB

39C3 85B3

39C3 85B3

CLK98M_FW_XI_R

CLK98M_FW_XI_R - @mlb_lib.MLB

39C2 85B3

39C2 85B3

COMP_1V8S0

COMP_1V8S0 - @mlb_lib.MLB

71C7

71C7

COMP_RC_1V8S0

CPUIMVP_ISUMG_IOUT

CPUIMVP_ISUMG_RN

CPUIMVP_ISUM_IOUT

CPUIMVP_ISUM_RN

CPUPTHMSNS_D2_N

CPUPTHMSNS_D2_P

CPUVCCIOS0_AGND

CPUVCCIOS0_CS_N

CPUVCCIOS0_CS_P

CPUVCCIOS0_DRVH

CPUVCCIOS0_DRVL

CPUVCCIOS0_FB

CPUVCCIOS0_FSEL

CPUVCCIOS0_LL

CPUVCCIOS0_OCSET

CPUVCCIOS0_PG00D

CPUVCCIOS0_RTN

CPUVCCIOS0_SREF

CPUVCCIOS0_VBST

CPUVCCIOS0_VBST_RC

CPUVCCIOS0_VO

CPUVCCIO_IOUT

CPUVCCSAS0_SNUB

CPUVSENSE_IN

CPU_AXG_SENSE_N

CPU_AXG_SENSE_P

CPU_AXG_VALSENSE_N

CPU_AXG_VALSENSE_P

CPU_CATERR_L

CPU_CLKREQ_L

CPU_CFG<0>

CPU_CFG<1..8>

CPU_CFG<1>

CPU_CFG<2>

CPU_CFG<3>

CPU_CFG<4>

CPU_CFG<5>

CPU_CFG<6>

CPU_CFG<7>

CPU_CFG<17..16>

CPU_COMP0

CPU_COMP1

CPU_COMP2

CPU_COMP3

CPU_DDR_VREF

CPU_EDP_COMP

CPU_MEM_RESET_L

CPU_MEM_VREFDQ_A

CPU_MEM_VREFDQ_B

CPU_PECI_L

CPU_PECI_R

CPU_PEG_COMP

CPU_PEG_RBIA5

CPU_PROCHOT_BUF

CPU_PROCHOT_L

CPU_PROCHOT_L_R

CPU_PROCHOT_R_L

CPU_PROX_SEL_L

CPU_PROX_THRM_N

CPU_PROX_THRM_P

CPU_PST_L

CPU_PWRGD

CPU_RESET_L

CPU_SM_RCMP<0>

CPU_SM_RCMP<2..0>

CPU_SM_RCMP<1>

CPU_SM_RCMP<2>

CPU_THERMD_N

CPU_THERMD_P

CPU_VCCIOSENSE_N

CPU_VCCIOSENSE_P

CPU_VCCIO_SEL

CPU_VCCSA_SENSE

CPU_VCCSA_VID<0>

CPU_VCCSA_VID<1>

CPU_VCCSENSE_N

CPU_VCCSENSE_P

CPU_VCC_VALSENSE_N

CPU_VCC_VALSENSE_P

CPU_VIDALERT_L

CPU_VIDALERT_L_R

CPU_VIDSCLK

CPU_VIDSCLK_R

CPU_VIDSOUT

CPU_VIDSOUT_R

CPU_VTTSELECT

CS4206_SPDIF_OUT

CS4206_VCOM

CS4206_VREF_ADC

DC1N_VSENSE_EN_L

DC1N_IOUT

DC1N_SS_VSENSE

DC_TEST_B3_C2

DC_TEST_B63_A64

DC_TEST_B65_C64

DC_TEST_B664_BH65

DC_TEST_BH1_B62

DC_TEST_BH3_B32

DC_TEST_B364_BH63

DORREG_IV8_VREF

DORREG_DRVH

DORREG_DRVL

DORREG_MODE

DORREG_PG00D

DORREG_REFIN

DORREG_SNUB

DORREG_SW

DORREG_TRIP

DORREG_VBST

DORREG_VBST_R

DORREG_VOVSNS

DORREG_VTTSNS

DIMM_PROX_THRM_N

DIMM_PROX_THRM_P

DMI_CLK100M_CPU_N

DMI_CLK100M_CPU_P

DMI_N2S_N<0>

DMI_N2S_N<3..0>

DMI_N2S_N<1>

DMI_N2S_N<2>

COMP_RC_1V8S0

CPUIMVP_ISUMG_IOUT

CPUIMVP_ISUMG_RN

CPUIMVP_ISUM_IOUT

CPUIMVP_ISUM_RN

CPUPTHMSNS_D2_N

CPUPTHMSNS_D2_P

CPUVCCIOS0_AGND

CPUVCCIOS0_CS_N

CPUVCCIOS0_CS_P

CPUVCCIOS0_DRVH

CPUVCCIOS0_DRVL

CPUVCCIOS0_FB

CPUVCCIOS0_FSEL

CPUVCCIOS0_LL

CPUVCCIOS0_OCSET

CPUVCCIOS0_PG00D

CPUVCCIOS0_RTN

CPUVCCIOS0_SREF

CPUVCCIOS0_VBST

CPUVCCIOS0_VBST_RC

CPUVCCIOS0_VO

CPUVCCIO_IOUT

CPUVCCSAS0_SNUB

CPUVSENSE_IN

CPU_AXG_SENSE_N

CPU_AXG_SENSE_P

CPU_AXG_VALSENSE_N

CPU_AXG_VALSENSE_P

CPU_CATERR_L

CPU_CLKREQ_L

CPU_CFG<0>

CPU_CFG<1..8>

CPU_CFG<1>

CPU_CFG<2>

CPU_CFG<3>

CPU_CFG<4>

CPU_CFG<5>

CPU_CFG<6>

CPU_CFG<7>

CPU_CFG<17..16>

CPU_COMP0

CPU_COMP1

CPU_COMP2

CPU_COMP3

CPU_DDR_VREF

CPU_EDP_COMP

CPU_MEM_RESET_L

CPU_MEM_VREFDQ_A

CPU_MEM_VREFDQ_B

CPU_PECI_L

CPU_PECI_R

CPU_PEG_COMP

CPU_PEG_RBIA5

CPU_PROCHOT_BUF

CPU_PROCHOT_L

CPU_PROCHOT_L_R

CPU_PROCHOT_R_L

CPU_PROX_SEL_L

CPU_PROX_THRM_N

CPU_PROX_THRM_P

CPU_PST_L

CPU_PWRGD

CPU_RESET_L

CPU_SM_RCMP<0>

CPU_SM_RCMP<2..0>

CPU_SM_RCMP<1>

CPU_SM_RCMP<2>

CPU_THERMD_N

CPU_THERMD_P

CPU_VCCIOSENSE_N

CPU_VCCIOSENSE_P

CPU_VCCIO_SEL

CPU_VCCSA_SENSE

CPU_VCCSA_VID<0>

CPU_VCCSA_VID<1>

CPU_VCCSENSE_N

CPU_VCCSENSE_P

CPU_VCC_VALSENSE_N

CPU_VCC_VALSENSE_P

CPU_VIDALERT_L

CPU_VIDALERT_L_R

CPU_VIDSCLK

CPU_VIDSCLK_R

CPU_VIDSOUT

CPU_VIDSOUT_R

CPU_VTTSELECT

CS4206_SPDIF_OUT

CS4206_VCOM

CS4206_VREF_ADC

DC1N_VSENSE_EN_L

DC1N_IOUT

DC1N_SS_VSENSE

DC_TEST_B3_C2

DC_TEST_B63_A64

DC_TEST_B65_C64

DC_TEST_B664_BH65

DC_TEST_BH1_B62

DC_TEST_BH3_B32

DC_TEST_B364_BH63

DORREG_IV8_VREF

DORREG_DRVH

DORREG_DRVL

DORREG_MODE

DORREG_PG00D

DORREG_REFIN

DORREG_SNUB

DORREG_SW

DORREG_TRIP

DORREG_VBST

DORREG_VBST_R

DORREG_VOVSNS

DORREG_VTTSNS

DIMM_PROX_THRM_N

DIMM_PROX_THRM_P

DMI_CLK100M_CPU_N

DMI_CLK100M_CPU_P

DMI_N2S_N<0>

DMI_N2S_N<3..0>

DMI_N2S_N<1>

DMI_N2S_N<2>

DMI_N2S_N<3>

DMI_N2S_P<0>

DMI_N2S_P<3..0>

DMI_N2S_P<1>

DMI_N2S_P<2>

DMI_N2S_P<3>

DMI_S2N_N<0>

DMI_S2N_N<3..0>

DMI_S2N_N<1>

DMI_S2N_N<2>

DMI_S2N_N<3>

DMI_S2N_P<0>

DMI_S2N_P<3..0>

DMI_S2N_P<1>

DMI_S2N_P<2>

DMI_S2N_P<3>

DPAACONN_20_RC

DPAPWRSW_CT

DPAPWRSW_HVEN_L_R

DPAPWRSW_IFLT

DPAPWRSW_ILIM

DPLL_REF_CLK

DPLL_REF_CLK_L

DPSDRVA_CEXT

DPSDRVA_I2C_ADDR0

DPSDRVA_I2C_ADDR1

DPSDRVA_I2C_CTL_EN

DPSDRVA_REXT

DP_AUXCH_ISOL

DP_AUX_CH_CK_N

DP_AUX_CH_CK_P

DP_AUX_CH_C_N

DP_AUX_CH_C_P

DP_A_BIAS

DP_A_BIAS0N

DP_A_BIAS0P

DP_A_BIAS2N

DP_A_BIAS2P

DP_A_CA_DET

DP_A_EXT_AUXCH_N

DP_A_EXT_AUXCH_P

DP_A_EXT_HPD

DP_A_PWRDWN

DP_A_PWRDWN_FET_R

DP_A_PWRDWN_INV

DP_A_PWRDWN_R

DP_EG_AUX_CH_N

DP_EG_AUX_CH_P

DP_EXT_A_AUXCH_C_N

DP_EXT_A_AUXCH_C_P

DP_EXT_A_AUXCH_N

DP_EXT_A_AUXCH_P

DP_EXT_A_DDC_CLK

DP_EXT_A_DDC_CLK_AUX_P

DP_EXT_A_DDC_DATA

DP_EXT_A_DDC_DATA_AUX_N

DP_EXT_A_HPD

DP_EXT_A_ML_C_N<0>

DP_EXT_A_ML_C_N<3..0>

DP_EXT_A_ML_C_N<1>

DP_EXT_A_ML_C_N<2>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

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DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

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DP_EXT_A_ML_C_N<3>

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DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

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DP_EXT_A_ML_C_N<3>

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DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

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DP_EXT_A_ML_C_N<3>

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DP_EXT_A_ML_C_N<3>

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DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_ML_C_N<3>

DP_EXT_A_M

	8	7	6	5	4	3	2	1
D	ENET_MDI_P<3..0>	ENET_MDI_P<3..0> - @m1b_lib.MLB	85D3	FDI_FSYNC<1>	FDI_FSYNC<1> - @m1b_lib.MLB	18C5		
	ENET_MDI_P<1>	ENET_MDI_P<1> - @m1b_lib.MLB	36C8 37C3	FDI_INT	=FDI_FSYNC<1..0> - @m1b_lib.MLB	8D8		
	ENET_MDI_P<2>	ENET_MDI_P<2> - @m1b_lib.MLB	36C8 37C3	FDI_INT - @m1b_lib.MLB	805 18C5 81D3			
	ENET_MDI_P<3>	ENET_MDI_P<3> - @m1b_lib.MLB	36B8 37C3	FDI_LSYNC<0>	FDI_INT - @m1b_lib.MLB	808 10C8		
	ENET_MDI_TRAN_N<0>	ENET_MDI_TRAN_N<0> - @m1b_lib.MLB	36B3	FDI_LSYNC<0> - @m1b_lib.MLB	18C5			
	ENET_MDI_TRAN_N<3..0	ENET_MDI_TRAN_N<3..0> - @m1b_lib.MLB	85D3	FDI_LSYNC<1..0> - @m1b_lib.MLB	808			
	>	@m1b_lib.MLB		FDI_LSYNC<1..0> - @m1b_lib.MLB	805 81D3			
	ENET_MDI_TRAN_N<1>	ENET_MDI_TRAN_N<1> - @m1b_lib.MLB	36C4	FDI_LSYNC<1>	18C5			
	ENET_MDI_TRAN_N<2>	ENET_MDI_TRAN_N<2> - @m1b_lib.MLB	36C4	=FDI_LSYNC<1..0> - @m1b_lib.MLB	808			
	ENET_MDI_TRAN_N<3>	ENET_MDI_TRAN_N<3> - @m1b_lib.MLB	36B3	FWOHCI_CLK98M_LCLK	FWOHCI_CLK98M_LCLK - @m1b_lib.MLB	39C3		
C	ENET_MDI_TRAN_P<0>	ENET_MDI_TRAN_P<0> - @m1b_lib.MLB	36C3	FWOHCI_LINKON_L	FWOHCI_LINKON_L - @m1b_lib.MLB	39C3		
	ENET_MDI_TRAN_P<3..0	ENET_MDI_TRAN_P<3..0> - @m1b_lib.MLB	85D3	FWOHCI_LPS	FWOHCI_LPS - @m1b_lib.MLB	39B3		
	>	@m1b_lib.MLB		FWOHCI_LREQ	FWOHCI_LREQ - @m1b_lib.MLB	39B3		
	ENET_MDI_TRAN_P<1>	ENET_MDI_TRAN_P<1> - @m1b_lib.MLB	36C4	FWPHY_BMODE	FWPHY_BMODE - @m1b_lib.MLB	39A6		
	ENET_MDI_TRAN_P<2>	ENET_MDI_TRAN_P<2> - @m1b_lib.MLB	36C4	FWPHY_CLK98M_PCLK	FWPHY_CLK98M_PCLK - @m1b_lib.MLB	39B3		
	ENET_MEDIA_SENSE	ENET_MEDIA_SENSE - @m1b_lib.MLB	37B1 38B5	FWPHY_CPS	FWPHY_CPS - @m1b_lib.MLB	39B3		
	ENET_MEDIA_SENSE_RDIV	ENET_MEDIA_SENSE_RDIV - @m1b_lib.MLB	17C8 38B3	FWPHY_LKON_DS2	FWPHY_LKON_DS2 - @m1b_lib.MLB	39B3		
	V	@m1b_lib.MLB		FWPHY_PINT	FWPHY_PINT - @m1b_lib.MLB	39B3		
	ENET_MSNS_EN	ENET_MSNS_EN - @m1b_lib.MLB	38A4	FWPHY_R0	FWPHY_R0 - @m1b_lib.MLB	39B3		
	ENET_MSNS_EN_L	ENET_MSNS_EN_L - @m1b_lib.MLB	38A4	FWPHY_R1	FWPHY_R1 - @m1b_lib.MLB	39B3		
B	ENET_PWR_EN	ENET_PWR_EN - @m1b_lib.MLB	19C1 24D4	FWPHY_RESET_L	FWPHY_RESET_L - @m1b_lib.MLB	39A3		
	ENET_RESET_L	ENET_RESET_L - @m1b_lib.MLB	27D1 34B8 85D3	FWPHY_TESTM	FWPHY_TESTM - @m1b_lib.MLB	39A6		
	ENET_SR_LX	ENET_SR_LX - @m1b_lib.MLB	37D4 38D5	FWPHY_TESTW	FWPHY_TESTW - @m1b_lib.MLB	39A6		
	ENET_SR_VFB	ENET_SR_VFB - @m1b_lib.MLB	37D4 38C5	FWXIO_CYCLEOUT	FWXIO_CYCLEOUT - @m1b_lib.MLB	39B6		
	ENET_WAKE_L	ENET_WAKE_L - @m1b_lib.MLB	37B8 38C1	FWXIO_JTAG_TCK	FWXIO_JTAG_TCK - @m1b_lib.MLB	39B6		
	EXCARD_CLKREQ_L	EXCARD_CLKREQ_L - @m1b_lib.MLB	17B4 17B5	FWXIO_JTAG_TRST	FWXIO_JTAG_TRST - @m1b_lib.MLB	39A6		
	FAN0_CTL_F	FAN0_CTL_F - @m1b_lib.MLB	60C3 91D5	FWXIO_REF0_PCIE	FWXIO_REF0_PCIE - @m1b_lib.MLB	39C6		
	FAN0_TACH_F	FAN0_TACH_F - @m1b_lib.MLB	60C3 91D5	FWXIO_REF1_PCIE	FWXIO_REF1_PCIE - @m1b_lib.MLB	39C6		
	FAN_S0_EN_L	FAN_S0_EN_L - @m1b_lib.MLB	60C6	FWXIO_REFCLK_SEL	FWXIO_REFCLK_SEL - @m1b_lib.MLB	39B6		
	FAN_S0_EN_L_G	FAN_S0_EN_L_G - @m1b_lib.MLB	60C5	FWXIO_REF_PCIE	FWXIO_REF_PCIE - @m1b_lib.MLB	39C7		
A	FAN_TACH0	FAN_TACH0 - @m1b_lib.MLB	60C7	FWXIO_SCL	FWXIO_SCL - @m1b_lib.MLB	39B6		
	FB_1V8S0	FB_1V8S0 - @m1b_lib.MLB	71C7	FWXIO_SDA	FWXIO_SDA - @m1b_lib.MLB	39B6		
	FB_B0_A<8..0>	FB_B0_A<8..0> - @m1b_lib.MLB	88D1	FWXIO_SNOOP_EN	FWXIO_SNOOP_EN - @m1b_lib.MLB	39A6		
	FB_B0_ABI_L	FB_B0_ABI_L - @m1b_lib.MLB	88D1	FWXIO_VDDI5COMB	FWXIO_VDDI5COMB - @m1b_lib.MLB	39C3		
	FB_B0_CAS_L	FB_B0_CAS_L - @m1b_lib.MLB	88D1	FWXIO_VDD33COMB	FWXIO_VDD33COMB - @m1b_lib.MLB	39C3		
	FB_B0_CKE_L	FB_B0_CKE_L - @m1b_lib.MLB	88D1	FWXIO_VDD33COMB0	FWXIO_VDD33COMB0 - @m1b_lib.MLB	39A6		
	FB_B0_CLK_N	FB_B0_CLK_N - @m1b_lib.MLB	88D1	FWXIO_VREG_PD33	FWXIO_VREG_PD33 - @m1b_lib.MLB	39A6		
	FB_B0_CLK_P	FB_B0_CLK_P - @m1b_lib.MLB	88D1	FW_CLKREQ_L	FW_CLKREQ_L - @m1b_lib.MLB	17B5 17C4 39C6		
	FB_B0_CS_L	FB_B0_CS_L - @m1b_lib.MLB	88D1	FW_CUTOFF	FW_CUTOFF - @m1b_lib.MLB	41A6 41C8		
	FB_B0_DBI_L<0>	FB_B0_DBI_L<0> - @m1b_lib.MLB	88C1	FW_EN_A	FW_EN_A - @m1b_lib.MLB	41C7		
B	FB_B0_DBI_L<1>	FB_B0_DBI_L<1> - @m1b_lib.MLB	88C1	FW_EN_B	FW_EN_B - @m1b_lib.MLB	41C7		
	FB_B0_DBI_L<2>	FB_B0_DBI_L<2> - @m1b_lib.MLB	88C1	FW_FETGATE	FW_FETGATE - @m1b_lib.MLB	4107		
	FB_B0_DBI_L<3>	FB_B0_DBI_L<3> - @m1b_lib.MLB	88C1	FW_FETV	FW_FETV - @m1b_lib.MLB	41C6		
	FB_B0_DQ<7..0>	FB_B0_DQ<7..0> - @m1b_lib.MLB	88C1	FW_FET_TEMP	FW_FET_TEMP - @m1b_lib.MLB	41A8 41B7		
	FB_B0_DQ<15..8>	FB_B0_DQ<15..8> - @m1b_lib.MLB	88C1	FW_FET_TREF	FW_FET_TREF - @m1b_lib.MLB	41A8		
	FB_B0_DQ<23..16>	FB_B0_DQ<23..16> - @m1b_lib.MLB	88C1	FW_IREF	FW_IREF - @m1b_lib.MLB	41C5		
	FB_B0_DQ<31..24>	FB_B0_DQ<31..24> - @m1b_lib.MLB	88C1	FW_LATEVG_L	FW_LATEVG_L - @m1b_lib.MLB	41A4		
	FB_B0_EDC<0>	FB_B0_EDC<0> - @m1b_lib.MLB	88D1	FW_LATEVG_PP2V4_RC	FW_LATEVG_PP2V4_RC - @m1b_lib.MLB	41A5		
	FB_B0_EDC<1>	FB_B0_EDC<1> - @m1b_lib.MLB	88D1	FW_LATEVG_REF	FW_LATEVG_REF - @m1b_lib.MLB	41A5		
	FB_B0_EDC<2>	FB_B0_EDC<2> - @m1b_lib.MLB	88D1	FW_OC_DR	FW_OC_DR - @m1b_lib.MLB	41C7		
A	FB_B0_EDC<3>	FB_B0_EDC<3> - @m1b_lib.MLB	88C1	FW_P0_TPA_C	FW_P0_TPA_C - @m1b_lib.MLB	40C3		
	FB_B0_RAS_L	FB_B0_RAS_L - @m1b_lib.MLB	88D1	FW_P0_TPA_L_N	FW_P0_TPA_L_N - @m1b_lib.MLB	40D2 85B3		
	FB_B0_WCLK_N<0>	FB_B0_WCLK_N<0> - @m1b_lib.MLB	88C1	FW_P0_TPA_L_P	FW_P0_TPA_L_P - @m1b_lib.MLB	40D3 85B3		
	FB_B0_WCLK_N<1>	FB_B0_WCLK_N<1> - @m1b_lib.MLB	88C1	FW_P0_TPBIA5	FW_P0_TPBIA5 - @m1b_lib.MLB	39A2 40D4		
	FB_B0_WCLK_P<0>	FB_B0_WCLK_P<0> - @m1b_lib.MLB	88C1	FW_P0_TPB_L_N	FW_P0_TPB_L_N - @m1b_lib.MLB	40C3 85B3		
	FB_B0_WCLK_P<1>	FB_B0_WCLK_P<1> - @m1b_lib.MLB	88C1	FW_P0_TPB_L_P	FW_P0_TPB_L_P - @m1b_lib.MLB	40D2 85B3		
	FB_B0_WE_L	FB_B0_WE_L - @m1b_lib.MLB	88D1	FW_P12V_FETPWR	FW_P12V_FETPWR - @m1b_lib.MLB	41C6		
	FB_B1_A<8..0>	FB_B1_A<8..0> - @m1b_lib.MLB	88D1	FW_PHY_DS0	FW_PHY_DS0 - @m1b_lib.MLB	39C2 40B7		
	FB_B1_ABI_L	FB_B1_ABI_L - @m1b_lib.MLB	88D1	FW_PHY_DS1	FW_PHY_DS1 - @m1b_lib.MLB	39C2 40B7		
	FB_B1_CAS_L	FB_B1_CAS_L - @m1b_lib.MLB	88D1	FW_PHY_PC0	FW_PHY_PC0 - @m1b_lib.MLB	39C2 40A7		
B	FB_B1_CKE_L	FB_B1_CKE_L - @m1b_lib.MLB	88D1	FW_PME_L	FW_PME_L - @m1b_lib.MLB	20B6 20D6 39B6		
	FB_B1_CLK_N	FB_B1_CLK_N - @m1b_lib.MLB	88D1	FW_PORT0_TPAL_N	FW_PORT0_TPAL_N - @m1b_lib.MLB	41C2 85B3		
	FB_B1_CLK_P	FB_B1_CLK_P - @m1b_lib.MLB	88D1	FW_PORT0_TPAL_P	FW_PORT0_TPAL_P - @m1b_lib.MLB	41C2 85B3		
	FB_B1_CS_L	FB_B1_CS_L - @m1b_lib.MLB	88D1	FW_PORT0_TPA_N	FW_PORT0_TPA_N - @m1b_lib.MLB	40C1 41C4 85B3		
	FB_B1_DBI_L<0>	FB_B1_DBI_L<0> - @m1b_lib.MLB	88C1	FW_PORT0_TPA_P	FW_PORT0_TPA_P - @m1b_lib.MLB	40C1 41C4 85B3		
	FB_B1_DBI_L<1>	FB_B1_DBI_L<1> - @m1b_lib.MLB	88C1	FW_PORT0_TPB_L_N	FW_PORT0_TPB_L_N - @m1b_lib.MLB	41C2		
	FB_B1_DBI_L<2>	FB_B1_DBI_L<2> - @m1b_lib.MLB	88C1	FW_PORT0_TPB_L_P	FW_PORT0_TPB_L_P - @m1b_lib.MLB	41C2 85B3		
	FB_B1_DBI_L<3>	FB_B1_DBI_L<3> - @m1b_lib.MLB	88C1	FW_PORT0_TPB_N	FW_PORT0_TPB_N - @m1b_lib.MLB	41C2 85B3		
	FB_B1_DQ<7..0>	FB_B1_DQ<7..0> - @m1b_lib.MLB	88C1	FW_PORT0_TPB_P	FW_PORT0_TPB_P - @m1b_lib.MLB	41C2 85B3		
	FB_B1_DQ<15..8>	FB_B1_DQ<15..8> - @m1b_lib.MLB	88C1	FW_PORT0_VP	FW_PORT0_VP - @m1b_lib.MLB	41D2 91B5		
B	FB_B1_DQ<23..16>	FB_B1_DQ<23..16> - @m1b_lib.MLB	88B1	FW_PORT0_VP_F	FW_PORT0_VP_F - @m1b_lib.MLB	41D2		
	FB_B1_DQ<31..24>	FB_B1_DQ<31..24> - @m1b_lib.MLB	88B1	FW_PORTPWR_DISABLE_L	FW_PORTPWR_DISABLE_L - @m1b_lib.MLB	41A3 41C8		
	FB_B1_EDC<0>	FB_B1_EDC<0> - @m1b_lib.MLB	88C1	FW_PWR_EN	FW_PWR_EN - @m1b_lib.MLB	20B6 20C6		
	FB_B1_EDC<1>	FB_B1_EDC<1> - @m1b_lib.MLB	88C1	FW_REF_A	FW_REF_A - @m1b_lib.MLB	41B5		
	FB_B1_EDC<2>	FB_B1_EDC<2> - @m1b_lib.MLB	88C1	FW_REF_B	FW_REF_B - @m1b_lib.MLB	41B5		
	FB_B1_EDC<3>	FB_B1_EDC<3> - @m1b_lib.MLB	88C1	FW_RESET_L	FW_RESET_L - @m1b_lib.MLB	27D1 39C6		
	FB_B1_RAS_L	FB_B1_RAS_L - @m1b_lib.MLB	88D1	FW_SNSN				

	8	7	6	5	4	3	2	1
D	MEM_A_DQ<45> MEM_A_DQ<46> MEM_A_DQ<47> MEM_A_DQ<48> MEM_A_DQ<55..48> MEM_A_DQ<49> MEM_A_DQ<50> MEM_A_DQ<51> MEM_A_DQ<52> MEM_A_DQ<53> MEM_A_DQ<54> MEM_A_DQ<55> MEM_A_DQ<56> MEM_A_DQ<63..56> MEM_A_DQ<57> MEM_A_DQ<58> MEM_A_DQ<59> MEM_A_DQ<60> MEM_A_DQ<61> MEM_A_DQ<62> MEM_A_DQ<63> MEM_A_DQS_N<0> MEM_A_DQS_N<1> MEM_A_DQS_N<2> MEM_A_DQS_N<3> MEM_A_DQS_N<4> MEM_A_DQS_N<5> MEM_A_DQS_N<6> MEM_A_DQS_N<7> MEM_A_DQS_P<0> MEM_A_DQS_P<1> MEM_A_DQS_P<2> MEM_A_DQS_P<3> MEM_A_DQS_P<4> MEM_A_DQS_P<5> MEM_A_DQS_P<6> MEM_A_DQS_P<7> MEM_A_ODT<0> MEM_A_ODT<3..0> MEM_A_ODT<1> MEM_A_RAS_L MEM_A_SA<0> MEM_A_SA<1> MEM_A_WE_L MEM_B_A<0> MEM_B_A<15..0> MEM_B_A<1> MEM_B_A<2> MEM_B_A<3> MEM_B_A<4> MEM_B_A<5> MEM_B_A<6> MEM_B_A<7> MEM_B_A<8> MEM_B_A<9> MEM_B_A<10> MEM_B_A<11> MEM_B_A<12> MEM_B_A<13> MEM_B_A<14> MEM_B_A<15> MEM_B_BA<0> MEM_B_BA<2..0> MEM_B_BA<1> MEM_B_BA<2> MEM_B_CAS_L MEM_B_CKE<0> MEM_B_CKE<3..0> MEM_B_CKE<1> MEM_B_CLK_N<0> MEM_B_CLK_N<5..0> MEM_B_CLK_N<1> MEM_B_CLK_P<0> MEM_B_CLK_P<5..0> MEM_B_CLK_P<1> MEM_B_CS_L<0> MEM_B_CS_L<3..0> MEM_B_CS_L<1> MEM_B_DQ<0> MEM_B_DQ<7..0> MEM_B_DQ<1> MEM_B_DQ<2> MEM_B_DQ<3> MEM_B_DQ<4> MEM_B_DQ<5> MEM_B_DQ<6> MEM_B_DQ<7> MEM_B_DQ<8> MEM_B_DQ<15..8> MEM_B_DQ<9> MEM_B_DQ<10> MEM_B_DQ<11> MEM_B_DQ<12> MEM_B_DQ<13> MEM_B_DQ<14> MEM_B_DQ<15> MEM_B_DQ<16> MEM_B_DQ<23..16> MEM_B_DQ<17> MEM_B_DQ<18> MEM_B_DQ<19> MEM_B_DQ<20> MEM_B_DQ<21> MEM_B_DQ<22> MEM_B_DQ<23> MEM_B_DQ<24> MEM_B_DQ<31..24> MEM_B_DQ<25> MEM_B_DQ<26> MEM_B_DQ<27> MEM_B_DQ<28> MEM_B_DQ<29> MEM_B_DQ<30> MEM_B_DQ<31> MEM_B_DQ<32> MEM_B_DQ<39..32> MEM_B_DQ<33> MEM_B_DQ<34> MEM_B_DQ<35> MEM_B_DQ<36> MEM_B_DQ<37> MEM_B_DQ<38> MEM_B_DQ<39> MEM_B_DQ<40>	MEM_A_DQ<45> MEM_A_DQ<46> MEM_A_DQ<47> MEM_A_DQ<48> MEM_A_DQ<55..48> MEM_A_DQ<49> MEM_A_DQ<50> MEM_A_DQ<51> MEM_A_DQ<52> MEM_A_DQ<53> MEM_A_DQ<54> MEM_A_DQ<55> MEM_A_DQ<56> MEM_A_DQ<63..56> MEM_A_DQ<57> MEM_A_DQ<58> MEM_A_DQ<59> MEM_A_DQ<60> MEM_A_DQ<61> MEM_A_DQ<62> MEM_A_DQ<63> MEM_A_DQS_N<0> MEM_A_DQS_N<1> MEM_A_DQS_N<2> MEM_A_DQS_N<3> MEM_A_DQS_N<4> MEM_A_DQS_N<5> MEM_A_DQS_N<6> MEM_A_DQS_N<7> MEM_A_DQS_P<0> MEM_A_DQS_P<1> MEM_A_DQS_P<2> MEM_A_DQS_P<3> MEM_A_DQS_P<4> MEM_A_DQS_P<5> MEM_A_DQS_P<6> MEM_A_DQS_P<7> MEM_A_ODT<0> MEM_A_ODT<3..0> MEM_A_ODT<1> MEM_A_RAS_L MEM_A_SA<0> MEM_A_SA<1> MEM_A_WE_L MEM_B_A<0> MEM_B_A<15..0> MEM_B_A<1> MEM_B_A<2> MEM_B_A<3> MEM_B_A<4> MEM_B_A<5> MEM_B_A<6> MEM_B_A<7> MEM_B_A<8> MEM_B_A<9> MEM_B_A<10> MEM_B_A<11> MEM_B_A<12> MEM_B_A<13> MEM_B_A<14> MEM_B_A<15> MEM_B_BA<0> MEM_B_BA<2..0> MEM_B_BA<1> MEM_B_BA<2> MEM_B_CAS_L MEM_B_CKE<0> MEM_B_CKE<3..0> MEM_B_CKE<1> MEM_B_CLK_N<0> MEM_B_CLK_N<5..0> MEM_B_CLK_N<1> MEM_B_CLK_P<0> MEM_B_CLK_P<5..0> MEM_B_CLK_P<1> MEM_B_CS_L<0> MEM_B_CS_L<3..0> MEM_B_CS_L<1> MEM_B_DQ<0> MEM_B_DQ<7..0> MEM_B_DQ<1> MEM_B_DQ<2> MEM_B_DQ<3> MEM_B_DQ<4> MEM_B_DQ<5> MEM_B_DQ<6> MEM_B_DQ<7> MEM_B_DQ<8> MEM_B_DQ<15..8> MEM_B_DQ<9> MEM_B_DQ<10> MEM_B_DQ<11> MEM_B_DQ<12> MEM_B_DQ<13> MEM_B_DQ<14> MEM_B_DQ<15> MEM_B_DQ<16> MEM_B_DQ<23..16> MEM_B_DQ<17> MEM_B_DQ<18> MEM_B_DQ<19> MEM_B_DQ<20> MEM_B_DQ<21> MEM_B_DQ<22> MEM_B_DQ<23> MEM_B_DQ<24> MEM_B_DQ<31..24> MEM_B_DQ<25> MEM_B_DQ<26> MEM_B_DQ<27> MEM_B_DQ<28> MEM_B_DQ<29> MEM_B_DQ<30> MEM_B_DQ<31> MEM_B_DQ<32> MEM_B_DQ<39..32> MEM_B_DQ<33> MEM_B_DQ<34> MEM_B_DQ<35> MEM_B_DQ<36> MEM_B_DQ<37> MEM_B_DQ<38> MEM_B_DQ<39> MEM_B_DQ<40>	1287 2988 1287 2988 1287 2988 1287 29A8 8203 1287 29A8 1287 29A8 1287 29A8 1287 29A8 1287 29A8 1287 29A8 1287 29A8 1287 29A8 8203 1287 29A8 1287 29A8 1287 29A8 1287 29A8 1287 29A8 1287 29A8 1287 29A8 1205 2908 82C3 1205 2908 82C3 1205 2908 82C3 1205 2908 82C3 1205 2988 82C3					

[illegible]

[illegible]

8				7				6				5				4				3				2				1			
D	USB_BT_CONN_P	USB_BT_CONN_P - @m1b_lib.MLB	35C6	83B3	91D7	VREFMRGN_MEMVREG_FBV	VREFMRGN_MEMVREG_FBVREF -	3305	XDP_PCH_TCK	XDP_PCH_TCK - @m1b_lib.MLB	17C8	24A4	24A8	24C7	81D1	D															
	USB_BT_N	USB_BT_N - @m1b_lib.MLB	25B4	35C4	83B3	REF	@m1b_lib.MLB		XDP_PCH_TDI	XDP_PCH_TDI - @m1b_lib.MLB	17C8	24A4	24B6	81D1																	
	USB_BT_P	USB_BT_P - @m1b_lib.MLB	25B4	35C4	83B3	VREFMRGN_SODIMMA_DQ	VREFMRGN_SODIMMA_DQ - @m1b_lib.MLB	3305	XDP_PCH_TDO	XDP_PCH_TDO - @m1b_lib.MLB	17C8	24A4	24B6	81D1																	
	USB_B_MUXED_N	USB_B_MUXED_N - @m1b_lib.MLB	43D3	83B3		VREFMRGN_SODIMMB_DQ	VREFMRGN_SODIMMB_DQ - @m1b_lib.MLB	3305	XDP_PCH_TMS	XDP_PCH_TMS - @m1b_lib.MLB	17C8	24A4	24B6	81D1																	
	USB_B_MUXED_P	USB_B_MUXED_P - @m1b_lib.MLB	43D3	83B3		VREFMRGN_SODIMMS_CA	VREFMRGN_SODIMMS_CA - @m1b_lib.MLB	3305	XDP_PRESENT_L	XDP_PRESENT_L - @m1b_lib.MLB	24B2	24C4	81C1																		
	USB_DEBUGPRT_EN_L	USB_DEBUGPRT_EN_L - @m1b_lib.MLB	43C3	46B8		VRT29_VIN	VRT29_VIN - @m1b_lib.MLB	7808	XDP_RES_RL	XDP_RES_RL - @m1b_lib.MLB	24C2	81C1																			
	USB_EXT_A_N	USB_EXT_A_N - @m1b_lib.MLB	25A4	43A6	83C3	VR_AXG_BOOT1_RC	VR_AXG_BOOT1_RC - @m1b_lib.MLB	69C6	XDP_SEL	XDP_SEL - @m1b_lib.MLB	24A8	24B1	24B8																		
	USB_EXT_A_OC_L	USB_EXT_A_OC_L - @m1b_lib.MLB	25A4	43C8		VR_AXG_COMP	VR_AXG_COMP - @m1b_lib.MLB	67C5	XDP_TDI	XDP_TDI - @m1b_lib.MLB	24B8	24C4	81D1																		
	USB_EXT_A_P	USB_EXT_A_P - @m1b_lib.MLB	25A4	43A6	83C3	VR_AXG_COMP_RC	VR_AXG_COMP_RC - @m1b_lib.MLB	67C7	XDP_TDO	XDP_TDO - @m1b_lib.MLB	24A8	24C4	81D1																		
	USB_EXTB_N	USB_EXTB_N - @m1b_lib.MLB	25C4	43D5	83B3	VR_AXG_DRV1_BOOT	VR_AXG_DRV1_BOOT - @m1b_lib.MLB	69C7	XDP_TMS	XDP_TMS - @m1b_lib.MLB	24A4	24C4	81D1																		
C	USB_EXTB_OC_L	USB_EXTB_OC_L - @m1b_lib.MLB	25C4	43D8		VR_AXG_DRV1_LGATE	VR_AXG_DRV1_LGATE - @m1b_lib.MLB	69C7	XDP_VR_READY	XDP_VR_READY - @m1b_lib.MLB	24C7	81C1	C																		
	USB_EXTB_P	USB_EXTB_P - @m1b_lib.MLB	25C4	43D5	83B3	VR_AXG_DRV1_PVCC	VR_AXG_DRV1_PVCC - @m1b_lib.MLB	69C7																							
	USB_EXTC_N	USB_EXTC_N - @m1b_lib.MLB	25C4	43C3	83B3	VR_AXG_DRV1_UGATE	VR_AXG_DRV1_UGATE - @m1b_lib.MLB	69C7																							
	USB_EXTC_OC_L	USB_EXTC_OC_L - @m1b_lib.MLB	25C4	43D8		VR_AXG_FB	VR_AXG_FB - @m1b_lib.MLB	67C4																							
	USB_EXTC_P	USB_EXTC_P - @m1b_lib.MLB	25C4	43B3	83B3	VR_AXG_HFREQ_COMP	VR_AXG_HFREQ_COMP - @m1b_lib.MLB	67C5																							
	USB_EXTD_N	USB_EXTD_N - @m1b_lib.MLB	25B4	43B6	83B3	VR_AXG_IMON	VR_AXG_IMON - @m1b_lib.MLB	56B4	67B1																						
	USB_EXTD_OC_L	USB_EXTD_OC_L - @m1b_lib.MLB	25A4	43B8		VR_AXG_ISNS_N	VR_AXG_ISNS_N - @m1b_lib.MLB	67C1	69B2																						
	USB_EXTD_P	USB_EXTD_P - @m1b_lib.MLB	25A4	43B6	83B3	VR_AXG_ISNS_N_R	VR_AXG_ISNS_N_R - @m1b_lib.MLB	69C4																							
	USB_HUB1_CFG_SEL0	USB_HUB1_CFG_SEL0 - @m1b_lib.MLB	25C6			VR_AXG_ISNS_P	VR_AXG_ISNS_P - @m1b_lib.MLB	67C1	69C2																						
	USB_HUB1_CFG_SEL1	USB_HUB1_CFG_SEL1 - @m1b_lib.MLB	25C6			VR_AXG_ISNS_RP	VR_AXG_ISNS_RP - @m1b_lib.MLB	67C3																							
B	USB_HUB1_NONREM0	USB_HUB1_NONREM0 - @m1b_lib.MLB	25C6			VR_AXG_PH1_SNUB	VR_AXG_PH1_SNUB - @m1b_lib.MLB	69C4						B																	
	USB_HUB1_NONREM1	USB_HUB1_NONREM1 - @m1b_lib.MLB	25C6			VR_AXG_PHASE1	VR_AXG_PHASE1 - @m1b_lib.MLB	69C7																							
	USB_HUB1_RBIAS	USB_HUB1_RBIAS - @m1b_lib.MLB	25C5			VR_AXG_PWM	VR_AXG_PWM - @m1b_lib.MLB	67D1	69C8																						
	USB_HUB1_TEST	USB_HUB1_TEST - @m1b_lib.MLB	25C6			VR_AXG_RGND	VR_AXG_RGND - @m1b_lib.MLB	67C5																							
	USB_HUB1_UP_N	USB_HUB1_UP_N - @m1b_lib.MLB	19D3	25C4	83C3	VR_AXG_SW_FREQ	VR_AXG_SW_FREQ - @m1b_lib.MLB	67B3																							
	USB_HUB1_UP_P	USB_HUB1_UP_P - @m1b_lib.MLB	19D3	25C4	83C3	VR_AXG_TCOMP	VR_AXG_TCOMP - @m1b_lib.MLB	67C4																							
	USB_HUB1_VBUS_DET	USB_HUB1_VBUS_DET - @m1b_lib.MLB	25C5			VR_AXG_TM	VR_AXG_TM - @m1b_lib.MLB	67D4																							
	USB_HUB1_XTAL1	USB_HUB1_XTAL1 - @m1b_lib.MLB	25C6			VR_AXG_TMN	VR_AXG_TMN - @m1b_lib.MLB	67C5																							
	USB_HUB1_XTAL2	USB_HUB1_XTAL2 - @m1b_lib.MLB	25C6			VR_AXG_VSEN	VR_AXG_VSEN - @m1b_lib.MLB	67C5																							
	USB_HUB2_CFG_SEL0	USB_HUB2_CFG_SEL0 - @m1b_lib.MLB	25A6			VR_AXG_VSNS_R_N	VR_AXG_VSNS_R_N - @m1b_lib.MLB	67D7																							
A	USB_HUB2_CFG_SEL1	USB_HUB2_CFG_SEL1 - @m1b_lib.MLB	25A6			VR_AXG_VSNS_R_P	VR_AXG_VSNS_R_P - @m1b_lib.MLB	67D7						A																	
	USB_HUB2_NONREM0	USB_HUB2_NONREM0 - @m1b_lib.MLB	25A6			VR_AXG_VSNS_XW_N	VR_AXG_VSNS_XW_N - @m1b_lib.MLB	67D7																							
	USB_HUB2_NONREM1	USB_HUB2_NONREM1 - @m1b_lib.MLB	25A6			VR_AXG_VSNS_XW_P	VR_AXG_VSNS_XW_P - @m1b_lib.MLB	67D8																							
	USB_HUB2_RBIAS	USB_HUB2_RBIAS - @m1b_lib.MLB	25A5			VR_CPU_BOOT1_RC	VR_CPU_BOOT1_RC - @m1b_lib.MLB	68D6																							
	USB_HUB2_TEST	USB_HUB2_TEST - @m1b_lib.MLB	25B6			VR_CPU_BOOT2_RC	VR_CPU_BOOT2_RC - @m1b_lib.MLB	68C6																							
	USB_HUB2_UP_N	USB_HUB2_UP_N - @m1b_lib.MLB	19D3	25A4	83C3	VR_CPU_BOOT3_RC	VR_CPU_BOOT3_RC - @m1b_lib.MLB	68B6																							
	USB_HUB2_UP_P	USB_HUB2_UP_P - @m1b_lib.MLB	19C3	25A4	83C3	VR_CPU_COMP	VR_CPU_COMP - @m1b_lib.MLB	67B5																							
	USB_HUB2_VBUS_DET	USB_HUB2_VBUS_DET - @m1b_lib.MLB	25A5			VR_CPU_DRV1_BOOT	VR_CPU_DRV1_BOOT - @m1b_lib.MLB	68D7																							
	USB_HUB2_XTAL1	USB_HUB2_XTAL1 - @m1b_lib.MLB	25B6			VR_CPU_DRV1_LGATE	VR_CPU_DRV1_LGATE - @m1b_lib.MLB	68D7																							
	USB_HUB2_XTAL2	USB_HUB2_XTAL2 - @m1b_lib.MLB	25A6			VR_CPU_DRV1_PVCC	VR_CPU_DRV1_PVCC - @m1b_lib.MLB	68D7																							
A	USB_HUB_RESET	USB_HUB_RESET - @m1b_lib.MLB	25B2			VR_CPU_DRV1_UGATE	VR_CPU_DRV1_UGATE - @m1b_lib.MLB	68D7						A																	
	USB_HUB_RESET_L	USB_HUB_RESET_L - @m1b_lib.MLB	25B1	25B6	25C6	VR_CPU_DRV2_BOOT	VR_CPU_DRV2_BOOT - @m1b_lib.MLB	68C7																							
	USB_HUB_SOFT_RESET_L	USB_HUB_SOFT_RESET_L - @m1b_lib.MLB	19C1	24D4	25B2	VR_CPU_DRV2_LGATE	VR_CPU_DRV2_LGATE - @m1b_lib.MLB	68B7																							
	USB_IR_N	USB_IR_N - @m1b_lib.MLB	25C4	44C3	83B3	VR_CPU_DRV2_UGATE	VR_CPU_DRV2_UGATE - @m1b_lib.MLB	68C7																							
	USB_IR_P	USB_IR_P - @m1b_lib.MLB	25C4	44C3	83B3	VR_CPU_DRV2_VCC	VR_CPU_DRV2_VCC - @m1b_lib.MLB	68C7																							
	USB_IR_R_N	USB_IR_R_N - @m1b_lib.MLB	44C4	83B3		VR_CPU_DRV3_BOOT	VR_CPU_DRV3_BOOT - @m1b_lib.MLB	68A7																							
	USB_IR_R_P	USB_IR_R_P - @m1b_lib.MLB	44C4	83B3		VR_CPU_DRV3_LGATE	VR_CPU_DRV3_LGATE - @m1b_lib.MLB	68A7																							
	USB_PORT0_N	USB_PORT0_N - @m1b_lib.MLB	43A4	83B3		VR_CPU_DRV3_UGATE	VR_CPU_DRV3_UGATE - @m1b_lib.MLB	68A7																							
	USB_PORT0_P	USB_PORT0_P - @m1b_lib.MLB	43A4	83B3		VR_CPU_DRV3_VCC	VR_CPU_DRV3_VCC - @m1b_lib.MLB	68B7																							
	USB_PORT1_N	USB_PORT1_N - @m1b_lib.MLB	43B5	83B3		VR_CPU_FB	VR_CPU_FB - @m1b_lib.MLB	67B5																							
A	USB_PORT1_P	USB_PORT1_P - @m1b_lib.MLB	43B5	83B3		VR_CPU_FB2	VR_CPU_FB2 - @m1b_lib.MLB	67B8						A																	
	USB_PORT2_N	USB_PORT2_N - @m1b_lib.MLB	43C2	83B3		VR_CPU_FB_RC	VR_CPU_FB_RC - @m1b_lib.MLB	67C8																							
	USB_PORT2_P	USB_PORT2_P - @m1b_lib.MLB	43B2	83B3		VR_CPU_FDVID	VR_CPU_FDVID - @m1b_lib.MLB	67C5																							
	USB_PORT3_N	USB_PORT3_N - @m1b_lib.MLB	43D2	83B3		VR_CPU_HFREQ_COMP	VR_CPU_HFREQ_COMP - @m1b_lib.MLB	67B5																							
	USB_PORT3_P	USB_PORT3_P - @m1b_lib.MLB	43D2	83B3		VR_CPU_IAUTO	VR_CPU_IAUTO - @m1b_lib.MLB	67B5																							
	USB_PWR_EN	USB_PWR_EN - @m1b_lib.MLB	43C8	62D1		VR_CPU_IMON	VR_CPU_IMON - @m1b_lib.MLB	56D4	67B5																						
	VBIAS_DAC	VBIAS_DAC - @m1b_lib.MLB	53D5			VR_CPU_ISNS1_N	VR_CPU_ISNS1_N - @m1b_lib.MLB	67C1	68C2																						
	VCCSA0_AGND	VCCSA0_AGND - @m1b_lib.MLB	64B6			VR_CPU_ISNS1_N_R	VR_CPU_ISNS1_N_R - @m1b_lib.MLB	68D3																							
	VCCSA0_BOOT_RC	VCCSA0_BOOT_RC - @m1b_lib.MLB	64C5			VR_CPU_ISNS1_P	VR_CPU_ISNS1_P - @m1b_lib.MLB	67C1	68D2																						
	VCCSA0_CS_N	VCCSA0_CS_N - @m1b_lib.MLB	56A8	64B2	89D3	VR_CPU_ISNS2_N	VR_CPU_ISNS2_N - @m1b_lib.MLB	67C1	68B2																						
A	VCCSA0_CS_P	VCCSA0_CS_P - @m1b_lib.MLB	56A8	64B2	89D3	VR_CPU_ISNS2_N_R	VR_CPU_ISNS2_N_R - @m1b_lib.MLB	68C3						A																	
	VCCSA0_DRVH	VCCSA0_DRVH - @m1b_lib.MLB	64C5			VR_CPU_ISNS2_P	VR_CPU_ISNS2_P - @m1b_lib.MLB	67C1	68C2																						
	VCCSA0_DRVL	VCCSA0_DRVL - @m1b_lib.MLB	64C5			VR_CPU_ISNS3_N	VR_CPU_ISNS3_N - @m1b_lib.MLB	67B1	68A2																						
	VCCSA0_FB	VCCSA0_FB - @m1b_lib.MLB	64C6			VR_CPU_ISNS3_N_R	VR_CPU_ISNS3_N_R - @m1b_lib.MLB	68A3																							
	VCCSA0_FSEL	VCCSA0_FSEL - @m1b_lib.MLB	64B6			VR_CPU_ISNS3_P	VR_CPU_ISNS3_P - @m1b_lib.MLB	67B1	68A2																						
	VCCSA0_LL	VCCSA0_LL - @m1b_lib.MLB	64C5			VR_CPU_N_PSI	VR_CPU_N_PSI - @m1b_lib.MLB	67C5																							
	VCCSA0_OCSET	VCCSA0_OCSET - @m1b_lib.MLB	64C6			VR_CPU_PH1_SNUB	VR_CPU_PH1_SNUB - @m1b_lib.MLB	68D4																							
	VCCSA0_SET0	VCCSA0_SET0 - @m1b_lib.MLB	64B6			VR_CPU_PH2_SNUB	VR_CPU_PH2_SNUB - @m1b_lib.MLB	68C4																							
	VCCSA0_SET1	VCCSA0_SET1 - @m1b_lib.MLB	64B6			VR_CPU_PH3_SNUB	VR_CPU_PH3_SNUB - @m1b_lib.MLB	68A4																							
	VCCSA0_SNUB	VCCSA0_SNUB - @m1b_lib.MLB	64C3			VR_CPU_PHASE1	VR_CPU_PHASE1 - @m1b_lib.MLB	68D7																							
A	VCCSA0_SREF	VCCSA0_SREF - @m1b_lib.MLB	64C6			VR_CPU_PHASE2	VR_CPU_PHASE2 - @m1b_lib.MLB	68C7						A																	
	VCCSA0_VBST	VCCSA0_VBST - @m1b_lib.MLB	64C5			VR_CPU_PHASE3	VR_CPU_PHASE3 - @m1b_lib.MLB	68A7																							
	VCCSA0_VO	VCCSA0_VO - @m1b_lib.MLB	64C6			VR_CPU_PSIComp	VR_CPU_PSIComp - @m1b_lib.MLB	67B5																							
	VCCSAVSENSE_IN	VCCSAVSENSE_IN - @m1b_lib.MLB	56A7			VR_CPU_PSIComp1	VR_CPU_PSIComp1 - @m1b_lib.MLB	67C8																							
	VCCSA_IOUT	VCCSA_IOUT - @m1b_lib.MLB	56A7			VR_CPU_PWM1	VR_CPU_PWM1 - @m1b_lib.MLB	67C1	68D8																						
	VDDQ_IOUT	VDDQ_IOUT - @m1b_lib.MLB	56D6			VR_CPU_PWM2	VR_CPU_PWM2 - @m1b_lib.MLB	67C1	68C8																						
	VDD_IVSS0	VDD_IVSS0 - @m1b_lib.MLB	71B7			VR_CPU_PWM3	VR_CPU_PWM3 - @m1b_lib.MLB	67B1	68A8																						
	VMON_3V3_DIV	VMON_3V3_DIV - @m1b_lib.MLB	62C8			VR_CPU_PWM4	VR_CPU_PWM4 - @m1b_lib.MLB	67C3																							
	VMON_Q2_BASE	VMON_Q2_BASE - @m1b_lib.MLB	62C7			VR_CPU_RAMP_ADJ	VR_CPU_RAMP_ADJ - @m1b_lib.MLB	67B3																							
	VMON_Q3_BASE	VMON_Q3_BASE - @m1b_lib.MLB	62C7			VR_CPU_RGND	VR_CPU_RGND - @m1b_lib.MLB	67A7																							
A	VMON_Q4_BASE	VMON_Q4_BASE - @m1b_lib.MLB	62B7			VR_CPU_SUTH	VR_CPU_SUTH - @m1b_lib.MLB	67C5						A																	
	VR3V3_G3H_BST	VR3V3_G3H_BST - @m1b_lib.MLB	65D5			VR_CPU_SW_FREQ	VR_CPU_SW_FREQ - @m1b_lib.MLB	67B3																							
	VR3V3_G3H_EN	VR3V3_G3H_EN - @m1b_lib.MLB	65D6			VR_CPU_TM	VR_CPU_TM - @m1b_lib.MLB	67B5																							
	VR3V3_G3H_ENL	VR3V3_G3H_ENL - @m1b_lib.MLB	65C5			VR_CPU_TMN	VR_CPU_TMN - @m1b_lib.MLB	67A7																							
	VR3V3_G3H_FB	VR3V3_G3H_FB - @m1b_lib.MLB	65C6			VR_CPU_VSEN	VR_CPU_VSEN - @m1b_lib.MLB	67A6																							
	VR3V3_G3H_ILIM	VR3V3_G3H_ILIM - @m1b_lib.MLB	65C5			VR_CPU_VSNS_R_N	VR_CPU_VSNS_R_N - @m1b_lib.MLB	67A7																							
	VR3V3_G3H_LXS	VR3V3_G3H_LXS - @m1b_lib.MLB	65C4			VR_CPU_VSNS_R_P	VR_CPU_VSNS_R_P - @m1b_lib.MLB	67A7																							
	VR3V3_G3H_SW	VR3V3_G3H_SW - @m1b_lib.MLB	65C5			VR_CPU_VSNS_XW_N	VR_CPU_VSNS_XW_N - @m1b_lib.MLB	67A8																							
	VR3V3_G3H_TON	VR3V3_G3H_TON - @m1b_lib.MLB	65C6			VR_CPU_VSNS_XW_P	VR_CPU_VSNS_XW_P - @m1b_lib.MLB	67A8																							
	VR3V3_G3H_V5V	VR3V3_G3H_V5V - @m1b_lib.MLB	65D6			VR_EN_PWR_OVP	VR_EN_PWR_OVP - @m1b_lib.MLB	67B3																							
A	VR3V3_G3H_VSEN	VR3V3_G3H_VSEN - @m1b_lib.MLB	65C5			VR_HOT_L	VR_HOT_L - @m1b_lib.MLB	67B5						A																	
	VR5V_G3H_SW	VR5V_G3H_SW - @m1b_lib.MLB	65B5			VR_RSET	VR_RSET - @m1b_lib.MLB	67B4																							
	VR5V_G3H_VCC	VR5V_G3H_VCC - @m1b_lib.MLB	65B5			VR_SEN_R2	VR_SEN_R2 - @m1b_lib.MLB	67D7																							
	VR5V_S4_BOOT	VR5V_S4_BOOT - @m1b_lib.MLB	65B5			VTTCLAMP_EN	VTTCLAMP_EN - @m1b_lib.MLB	28B3																							
	VR5V_S4_COMP	VR5V_S4_COMP - @m1b_lib.MLB	65A5			VTTCLAMP_L	VTTCLAMP_L - @m1b_lib.MLB	28B2																							
	VR5V_S4_COMP_RC	VR5V_S4_COMP_RC - @m1b_lib.MLB	65A5			WIFI_EVENT_L	WIFI_EVENT_L - @m1b_lib.MLB	46B8	47A2																						
	VR5V_S4_EN_IC	VR5V_S4_EN_IC - @m1b_lib.MLB	65B6			WIFI_PROX_THRM_N	WIFI_PROX_THRM_N - @m1b_lib.MLB	52C3	52D8																						
	VR5V_S4_FB	VR5V_S4_FB - @m1b_lib.MLB	65A5			WIFI_PROX_THRM_P	WIFI_PROX_THRM_P - @m1b_lib.MLB	52C3	52D8																						
	VR5V_S4_FB_RC	VR5V_S4_FB_RC - @m1b_lib.MLB	65A4			WLAN_THROTTLE_L	WLAN_THROTTLE_L - @m1b_lib.MLB	35D7	91D7																						
	VR5V_S4_FB_SNS	VR5V_S4_FB_SNS - @m1b_lib.MLB	65A3			WOL_EN	WOL_EN - @m1b_lib.MLB	20B8	26C6	62A4																					
A	VR5V_S4_OCSET	VR5V_S4_OCSET - @m1b_lib.MLB	65A5			XDP_BPM_L<0>	XDP_BPM_L<0> - @m1b_lib.MLB	11B2	24D7					A																	
	VR5V_S4_RT	VR5V_S4_RT - @m1b_lib.MLB	65A6			XDP_BPM_L<3..0>	XDP_BPM_L<3..0> - @m1b_lib.MLB	81D1																							
	VR5V_S4_SS	VR5V_S4_SS - @m1b_lib.MLB	65A6			XDP_BPM_L<1>	XDP_BPM_L<1> - @m1b_lib.MLB	11B2	24D7																						
	VREFMRGN_CA_SODIMMA</																														

D	Title: Cref Part Report Design: mlb Date: Apr 8 14:52:14 2011			C1706 CAP_402-1 mlb[1605] C1707 CAP_402-1 mlb[1605] C1708 CAP_402-1 mlb[1605] C1709 CAP_402-1 mlb[1604] C1710 CAP_402-1 mlb[1604] C1711 CAP_402-1 mlb[1604] C1712 CAP_402-1 mlb[1603] C1713 CAP_402-1 mlb[1603] C1714 CAP_402-1 mlb[1603] C1715 CAP_402-1 mlb[1602] C1716 CAP_402-1 mlb[1602] C1717 CAP_402-1 mlb[1602] C1718 CAP_0402-1 mlb[1607] C1719 CAP_0402-1 mlb[1607] C1720 CAP_0402-1 mlb[1606] C1721 CAP_0402-1 mlb[1606] C1722 CAP_0402-1 mlb[1606] C1723 CAP_0402-1 mlb[1605] C1724 CAP_0402-1 mlb[1605] C1725 CAP_0402-1 mlb[1605] C1726 CAP_0603 mlb[16C7] C1727 CAP_0603 mlb[16C7] C1728 CAP_0603 mlb[16C6] C1729 CAP_0603 mlb[16C6] C1730 CAP_0603 mlb[16C6] C1731 CAP_0603 mlb[16C5] C1732 CAP_0603 mlb[16C5] C1733 CAP_0603 mlb[16C5] C1738 CAP_402-1 mlb[16B7] C1739 CAP_402-1 mlb[16B7] C1740 CAP_402-1 mlb[16B6] C1741 CAP_402-1 mlb[16B6] C1742 CAP_402-1 mlb[16B6] C1743 CAP_402-1 mlb[16B6] C1744 CAP_402-1 mlb[16B5] C1745 CAP_402-1 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CAP_402 mlb[23B7] C2439 CAP_402-1 mlb[23B7] C2440 CAP_402 mlb[23D3] C2441 CAP_402 mlb[23D3] C2442 CAP_402-1 mlb[23C3] C2444 CAP_402-1 mlb[23D2] C2446 CAP_402-1 mlb[23D1] C2450 CAP_603 mlb[23C7] C2451 CAP_402 mlb[23C7] C2452 CAP_402-1 mlb[23D2] C2453 CAP_603 mlb[23C4] C2454 CAP_402-1 mlb[23C3] C2455 CAP_402 mlb[23C7] C2456 CAP_402-1 mlb[23B2] C2460 CAP_603 mlb[23C1] C2463 CAP_402-1 mlb[23B1] C2469 CAP_402-1 mlb[23C1] C2475 CAP_402-1 mlb[23D2] C2476 CAP_402-1 mlb[23D1] C2481 CAP_402-1 mlb[23C2] C2482			

D	C4460	CAP_0201	m1b[42C3]	C6503	CAP_402	m1b[60C4]	C7523	CAP_402	m1b[68C5]	C9109	CAP_402-1	m1b[75D5]	D
	C4461	CAP_0201	m1b[42B3]	C6505	CAP_603	m1b[60C4]	C7527	CAP_P_SM	m1b[68B3]	C9110	CAP_402-1	m1b[75D7]	
	C4462	CAP_0201	m1b[42B3]	C6506	CAP_603	m1b[60C3]	C7528	CAP_402	m1b[68B4]	C9111	CAP_402-1	m1b[75D6]	
	C4463	CAP_0201	m1b[42B3]	C6507	CAP_402	m1b[60C3]	C7529	CAP_P_SM	m1b[68B2]	C9112	CAP_402-1	m1b[75D6]	
	C4600	CAP_P_CASE-B2-SM	m1b[43B7]	C6520	CAP_402	m1b[60C4]	C7530	CAP_402	m1b[68C4]	C9113	CAP_402-1	m1b[75D6]	
	C4601	CAP_402	m1b[43B8]	C6700	CAP_402	m1b[61D2]	C7531	CAP_402	m1b[68C4]	C9114	CAP_402-1	m1b[75D5]	
	C4603	CAP_402	m1b[43D6]	C6701	CAP_402	m1b[61B7]	C7540	CAP_402	m1b[68A3]	C9120	CAP_402-1	m1b[75C6]	
	C4604	CAP_402	m1b[43D6]	C6702	CAP_603	m1b[61B7]	C7541	CAP_402	m1b[68B6]	C9121	CAP_402-1	m1b[75C6]	
	C4605	CAP_402	m1b[43D2]	C6709	CAP_402	m1b[61D3]	C7542	CAP_P_SM	m1b[68B1]	C9122	CAP_402-1	m1b[75C5]	
	C4606	CAP_P_CASE-B2-SM	m1b[43D8]	C6710	CAP_402	m1b[61D6]	C7543	CAP_402	m1b[68B5]	C9130	CAP_402-LF	m1b[75C5]	
C	C4612	CAP_402	m1b[43D6]	C6711	CAP_402	m1b[61D6]	C7546	CAP_P_SM	m1b[68B2]	C9143	CAP_402-1	m1b[75D3]	C
	C4615	CAP_402	m1b[43C2]	C6730	CAP_402	m1b[61C2]	C7547	CAP_P_SM	m1b[68B2]	C9144	CAP_402-1	m1b[75D4]	
	C4623	CAP_402	m1b[43B6]	C6731	CAP_402	m1b[61C3]	C7548	CAP_402	m1b[68A4]	C9145	CAP_402-1	m1b[75D3]	
	C4625	CAP_402	m1b[43B4]	C6740	CAP_402	m1b[61B2]	C7549	CAP_P_SM	m1b[68B2]	C9146	CAP_603	m1b[75D3]	
	C4632	CAP_402	m1b[43B6]	C6741	CAP_402	m1b[61C3]	C7550	CAP_402	m1b[68B4]	C9147	CAP_603	m1b[75D3]	
	C4635	CAP_402	m1b[43A4]	C6760	CAP_402	m1b[61B2]	C7551	CAP_402	m1b[68B4]	C9150	CAP_402-1	m1b[75C3]	
	C4650	CAP_402	m1b[43D5]	C6761	CAP_402	m1b[61B3]	C7560	CAP_P_CASE-B4-SM	m1b[68A7]	C9151	CAP_402-1	m1b[75C3]	
	C4790	CAP_402-1	m1b[44C4]	C6810	CAP_402	m1b[62D2]	C7561	CAP_P_CASE-B4-SM	m1b[68A7]	C9152	CAP_402-1	m1b[75C3]	
	C4791	CAP_402	m1b[44C4]	C6812	CAP_402	m1b[62D3]	C7562	CAP_P_CASE-B4-SM	m1b[68A7]	C9153	CAP_402-1	m1b[75C4]	
	C4792	CAP_402	m1b[44C5]	C6821	CAP_402	m1b[62B2]	C7563	CAP_P_CASE-B4-SM	m1b[68A6]	C9160	CAP_402-LF	m1b[75C4]	
B	C4793	CAP_402-1	m1b[44B4]	C6822	CAP_402	m1b[62B1]	C7564	CAP_P_CASE-B4-SM	m1b[68A6]	C9170	CAP_402-LF	m1b[75C4]	B
	C4794	CAP_402	m1b[44B5]	C6842	CAP_402	m1b[62D7]	C7565	CAP_P_CASE-B4-SM	m1b[68A5]	C9200	CAP_402	m1b[76C7]	
	C4800	CAP_402	m1b[45C3]	C6860	CAP_402	m1b[62A7]	C7566	CAP_P_CASE-B4-SM	m1b[68A5]	C9210	CAP_402-1	m1b[76B7]	
	C4801	CAP_402	m1b[45D6]	C6881	CAP_402	m1b[62C3]	C7580	CAP_402	m1b[68D5]	C9215	CAP_402-1	m1b[76A7]	
	C4802	CAP_402	m1b[45A3]	C6886	CAP_402	m1b[62C2]	C7581	CAP_603	m1b[68D5]	C9221	CAP_0201	m1b[76B2]	
	C4803	CAP_402	m1b[45C2]	C6887	CAP_402	m1b[62C3]	C7582	CAP_603	m1b[68C4]	C9222	CAP_0201	m1b[76B2]	
	C4806	CAP_402	m1b[45C6]	C6888	CAP_402	m1b[62C2]	C7583	CAP_603	m1b[68B4]	C9223	CAP_0201	m1b[76B2]	
	C4809	CAP_402	m1b[45C4]	C7101	CAP_603	m1b[64C5]	C7601	CAP_402	m1b[69C7]	C9224	CAP_0201	m1b[76B2]	
	C4902	CAP_895	m1b[46D4]	C7102	CAP_402	m1b[64B6]	C7602	CAP_402	m1b[69C3]	C9225	CAP_0201	m1b[76A4]	
	C4903	CAP_402	m1b[46D4]	C7103	CAP_402	m1b[64B6]	C7603	CAP_402	m1b[69C6]	C9226	CAP_0201	m1b[76A2]	
A	C4904	CAP_402	m1b[46D3]	C7105	CAP_402	m1b[64B7]	C7607	CAP_402	m1b[69D4]	C9227	CAP_0201	m1b[76A2]	A
	C4905	CAP_402	m1b[46D3]	C7119	CAP_402	m1b[64B3]	C7608	CAP_402	m1b[69C4]	C9228	CAP_0201	m1b[76B4]	
	C4906	CAP_402	m1b[46D3]	C7120	CAP_P_SM	m1b[64C3]	C7610	CAP_402	m1b[69D4]	C9230	CAP_0201	m1b[76A4]	
	C4907	CAP_402	m1b[46D2]	C7121	CAP_402	m1b[64C3]	C7615	CAP_603	m1b[69D4]	C9231	CAP_0201	m1b[76B4]	
	C4910	CAP_402	m1b[46A5]	C7122	CAP_402	m1b[64C3]	C7620	CAP_P_CASE-B4-SM	m1b[69C3]	C9300	CAP_402	m1b[77D7]	
	C4920	CAP_402	m1b[46D3]	C7123	CAP_402	m1b[64B2]	C7621	CAP_P_CASE-B4-SM	m1b[69C2]	C9301	CAP_402	m1b[77D7]	
	C5001	CAP_402	m1b[47D8]	C7130	CAP_402	m1b[64C4]	C7622	CAP_P_CASE-B4-SM	m1b[69C2]	C9302	CAP_402	m1b[77D7]	
	C5010	CAP_402	m1b[47B7]	C7140	CAP_402	m1b[64B3]	C7623	CAP_402	m1b[69C1]	C9303	CAP_402	m1b[77D7]	
	C5011	CAP_402	m1b[47B7]	C7201	CAP_402	m1b[65D5]	C7633	CAP_402	m1b[69D3]	C9304	CAP_402	m1b[77D7]	
	C5020	CAP_402	m1b[47D7]	C7202	CAP_402	m1b[65D6]	C7634	CAP_603	m1b[69D3]	C9305	CAP_402	m1b[77D7]	

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D	R2069	RES_201	m1b[19C2]	R2070	RES_201	m1b[19B4]	R2111	RES_201	m1b[2008]	R2112	RES_201	m1b[2007]	R2113	RES_201	m1b[2007]	R2114	RES_201	m1b[20A7]	R2115	RES_201	m1b[20A7]	R2116	RES_201	m1b[20A3]	R2117	RES_201	m1b[20A6]	R2118	RES_201	m1b[20A2]	R2130	RES_201	m1b[20A3]	R2140	RES_201	m1b[2003]	R2150	RES_201	m1b[2003]	R2155	RES_201	m1b[2003]	R2156	RES_201	m1b[20C3]	R2160	RES_201	m1b[20B8]	R2170	RES_201	m1b[20B4]	R2172	RES_201	m1b[20A7]	R2173	RES_201	m1b[20A7]	R2174	RES_201	m1b[20A7]	R2175	RES_201	m1b[20A8]	R2184	RES_201	m1b[20B7]	R2185	RES_201	m1b[20B7]	R2186	RES_201	m1b[20B7]	R2190	RES_201	m1b[2006]	R2191	RES_201	m1b[20A6]	R2192	RES_201	m1b[20C7]	R2193	RES_201	m1b[20C8]	R2194	RES_201	m1b[20C7]	R2196	RES_201	m1b[20A5]	R2197	RES_201	m1b[20A4]	R2198	RES_201	m1b[20A3]	R2199	RES_201	m1b[20A3]	R2404	RES_402	m1b[23B7]	R2405	RES_402	m1b[23C7]	R2415	RES_402	m1b[23D7]	R2450	RES_201	m1b[23C7]	R2451	RES_402	m1b[23C5]	R2490	RES_402	m1b[23B5]	R2491	RES_402	m1b[23B5]	R2500	RES_402	m1b[24C8]	R2501	RES_402	m1b[24C8]	R2502	RES_402	m1b[24C7]	R2504	RES_402	m1b[24C4]	R2505	RES_402	m1b[24C4]	R2510	RES_402	m1b[24D2]	R2511	RES_402	m1b[24D1]	R2512	RES_402	m1b[24D1]	R2513	RES_402	m1b[24D2]	R2514	RES_402	m1b[24D1]	R2515	RES_402	m1b[24C4]	R2516	RES_402	m1b[24C4]	R2520	RES_402	m1b[24B7]	R2521	RES_402	m1b[24B7]	R2522	RES_402	m1b[24A7]	R2523	RES_402	m1b[24A7]	R2524	RES_402	m1b[24A5]	R2526	RES_402	m1b[24A6]	R2530	RES_402	m1b[24C2]	R2531	RES_402	m1b[24B2]	R2532	RES_402	m1b[24B1]	R2540	RES_402	m1b[24D6]	R2550	RES_402	m1b[24A3]	R2551	RES_402	m1b[24A3]	R2552	RES_402	m1b[24A2]	R2553	RES_402	m1b[24B3]	R2556	RES_402	m1b[24A2]	R2557	RES_402	m1b[24B3]	R2600	RES_402	m1b[25C4]	R2601	RES_402	m1b[25C8]	R2602	RES_402	m1b[25C8]	R2603	RES_402	m1b[25C8]	R2604	RES_402	m1b[25C9]	R2605	RES_402	m1b[25C7]	R2606	RES_402	m1b[25C7]	R2607	RES_402	m1b[25C6]	R2620	RES_402	m1b[25C4]	R2621	RES_402	m1b[25C3]	R2622	RES_402	m1b[25C3]	R2630	RES_402	m1b[25C7]	R2640	RES_402	m1b[25C2]	R2641	RES_402	m1b[25B1]	R2642	RES_402	m1b[25B3]	R2650	RES_402	m1b[25A4]	R2651	RES_402	m1b[25B8]	R2652	RES_402	m1b[25A8]	R2653	RES_402	m1b[25B8]	R2654	RES_402	m1b[25A8]	R2655	RES_402	m1b[25B7]	R2656	RES_402	m1b[25A7]	R2657	RES_402	m1b[25A6]	R2670	RES_402	m1b[25A4]	R2671	RES_402	m1b[25A3]	R2672	RES_402	m1b[25A3]	R2680	RES_402	m1b[25B7]	R2705	RES_402	m1b[26C6]	R2706	RES_402	m1b[26C6]	R2740	RES_402	m1b[26D3]	R2741	RES_402	m1b[26D2]	R2743	RES_402	m1b[26C3]	R2745	RES_402	m1b[26C3]	R2800	RES_402	m1b[27D7]	R2810	RES_402	m1b[27C7]	R2826	RES_402	m1b[27B2]	R2827	RES_402	m1b[27B2]	R2829	RES_402	m1b[27B2]	R2840	RES_402	m1b[27C3]	R2850	RES_402	m1b[27A7]	R2860	RES_402	m1b[27A6]	R2861	RES_402	m1b[27A5]	R2862	RES_402	m1b[27A5]	R2863	RES_402	m1b[27A6]	R2871	RES_402	m1b[27C2]	R2875	RES_402	m1b[27D2]	R2876	RES_402	m1b[27D2]	R2877	RES_402	m1b[27D3]	R2879	RES_402	m1b[27D3]	R2880	RES_402	m1b[27B3]	R2881	RES_402	m1b[27C2]	R2883	RES_402	m1b[27C2]	R2888	RES_402	m1b[27D2]	R2895	RES_402	m1b[27A3]	R2896	RES_402	m1b[27A4]			
	R2897	RES_402	m1b[27A3]	R2901	RES_402	m1b[28C6]	R2902	RES_402	m1b[28B6]	R2905	RES_402	m1b[28D6]	R2910	RES_402	m1b[28C6]	R2915	RES_402	m1b[28B7]	R2916	RES_402	m1b[28B6]	R2917	RES_402	m1b[28A6]	R2920	RES_402	m1b[28C4]	R2921	RES_402	m1b[28C4]	R2922	RES_402	m1b[28D3]	R2923	RES_402	m1b[28C4]	R2950	RES_603	m1b[28B2]	R2951	RES_402	m1b[28B3]	R3140	RES_402	m1b[30B2]	R3141	RES_402	m1b[30B2]	R3240	RES_402	m1b[31B2]	R3241	RES_402	m1b[31B2]	R3401	RES_402	m1b[33D4]	R3402	RES_402	m1b[33C4]	R3403	RES_402	m1b[33D2]	R3404	RES_402	m1b[33D2]	R3405	RES_402	m1b[33D2]	R3406	RES_402	m1b[33C2]	R3407	RES_402	m1b[33B4]	R3408	RES_402	m1b[33B4]	R3409	RES_402	m1b[33C2]	R3410	RES_402	m1b[33C2]	R3411	RES_402	m1b[33C2]	R3412	RES_402	m1b[33B2]	R3413	RES_402	m1b[33B4]	R3414	RES_402	m1b[33B2]	R3415	RES_402	m1b[33A4]	R3418	RES_402	m1b[33D7]	R3419	RES_402	m1b[33C7]	R3502	RES_402	m1b[34D4]	R3505	RES_402	m1b[34D2]	R3510	RES_402	m1b[34B7]	R3511	RES_402	m1b[34B7]	R3512	RES_402	m1b[34B2]	R3514	RES_402	m1b[34B3]	R3530	RES_402	m1b[34B6]	R3531	RES_402	m1b[34B3]	R3535	RES_402	m1b[34A3]	R3555	RES_402	m1b[34C4]	R3560	RES_402	m1b[34C4]	R3561	RES_402	m1b[34C4]	R3562	RES_402	m1b[34C4]	R3563	RES_402	m1b[34C4]	R3564	RES_402	m1b[34C4]	R3565	RES_402	m1b[34C4]	R3566	RES_402	m1b[34C4]	R3567	RES_402	m1b[34C4]	R3601	RES_402	m1b[35D7]	R3640	RES_402	m1b[35B4]	R3650	RES_402	m1b[35C2]	R3651	RES_402	m1b[35C2]	R3653	RES_402	m1b[35B4]	R3654	RES_402	m1b[35A4]	R3800	RES_402	m1b[36B4]	R3801	RES_402	m1b[36B4]	R3802	RES_402	m1b[36B4]	R3803	RES_402	m1b[36B3]	R3910	RES_402	m1b[37C5]	R3930	RES_402	m1b[37B2]	R3940	RES_402	m1b[37C6]	R3941	RES_402	m1b[37C6]	R3942	RES_402	m1b[37C6]	R3943	RES_402	m1b[37B7]	R3950	RES_402	m1b[37B2]	R3951	RES_402	m1b[37B2]	R3952	RES_402	m1b[37B2]	R3953	RES_402	m1b[37B2]	R3965	RES_402	m1b[37A6]	R3971	RES_402	m1b[37B2]	R3974	RES_402	m1b[37B2]	R3975	RES_402	m1b[37B2]	R3976	RES_402	m1b[37B2]	R3977	RES_402	m1b[37B2]	R3978	RES_402	m1b[37B2]	R3980	RES_402	m1b[37B2]	R3990	RES_402	m1b[37A7]	R3997	RES_402	m1b[37B2]	R4010	RES_402	m1b[38B4]	R4011	RES_402	m1b[38A4]	R4012	RES_402	m1b[38A4]	R4019	RES_402	m1b[38B4]	R4050	RES_402	m1b[38C2]	R4110	RES_603	m1b[39D7]	R4117	RES_603	m1b[39D7]	R4119	RES_603	m1b[39D8]	R4125	RES_603	m1b[39D2]	R4129	RES_402	m1b[39D2]	R4135	RES_603	m1b[39D1]	R4140	RES_402	m1b[39C6]	R4141	RES_402	m1b[39C7]	R4150	RES_402	m1b[39B7]	R4151	RES_402	m1b[39B7]	R4152	RES_402	m1b[39B7]	R4153	RES_402	m1b[39B7]	R4160	RES_402	m1b[39B7]	R4162	RES_402	m1b[39A7]	R4170	RES_402	m1b[39C2]	R4171	RES_402	m1b[39C2]	R4175	RES_402	m1b[39B2]	R4180	RES_402	m1b[39A7]	R4181	RES_402	m1b[39A8]	R4182	RES_402	m1b[39A8]	R4183	RES_402	m1b[39A7]	R4184	RES_402	m1b[39A7]	R4185	RES_402	m1b[39B2]	R4186	RES_402	m1b[39B2]	R4190	RES_402	m1b[39C1]	R4191	RES_402	m1b[39C2]	R4250	RES_402	m1b[40C3]	R4251	RES_402	m1b[40C3]	R4252	RES_402	m1b[40C3]	R4253	RES_402	m1b[40C3]	R4254	RES_402	m1b[40B3]	R4255	RES_402	m1b[40B7]	R4256	RES_402	m1b[40B6]	R4257	RES_402	m1b[40A7]	R4258	RES_402	m1b[40A7]	R4300	RES_2512	m1b[41D7]
	R4310	RES_402	m1b[41C2]	R4311	RES_402	m1b[41B2]	R4312	RES_402	m1b[41C2]	R4313	RES_402	m1b[41B2]	R4320	RES_402	m1b[41D7]	R4321	RES_402	m1b[41D7]	R4322	RES_402	m1b[41C7]	R4323	RES_402	m1b[41D7]	R4324	RES_402	m1b[41C5]	R4325	RES_402	m1b[41C6]	R4326	RES_402	m1b[41C6]	R4327	RES_402	m1b[41B5]	R4328	RES_402	m1b[41B5]	R4329	RES_402	m1b[41B6]	R4335	RES_402	m1b[41B2]	R4340	RES_402	m1b[41A8]	R4341	RES_402	m1b[41A7]	R4342	RES_402	m1b[41A8]	R4343	RES_402	m1b[41A7]	R4350	RES_402	m1b[41A5]	R4351	RES_402	m1b[41A5]	R4352	RES_402	m1b[41A5]	R4353	RES_402	m1b[41A4]	R4354	RES_402	m1b[41A3]	R4390	RES_402	m1b[41B3]	R4400	RES_201	m1b[42C7]	R4451	RES_201	m1b[42B7]	R4452	RES_201	m1b[42B7]	R4453	RES_201	m1b[42A7]	R4460	RES_201	m1b[42C3]	R4461	RES_201	m1b[42B3]	R4462	RES_201	m1b[42B3]	R4463	RES_201	m1b[42A3]	R4600	RES_402	m1b[43B6]	R4601	RES_402	m1b[43D6]	R4650	RES_402	m1b[43D4]	R4651	RES_402	m1b[43C4]	R4652	RES_402	m1b[43C4]	R4700	RES_402	m1b[44D5]	R4790	RES_402	m1b[44C7]	R4794	RES_402	m1b[44B7]	R4798	RES_402	m1b[44C3]	R4799	RES_402	m1b[44C3]	R4800	RES_402	m1b[44C5]	R4801	RES_402	m1b[45D5]	R4802	RES_402	m1b[45D5]	R4803	RES_402	m1b[45C6]	R4805	RES_402	m1b[45D3]	R4808	RES_402	m1b[45D3]	R4820	RES_402	m1b[45B2]	R4821	RES_805	m1b[45A3]	R4822	RES_805	m1b[45A3]	R4823	RES_402	m1b[45A2]	R4901	RES_201	m1b[46D1]	R4902																																																																																																																																																																																																																	

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ZH0933	HOLE_VIA	m1b[9A8]
ZH0934	HOLE_VIA	m1b[9A8]
ZH0935	HOLE_VIA	m1b[9B8]
ZH0936	HOLE_VIA	m1b[9A8]
ZH0937	HOLE_VIA	m1b[9A8]
ZH0938	HOLE_VIA	m1b[9A8]
ZH0940	HOLE_VIA	m1b[9B7]
ZH0941	HOLE_VIA	m1b[9B7]
ZH0942	HOLE_VIA	m1b[9A7]
ZH0943	HOLE_VIA	m1b[9A7]
ZH0944	HOLE_VIA	m1b[9A7]
ZH0945	HOLE_VIA	m1b[9B7]
ZH0946	HOLE_VIA	m1b[9A7]
ZH0947	HOLE_VIA	m1b[9A7]
ZH0948	HOLE_VIA	m1b[9A7]
ZH0949	HOLE_VIA	m1b[9A7]
ZH0950	HOLE_VIA	m1b[9B7]
ZH0951	HOLE_VIA	m1b[9B7]
ZH0952	HOLE_VIA	m1b[9A7]
ZH0953	HOLE_VIA	m1b[9A7]
ZH0954	HOLE_VIA	m1b[9A7]
ZH0955	HOLE_VIA	m1b[9B6]
ZH0956	HOLE_VIA	m1b[9B6]
ZH0957	HOLE_VIA	m1b[9A6]
ZH0958	HOLE_VIA	m1b[9A6]
ZP0900	POGO_PIN_1P_SM	m1b[9D4]
ZP0901	POGO_PIN_1P_SM	m1b[9D4]
ZP0902	POGO_PIN_1P_SM	m1b[9D3]
ZP0903	POGO_PIN_1P_SM	m1b[9D3]
ZP0907	POGO_PIN_1P_SM	m1b[9D1]
ZP0920	POGO_PIN_1P_SM	m1b[9C3]
ZP0921	POGO_PIN_1P_SM	m1b[9C3]
ZP0922	POGO_PIN_1P_SM	m1b[9C2]
ZT0840	PCB_STANDOFF	m1b[8C1]
ZT0841	PCB_STANDOFF	m1b[8C1]
ZT0910	BOSS	m1b[9C8]
ZT0911	BOSS	m1b[9B8]
ZT0912	BOSS	m1b[9C7]
ZT0913	BOSS	m1b[9B7]
ZT0914	BOSS	m1b[9B8]
ZT0915	BOSS	m1b[9B7]
ZT0940	PCB_STANDOFF	m1b[9C8]
ZT0941	PCB_STANDOFF	m1b[9C6]
ZT0960	PCB_STANDOFF	m1b[9D8]
ZT0961	PCB_STANDOFF	m1b[9D6]
ZT0962	PCB_STANDOFF	m1b[9D8]
ZT0966	PCB_STANDOFF	m1b[9D7]
ZT0970	PCB_STANDOFF	m1b[9C3]

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